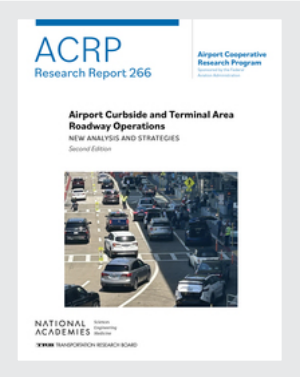
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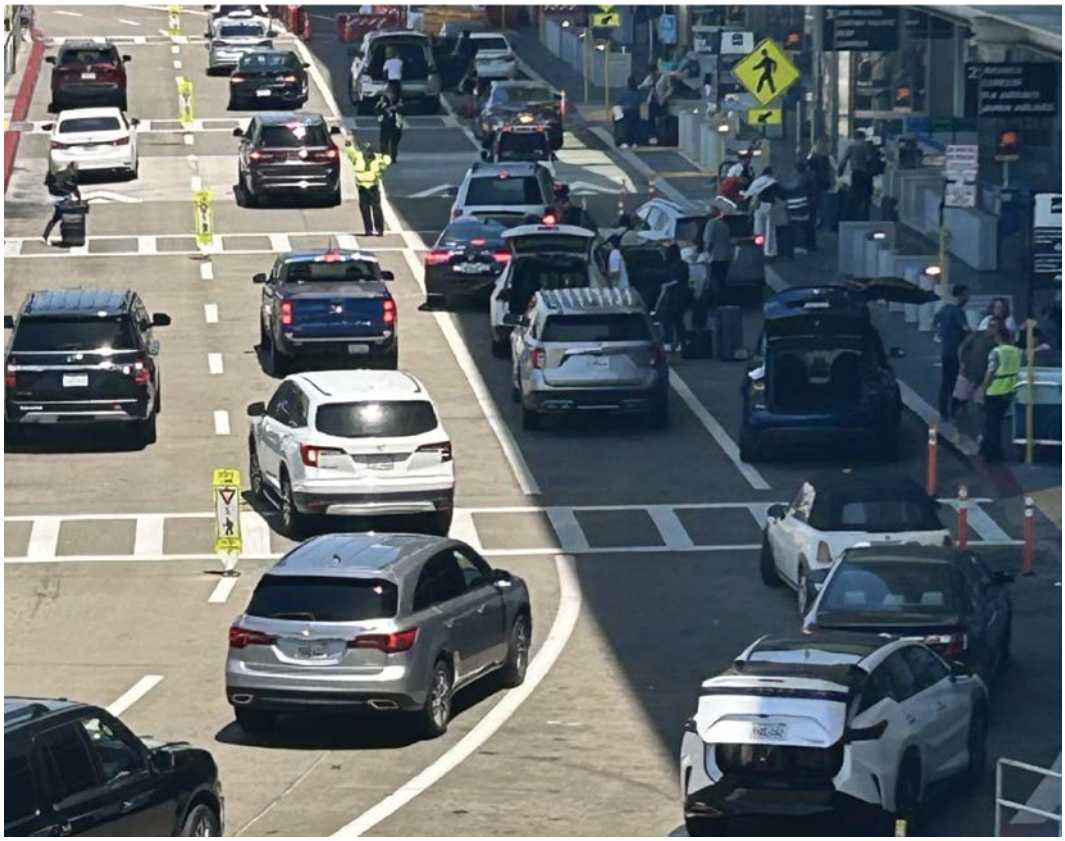
ACRP
Research Report 266

Airport Cooperative Research Program
Sponsored by the Federal Aviation Administration

Airport Curbside and Terminal Area Roadway Operations

NEW ANALYSIS AND STRATEGIES

Second Edition



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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 266

Airport Curbside and Terminal Area Roadway Operations

NEW ANALYSIS AND STRATEGIES

Second Edition

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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Cover photo shows curbside congestion and enforcement, San Francisco International Airport, San Francisco, California.

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The other authors were Ernest Choi of InterVISTAS Consulting, who conducted the traffic simulations serving as inputs to the update of the Quick Analysis Tool for Airport Roadways (QATAR) and assisted with the report preparation; Kush Bhagat, Lake Trask, Paul Ryus, and Richard Dowling of Kittelson & Associates, Inc., who led the update and refinement of QATAR; Jacob Sotsky, Kirkland Ellens, Kurt Schwager, and Andrew Blaisdell of Jacobsen|Daniels, who conducted the literature review and assisted with documentation of innovative strategies; and Jeff Cushman of ABM Aviation, who reviewed the innovative strategies. Steven Derengowski and Sonya Sula at InterVISTAS Consulting assisted with the editing and production of the report.

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FOREWORD

By Matthew J. Griffin
Staff Officer
Transportation Research Board

ACRP Research Report 266: Airport Curbside and Terminal Area Roadway Operations: New Analysis and Strategies, Second Edition builds on research published in *ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations* by providing airport operators with new tactics for managing their curbsides. Landside managers and planners at airports of all sizes, and the consultants that support them, will be able to leverage this resource, which includes a number of strategies, and the updated Quick Analysis Tool for Airport Roadways (QATAR) to better manage their access roadways and curbside, in part, by reducing traffic demand and mitigating congestion.

ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations was published in 2010. Since it was published, the inception of ride-sharing services has brought a paradigm shift, not only to the airport curb but to society in general. Transportation network companies have demonstrated the effectiveness of peer-to-peer mobile services and inspired new services that embrace this same model. There are opportunities and challenges in managing traditional and new types of ground transportation service providers. Airports needed new strategies for managing the changing marketplace, and QATAR needed updating to remain valid. Additionally, airports needed tactics for anticipating future ground service providers along with scalable strategies for traffic demand and congestion management.

Under ACRP Project 07-17, "Update to ACRP Report 40: Airport Roadway Analysis and Curbside Congestion Management Strategies," InterVISTAS Consulting was asked to produce a guide that would be a thorough, user-friendly update of *ACRP Report 40*. They were supported by Kittelson & Associates, Inc. *ACRP Research Report 266* and the updated QATAR should help airports of all sizes better manage their access roadways and curbside. A robust data collection plan was developed and case studies illustrating selected demand and traffic-congestion management strategies were reviewed. To support the update of QATAR, more than 10,000 tests or simulations of various curbside roadway configurations were completed to determine their performance under varying traffic demand. In addition to the published report, QATAR and a user guide for it are available on the National Academies Press website (nap.nationalacademies.org) by searching for *ACRP Research Report 266: Airport Curbside and Terminal Area Roadway Operations: New Analysis and Strategies, Second Edition*.

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SUMMARY

Airport Curbside and Terminal Area Roadway Operations: New Analysis and Strategies, Second Edition

ACRP Research Report 266: Airport Curbside and Terminal Area Roadway Operations: New Analysis and Strategies, Second Edition (the Guide) is intended to assist airport operators, planners, consultants, and individuals tasked with analyzing airport terminal area roadway and curbside operations and to select strategies to mitigate congestion. The Guide, which is an update of *ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations*, presents guidelines for estimating airport roadway requirements and capacities, recommended performance measures, valid and useful analytical methods, and innovative strategies to reduce traffic demands and mitigate congestion on terminal area and curbside roadway operations.

Operating Characteristics of Airport Roadways

Airport access and circulation roadways, curbside roadways, and service roads are used by a wide variety of vehicles. Vehicles using airport roadways include private vehicles, rental cars, commercial ground transportation vehicles [e.g., taxicabs, transportation network company (TNC) vehicles, limousines or town cars, shared-ride vehicles, courtesy vehicles, and charter and scheduled buses and vans] and service and delivery vehicles, with autonomous vehicles (AVs) expected in the future.

The operating characteristics of airport terminal area roadways differ from those of non-airport roadways because of the high proportion of motorists who are unfamiliar with the airport (as well as the presence of many professional drivers who use the airport frequently). The relatively complex wayfinding, closely spaced decision points, and high proportion of large vehicles add significant stress for a motorist. The operation of weaving and merging areas on airport roadways differs from the operation of these areas on non-airport roadways because these operations occur at slower speeds on airport roadways than they do on freeways and arterial streets.

The operating characteristics of airport terminal curbsides differ significantly from those of most other roadways because of the interactions between vehicles maneuvering into and out of curbside spaces and vehicles traveling in the through or bypass lanes. Compared to vehicles parked on city streets, vehicles remain stopped at the curbside for less time, occupy longer spaces to ensure they can easily exit, frequently double-park or park in an orientation not parallel to the curb, and tend to stop near specific doors. Curbside and terminal area roadway traffic peaking is influenced by airline schedules and the length of time passengers arrive prior to their scheduled departure and leave after their flight has landed. Curbside spaces are frequently reserved for specific uses such as commercial ground transportation vehicles. See Chapter 2 of the Guide for further details about the unique operating characteristics of airport terminal area and curbside roadways.

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Estimating Airport Roadway Requirements

Estimates of traffic volumes or requirements for terminal area and curbside roadways can be prepared using the traditional four-step approach, commonly used by transportation planners, or the growth factor method. The four-step process yields separate estimates of the traffic volumes generated by airline passengers, visitors, employees, air cargo handlers, and other major airport land uses. However, this method requires extensive input data describing the characteristics of each of these traffic generators. The growth factor method yields acceptable, but less precise, results while requiring much less input data. However, this simpler method is less sensitive to changes in future conditions or travel patterns. Both methods rely upon observations of existing vehicular activity—traffic volumes, vehicle mix, dwell times, and other data that can be gathered using traffic recording devices, video cameras, Bluetooth sensors, data from connected vehicles, vehicle detectors, and other sources.

Estimates prepared using the traditional four-step approach begin with trip generation (the first step). Estimates of trip generation require data on the (1) number of originating and terminating airline passengers (existing and forecast) and their trip purposes, places of residence, choice of travel modes, circulation patterns, and peaking characteristics; (2) the number of persons employed by the airport, airlines, and other tenants (existing and forecast) and their work schedules, choice of travel modes, and circulation patterns; (3) trips generated by trucks transporting air cargo and persons working at air cargo facilities; and (4) traffic generated by service and delivery vehicles bringing goods and materials to/from the terminals, airport operations and maintenance vehicles, and other sources. The second step (trip distribution) yields an estimate of the proportion of airport-generated traffic using each airport access point and the regional paths these vehicles follow. The third step (trip assignment) results in estimates of the airport-generated traffic volumes on each segment of the airport roadway network, by time of day and type of vehicle. The fourth step compares the estimated traffic volume (airport plus non-airport) on each roadway segment with the capacity of the segment.

Estimates prepared using the growth factor method can apply the ratio between the existing and forecast annual (or peak month or peak hour, as available) airline passenger volumes (originating plus terminating passengers) to the traffic volumes on each roadway segment. The growth factor method can further potentially adjust these volumes to reflect anticipated changes in vehicle mix (resulting from the introduction of transit service or use of AVs, for example) or circulation patterns (resulting from regional transportation improvements or expected congestion).

Chapter 3 provides further detail on estimating airport roadway traffic requirements.

Analyzing Airport Terminal Area Roadway Operations

Airport roadways, weaving sections, and curbside areas can be analyzed using

- **Microsimulation**, a method that uses sophisticated computer programs to simulate the operation of individual vehicles on simulated roadway networks. Each vehicle and roadway network segment is assigned characteristics reflecting performance capabilities, driver behavior, number of lanes, operating speeds, and pedestrian activity. As each simulated vehicle travels through the computerized network, its performance can be recorded based on its interaction with other vehicles and traffic controls. Of the three methods listed, microsimulation is the most complex and requires the most effort and a specialized set of skills on the part of the user; however, this method provides the most detailed and reliable results. Chapter 4 presents guidelines on the use of microsimulation on airport roadways.

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- **Macroscopic methods**, which consider the flow of vehicle streams rather than the operations of individual vehicles. These methods approximate the interactions between individual vehicles, the behavior of individual drivers, and detailed characteristics of the roadways (or other facilities). Adjustment factors are used to account for atypical vehicles or driver characteristics, traffic flow constraints, or other operational characteristics. These methods produce results that are considered acceptable and more accurate than quick-estimation methods, and macroscopic methods can be used with less effort and training than microsimulation methods. Macroscopic methods are best for determining the refinements to a proposed design (or existing facility) that would eliminate capacity and congestion problems but are less satisfactory for evaluating operations under heavy congestion conditions or on atypical configurations. Chapter 4 reviews macroscopic methods for analyzing airport roadway weaving areas.
- **Quick-estimation methods**, which can be used to produce preliminary analyses of roadway operations (or other facilities) simply and rapidly. They generally consist of look-up tables, simple formulas using regression analysis of databases, or principles based on experience or practice and are built on broad-based assumptions about the characteristics of the facility being analyzed. Quick-estimation methods are ideal for quickly sizing a roadway or other facility and providing a first test of the facility's ability to properly accommodate the estimated requirements (existing or future) or the adequacy of a potential improvement measure.

Table 4-1 of the Guide presents the maximum service flow rate and adjusted flow rates for multilane airport roadways with uninterrupted flows. Chapter 4 describes the assumptions used to prepare this table, limitations to its use, and its application. Methods for analyzing signalized intersections and airport roadway weaving sections are also described.

Estimating Roadway Capacities

Roadway capacities are a function of the roadway characteristics and the vehicles and drivers using the roadway. Knowing the characteristics of a roadway section and the vehicles (and drivers) using the roadway, it is possible to calculate its *capacity*—the maximum hourly rate of vehicles flowing past a point. It is not possible or desirable for a roadway to operate at its capacity for sustained periods, as any minor disruption will cause congestion and undesirable levels of safety and driver comfort. Also, demand is rarely sustained at a constant level over an hour.

Roadway operations are characterized in terms of *quality-of-service* measures and *service flow rates*—the maximum flow rates that can be accommodated while maintaining a desired quality of service. Quality of service reflects how travelers perceive the performance of roadways and other facilities. While quality of service is often stratified quantitatively into levels of service indicated using the acronym "LOS" (e.g., LOS A to LOS F), the concept is broader and includes system-oriented metrics such as the sufficiency of the roadway to provide adequate capacity.

On airport roadways, a sufficiency of "under capacity" is typically considered to be the minimum "acceptable" level of performance because of the lack of alternative travel paths, the significant negative consequences resulting from travel delays (e.g., passengers missing their flights), and the ability to accommodate higher volumes during short periods and unusually busy days (such as during holidays) without reaching failure. In comparison, on regional freeways and arterials and in densely developed urban areas, a sufficiency of "near capacity" is often considered "acceptable" because motorists traveling on these roads can select alternative travel paths if desired. Chapter 4 presents further details on quality-of-service and sufficiency metrics.

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Evaluating Airport Curbside Operations

Evaluations of airport curbside roadway operations require analyses of both the curbside lanes (where motorists pick up or drop off passengers) and the adjacent through lanes. The recommended performance measure for airport curbside roadways is curbside utilization—the ability of a roadway to accommodate vehicles loading or unloading at the curbside during the design hour that indicates a curbside roadway's level of service. LOS C (illustrated in Figure 5-1 and presented in Table 5-2) is the suggested planning target.

Curbside roadway traffic volumes can be estimated using the same methods as those used for estimating airport terminal area roadway traffic, allowing for separate estimates of through traffic, vehicles dropping off and picking up passengers (by type of vehicle), vehicle dwell times, and pedestrian crossings.

Chapter 5 presents procedures for calculating curbside roadway requirements and factors influencing the requirements. It also describes the use of the Quick Analysis Tool for Airport Roadways (QATAR) for analysis of airport curbside roadway operations and testing how variations in input assumptions affect level of service. Appendix C provides an overview of the curbside analysis methodology in QATAR. QATAR and a user guide for QATAR are available on the National Academies Press website (nap.nationalacademies.org) by searching on *ACRP Research Report 266: Airport Curbside and Terminal Area Roadway Operations: New Analysis and Strategies, Second Edition*.

Improving Airport Terminal Area and Curbside Roadway Operations

Roadway operations can be adversely impacted by numerous deficiencies, including insufficient roadway length or merging capacity, lane imbalances, wayfinding challenges, insufficient decision-making distances and queuing space, insufficient curbside capacity (i.e., insufficient curb length or travel lanes), unusable curb space, and excessive dwell times. These deficiencies are detailed in Chapter 6.

Chapter 7 presents innovative strategies to improve airport terminal area and curbside roadway operations. These strategies are grouped into three primary categories and four complementary categories (see Figure S-1). The three primary categories are

1. **Enforcement strategies**, which underpin all other strategies. Without effective enforcement, it is difficult to implement other strategies efficiently and effectively. Enforcement strategies include the appropriate use of enforcement officers, crosswalk controls, regulatory signs, and other strategies.
2. **Traffic demand management strategies**, which include promoting the use of alternative passenger drop-off and pickup locations, promoting transit use, and charging motorists fees for curbside use.
3. **Congestion management strategies**, which are intended to improve curbside capacities and traffic flows, improve roadway operations, and control traffic flows.

Four strategy categories complement the three primary strategy categories:

1. **Customer service strategies** include providing customers with comfortable waiting areas and transit information and having staff available to help at drop-off and pickup areas.
2. **Sustainability strategies**, which reduce carbon emissions by encouraging the use of long-duration parking rather than drop-off and pickup alternatives and promoting the use of electric vehicles.

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Figure S-1. Hierarchy of strategies to improve airport roadway operations.

3. **Accommodating TNCs**, which includes operational measures that can be implemented by the TNCs (use of geofences, pooling, and dispatching procedures) and by the airport (providing subzones to help customers find their driver and alternative/remote boarding areas).
4. **Accommodating AVs**, which includes separating AVs from non-AVs and updating airport regulations and infrastructure in anticipation of the widespread adoption of AVs.

Furthermore, numerous supporting technologies are available to enhance these primary and complementary strategies. These supporting technologies can monitor traffic volumes and locations (by type of vehicle or individual vehicle), automatically dispatch taxicabs and other on-demand services, support curbside enforcement by directing officers to vehicles needing attention, guide motorists by displaying changeable directional or parking information, and provide other capabilities.

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CHAPTER 1

Purpose, Methodology, and Organization of This Guide

ACRP Research Report 266: Airport Curbside and Terminal Area Roadway Operations: New Analysis and Strategies, Second Edition (the Guide) is intended to assist airport operators and others in analyzing airport terminal area roadway and curbside operations and selecting strategies to mitigate congestion. The Guide, which is an update of *ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations*, presents guidelines for estimating airport roadway requirements and capacities, recommended performance measures, valid and useful analytical methods, and innovative strategies to reduce traffic demands and mitigate congestion on terminal area and curbside roadway operations.

1.1 Purpose of the Guide

This Guide is intended for use by three primary user groups. The first user group consists of airport operators, including the staff responsible for planning, designing, and overseeing the day-to-day operations of airport terminal area and curbside roadways. The second user group includes city, regional, and state transportation planners who may not be familiar with airport roadway operations. A third user group consists of airport consultants engaged to conduct planning, environmental, design, and other projects on behalf of airport operators and other sponsors. Some users may have significant experience with airports or aviation, but little familiarity with traffic engineering or transportation planning principles. Other users may have experience in traffic engineering and transportation planning, but little knowledge of airport operations. This Guide is intended to assist both groups.

For users unfamiliar with airports, the Guide describes the unique operating characteristics of airport terminal area roadways and curbside areas and how their operations differ from those of urban streets and regional highways. The Guide presents methods for estimating existing and future requirements and provides alternative methods for analyzing operations on airport roadways.

1.2 Methodology

The Guide reflects information gathered through an extensive literature review, a survey of airport operators, and follow-on interviews with airport staff who have implemented innovative strategies to reduce traffic demand and relieve roadway congestion. The scope of this research included updating the Quick Analysis Tool for Airport Roadways (QATAR) macrosimulation model presented in *ACRP Report 40* and used to analyze and evaluate airport curbside roadway operations and low-speed weaving. [The updated QATAR and a user guide are available on the National Academies Press website (nap.nationalacademies.org) by searching on *ACRP Research Report 266: Airport Curbside and Terminal Area Roadway Operations: New Analysis and Strategies, Second Edition*.] Due to the COVID-19 pandemic, it was not feasible to conduct field measurements or traffic surveys of airport roadway operations. Instead, to support the update of

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the model, the research team conducted more than 10,000 tests or simulations of various curbside roadway configurations to determine their performance under varying traffic demands, as described in this Guide.

1.3 Organization of the Guide

This Guide consists of seven chapters and three appendices. Chapters 2 through 7 and Appendices A through C are briefly described in the following:

- **Chapter 2: Framework for Analysis of Airport Roadways and Curbsides** describes the types of vehicles and roadways typically found at airports and their unique operating characteristics. An overview is provided of (1) the hierarchy of analytical methods presented in Chapters 4 and 5 of this Guide (quick-estimation methods, macroscopic models, and microsimulation models) and (2) the concepts of capacity and level of service, as presented in the *Highway Capacity Manual* (HCM). [Note that as used in this report, "HCM" indicates *Highway Capacity Manual 7th Edition: A Guide for Multimodal Mobility Analysis* (2022) and previous editions of the HCM.]
- **Chapter 3: Estimating Airport Roadway Requirements** describes the data required to analyze existing roadway traffic operations and ways to gather these data. Two alternative methods for estimating future roadway requirements are presented, along with the challenges inherent in each method.
- **Chapter 4: Analyzing Airport Terminal Area Roadways** presents definitions of levels of service for airport terminal area roadway operations and a hierarchy of analytical methods and models for analyzing terminal area roadway operations, including low-speed weaving areas, capacities, and levels of service.
- **Chapter 5: Evaluating Airport Curbside Operations** presents definitions of levels of service for airport curbside roadway operations and a hierarchy of analytical methods and models for quickly estimating curbside roadway capacities and levels of service, including an update of the QATAR model that incorporates changes requested by frequent users of this model.
- **Chapter 6: Improving Airport Curbside and Terminal Area Roadway Operations** presents examples of commonly occurring airport curbside and roadway operational problems and overviews steps for analyzing and evaluating airport roadway improvements.
- **Chapter 7: Innovative Strategies** presents strategies to improve enforcement, reduce traffic demands, and manage roadway congestion that apply to all commercial-service airports. It also presents information allowing airport managers to evaluate which strategies are best suited for their airport.
- **Appendix A: Glossary** defines technical terms used in the Guide.
- **Appendix B: Bibliography** lists resource documents reviewed for this research project.
- **Appendix C: Overview of the Curbside Analysis Methodology in the Quick Analysis Tool for Airport Roadways (QATAR)** details the basic assumptions used in developing the QATAR curbside queuing model.

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CHAPTER 2

Framework for Analysis of Airport Roadways and Curbsides

On-airport roadways are a unique class of roadways. Drivers who are unfamiliar with the airport mix with significant numbers of professionally driven large vans and buses. Entrances and exits at major airports operate at near-freeway speeds, while curbside roadways operate at much slower speeds, as drivers attempt to maneuver into and out of curbside spaces. Double- and triple-parking and jaywalking frequently occur on curbside roadways despite the visible presence of traffic enforcement officers.

The standard highway capacity analysis procedures can address some aspects of these conditions, but not the full spectrum of operating conditions that exist on airport terminal area and curbside roadways. The various users and types of airport roadways and curbsides and their unique operating characteristics are described in this chapter. Overviews are also presented of (1) the hierarchy of methods for analyzing airport roadway and curbside operations and (2) roadway capacity and quality-of-service concepts.

2.1 Users of Airport Roadways

Airport roadways provide access to and from the multiple land uses at an airport. These roadways serve vehicles transporting airline passengers and visitors (in this Guide, "visitors" refers to meeters/greeters and well-wishers accompanying or greeting airline passengers); employees of the airport, airlines, and other airport tenants; air cargo and mail; as well as vehicles used for the delivery of goods and services, maintenance, support of airport operations or construction, and other purposes.

A multitude of vehicle types now use airport roadways. They include private vehicles, rental cars, on-demand and pre-reserved taxicabs, transportation network company (TNC) vehicles, pre-arranged and on-demand limousines or town cars, door-to-door vans, courtesy vehicles, charter buses, scheduled buses, and service and delivery vehicles. Each vehicle/user type has its own special characteristics and affects airport roadway operations differently, as described below.

2.1.1 Private Vehicles

Privately owned and operated vehicles consist of automobiles, vans, pickup trucks, and motorcycles used to transport airline passengers; visitors; and employees of the airport operator, airlines, and other airport tenants. Motorists transporting airline passengers in private vehicles may use the curbside areas or other passenger drop-off and pickup areas, parking facilities (including cell phone waiting areas), or both.

2.1.2 Rental Cars

Rental car vehicles, including automobiles and vans, are used to transport airline passengers or visitors. These vehicles may be rented by a customer for the duration of their trip from either

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rental car companies doing business on or near the airport (traditional rental car businesses) or peer-to-peer businesses that connect customers with individual car owners. Rental car customers may use the terminal curbside areas, rental car ready and return areas, or both. At most airports, the on-airport companies lease sites where customers can return and pick up their cars. Some airports require peer-to-peer companies to lease parking spaces where customers can interact with vehicle owners to return and pick up their cars.

2.1.3 On-Demand Taxicabs

On-demand taxicabs provide door-to-door service without prior reservations. This service is typically exclusive (i.e., for a single party) and provided in vehicles capable of transporting a driver plus four customers and their baggage in most developed nations (three-wheeled taxicabs and even motorcycles may be used to provide taxi service in developing nations). On-demand taxicabs are typically licensed and regulated by a municipal taxicab authority. Typically, on-demand taxicabs drop off passengers in the departures area and wait for deplaning passengers at a taxicab stand (or in a taxicab queue) at the curbside area next to the baggage claim area or other designated boarding area. At large airports, taxicabs may wait for customers in a remotely located taxicab holding or staging area until they are dispatched to the curbside taxicab stand or customer boarding area in response to customer demand.

2.1.4 Pre-reserved Taxicabs

Pre-reserved taxicab service is exclusive, door-to-door transportation provided in vehicles capable of transporting up to five customers plus their baggage. Rather than being provided on demand, as is traditional taxicab service, pre-reserved taxicabs are provided in response to pre-arrangements (i.e., prior reservations) made by airline passengers seeking to be picked up by a specific company or driver, including suburban taxicabs not regulated by the local municipal taxicab authority responsible for regulation of on-demand taxicab activity at the airport. Passengers with special needs, such as those with skis, golf clubs, large amounts of baggage, or physical disabilities, may request service by specific vehicles or companies. Typically, pre-reserved taxicabs or specially requested taxicabs are not allowed to wait at the curbside taxicab stand with on-demand taxicabs but instead wait for their customers at a separate curb space or alternative location.

2.1.5 Pre-arranged Limousines

Pre-arranged limousine service is exclusive door-to-door transportation provided in luxury vehicles capable of transporting a single party consisting of up to five customers (or more in stretch limousines) regulated by a local or state agency. Generally, limousine service is only available to customers who have made prior reservations (i.e., pre-arranged) and are greeted (or picked up) by a driver having a waybill or other evidence of the reservation. Some airport operators allow limousine drivers to park at the curbside and wait for customers; others require that the drivers park in a parking lot or other designated zone and accompany their customers from the terminal to the parking area.

2.1.6 On-Demand Limousines or Town Cars

Privately operated, on-demand, door-to-door transportation is also provided by exclusive luxury vehicles or "town cars" capable of transporting up to five passengers and their baggage. These services are similar to on-demand taxicab services but are provided in luxury vehicles with higher fares than those charged for taxicab services.

2.1.7 Transportation Network Company

A TNC offers door-to-door, non-stop transportation for a single party as well as other transportation options (e.g., point-to-point, shared-ride transportation in a vehicle transporting other

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parties). Customers connect with and pay drivers, who are using personal vehicles, via the company's smartphone application. Typically, TNCs drop off customers and meet waiting customers at the departures and arrivals curbsides, although at some airports alternate locations or a single curbside are used for both activities. Generally, TNCs are required to wait for customers in a remotely located hold or staging area until their affiliated company dispatches the driver to the customer boarding area to pick up a waiting customer. While some companies offer service in licensed limousines, these vehicles are classified and regulated as pre-arranged limousines rather than TNC vehicles.

2.1.8 Door-to-Door Vans

Door-to-door or shared-ride van services are typically provided in vans capable of transporting 8 to 10 passengers and their baggage. The service is available on both an on-demand and pre-arranged basis. Passengers, who may share the vehicle with other traveling parties, are provided door-to-door service between the airport and their homes/offices or other locations but may encounter several (typically four or fewer) en route stops. Typically, door-to-door vans wait for deplaning passengers at the curbside next to the baggage claim area. Similar to taxicabs, vans may be required to wait in hold or staging areas until they are dispatched to the curbside in response to customer demand.

2.1.9 Courtesy Vehicles

Door-to-door courtesy vehicle service is shared-ride transportation provided by the operators of hotels, motels, rental car companies, parking lot operators (both privately owned and airport-operated parking facilities), and others solely for their customers. Typically, no fare is charged because the cost of the transportation is considered part of (or incidental to) the primary service being provided. Courtesy vehicle service is provided in shuttle vehicles, including 8- to 12-passenger vans (e.g., those operated by small motels), minibuses, and full-size buses (e.g., those operated by rental car companies at large airports). Typically, courtesy vehicles pick up customers at designated curbside areas that have been reserved or allocated for their use.

2.1.10 Charter Buses and Vans

Charter bus and van service (also referred to as tour bus or cruise ship bus service) is door-to-door service provided to a party (or group of passengers) that has made prior reservations or arrangements for the service. Charter bus and van service is provided using over-the-road coaches, full-size buses, minibuses, and vans seating more than five passengers. As charter bus service is sporadically provided at most airports, curb space (or other passenger pickup areas) is either not allocated for charter buses or is shared with other transportation services. Exceptions include airports regularly serving large volumes of charter or cruise ship passengers. Typically, charter buses and vans unload passengers on the deplaning curbside. Charter buses and vans picking up passengers are required to wait in a remotely located hold area until the arrival or assembly of the party being provided the service. In some cases, such as charter buses carrying passengers to and from cruise ships, buses may be accompanied by small trucks carrying passenger luggage.

2.1.11 Scheduled Buses

Scheduled buses provide shared-ride service at established stops along a fixed route and operate on a fixed scheduled basis. Typically, scheduled buses are operated by a public agency and make multiple stops along a designated route, but in some communities, express or semi-express service is operated by a private operator or public agency. The location and amount of curb space allocated to scheduled buses depends on the volume of such service and the policy of the airport operator. Scheduled buses do not use hold lots or staging areas at most airports.

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2.1.12 Active Transportation

Active transportation includes methods of travel such as walking, bicycling, kick scooters, and dockless electric scooters (or e-scooters). At airports in North America, these methods of travel have historically been rarely used by passengers for airport access or by the employees of the airport, airlines, or other tenants. However, their use may increase over time, particularly at locations where the communities surrounding the airport are well served by and promote active transportation modes. Normally, sidewalks or separate pathways are provided for pedestrians, while the operators of bicycles and scooters share roadway lanes with other vehicles.

2.1.13 Service and Delivery Vehicles

Service vehicles include a wide range of trucks, vans, semi-trailers, and other delivery vehicles used to transport goods, air cargo, mail, contractors, and refuse to and from the airport. Generally, deliveries are made at designated loading docks or warehouses, not at the terminal curbside. However, the pickup and drop-off locations for airline-operated small package delivery services, which are provided by small vans and light trucks, are at the terminal curbside at some airports.

2.1.14 Autonomous Vehicles

Autonomous cars and trucks are self-driving vehicles in which operation occurs without direct driver input into the control of steering, accelerating, and braking, and which are designed so that the driver is not expected to constantly monitor the roadway while the vehicle is operating in a self-driving mode. At present, AVs are not operating (or permitted to operate) on the airport curbsides or terminal area roadways of any North American airport. Limited tests have been conducted on AVs operating within parking lots and on secured airfields. As discussed in subsequent sections of this Guide, AVs are expected to operate on airport roadways in the future.

2.2 Types of Airport Roadways

While the airport passenger terminal building and surrounding area (the terminal area) is the most prominent location at an airport, depending on the size, type, and distribution of airport land uses, less than half of all traffic at an airport may be associated with passengers and visitors proceeding to/from the terminal area; the remaining traffic is generated by non-airline-passenger activities, including employees and tenants. Regardless of airport size, the variety of land uses found at an airport requires a network of roadways to provide for inbound and outbound traffic and the internal circulation of traffic between land uses. The roadway network consists of the types of roadways depicted in Figure 2-1.

2.2.1 Access Roadways

For purposes of this Guide, airport access roadways are defined as the roadways linking the regional highway and roadway network with the airport terminal and other areas of the airport that attract large volumes of airline-passenger-generated traffic, such as parking and rental car facilities. Access roadways provide for the free flow of traffic between the regional network and the passenger terminal building or other major public facilities and typically have a limited number of decision points (i.e., entrances or exits). At large airports, access roadways are often limited-access roadways with both at-grade intersections and grade-separated interchanges. At smaller airports, access roadways often have at-grade intersections that may be signalized, controlled by stop signs, or have roundabouts (yield-sign controlled).

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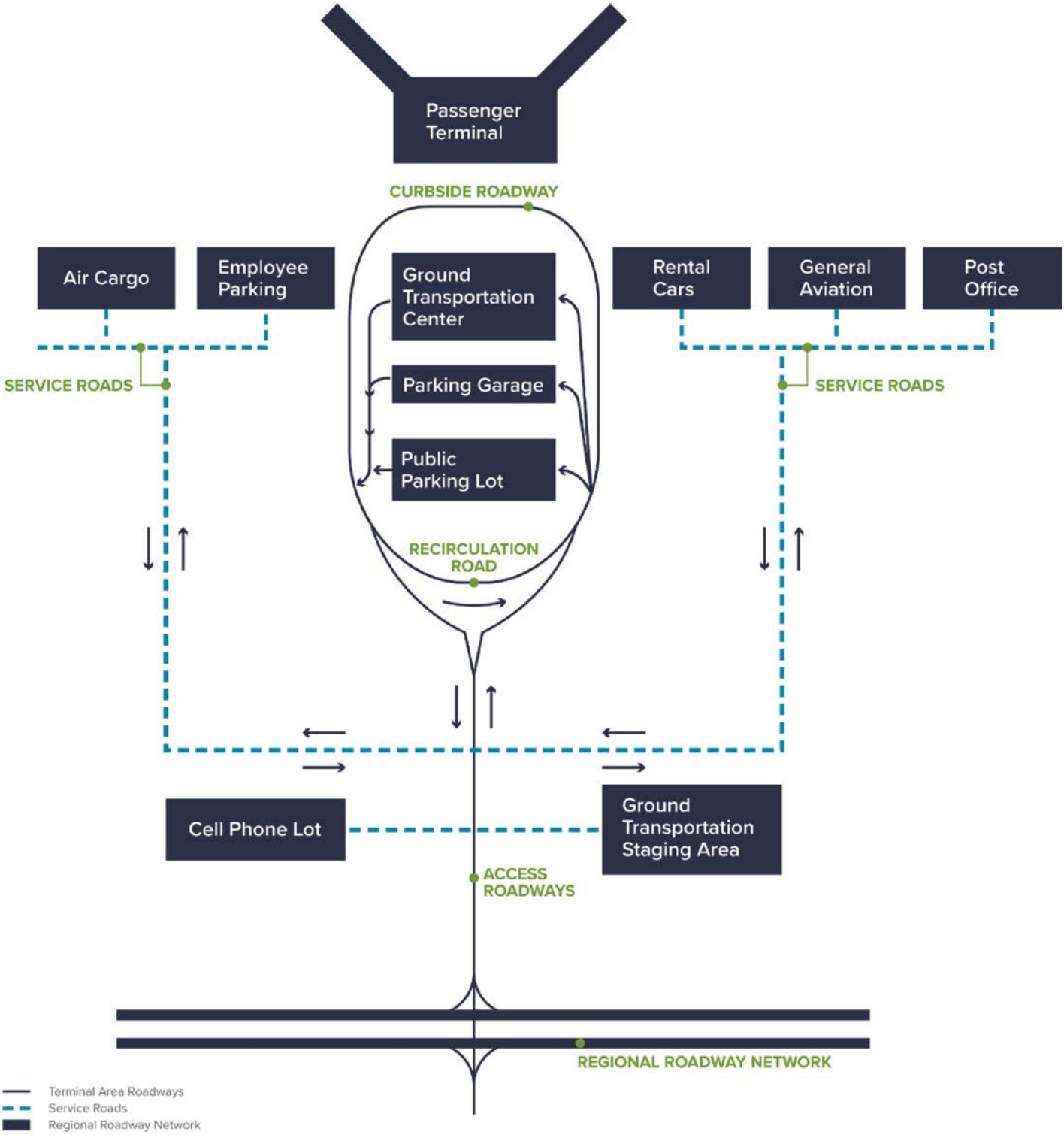


Figure 2-1. Airport land use and roadway types.

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2.2.2 Curbside Roadways

Curbside roadways are one-way roadways located immediately in front of the passenger terminal buildings where vehicles stop to pick up and drop off airline passengers and their baggage. Curbside roadways typically consist of (1) one or two inner lanes where vehicles stop or stand in a nose-to-tail manner while passengers board and alight, (2) an adjacent maneuvering lane, and (3) one or more through or bypass lanes. Curb space is often allocated or reserved along the inner lane for specific vehicles or classes of vehicles (e.g., taxicabs, TNCs, shuttle buses, courtesy vehicles, or curbside valet storage), particularly at the curbside areas serving baggage claim or passenger pickup.

As shown in Figure 2-2, depending on the configuration of the adjacent passenger terminal building, curbside roadways may include one, two, or more vertical levels and/or one, two, or more parallel roadways separated by raised medians (often called islands). At airports with dual-level curbsides, the upper-level curbside area is typically at the same level as airline passenger ticketing and check-in facilities inside the terminal building and is intended for passenger drop-off. The lower-level curbside area is typically at the same level as the baggage claim area and is intended for passenger pickup. At airports with multiple terminal buildings where one of the parallel roadways serves as a bypass roadway, cut-through roadways may be provided to allow vehicles to circulate between the inner and outer parallel roadways (and curbside roads).

2.2.3 Circulation Roadways

Circulation roadways generally serve a lower volume of traffic and may be more circuitous or less direct than terminal access roadways. Circulation roadways often provide a variety of paths for the movement of vehicles between the passenger terminal buildings, parking, cell phone waiting areas, and rental car facilities. Examples include return-to-terminal roadways that (1) allow motorists to proceed to parking after having dropped off airline passengers (or proceed from parking to the terminals) and (2) allow courtesy or other vehicles to return to the terminal (e.g., after having dropped off enplaning airline passengers and returning to pick up deplaning passengers on a different curbside roadway). Compared to access roadways, circulation roadways typically operate at lower speeds and allow for multiple decision points.

Access roadways, curbside roadways, and circulation roadways are considered "curbside and terminal area" roadways and are the focus of this Guide. Other airport roads include service and airfield roads, as described below.

2.2.4 Service Roads

Service roads link the airport access roadways with on-airport hotels, employee parking areas, employment centers (e.g., aircraft maintenance facilities or hangars), air cargo/air freight buildings and overnight parcel delivery services, loading docks/trash pickup areas, post offices, fixed-base operator (FBO)/general aviation areas, airport maintenance buildings and garages, military bases, and other non-secure portions of the airport that generate little airline passenger traffic.

The traffic generated by these land uses differs from that generated by the passenger terminal building in several respects. First, the traffic on service roads includes a higher proportion of trucks, semi-trailers, and other heavy vehicles than the traffic on curbside and terminal area roadways, which rarely serve trucks or delivery vehicles. Second, most drivers on the service roads (e.g., employees and drivers of cargo vehicles) use these roads frequently and are familiar with the roads and their destinations, unlike many drivers using the curbside and terminal area roadways.

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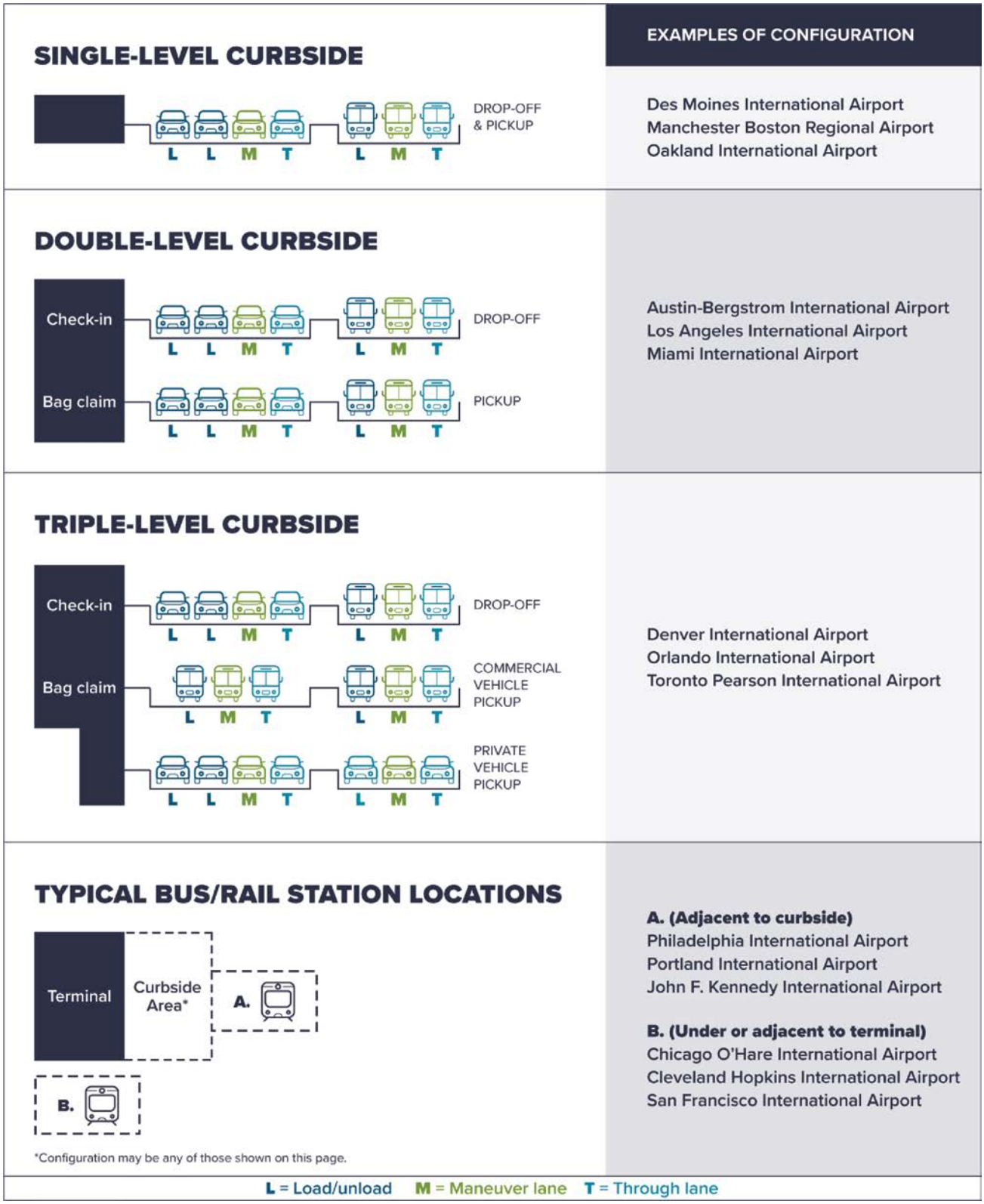


Figure 2-2. Airport curbside configurations.

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For purposes of operational analyses, the service roads are similar to those found in an industrial park. Typically, they consist of two- to four-lane roads with generous provision for the turning paths of large trucks and semi-trailers and for entering and exiting vehicles, including separate or exclusive turning lanes.

2.2.5 Airfield Roads

A separate network of roads, located within the aircraft operating area or the airfield, is used by ground service equipment, including vehicles servicing aircraft, towing aircraft, or towing baggage carts and vehicles used for runway maintenance or emergency response. Often these vehicles are not licensed to operate on public streets. Only drivers who have received formal training from the airport are permitted to operate vehicles with an airport permit in secure or restricted areas. The design and operation of these roads are addressed in guidelines issued by the FAA Series 150 Advisory Circulars as well as *ACRP Report 96: Apron Planning and Design Guidebook*.

The remainder of this Guide addresses curbside and terminal area roadways only.

2.3 Operating Characteristics of Airport Terminal Area Roadways

The operating characteristics of airport terminal area roadways differ from those of other public roadways. This section describes the distinguishing operating characteristics of airport terminal area roadways, weaving sections, and curbside areas.

2.3.1 What Makes Airport Roadway Operations Unique

The main differences between the operating characteristics of airport terminal area access and circulation roadways and those of non-airport roadways include the following:

- **A high proportion of motorists unfamiliar with airport roadways.** Because most airline passengers fly infrequently (e.g., fewer than four times per year), they (and the drivers who drop them off/pick them up) are not familiar with the roadways at their local airport, much less the roadways at their destination airport(s). Unlike commuters, who rarely need to refer to roadway signs, airline passengers rely upon signs (or other visual cues) to guide them into and out of an airport and to/from their destinations at the airport. Picking up passengers may be particularly challenging for unfamiliar motorists, who must follow the appropriate signs, be aware of all the traffic and pedestrian activity at the curbside areas, and also identify their party among crowds of other passengers waiting to be picked up.
- **Complex wayfinding.** Motorists entering an airport are provided with information concerning the locations of airlines (especially at airports with multiple terminals), parking products (e.g., hourly, daily, economy, and cell phone waiting area parking), rental car companies, and the entry to the departures/ticketing and arrivals/baggage claim levels (see Figure 2-3). Often the directional signs presenting this information to motorists (1) are closely spaced, limiting the time drivers have to comprehend and react to the messages, and (2) contain more information (i.e., more text) than those on comparable public roadways. For example, the general policy at U.S. airports is to display the name of every airline serving an airport, even those operating only a few times a week. Airport directional signs often include colors, fonts, symbols, and messages not commonly used on roadway signs governed by the *Manual on Uniform Traffic Control Devices for Streets and Highways* (published by FHWA, U.S. DOT) or equivalent guidelines issued by other transportation agencies.

  Because of the number, size, and complexity of signs at airports, motorists may not see regulatory or warning signs concerning height restrictions, speed limits, parking rates, security

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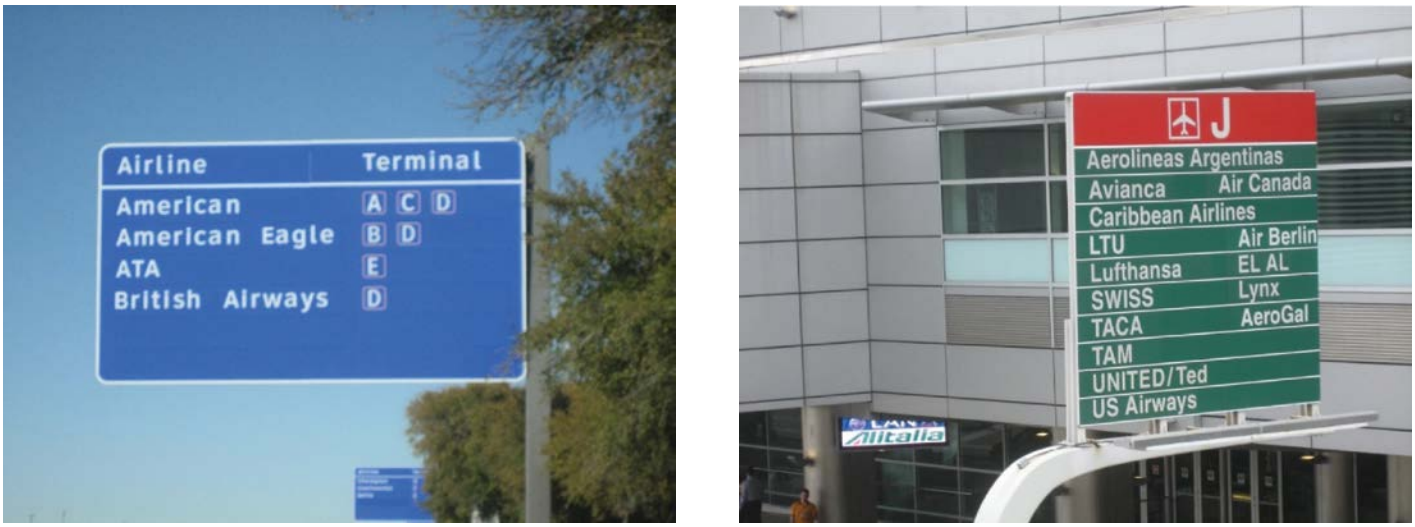


Figure 2-3. Complex airport roadway signs.

regulations, use restrictions (e.g., authorized vehicles only), and other messages. Advertising and promotional signs visible to motorists (including column wraps) may be further distractions. These signs may result in an overload of information and cause motorists to decelerate in an attempt to read the signs.

- **Stressful conditions.** Motorists operating on airport roadways are under more stress than typical motorists. This additional stress results from knowing that minor delays or wrong turns may cause a person to arrive too late to check baggage; claim a pre-reserved seat; greet an arriving passenger; or, in an extreme case, catch their flight. Congested airport roadways, closely spaced decision points, and complex signs can add to this stress and discomfort.

  Factors adding to passenger stress at an airport include the need to connect from a car to a plane; from a car to a shuttle bus; find a parking place; find a passenger (Where is Aunt Meg?); find the correct place to drop off or pick up a passenger; locate a TNC, courtesy vehicle, or city bus stop; and so forth. Passengers realize the importance of making correct decisions so that they do not miss their flights or rides in an environment that is more complicated and anxiety-filled than a typical roadway situation. Each action at an airport is part of a chain of events, any one of which can go wrong and disrupt or delay a vacation, business meeting, or other important event.
- **A high proportion of heavy vehicles.** More than 10 types of ground transportation service regularly operate on airport roadways, excluding maintenance vehicles, snowplows, or other vehicles used by airport operations. The characteristics of each service, the needs of the customers using the services, and the operating characteristics of the vehicles used to provide these services must be considered when developing physical and operational plans for airport curbside and terminal area roadways.

  Courtesy vehicles, door-to-door vans, scheduled buses, and other large vehicles may represent 10% to 20% of the traffic volume on a terminal area roadway. On a typical public street, heavy vehicles (including single-unit and trailer trucks) may comprise less than 5% of the traffic volume. Capacity calculation procedures reduce the capacity of public highways with a high percentage of heavy vehicles reflecting the larger size and slower acceleration/deceleration rates of these vehicles.

  However, the use of a capacity adjustment factor may not be necessary on airport terminal area roadways because courtesy vehicles, vans, and buses operating on those roadways do not interfere with the flow of other traffic to the extent that they do on public highways. On airport terminal area roadways, heavy vehicles (1) can operate at the range of prevailing speeds

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typically found on airport roadways [i.e., 25 miles per hour (mph) to 45 mph] and (2) have sufficient power to accelerate and decelerate at rates that are comparable to those of private vehicles, as most airport roadways are level or have gentle grades, and are able to do so unless they are transporting standing passengers. In some locations, it is possible that large vehicles, such as courtesy vans or shuttle buses, may (1) obstruct motorists' views of wayfinding and other signs and (2) interfere with traffic operations in the curbside through lanes as these courtesy vehicles maneuver into and out of curbside areas. In such cases, capacity adjustment factors may be warranted.

- **A mix of experienced and inexperienced drivers.** While most private vehicle drivers use an airport infrequently, 20% to 30% of the vehicles on airport roadways are operated by professional drivers who are thoroughly familiar with the on-airport roadways, as they use them frequently—perhaps several times each day. This difference contributes to vehicles operating at a range of speeds on the same roadway segment—slow-moving vehicles (e.g., drivers of private vehicles who are unfamiliar with the airport roadways attempting to read signs or complete required turns and maneuvers) and faster vehicles (e.g., taxicabs, TNCs, limousines, and courtesy vehicles operated by professional drivers familiar with the airport roadways who may ignore posted speed limits).
- **Recirculating traffic.** Motorists, having dropped off a passenger, may recirculate along the airport's roadways in order to enter a parking facility or return to the terminal. Motorists may be required by traffic control officers to exit the curbside area if they are not actively loading or unloading passengers, are unable to find an empty curbside space, or are waiting for an arriving passenger who is not yet at the curbside. Motorists exiting the curbside area may either wait in a cell phone or waiting area until their passenger arrives (which is encouraged by airport operators) or recirculate and return to the curbside. Table 2-1 indicates the percentage of private vehicle traffic found to be recirculating past the terminal, with some vehicles doing so multiple times.

  These recirculating vehicles contribute to roadway congestion and represent unnecessary traffic volumes. Factors contributing to the volume and proportion of recirculating curbside roadway traffic include (1) an airport operator's tolerance of vehicles lingering unnecessarily at the curbsides, (2) motorists not recognizing the difference between the published flight arrival time and the time when a passenger will actually arrive at the curbside, (3) motorists attempting to wait at the curbsides for passengers whose flights have been delayed, and (4) drivers of commercial vehicles who, in violation of airport regulations, are improperly soliciting customers along the curbside roadway. A lack of available close-in public parking spaces may cause motorists, seeking such spaces, to exit the terminal area and travel to remotely located parking areas.
- **Atypical operations.** The operation of airport roadways—particularly the primary entry and exit roads, terminal area roadways, and curbsides—differs from arterial roads in urban and rural areas. For example, airport roadways generally operate at slower speeds (35 mph or less), rarely have signalized intersections along the major entry and exit roads, primarily consist of one-way roadways, have fewer driveways or entry/exit points, and have no street parking.

Table 2-1. Percentage of private vehicles recirculating to the arrivals curbside.

Airport	Recirculating (%)
Baltimore/Washington International Thurgood Marshall Airport	50%
San Francisco International Airport	43%
Seattle-Tacoma International Airport	30%
Dallas Love Field	26%
Reagan Washington National Airport	15%

Source: Based on data provided by Ricondo & Associates, Inc., June 2009.

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2.3.2 What Makes Airport Roadway Weaving Section Operations Unique

Weaving is defined as the crossing of two or more traffic streams traveling in the same direction along a length of highway without the aid of a traffic signal or other control device. A weaving maneuver occurs when vehicles enter a roadway segment from one side and exit the segment on the other while other vehicles do the opposite at the same time. The most common example of weaving occurs on freeways where an on-ramp is followed by an off-ramp a short distance later, and those two ramps are connected by an auxiliary lane. The weaving movement occurs when vehicles on the freeway move into the auxiliary lane to exit via the off-ramp, while vehicles from the on-ramp move from the auxiliary lane onto the freeway.

The operation of weaving and merging areas on airport roadways differs from the operation of these areas on non-airport roadways primarily because, on airport roadways, these operations occur at slower speeds than they do on freeways and arterial streets. Weaving analyses are generally conducted for freeways and arterial streets on which vehicles operate at higher speeds than on most airport roadways. At higher speeds, drivers require larger gaps between successive vehicles in order to merge into or weave across a traffic stream. The HCM provides a methodology for assessing weaving segments on freeways and multilane highways or collector-distributor roadways. These analyses are based on roadway free-flow speeds of 50 mph or higher. As such, the methodologies in the HCM for weaving sections are not suitable as is for airport roadways, which operate at lower free-flow speeds. Chapter 4 of this Guide presents alternative metrics and analysis methods for use on airport roadway weaving segments.

Upon entering an airport, motorists typically encounter a series of exits or turns leading to (1) non-terminal areas (e.g., economy parking, air cargo, general aviation), (2) close-in parking (e.g., hourly, daily, or cell phone waiting areas) and rental car return (by company), and (3) ticketing/departures and baggage claim/arrivals curbside areas. Upon exiting the airport, motorists may encounter a similar series of exits as well as roads leading back to the terminal and alternative regional destinations.

Often the distance between successive decision points is much less than that suggested by highway design standards established for limited-access highways because of the relatively short distances available between an airport entrance and the terminal area. Unlike a regional highway, where decision points may be separated by a mile or more, successive decision points on an airport roadway may be separated by 500 feet or less. Even though motorists on airport roadways are traveling at speeds that are slower than the speed of travel on freeways or arterial roadways (e.g., 35 mph or less), the limited distances between decision points, coupled with a driver's unfamiliarity with the airport, may compromise the ability of a driver to recognize, read, and react to roadway directional signs, or cause the driver not to allow adequate time to complete required merging and weaving maneuvers.

2.3.3 What Makes Airport Curbside Operations Unique

As noted, curbside roadways consist of an inner curbside lane(s) where vehicles stop or stand typically in a nose-to-tail arrangement while passengers board and alight, an adjacent maneuvering lane that vehicles may occupy while decelerating or accelerating to enter or exit the curbside lane, and one or more "through" or bypass lanes. The operating characteristics of airport terminal curbsides differ significantly from those of most other roadways because of the interactions between vehicles maneuvering into and out of curbside spaces and vehicles traveling in the through or bypass lanes.

The capacity of a curbside roadway is defined both by the number of vehicles that can be accommodated while stopping to pick up or drop off passengers and the number of vehicles that

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can be accommodated while traveling past the curbside in the through lanes. The capacity of the through lanes is restricted by vehicles that are double-parked (which is often tolerated on airport curbside roadways) or triple-parked as well as by pedestrian crosswalks. These capacity restrictions can cause traffic delays and the formation of queues that block vehicles trying to maneuver around stopped vehicles or attempting to enter and exit curbside spaces. (Additional information on the operating characteristics of curbside roadways is presented in Chapter 5.)

The length (or capacity) of a curbside area must be in balance with the capacity of the through lanes drivers use to enter and exit the curbside area. For example, a mile-long curbside served by only two lanes (one curbside lane and one through lane) would be imbalanced because, even though the curb length could accommodate a large number of vehicles, traffic flow in the single through lane would be delayed every time a vehicle maneuvered into and out of a curbside space or double-parked while waiting for an empty space. The reverse analogy or imbalance would occur with a very short curbside area and multiple through lanes.

Other operating characteristics of airport curbside roadways that differ from public roads, as further described in Chapter 5, include the following:

- **Dwell times.** The length of time a vehicle remains stopped at the curbside area is referred to as "dwell time." Generally, vehicles transporting a larger number of passengers and baggage require longer dwell times. The number of vehicles that can be accommodated along a given curbside length is determined by the size of the vehicles (i.e., the length of the stall each vehicle occupies, including maneuvering space in front of and behind the vehicle) and the amount of time each vehicle remains stopped at the curbside (i.e., the dwell time). Dwell times at a particular airport are affected by (1) enforcement policies (i.e., strict enforcement leads to shorter dwell times) and (2) local driver behavior (i.e., do drivers double-park in a way that allows other motorists to easily enter and exit the lane immediately adjacent to the terminal).

  At most airports, private vehicles dropping off passengers typically have shorter dwell times than those vehicles picking up passengers (unless motorists are strictly prohibited from parking at the curbside while waiting for the passengers who have not yet arrived at the curbside). Thus, although airports generally have equivalent volumes of enplaning and deplaning airline passengers (and associated traffic volumes), the required capacity or length of an arrivals (pickup) curbside area is typically greater than that of the departures (drop-off) curbside area.
- **Maneuvering traffic and parking preferences.** Unlike motorists on city streets, motorists parallel parking at airports rarely back into a curbside space. Motorists frequently stop with their vehicles askew to the travel lanes or sidewalk areas rather than maneuvering their vehicles into positions parallel to the curbside. By doing so, they may block or interfere with the flow of traffic in other lanes. Motorists leave space between successive vehicles to ensure that they are not blocked and to allow access to their trunk or baggage storage area.

  Motorists using airport curbside roadways may stop in the second lane even if there is an empty space in the curbside lane to (1) avoid being blocked in by other motorists and (2) reduce the walking distances of passengers being dropped off (e.g., stop near a desired door or skycap position) or being picked up (e.g., stop at a point near where the person is standing or expected to exit the terminal building). Thus, motorists frequently stop in the second lane in front of the door serving the desired airline even though there may be an empty curbside space located downstream. The propensity to avoid inner lanes and double-park reflects local driver behavior or courtesy.
- **Capacity of adjacent through lanes.** Through-lane capacity is reduced by traffic entering and exiting curbside spaces, high proportions of vehicles double- and triple-parking, the use of the maneuver lanes, and other factors. As such, the capacity analysis procedures presented in the HCM are not applicable. Chapter 5 of this Guide presents suggested methods for calculating the capacities of curbside lanes and through lanes at airports.

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- **Uneven distribution of demand.** Curbside demand is not uniformly distributed during peak periods, reflecting (1) airline schedules and (2) the uneven distribution of the times passengers arrive at the enplaning curbside prior to their scheduled departures (lead time) or the times passengers arrive at the deplaning curbside after their flights have landed (lag times). Furthermore, stopped vehicles are not uniformly distributed along the length of a curbside area, reflecting motorist preferences for spaces near specific doors and skycap positions and their aversion to spaces near columns or without weather protection, if weather-protected spaces are available.

  An aerial view of a busy terminal curbside area would show vehicles stopped adjacent to the door(s) serving major airlines. The airline with the largest market share generally will get its first choice of ticket counter and baggage claim area locations. Often, this airline selects the most prominent location, which generally is the area nearest the entrance to the curbside area. Thus, curbside demand is often heaviest at the entrance to the curbside area, causing double-parked vehicles and congestion in this area, while downstream areas remain unoccupied.
- **Allocation of space for commercial vehicles and other uses.** At most airports, curb space is allocated to commercial vehicles at the pickup curbside area. In the allocation of commercial vehicle curb space, multiple factors must be considered in addition to calculated space requirements, such as customer service, operational needs, airport policies, revenues, and perceived or actual competition among ground transportation services. Curb space may also be allocated for disabled parking, police vehicles, airport vehicles, valet parking drop-off/pickup, tow trucks, and other users.
- **Allocation of private vehicle traffic to multiple curbside roadways.** At airports having inner and outer curbside roadways, one curbside roadway is generally reserved for private vehicles and the other curbside roadway(s) is reserved for commercial ground transportation and other vehicles. It is beneficial to balance demand among all curbside roadways, but in practice, it is difficult to do so. This is because, if given a choice of terminal building curbside areas, private vehicle motorists—especially those unfamiliar with the airport—prefer to use the curbside area they perceive as being most convenient. Few motorists will respond to suggestions to use an alternate, less congested curbside area. For this reason, it is unusual to have multiple private vehicle drop-off (or pickup) curbside roadways. However, it is fairly common to direct commercial vehicles to multiple curbside areas.
- **Crosswalk location, frequency, and controls.** Crosswalks provide for the safe movement of pedestrians between the terminal building and center-island curbside areas or a parking facility located opposite the terminal building. The use of crosswalks can be encouraged—and crossing between designated crossing locations discouraged—by providing numerous crosswalks at convenient (e.g., closely spaced) locations and/or fences or other barriers to pedestrians along the outer island.

  However, providing multiple crosswalks has the potential to have an adverse effect on the flow of through traffic, thus increasing overall delay for arriving and departing motorists. Motorists are often required to stop at more than one crosswalk because traffic controls at the crosswalks (whether traffic officers or signals) may not be coordinated in such a way as to allow a continuous flow of through vehicles, such as what commonly occurs on an urban street. Multiple crosswalks also reduce the available length of curb space. A single crosswalk has less impact on through traffic and available curb length than multiple, unsignalized crosswalks, although multiple crosswalks are more convenient.
- **Curbside lane widths.** At most airports, curbside roadway lane widths are the same as those on public streets (e.g., 10 to 12 feet). Recognizing the tendency of drivers to double-park, some airport operators have elected to delineate one double-wide (e.g., 20 to 24 feet) curbside lane rather than two adjacent 10- to 12-foot lanes. (See Figure 2-4.)
- **Availability of short-duration parking.** Curbside demand can be influenced by the availability and price of conveniently located, short-duration (e.g., hourly) parking. If such parking is available, easily accessible, and reasonably priced, some motorists may choose to park and thus reduce

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Figure 2-4. Double-wide curbside lane at Washington Dulles International Airport.

the volume using the curbside areas. Similarly, the availability of cell phone waiting areas can help to reduce the volume of vehicles circulating past the pickup curbside areas and reduce average curbside dwell times. Conversely, the perceived lack of empty space, inconvenient access, and high costs for short-duration parking can discourage motorists from parking. A lack of empty spaces and high costs can increase curbside demands.

- **Multi-terminal airports.** Large airports may have multiple terminals, each with separate curbside areas, or with continuous curbsides that extend between terminal buildings. Curbside operations at each terminal may differ, reflecting the characteristics of the dominant passenger groups and airlines (e.g., international vs. domestic passengers or legacy vs. low-cost carriers).
- **Recirculating or bypass traffic.** At many airports, there is a significant proportion of non-stopping or bypass traffic on the terminal curbsides. This bypass traffic includes (1) recirculating traffic that, because of police enforcement or other reasons, passes the terminal curbside (particularly the deplaning curbside) more than once; (2) curbside traffic destined for a terminal or adjacent curbside section, which must bypass another curbside area on the way; and (3) non-curbside traffic traveling past the curbside (e.g., cut-through vehicles, employee vehicles, or airport service or maintenance vehicles). At some airports, large volumes of non-airport traffic use airport roadways to avoid congestion on parallel off-airport roads, thus contributing to peak-period roadway volumes and congestion.
- **Non-standard curbside configurations.** While most airports have linear curbsides where vehicles stop bumper-to-bumper or nose-to-tail, a few airports have non-standard curbside configurations such as the following:
  - **Pull-through private vehicle spaces.** As shown in Figure 2-5, the curbside areas at some overseas airports (as well as at some U.S. airports, e.g., St. Louis Lambert International and Little Rock National airports) have (or had) pull-through spaces arranged at 45-degree angles that allow motorists to pull through, similar to the way they would at a drive-through window.
  - **Angled commercial vehicle spaces.** The commercial vehicle curbside areas at the airports serving Atlanta, Newark, and Orlando, among others, have angled spaces that require vehicles to back up to exit.
  - **Driver-side loading.** As shown in Figure 2-6, at a few airports (e.g., Bush Intercontinental Airport/Houston and Mineta San Jose International Airport), the deplaning curbsides are located on the driver's side of the vehicle, requiring private vehicle passengers to open the door and enter or exit the vehicle on the side away from the terminal building while standing in

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Figure 2-5. Pull-through spaces at Brussels International Airport.

a traffic lane. Driver-side loading is used at some airports for taxicabs and TNCs because passengers may enter the cabin from either side of the vehicle.

- **Brief parking zones—pay for curbside use.** Some European airports do not provide free curb space but instead provide parking areas adjacent to the terminals that motorists can use for a fee. These areas can be configured parallel to the curbside (see Figure 2-7) or in a traditional parking lot adjacent to the terminal building (see Figure 2-8). In Europe, unattended vehicles are permitted in these zones, but in the United States, current security regulations prohibit unattended vehicles at the terminal curbsides.
- **Supplemental curbsides.** Some airports provide supplemental curbsides in or near parking structures or at remotely located sites. The airports serving Los Angeles and St. Louis have private vehicle curbside areas within parking structures (see Figure 2-9). The airports serving Boston, Las Vegas, San Francisco, and Seattle have commercial ground transportation vehicle curbside areas within parking structures.

The analytical procedures described in this Guide are most relevant for airports with traditional curb space configurations or layouts because of the differing dwell times and through-lane operations that occur with other configurations.



Figure 2-6. Driver-side loading at Norman Y. Mineta San Jose International Airport.

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Figure 2-7. Brief curbside parking zone at Munich Airport.



Figure 2-8. Pay-for-use curbside at Charles de Gaulle Airport.



Figure 2-9. Supplemental curbside in P3 garage at Los Angeles International Airport.

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2.4 Overview of Analytical Framework Hierarchy

Subsequent chapters of this Guide present alternative methods for analyzing airport roadways, weaving sections, and curbside areas, recognizing the unique characteristics of these facilities. The alternative analysis methods or hierarchy differ in terms of (1) the level of effort or time needed to conduct the analysis, (2) the expected level of accuracy or reliability of the results, and (3) the necessary level of user skill or experience. The three methods—quick-estimation methods, macroscopic methods, and microsimulation methods—are described in the following paragraphs.

2.4.1 Quick-Estimation Methods

Quick-estimation methods, as the name suggests, can be used to produce preliminary analyses of roadway operations (or other facilities) simply and rapidly. They generally consist of look-up tables, simple formulas based on regression analysis of databases, or principles based on experience or practice, and are based on broad assumptions about the characteristics of the facility being analyzed. As such, they provide a first test of the ability of a roadway or other facility to properly accommodate the estimated requirements (existing or future) or the adequacy of a potential improvement measure.

Quick-estimation methods are ideal for quickly sizing a facility. The analyst can easily check which of many possible roadway design options is sufficient to serve the forecast demand. These methods, however, are less than satisfactory for estimating the operating performance of a given roadway or for refining a given design. If information on the actual performance of a given facility or how to refine a particular design is desired, then macroscopic methods should be used.

2.4.2 Macroscopic Methods

Macroscopic methods are used to consider the flows of vehicle streams, rather than the flows or operations of individual vehicles. The HCM and *NCHRP Report 825: Planning and Preliminary Engineering Applications Guide to the Highway Capacity Manual*, are examples of a set of macroscopic methods for evaluating roadway operations. As such, these methods approximate the interactions between individual vehicles, the behavior of individual drivers, and detailed characteristics of the roadways (or other facilities). Adjustment factors, typically developed through empirical observations or microsimulation methods, are often used to account for atypical vehicles or driver characteristics, traffic flow constraints, or other operational characteristics. These methods produce results that are considered acceptable and more accurate than quick-estimation methods and can be used with less effort and training than microsimulation methods.

Macroscopic methods can provide reliable estimates of the steady-state performance of a roadway averaged over a given analysis period. They are best for determining the refinements to a proposed design (or existing facility) that would eliminate capacity and congestion problems. These methods are less satisfactory for quantifying facility operations under heavy congestion conditions or on atypical roadway configurations.

Macroscopic methods are generally unsatisfactory for comparing alternative improvements that reduce but do not eliminate congestion. Under heavily congested conditions (hourly demand exceeding capacity), queuing vehicles from one part of the roadway affect both upstream and downstream operations in a manner that cannot be easily estimated using macroscopic methods. Macroscopic methods also cannot be used for unusual facility types or situations for which they were not designed. In those situations, microsimulation methods must be used.

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2.4.3 Microsimulation Methods

Microsimulation methods consist of the use of sophisticated computer programs to simulate the operation of individual vehicles on simulated roadway networks. Each vehicle is assigned characteristics, such as a destination, performance capabilities, and driver behavior. Each roadway network is defined using characteristics such as number, length, and width of lanes; operating speeds; traffic controls; and pedestrian activity. As each simulated vehicle travels through the computerized roadway network, various aspects of its performance can be recorded based on its interaction with other vehicles and traffic controls. These performance statistics can be summarized in many ways, including commonly used performance measures, such as travel time and delays, travel speeds, and queue lengths. Also, some microsimulation models produce a visual display of the simulated roadway operations, which can be helpful when evaluating operations or presenting results.

Of the three methods for analyzing airport roadway conditions, microsimulation methods are the most complex and require the most effort and skill on the part of the user, but they also can produce the most detailed and reliable results. The use of microsimulation methods is suggested when macroscopic methods do not yield reasonable results, do not provide sufficient detail, or when the conditions being analyzed are outside the ranges addressed by macroscopic methods.

Additional information regarding the application of these three analysis methods is presented in subsequent chapters of this Guide.

2.5 Overview of Capacity and Quality-of-Service Concepts

The concepts of capacity and quality of service, as presented in the HCM, are fundamental to analyses of roadway and other transportation facilities and well understood by traffic engineers and transportation planning professionals. This section is intended to provide an overview of capacity and quality-of-service concepts for users not familiar with the HCM.

2.5.1 Capacity Concept

The capacity of a rectangle or a box can be easily defined by its size (i.e., its area or volume) as the maximum amount the object can accommodate is fixed. This is not true with objects that serve as "processors," such as roadways, ticket counters, or runways. The capacity of a roadway, for example, depends not only on its size (e.g., the number of lanes and other geometric design aspects), but also on the characteristics of the vehicles using the roadway (e.g., their size, performance, spacing, speed, and many other operating characteristics). If all the vehicles on a roadway had identical sizes, distances apart, speeds, performance capabilities, driver skills, and other characteristics, then the capacity of the roadway (number of vehicles traversing a point or section during a unit of time) would be expected to be substantially higher than the capacity of the same roadway if it were serving a mix of vehicles with differing sizes, speeds, capabilities, and driver characteristics.

Accordingly, the capacity of a roadway—even roadways with the same number of lanes—varies based on the characteristics of the roadway (e.g., lane and shoulder widths, level or rolling terrain, intersection and driveway spacing, and traffic control types) and the characteristics of the vehicles and drivers using the roadway (e.g., the proportion of buses, single-unit or tractor-trailer trucks, daily and hourly variations in use, and familiarity of the typical drivers with the roadway). With knowledge of the characteristics of a roadway section and the vehicles (and drivers) using the roadway, it is possible to calculate its capacity—the maximum hourly rate of vehicles flowing past a point.

However, it is not possible or desirable for a roadway to operate at its capacity for sustained periods, as any minor disruption will cause congestion, which results in delays or lengthy queues and undesirable levels of safety and driver comfort. Also, demand rarely is sustained at a constant level over an hour; it frequently varies within the hour. Thus, roadway capacity, while stated in

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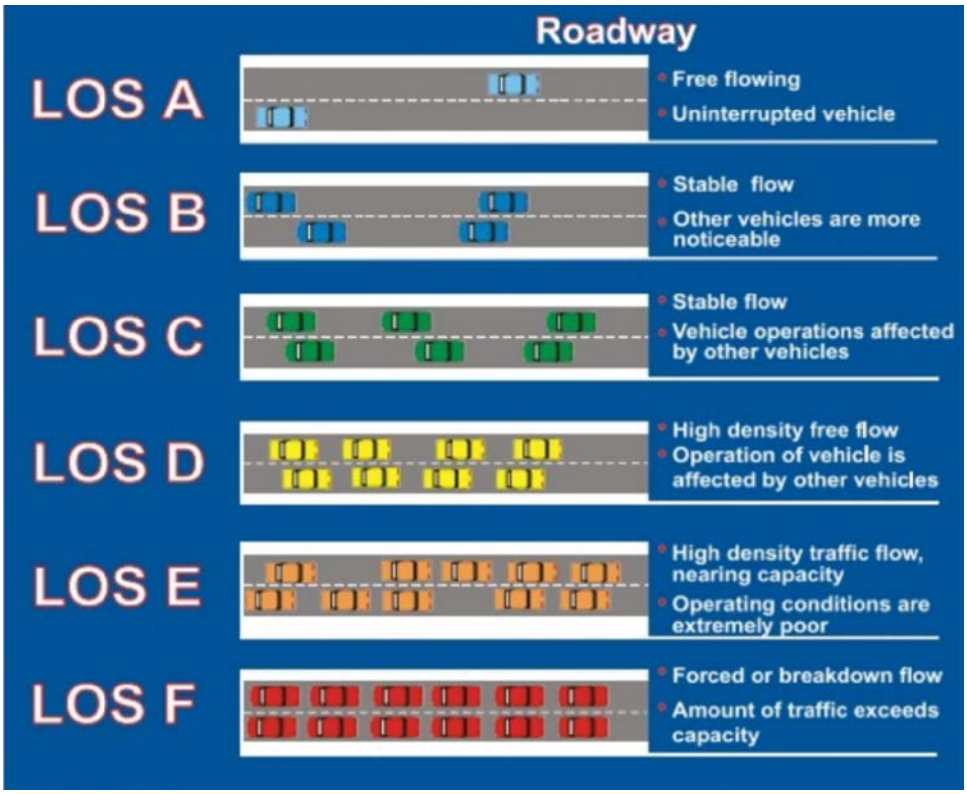
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terms of "base" vehicles (e.g., passenger-car equivalents) per hour, is sometimes computed for only the peak 15-minute flow rate within that hour. In addition, roadway operations are characterized in terms of quality-of-service measures and service flow rate—the maximum flow rate that can be accommodated while maintaining a desired quality of service.

2.5.2 Quality-of-Service Concept

Quality of service is a qualitative measure of how travelers perceive the performance of freeways, urban streets, and signalized and unsignalized intersections. Quality of service is often stratified quantitatively into levels of service indicated using the acronym LOS (e.g., the A-to-F LOS scale common within the HCM, as shown in Figure 2-10). However, the concept is broader than levels of service and can include system-oriented metrics such as the sufficiency of the roadway to provide adequate capacity. Further detail on levels of service can be found in Chapter 5, "Quality and Level-of-Service Concepts," in the HCM.

Performance measures from the perspective of the system provider are also useful in characterizing the performance of roadways. One of these measures, sufficiency, is particularly well suited to airport terminal roadways where the complexities of the travel environment preclude accurate prediction of service measures such as delay. Quality of service is defined in terms of parameters that can be perceived by the users of a transportation facility and that can be measured and predicted. Common performance measures used to characterize quality of service include (1) the density of the traffic flow (passenger cars per mile per travel lane) for a freeway or other unsignalized multilane roadway and (2) control delay (seconds per vehicle) for signalized and unsignalized intersections. Sufficiency is often characterized by capacity-related performance measures such as the volume-to-capacity or demand-to-capacity ratio.



Source: Maryland Department of Transportation

Figure 2-10. Roadway LOS scale.

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2.5.3 Acceptable Performance for Airport Roadways

As noted, performance measures are typically used to determine if a roadway can properly accommodate existing or future traffic operations or to compare alternative improvement options. As noted previously, the metric of "sufficiency" is recommended for airport roadways. On airport roadways, a sufficiency of "under capacity" is typically considered to be the minimum "acceptable" level of performance because of the lack of alternative travel paths, the significant negative consequences resulting from travel delays (e.g., passengers missing their flights), and the ability to accommodate higher volumes during short periods and unusually busy days (such as during holidays) without reaching failure. In comparison, on regional freeways and arterials and in densely developed urban areas, a sufficiency of "near capacity" is often considered "acceptable" because motorists traveling on regional roadway networks can select alternative travel paths should their preferred path be congested. These metrics are discussed further in the following paragraphs and in detail in Chapter 5.

2.5.4 Commonly Used Definitions for Planning Applications

NCHRP Report 825: Planning and Preliminary Engineering Applications Guide to the Highway Capacity Manual provides a guide to the assessment of intersection capacity and the use of sufficiency as a performance measure. It is proposed that similar performance measures be used for the assessment and planning of airport curbsides and terminal area roadways. The sufficiency categories proposed for airport curbsides and terminal area roadways are as follows:

- **Under capacity**—Operating beneath the capacity of a roadway or curbside. All vehicular traffic demands can be accommodated, delays are low to moderate, and maneuverability is acceptable. For comparison with the level-of-service grades used in the HCM, this is the equivalent of LOS A through LOS C. As previously noted, it is proposed that airport curbsides and roadways be planned and designed for a sufficiency of under capacity.
- **Near capacity**—Operating near the capacity of a roadway or curbside. Demand is approaching the roadway capacity, delays are moderate to high, and maneuverability is somewhat restricted. For comparison with the level-of-service grades used in the HCM, this is the equivalent of LOS D. As previously described, when analyzing existing operations or designing new facilities, especially at large-hub airports, airport management may elect to accept near capacity depending on customer expectations, incremental costs, expected remaining useful life, and other factors, realizing that the service level experienced may decrease to at capacity or over capacity on some peak days of the year.
- **At capacity**—Operating at or very close to the capacity of a roadway or curbside. Demand is near or at the roadway capacity, delays are high, and maneuverability is highly restricted. For comparison with the level-of-service grades used in the HCM, this is the equivalent of LOS E.
- **Over capacity**—Operating over the capacity of a roadway or curbside. Demand for roadways cannot be accommodated, delays are high and growing, maneuverability is highly restricted, and queues extend back from the terminal area. For comparison with the level-of-service grades used in the HCM, this is the equivalent of LOS F. Regularly occurring operations that are over capacity are not acceptable to airport users.

The International Air Transport Association (IATA) defines four comparable sufficiency categories (or level-of-service classifications) for analysis of passenger terminal facilities: over-design, optimum, sub-optimum, and under-provided. IATA recommends that airport planners strive to reach the optimum range for each passenger terminal facility to strike a reasonable balance between service quality and costs.

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CHAPTER 3

Estimating Airport Roadway Requirements

This chapter presents methods for estimating existing and future airport roadway traffic volumes or requirements. The data needed to analyze existing roadway traffic volumes and operations are described, and two alternative methods for estimating future roadway traffic volumes are presented. One method, the traditional four-step approach commonly used by transportation planners, incorporates estimates of the roadway traffic volumes generated by airline passengers, visitors, employees, air cargo handlers, and major airport land uses. This method requires extensive data describing the characteristics of each of these traffic generators. The second method, the growth factor method, yields acceptable, but less precise, results while requiring much less input data. However, this simpler method is less sensitive to changes in future conditions or travel patterns.

3.1 Establishing Existing Airport Roadway Traffic Volumes and Patterns

Analyses of existing conditions and estimates of future conditions are based on observed vehicular activity. Details describing the volume, mix, and pattern of roadway traffic can be determined using data recently gathered for other purposes, if suitable, or by conducting special-purpose surveys. Additional information about traffic surveys can be found in the ITE *Manual of Transportation Engineering Studies*, *ACRP Research Report 235: Guidebook for Conducting Airport User Surveys and Other Customer Research*, and other references listed in the bibliography provided in Appendix B to this Guide.

3.1.1 Methods of Determining Existing Roadway Activity

The initial step in planning a special-purpose traffic survey is to determine what data—traffic volumes, roadway operations, vehicle characteristics, or other data—are required to support the planned analysis, whether analyzing existing operations or forecasting future operations. The traffic data required to support analyses of existing and future conditions include

- **Traffic volumes.** The volume of vehicles, including non-motorized vehicles, using a roadway or roadway segment forms the basis for analyses of traffic operations and for determining roadway requirements. Documentation of traffic volumes on all major airport roadways may be required to conduct master plans and comprehensive landside analyses, while the traffic volumes on only a few roadway segments (e.g., a curbside roadway) may be needed to conduct a more focused analysis.

  Traffic volumes are normally recorded in 15-minute increments to identify the peak period of activity on individual roadway segments. If peak airport traffic periods are known, it may be possible to record the traffic volumes during a 3-hour window that includes this peak period rather than conducting day-long, 48-hour, or 7-day surveys.

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- **Motorized vehicles.** Most airport roadway traffic is comprised of motorized vehicles such as private vehicles, taxicabs, TNCs, shuttles, courtesy vans, and buses. As such, analyses of airport roadways—including access, circulation, and curbside roadways—focus on the volume, mix, and operations of these vehicles.
- **Non-motorized vehicles.** Bicycles, scooters, and other active modes typically represent a very small proportion of airport roadway traffic. While bicycles or other active modes have historically been used rarely by airline passengers or visitors, they are used by a small and increasing number of employees for their journey to work or for their last mile of travel (i.e., from a transit stop to their place of work). The ITE *Manual of Transportation Engineering Studies* and other sources describe best practices for conducting surveys of bicycles and other active modes.
- **Pedestrians.** The volume of pedestrians, such as those on the sidewalks adjacent to a terminal or crossing a curbside roadway, forms the basis for analysis of circulation needs/sidewalk capacities, crosswalk operations, and the need for grade-separated roadway crossings. Like vehicle traffic volumes, pedestrian traffic volumes are normally recorded in 15-minute increments. The surveys may also monitor the amount and types of baggage accompanying these pedestrians.
- **Transit services.** The number of transit riders boarding or alighting from bus or rail at an airport can often be obtained from the transit operator. Supplemental surveys may be required to determine the amount and types of baggage accompanying transit riders using an airport bus stop/rail station and whether the riders are airline passengers or employees working at the airport.
- **Vehicle mix.** In an airport environment, vehicle mix (or vehicle classification) refers to the portion of the traffic volume accounted for by individual modes, as defined by the type of service each mode provides (e.g., taxicab, TNC, courtesy vehicle, charter bus) and the type of vehicle used (e.g., sedan, passenger van, minibus, full-size bus). The proportion of large or heavy vehicles impacts roadway capacities and curbside operations.
- **Vehicle occupancies.** The number of passengers in vehicles (or vehicle occupancies) is used to determine the proportion of vehicles transporting multiple passengers (e.g., high-occupancy vehicles) and using the roadways efficiently. In theory, vehicle occupancies can be determined by observing or recording the number of passengers (excluding the driver of commercial vehicles) in passing vehicles. However, in practice, it is difficult to observe the number of occupants in a passing vehicle, particularly large vehicles, or those with tinted windows.
- **Dwell times.** The length of time a vehicle spends stopped in a curbside lane (or other location) while loading or unloading passengers and their baggage is referred to as its dwell time. Typically, the dwell time is defined as the length of time between when the driver parks (i.e., the vehicle comes to a complete stop) and when the driver first attempts to rejoin the traffic stream. It does not include any time during which the driver may be ready to depart but is prevented from doing so by other vehicles. For some analyses, it is also helpful to measure "active" dwell times (i.e., the length of time a vehicle remains at a curbside while actively loading/unloading passengers and their baggage) as opposed to the "total" dwell time, which reflects the time difference between when a vehicle first stops at a curbside until it leaves the curbside. Dwell time data are required to analyze curbside roadway operations.
- **Curbside occupancies.** This is the number of vehicles occupying or stopped at a curbside roadway or roadway section and is an indication of the number of double- or triple-parked vehicles and which portions of the available curbside are used by motorists. These data can be determined through observations of real-time operations or video recordings of curbside operations.
- **Queue length.** Queue length is the distance of, or number of vehicles in, a line of vehicles waiting to proceed along a roadway where (1) the flow rate of the front of the queue determines the average speed within the queue and (2) the rate of vehicles arriving in the queue is greater than the rate of vehicles leaving the queue. Queues form when a group of vehicles is delayed because of downstream congestion or bottlenecks. The length of a queue can be measured by observing, at fixed intervals, the length of slow-moving or stopped vehicles, and the time of a

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queue can be measured by observing how long it takes a vehicle to travel from the back to the front of a queue. The number of vehicles in a queue and the duration, or persistence, of the queue, also can be determined through observations. These data are used to support evaluations of airport roadway operations, particularly at the entries or exits of parking and other facilities.

- **Travel speeds.** Average travel speeds can be measured by recording the time it takes random vehicles to travel a known distance, such as between two fixed objects or points. Average travel speeds—particularly along a roadway segment having a length of 1,000 feet or more—can be used to support evaluations of airport roadway operations. Measuring instantaneous speeds (also known as spot speeds) is not useful in airport roadway analyses because the speeds of individual vehicles tend to vary significantly on the roadway network.
- **Weaving patterns.** The proportion and volume of weaving traffic (i.e., traffic crossing two or more traffic streams without the aid of a traffic signal or other control device) can be determined by manual surveys of traffic entering and exiting a weaving segment or observing video recordings of a weaving segment.
- **Recirculating traffic volumes.** The proportion of vehicles passing the curbside or other location multiple times can be determined by recording and matching license plate numbers or other methods of identifying individual vehicles (e.g., monitoring the Bluetooth signal of mobile phones inside passing vehicles).
- **Other data.** Depending on the nature of the traffic operations problem being addressed, intercept and other surveys of airline passengers, visitors, and employees are used to gather data such as vehicle mode-choice patterns; passenger arrival patterns; passenger regional approach/departure routes; place of trip origin/destination; use of airport parking facilities; and, for employees, work schedules and location of places of work at the airport. *ACRP Report 26* provides guidelines on the conduct of surveys of airline passengers, visitors, and employees.

The data needed to support analyses of airport roadway and curbside operations are most frequently gathered using manual surveys or automatic traffic recorders. In some communities, the required traffic volume data—especially the volumes on airport access roads or adjacent highways—can be obtained from local public works or traffic engineering departments. Increasingly, manual surveys are being supplemented by transportation analytics, with the expectation that in the near future these sources will reduce the extent of required manual traffic surveys. Examples of the data now available using transportation analytics and the potential application of these data for analyses of airport roadway operations include the following:

- **Video recording.** Video recordings of airport roadway traffic, after subsequent manual extraction, can furnish data on pedestrian and vehicle volumes, vehicle mix, and turning and weaving movements. Video recordings can be obtained from cameras installed temporarily to record specific roadway operations or permanently mounted airport-controlled security cameras. It is anticipated that ultimately the extraction of data will be automated, reducing the time and costs of gathering the desired information. Currently, private vendors offer data obtained from videos defining the trajectories of individual vehicles through an intersection or roadway segments, which can support roadway safety analyses and turning movement counts, among other analyses. When available, such data will be useful in analyzing curbside roadway operations.
- **Bluetooth sensors.** The signals emitted from the mobile telephones, smart watches, fitness devices, and other personal devices worn or carried by passengers in a vehicle, and even signals emitted by the vehicles themselves (e.g., tire pressure sensors) can be detected as vehicles pass sensors located adjacent to a roadway or other site. By matching the signals at multiple points, the speeds, travel paths, or routes of individual vehicles can be determined. It is estimated that 20% to 30% of the vehicles passing a sensor can be detected, leading to reliable samples. At airports, the use of Bluetooth sensors can help determine travel speeds (or delays) on the primary access roads and the proportion of non-airport (or cut-through) traffic using airport roadways.

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- **Connected vehicles.** With an expected increase in the proportion of connected vehicles (i.e., vehicles capable of communicating and sharing data with roadway infrastructure and other vehicles), it will be possible to track and aggregate the speeds, routes, dwell times, turning movements, and other data obtained from individual vehicles. Currently, one vendor offers detailed data (e.g., use of seat belts and headlights, as well as vehicle paths, speeds, and points of origin and destination) obtained from vehicles equipped with General Motors' OnStar service. It is estimated that this sample now represents less than 1% of the vehicles on the road. As the sample increases, it is expected that data obtained from connected vehicles may replace Bluetooth data because of the superior granularity and availability of the data.
- **Vehicle detectors.** Induction loops, video detection systems, and microwave sensors, embedded in roadway pavements or mounted overhead, produce accurate counts of traffic volumes and can detect the size/type of individual vehicles. Permanent detectors are frequently installed along major highways and at intersections by state and local transportation agencies. The resulting data are used to monitor traffic volumes, speeds, and patterns and determine hourly, daily, and annual use of roadways. Permanent detectors have been installed along the access roads of some airports (e.g., Boston Logan International Airport) to monitor traffic volumes and patterns. Temporary vehicle detectors (e.g., video or microwave sensors) are regularly used to gather traffic volume and speed data.
- **Cell tower data/probe vehicles.** Origin-destination data (e.g., trips from one zip code to another) are available from vendors that obtain and summarize data generated by trucks and other vehicles as they pass between cell towers. At present, these data are suitable for regional or corridor planning analyses, but they are not yet considered sufficiently granular or accurate to support analyses of airport roadway operations. FHWA provides free access to the National Performance Management Research Data Set (NPMRDS) data obtained from individual sample or probe vehicles, intended for transportation system performance management and to monitor and report congestion and freight performance. Again, these data are not yet sufficiently detailed to support analyses of airport roadway operations.
- **Parking access control systems.** Modern parking access control systems report the hourly (or smaller time increment) volume of entering and exiting vehicles by individual parking facility entry/exit lane or access point. At airports where many passengers and visitors park, these data can be used to analyze seasonal and daily patterns of airport roadway traffic and to identify peak periods of roadway traffic.

3.1.2 Selecting Survey Dates

Ideally, the traffic volume and curbside surveys should be conducted during the peak hours on a typical busy day (ideally during a peak month). Typically, the peak days occur during the months with the largest volumes of airline passenger traffic. At many airports, the busiest days are Mondays and Fridays, but at some airports—especially those serving large volumes of non-business passengers—the busiest days may be Sundays. When it is not possible to conduct a survey during the busiest month, traffic volume data need to be adjusted to replicate peak-month conditions based on monthly airline passenger data (particularly originating and terminating passenger volumes), public parking transactions, volumes of passengers passing through security screening positions, or other indicators of traffic activity.

3.1.3 Selecting Survey Hours

The peak hours for roadway traffic precede the peak hour for originating airline passenger departures and follow the peak hour for terminating airline passenger arrivals. Peak-hour traffic volumes can be determined by recording the number of vehicles on the roadway by type of vehicle (for curbside surveys), recording the number of vehicles on the roadway during each

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15-minute increment, and then either identifying the four consecutive 15-minute increments with the largest traffic volumes or the busiest 15-minute increment. It is suggested that surveys of the departures area (passenger drop-off area) roadways be conducted during the 3 hours prior to and including the 60-minute period with the most departing seats and that surveys of the arrivals area (passenger pickup area) roadways be conducted during the 3 hours including and after the 60-minute period with the most arriving flights. The 60-minute departures and arrivals flight peaks do not necessarily coincide, as noted previously. At airports serving high volumes of connecting passenger traffic, roadway peak hours may not coincide with the airport's busiest hours for departing and/or arriving flights. In such situations, existing vehicle data (such as commercial ground transportation trips recorded by the airport's automatic vehicle identification system) or TSA security checkpoint screening data may be used to identify potential survey periods.

3.2 Estimating Future Airport Roadway Traffic Volumes—Traditional Four-Step Approach

Developing a comprehensive estimate of future traffic volumes on airport roadways using the traditional four-step approach involves the following:

- **Step 1: Trip generation.** Estimating the traffic volume generated by each on-airport land use during the future airport-wide peak hour(s) as well as the peak hour(s) of activity for each land use.
- **Step 2: Trip distribution.** Determining the points where trips generated by each airport land use enter or exit the airport roadway network.
- **Step 3: Mode-choice analysis.** Analyzing the travel mode-choice patterns of passengers and employees.
- **Step 4: Trip assignment.** Assigning the estimated traffic volumes to the on-airport and regional roadway networks.

In regional transportation planning, the third step—mode-choice analysis—is conducted using sophisticated travel demand forecasting models. These models are used to estimate future mode-choice patterns or changes in existing patterns caused by the introduction of new travel modes (e.g., rail service) or expected changes in passenger travel time or travel cost. Such models are rarely required in an airport setting. A possible exception, triggering the need to analyze future travel mode-choice patterns, could be the introduction of new rail service or a major expansion of existing service, if this service were expected to attract significant numbers of airline passengers or employees who currently travel by private vehicles.

Expected increased use of TNCs, for example, does not warrant developing a mode-choice model. This is because, prior to the introduction of TNCs, those airport passengers choosing TNCs typically traveled in private vehicles, taxicabs, or shuttles, with few if any previously using public bus or rail transit. Increased use of TNCs would be expected to have a minor impact on the total number of vehicles entering/exiting the airport, but instead divert passengers from one type of vehicle to another, a level of detail not suitable for a typical mode-choice model. However, if TNCs do not pick up customers at the curbside, the roadways used by TNCs could differ from those of the vehicles they replaced.

Expected use of AVs could warrant use of mode-choice models if reliable data describing the operations and use of such vehicles were available. At the time this Guide was prepared, reliable data describing the operation and use of AVs at airports were not available. Until data gathered from observations of AVs operating in similar environments are available, it is suggested that the effort required to develop and use mode-choice models to predict the impact of AVs is not justified.

The remaining three steps, those applicable to airport roadway operations, as well as challenges to using this approach, are described in the following.

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3.2.1 Estimating Traffic Volumes (Trip Generation)

The key generators of airport roadway traffic are (1) airline passengers and accompanying visitors, (2) employees working at the airport, air cargo and airmail services, airlines, in-terminal concessionaires, and other building tenants, and (3) airport tenants with service or delivery needs. At most airports, the data required to estimate the volume of traffic generated by airline passengers are more readily available than comparable data for employees, air cargo, or service and delivery vehicles.

Reliable statistics documenting existing monthly and annual volumes of airline passengers and air cargo tonnage and forecasts of airline passengers and air cargo tonnage are available for virtually all commercial-service airports. However, as described in greater detail in subsequent paragraphs, most airport operators have limited-to-no data available on the number of employees working at their airports at any point in time or the types of air cargo shipments (e.g., overnight deliveries, small parcels, international, or other types of freight). As a result, forecasts of traffic generated by airline passengers are often developed in substantially more detail than forecasts of traffic generated by employees, air cargo, or services and deliveries. However, traffic generated by airline passengers may represent less than half of the total (daily) vehicular traffic generated at an airport.

Traffic Generated by Airline Passengers

Estimating the volume of traffic generated by airline passengers requires the following inputs.

Number of Originating and Terminating Airline Passengers. Roadway traffic operations are analyzed considering the peak-hour volume (i.e., the traffic volume occurring during the busiest 60 consecutive minutes). Analyses of airport roadway traffic begin with the hourly numbers of originating and terminating airline passengers (or preferably the numbers occurring in 15-minute increments). Originating and terminating airline passenger numbers (rather than enplaned and deplaned passenger numbers) are used to generate traffic volumes because these volumes exclude those passengers transferring between flights who do not use the airport curbsides or roadways.

Analyses of hour-by-hour airline passenger numbers indicate when the largest numbers of originating passengers, terminating passengers, and total passengers (originating plus terminating) arrive at, or depart from, the airport. Separate analyses of these three peak periods (originating, terminating, and total) are required because peak periods of demand on some roadway segments coincide with the originating passenger peak periods (e.g., the departures curbside area), and some coincide with the terminating passenger peak periods (e.g., the arrivals curbside area). The total peak-period traffic volume may not coincide with the peak period of either the originating or terminating passengers but may instead reflect the busiest overall period at the airport (e.g., the hour with the largest traffic volumes on the airport entry and exit roadways).

At airports with significant numbers of connecting passengers, the peak hours of airline passenger activity may not correlate with the peak hour of roadway traffic volumes. For airports with multiple terminals or multiple large concourses, it may be necessary to gather these hourly data for each terminal or each concourse. Existing originating and terminating airline passenger numbers are available through two Bureau of Transportation Statistics products: the Airline Origin and Destination Survey (DB1B) (https://www.bts.gov/topics/airlines-and-airports/origin-and-destination-survey-data) and Data Bank 28IS - T-100 and T-100(f) International Segment Data, U.S. and Foreign Air Carriers Traffic and Capacity Data (World Area Code) (https://www.bts.gov/browse-statistical-products-and-data/bts-publications/%E2%80%A2-data-bank-28is-t-100-and-t-100f). These data are based on a 10% sample of all airline tickets collected by U.S. airlines. Since foreign flag airlines are not required to participate in this ticket sample, the published originating-terminating airline passenger data may underreport passenger numbers at major international gateway airports.

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Future peak-hour airline passenger numbers are a function of the future flight schedules of each airline, the anticipated size of aircraft operated (i.e., number of seats), and anticipated passenger load factors. Forecasts of airline passengers can be obtained from recent airport master plans, the FAA Terminal Area Forecast (see https://www.faa.gov/data_research/aviation/taf/), and other sources. Master plans may present forecasts of annual or daily airline passenger numbers, as determined using an average day of the peak month or standard busy-day rate. Such forecasts (particularly at small and medium commercial-service airports) may assume that the existing relationship between peak-hour and daily airline passenger numbers will remain constant through the forecast period unless a significant change in airline operations is expected.

Passenger Characteristics. When possible, it is helpful to disaggregate the numbers of originating and terminating airline passengers by trip purpose (business vs. non-business or leisure) and place of residency (local residents vs. visitors or non-residents), rather than just considering the total passenger numbers. This is because airline passenger travel patterns (e.g., vehicle occupancies, circulation, and mode-choice patterns) are a function of the passenger's trip purpose, place of residence, and type of flight (short-haul domestic, long-haul, transborder, overseas, or other). For example, resident travelers are more likely to use private vehicles and park for the duration of their trips, while non-residents are more likely to travel to the airport in rental cars or hotel/motel courtesy vehicles and not use parking facilities. Business travelers are generally more time sensitive and less cost-sensitive than leisure or non-business travelers, which influences their choice of parking facilities, and use and familiarity with airport roadways. Typically, these data are obtained from surveys of airline passengers or from data at peer airports.

Lead and Lag Times. Airline passenger numbers are reported by the airlines according to the time aircraft are scheduled to depart (push away from the gate) and arrive (touch down). Since these times do not coincide with the times motorists enter and exit airport roadways, to analyze airport roadway traffic operations it is necessary to adjust these times to reflect how far in advance of their scheduled flight departure times passengers enter an airport (lead time) and how long after their scheduled flight arrival times passengers exit an airport (lag time). International passengers typically have longer lead and lag times than domestic passengers (because of the 2-hour advance check-in required by most airlines for international travel and the time required by passengers to clear immigration and customs processing). Leisure travelers typically have longer lead and lag times than business travelers (because they are more likely to have checked baggage and allow for an extra time buffer). Typically, these data are obtained from surveys of airline passengers or from data at peer airports. Lead time data may be aggregated to form a representative distribution (sometimes referred to as an earliness distribution). Similarly, a representative distribution of lag times is sometimes referred to as a lateness distribution. The TSA reports passenger earliness distribution times (minutes before scheduled flight departure time) based on the time passengers check their baggage, which can serve as a proxy or indicator of lead time distributions.

Travel Mode Choices. To convert person trips into vehicle trips, it is necessary to first determine the travel modes used by airline passengers (or the percentage of passengers using each available travel mode). Regional transportation planning often considers just two travel modes—private vehicles and public transit—whereas airport roadway planning requires consideration of taxicabs, TNCs, limousines, courtesy vehicles, rental cars, scheduled buses, and other travel modes.

As noted, travel modes are a function of trip purpose and place of residency. Airports serving a large proportion of leisure passengers have distinctly different travel-mode-choice patterns than those serving business markets. However, at most U.S. airports, 70% to 80% of all airline passengers arrive and depart in private vehicles or rental cars. Typically, fewer than 5% to 10% of all passengers use public transportation (e.g., scheduled buses or trains, or door-to-door shared-ride

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vans). The remaining passengers typically use taxicabs, TNCs, courtesy vehicles serving hotels/motels, parking facilities, rental cars (e.g., if the rental car facilities are remotely located), or transportation services that require prior reservations (e.g., limousines, charter or tour buses/vans). Table 3-1 presents the mode-choice patterns for airports serving Grand Rapids, Los Angeles, San Francisco, Seattle-Tacoma, and New York. Using the format shown in Table 3-1, some airline passengers are counted twice (e.g., a private vehicle driver who parks in an economy lot and rides a courtesy vehicle or a rental car customer who also uses a courtesy vehicle).

Vehicle Occupancies. Vehicle occupancies (the number of passengers per vehicle) are used to translate or convert "person trips" by travel mode into vehicle trips. When analyzing airport roadways, vehicle occupancies represent the number of airline passengers in each vehicle (i.e., excluding visitors accompanying airline passengers or the drivers of commercial vehicles). Typically, these data are obtained from surveys of airline passengers (for single-occupancy vehicles, such as private vehicles, taxicabs, TNCs, and limousines) or visual observations for multi-party vehicles, such as courtesy vehicles, buses, and vans. The average occupancy of private vehicles operating at airports is higher than the average occupancy of private vehicles operating on

Table 3-1. Typical vehicle mode choice and occupancies at selected airports—originating airline passengers.

Mode	Grand Rapids[a]	Los Angeles[b]	San Francisco[c]	Seattle-Tacoma[d]	John F. Kennedy[e]	Newark[e]	LaGuardia[e]	Typical Vehicle Occupancy (number of people)
Private Vehicles								
Curbside	42.7%	–	–	42.9%	–	–	–	1.2
Short-term parking	26.5%	–	–	2.0%	–	–	–	1.3
Long-term parking	–	–	–	4.6%	–	–	–	1.3
Off-airport parking	–	–	–	10.6%	–	–	–	1.3
Subtotal (private vehicles)	**69.2%**	**52.8%**	**44.9%**	**60.2%**	**30.3%**	**26.5%**	**35.8%**	
Rental cars	20.3%	–	–	14.3%	3.8%	2.7%	7.0%	1.4
Subtotal	**89.5%**	**52.8%**	**44.9%**	**74.5%**	**34.1%**	**29.2%**	**42.8%**	
Commercial vehicles								
Taxicabs	1.2%	5.8%	4.6%	11.4%	31.5%	44.4%	23.9%	1.4
Limousines	–	4.9%	9.9%	0.6%	–	–	–	1.3
TNCs	7.0%	–	24.8%	–	13.1%	14.4%	15.2%	1.2
Door-to-door vans	–	3.7%	4.9%	3.5%	–	–	–	5.0
Hotel/motel courtesy shuttles	2.1%	21.6%	9.6%	3.8%	7.9%	10.0%	10.1%	3.2
Public transit	0.3%	0.8%	–	5.5%	13.4%	2.0%	8.0%	6.4
Charter/other bus	–	10.5%	1.2%	0.7%	–	–	–	20.5
Subtotal (commercial vehicles)	**10.5%**	**47.2%**	**55.1%**	**25.5%**	**65.9%**	**70.8%**	**57.2%**	
Total	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	

Note: Percentages may not total 100 due to rounding.
[a]Gerald R. Ford International Airport, airport staff, based on July 2019 data.
[b]Los Angeles International Airport, Draft Environmental Impact Report, September 2016.
[c]San Francisco International Airport, 2016–2017 Curbside Congestion Study.
[d]Seattle-Tacoma International Airport, airport staff, based on 2019 data.
[e]Port Authority of New York & New Jersey, 2019 Air Traffic Report.

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public streets (particularly during commute hours) because vehicles at airports typically transport a group of airline passengers rather than just a single occupant.

On-Airport Traffic Circulation Patterns. The locations at an airport where motorists begin or end their trips and the paths they follow vary according to their choice of travel mode (and parking facilities), and the on-airport roadway network configuration. Airline passengers follow numerous travel paths at an airport. For example, a private vehicle driver may enter an airport and then do one or more of the following:

- Go directly to the enplaning (or deplaning) curbside area and then immediately exit the airport (e.g., a motorist dropping off an airline passenger who does not park), or recirculate and return to the curbside (e.g., a motorist attempting to pick up a passenger who was not allowed to remain stopped at the curbside).
- Go first to a cell phone waiting area, proceed to the deplaning curbside to pick up an arriving airline passenger, and then immediately exit the airport.
- Go directly to a parking facility and park for the trip's duration (e.g., a long-term parking patron).
- Go directly to the curbside area, drop off passenger(s), and then continue to a parking facility and park for the trip's duration (e.g., a long-term parking patron).
- Go directly to a parking facility, accompany a passenger into the terminal (or greet an arriving passenger at the baggage claim area), and then exit the airport (e.g., a short-term parking patron).
- After landing at the airport, a passenger could go directly to a parking facility, retrieve his/her vehicle (which has been parked for the trip duration), drive back to the terminal to pick up passengers, and then exit the airport (e.g., a long-term parking patron).

Similarly, rental car customers may go to the curbside area before they drop off rental cars or after they pick up rental cars. After dropping off customers, commercial vehicle drivers may either immediately exit the airport or wait in a holding or staging area, and then recirculate back to the terminal to pick up additional customers. Table 3-2 presents the travel paths and proportion of airline passengers using these paths for a typical large-hub airport. Medium- and small-hub airports have similar patterns, but at these airports, there may be greater use of private vehicles and

Table 3-2. Vehicle circulation patterns.

Travel Mode	Circulation Pattern	Percentage
Private Vehicles		
	Drop off at curb, then exit	31
	Drop off at curb, then park—Hourly, remain	9
	Drop off at curb, then park—Hourly, then exit	4
	Drop off at curb, then park—Daily Parking	7
	Drop off at curb, then park—Economy Parking	4
	Direct to park—Hourly, remain for duration	4
	Direct to park—Hourly, exit immediately	14
	Direct to park—Daily	14
	Direct to park—Economy	9
	Direct to off-airport	4
	Total	**100**
Rental Cars		
	Direct to rental car return	73
	Drop off at curb, then rental car return	23
	Direct to off-airport	4
	Total	**100**
Taxicabs		
	Drop off, then exit	83
	Drop off, then hold area	17
	Total	**100**

Source: Based on information provided by individual airport operators.

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less use of taxicabs, TNCs, limousines, courtesy vehicles, and public transit vehicles and fewer short-term parking patrons, especially if there are few international passengers. Again, these data are typically obtained from surveys of airline passengers.

Peak-Hour Factors. Airport roadway traffic is not uniformly distributed over a typical peak hour or other peak period. At small airports in particular, much larger volumes of traffic may occur during one 15-minute period than during the preceding or subsequent 15-minute period. Peak-hour (adjustment) factors are used to translate non-uniform flows into equivalent hourly flows to allow for the analyses of roadways exhibiting such non-uniform peaks. This translation is required because flow rates of vehicles per hour are used to define roadway capacities and analyze roadway operations. These peak-hour factors can be determined from airport roadway traffic surveys or indirectly from analyses of airline schedules. Traffic volumes generated by airline passengers can be estimated by the following:

- Multiplying the number of originating (or terminating) airline passengers during the peak 60-minute period by the percentage of passengers selecting each travel mode, adjusted using lead (or lag) times, and
- Dividing each volume by the corresponding vehicle occupancy, taking care not to double count the same passengers (e.g., those in courtesy vehicles transporting parking patrons). Exceptions are required for vehicles that may operate on a scheduled basis rather than in direct response to passenger demand (e.g., courtesy vehicles and scheduled buses).

Regression equations that correlate vehicle trips generated to airline passengers, to acres of airport property, or to other measures are provided in *Intermodal Ground Access to Airports: A Planning Guide*, the ITE *Trip Generation Handbook*, and other reference documents. Traffic volume estimates at commercial-service airports developed using such equations are not considered reliable because of the significant differences in the characteristics of each airport, including differences in airline activity peaking patterns and volumes; airline passenger demographics (e.g., trip purpose, place of residency, travel mode preferences); passenger circulation patterns on and off the airport; airport layouts; the availability of parking, public transit, and commercial ground transportation services; and other factors influencing traffic volumes. Furthermore, these metrics ignore trips generated by employees, air cargo, and other on-airport land uses.

Traffic Generated by Visitors

The volume of traffic generated by visitors accompanying departing airline passengers (i.e., well-wishers) and arriving airline passengers (i.e., commonly referred to as meeters and greeters) can be determined by establishing the average number of visitors accompanying each airline passenger or group of airline passengers. The number of visitors accompanying a passenger is a function of the airline passenger's trip destination and/or trip purpose and the demographics of the local community. For example, more visitors generally accompany airline passengers traveling overseas for leisure purposes than those accompanying business passengers traveling on domestic flights. In some cities, passengers are greeted by a large extended family group, rather than one or two persons. Typically, visitors either (1) use only the curbside areas; (2) park (for a short period) while they accompany the airline passenger group to/from the terminal building; (3) park (for a short period) in a parking lot, and, having met the deplaning passenger in the terminal building, return to their vehicle, drive to the curbside area to pick up the passenger, and then exit the airport; (4) wait in a cell phone waiting area for the arrival of a deplaning airline, drive to the curbside area to pick up the passenger, and then exit the airport; or (5) drop off enplaning passengers, park, and then return to the terminal to accompany the passengers to/from the gate (e.g., a passenger with special needs, such as an unaccompanied minor or a disabled passenger). Similar to the lead and lag times for airline passengers, visitor arrival times at the airport can lead or lag from the scheduled aircraft

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departure and arrival times (See Figure 3-1). By far, most visitors travel to and from an airport in private vehicles. They rarely (i.e., less than 5%) use public transportation or other travel modes.

Traffic Generated by Employees

Estimating the volume of traffic generated by airport employees requires the following inputs.

Volume of Employees and Their Work Schedules. On an average day, more than 10,000 people work at many large-hub airports, and more than 1,000 people work at typical medium-hub airports (see Table 3-3). These people are employed by the numerous employers located at an airport, as follows:

- The airport operator, including third-party contractors working directly for the airport operator (e.g., janitorial, parking operators, and bus operators) providing services that have been outsourced.
- The airlines, including flight crew, counter agents and baggage handlers, aircraft maintenance, and other employees who may not be working in the terminal building.
- Third-party service providers contracted by airlines to provide maintenance, ground handling, and other services.
- Concessionaires and other terminal building tenants, such as operators of newsstands, restaurants, and other retail establishments and rental car companies whose employees may not be working in the terminal building.



Source: TSA, August 2020, Planning Guidelines and Design Standards for Checked Baggage Inspection Systems.

Figure 3-1. Sample airport visitor lead and lag time.

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Table 3-3. Estimated number of employees at airports.

Airport	Hub Size	Total Employees[a]	Parking Permits	Estimated Average Daily Employees[b]
Boston Logan International	Large	–	–	14,600
Bush Intercontinental/Houston	Large	–	–	14,406
Chicago O'Hare International	Large	–	–	41,000
Dallas/Fort Worth International	Large	60,000	–	–
Denver International	Large	–	–	17,400
Fort Lauderdale-Hollywood International	Large	14,000	–	4,700
John F. Kennedy	Large	20,000	7,920	–
St. Louis Lambert International	Large	–	–	19,000
Las Vegas McCarran	Large	–	–	8,000
Los Angeles International	Large	–	–	40,000
Minneapolis–Saint Paul International	Large	21,200	–	–
Phoenix Sky Harbor	Large	32,000	16,019	8,000
Ronald Reagan Washington National	Large	16,184	–	–
Salt Lake City	Large	–	–	13,026
San Diego International	Large	–	–	3,000
San Francisco International	Large	12,500	17,425	–
Seattle-Tacoma International	Large	19,100	9,595	11,375
Tampa International	Large	6,000	–	–
Washington Dulles International	Large	19,850	–	–
John Wayne (Orange County, CA)	Medium	6,100	–	1,000
Louis Armstrong New Orleans International	Medium	4,825	–	–
Mineta San Jose International	Medium	4,750	–	–
Oakland International	Medium	–	–	10,500
Omaha Eppley Airfield	Medium	–	–	2,500
Portland International	Medium	14,500	–	5,000
Sacramento International	Medium	–	–	1,500

Source: Based upon information provided by individual airport operators.
[a]Includes badged and unbadged.
[b]Number of people working at the airport on an average day.

- Government agencies, including (at U.S. airports) the FAA, TSA, Customs and Border Protection, Immigration and Customs Enforcement, U.S. Postal Service, and (at some airports) the military.
- Air cargo shippers and forwarders.
- FBOs.
- Construction contractors, including construction workers and subcontractors.

Airport-based employees, particularly those employed by the airlines and cargo handlers, work unusual hours, because all commercial airports operate 365 days per year, and many operate 24 hours per day. Typically, the arrival and departure hours of employees at an airport do not coincide with regional commute hours or with an airport's peak enplaning or deplaning hours. For instance, major shift changes for airline employees often occur between 5 A.M. and 6 A.M. and between 2 P.M. and 3 P.M. Another complicating factor is the presence of flight crews, who may only travel to/from the airport a few days per month. The trips made by flight crews at an origin-destination airport are sporadic, but while on an assignment, they become like passengers at destination airports—requiring courtesy vehicle service or flight crew transportation services (i.e., chartered vans).

Generally, employers are required to report the total number of their employees requiring security badges but do not report the number of employees working on each shift, the starting/ending times of each shift, or the travel modes used by their employees. Other than at airports with transportation management programs or rideshare promotional programs, few airport operators have accurate data indicating the number of individuals working at the airport at any given time of day or the travel modes used by these individuals.

Surveys of the employers located at an airport are necessary to determine the number of people working at the airport, their work schedules, travel modes, and circulation patterns. Without

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such data (or traffic surveys conducted at the entry/exit to employee parking lots), it is difficult to determine the number and pattern of employee vehicle trips.

Employee Travel Mode Choices. As noted, little data are available describing the travel modes used by employees at an airport. Data presented in *ACRP Report 4: Ground Access to Major Airports by Public Transportation* (2008), indicated that, at 14 airports for which data were available, about 98% of all employees working at the airport arrived and departed in private vehicles (with the exception of Boston Logan, Chicago O'Hare, and Denver International Airports).

Employee reliance on private vehicles is a result of (1) employees working non-traditional hours that do not coincide with the operations or the schedules of public transportation, (2) employees residing in locations not well served by public transportation (i.e., outside the central business district), (3) employees working in locations outside of the terminal area that are not well served by public transportation, and (4) the availability of free or very-low-cost parking for employees on airport property.

One indicator of the number of vehicles driven by employees at an airport is the number of parking permits or identification badges issued by the airport operator to these individuals. For example, in 1996, it was determined that 61% of the employees who were issued security badges at Los Angeles International Airport had also been issued parking permits. The surveys indicated that, on a typical day, 29% of all employees were absent due to staff schedules, vacation, illness, or working away from the office. Of those employees traveling to work on a typical day, it was determined that 64% drove alone, 33% participated in a rideshare program, and 3% rode public transit, biked, or walked. The average vehicle occupancy for those individuals traveling to work at Los Angeles International was 1.38 employees per vehicle. Because most of the large employers operate multiple shifts, about 25% of the daily employee-generated vehicle trips occurred during a single hour. These data are similar to those reported at Boston Logan International Airport, where about 40% of all employees are absent on a given weekday and about 25% of those working on a given day arrive between 6 A.M. and 10 A.M.

Employee Circulation Patterns. The use of regional access roads and airport access roads by on-airport employees can be estimated by determining the minimum time path or minimum cost path between their places of residence and place of employment. Place of residence data, summarized at a zip-code level, can be obtained from parking permit applications or databases of airport-issued security badges. The minimum travel routes between these locations and points of access to the airport can be determined using regional planning models, from publicly available website-based mapping programs and services, or by planners familiar with the regional highway network.

Future Employment and Employee Work Schedules. Forecasts of employment and employee trips tend to be imprecise because reliable estimates of future employment generally are not available, and changes in future employment do not correlate well with changes in airline passenger numbers. Historically, planners have estimated future employment assuming that the rate of employment growth represents the average of the rate of growth in airline passenger and aircraft operations numbers. However, anecdotal information suggests that this assumption is no longer correct because the airlines appear to be reducing their numbers of employees in order to improve productivity levels and reduce costs. For example, the increasing share of passengers who obtain their boarding passes via the Internet or check their bags using electronic ticketing kiosks has reduced the need for ticket counter agents. It is suggested that additional research is required to develop methods for estimating the volume of traffic generated by employees at airports.

Sample Results. Using the steps presented above, the employee trip generation rates presented in Table 3-4 were developed as part of the Los Angeles International Airport Master Plan

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Table 3-4. Example of vehicle trips per employee working at Los Angeles International Airport.

	Employee trip generation rate (vehicle trips per employee)			
	Daily	Morning peak (8 A.M. to 9 A.M.)	Airport peak (11 A.M. to 12 P.M.)	Afternoon peak (5 P.M. to 6 P.M.)
Inbound	0.59	0.15	0.03	0.01
Outbound	0.59	0.01	0.03	0.15

Source: Leigh Fisher Associates, January 1996, using Los Angeles World Airports' rideshare database representing a typical weekday, Los Angeles International Airport Master Plan—Phase I, On-Airport Existing Transportation Conditions.

Update. These data are presented as an example of how employee trip generation rates can vary for a day or over specific hours, and this example is not intended as a suggested proxy for another application.

Traffic Generated by Air Cargo

Air cargo (including airmail) traffic includes the trucks transporting the cargo, the private vehicles driven by the employees in the air cargo terminals, and customer trips. This traffic is generated by air cargo facilities (cargo terminals) located away from the passenger terminal area, freight consolidators or forwarders, and small package deliveries made directly to the terminal area.

It is recommended that the volumes of trips generated by trucks, delivery vans, and air cargo employees be estimated separately. Employee vehicle trips are the largest component of the traffic generated by an air cargo facility (more than 70% of the total traffic volume, according to surveys conducted at Memphis and Los Angeles International Airports and other locations).

The volumes of truck and delivery van trips generated by an air cargo facility (i.e., the trip generation rate) are unique to an individual airport and not transferable to other airports. The two measures (or dependent variables) related to air cargo that are most readily available—air cargo tonnage and the size of air cargo buildings—are not reliable indicators of the volume of cargo-related truck or total vehicle trips, largely because there are many different forms of air cargo service, including integrated cargo handlers, all-cargo or heavy freight carriers, as well as import, export, and shipments that require special handling (e.g., flowers or fresh fish). Each form of air cargo may generate a different number of truck trips, operate at different truck arrival/departure times, and use different vehicle sizes. The use of cargo facilities located at an airport may also be used for truck-to-truck transfer by overnight express carriers.

For example, a local overnight delivery service operation might have multiple tractor-trailers picking up and dropping off containers, as well as dozens of local single-unit delivery vehicles distributing packages locally. Conversely, a large import/export freight operation may only generate a few tractor-trailer trips. Thus, although airport operators have reliable statistics on air cargo tonnage transported, tonnage is not a reliable indicator of the volume of truck trips because the volume of trips is a function of the type of cargo service and freight activity, not cargo tonnage (or the size of the air cargo terminal). *ACRP Synthesis 80: Estimating Truck Trip Generation for Airport Air Cargo Activity* provides a comprehensive review of this topic and the difficulty in estimating truck trips generated by air cargo activity.

Although not considered applicable to all airports, the data shown in Table 3-5, developed for Los Angeles International Airport, present the estimated vehicle trips generated by different cargo facilities (including employee trips) per ton of air cargo. Data from Chicago O'Hare International Airport, circa 2004, indicate that a general-purpose cargo facility generated about 0.13 daily truck trips per 1,000 annual cargo tons.

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Table 3-5. Estimated airport cargo trips per daily cargo tonnage at Los Angeles International Airport.

Cargo Shipper	Daily trips (in and out)	Facility peak hour				Commuter peak hour			
		Morning		Afternoon		Morning		Afternoon	
		In	Out	In	Out	In	Out	In	Out
International airline	25.2	0.39	0.13	0.19	0.29	0.23	0.13	0.16	0.163
Domestic airline	6.9	0.21	0.2	0.3	0.18	0.17	0.08	0.17	0.13
Overnight delivery service	3	0.3	0.24	0.77	0.27	0.3	0.03	0.55	0.26

Source: Leigh Fisher Associates, January 1996. Los Angeles International Airport Master Plan—Phase I, On-Airport Existing Transportation Conditions.

As noted, air cargo is transported by a wide variety of cargo shippers, each having different trip generation rates. Additional research is required to develop methods for estimating the volume of traffic generated by air cargo terminals at airports and the employees working in these terminals.

Traffic Generated by Service and Delivery Vehicles

Service and delivery vehicles include those vehicles (1) bringing goods and materials (other than air cargo) to/from terminal building loading docks, consolidated warehouses, and other sites on an airport; (2) transporting individuals performing airport maintenance and construction; (3) being used by airport police, fire, and emergency response staff; and (4) making trips not directly generated by airport passengers, employees, or air cargo. At most airports, little to no data are available on the current volume of service and delivery vehicle trips or the activities generating these trips (i.e., the extent of goods and material deliveries, trash removal, emergency responses, or construction deliveries and traffic).

Generally, no data are available to guide estimates of the future volume of service/delivery vehicle trips or the extent of future activities generating these trips. Additional research is required on this topic.

Traffic Generated by Other Airport Land Uses

Other land uses commonly found at public airports include general aviation/FBO facilities and military bases. At most commercial-service airports, these other land uses do not generate significant volumes of traffic during the peak hours for the airport or regional highway network. When the analysis is focused on the airport terminal area and primary airport access roadways, the traffic volumes generated by these land uses are often ignored or considered to be "background" traffic and combined with that of service and delivery vehicles.

Traffic volumes generated by general aviation are a function of the number of general aviation aircraft operations and the type of aircraft (business jets, air taxis, or small propeller aircraft). Traffic volumes generated by military bases vary according to the type of base and its function. Traffic volumes generated by non-aviation land uses that are not related to airport or aviation activity (e.g., industrial parks or large retail centers) can be estimated using the ITE *Trip Generation Handbook*.

The employees of airport rental car companies shuttle returned (or dirty) vehicles to a service area where the vehicles are washed, fueled, and serviced, and then stored until they are ready to be rented to another customer. Typically, the movement of vehicles occurs securely within the rental car company's property and does not require the use of public roadways. However, at a few airports, where the vehicle service area is located apart from the site where vehicles are returned and rented, the movement occurs on public roadways. In these instances, the shuttling of rental car vehicles should be included in estimates of traffic generated by other land uses.

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Traffic Generated by Non-Airport Vehicles Using Airport Roadways

Vehicles not related to the airport or airport land uses may use airport roadways as a shortcut to bypass congestion or delays on the regional roadway network. This traffic, commonly referred to as cut-through traffic, adds to airport roadway requirements and contributes to airport roadway congestion. Cut-through traffic occurs at airports having multiple entrance and exit points (e.g., Dallas/Fort Worth, Phoenix Sky Harbor, Washington Dulles International Airports, and Bush Intercontinental Airport/Houston) and where the roadway network configuration allows non-airport traffic to share the airport roadways with airport-generated traffic. Most airport operators discourage such cut-through traffic.

Determining the volume or proportion of existing cut-through traffic may require the use of Bluetooth surveys or other survey methods. It is not possible to identify cut-through vehicles or traffic volumes using simple traffic volume counts.

Estimating the volume of future cut-through traffic requires an understanding of future regional land uses and expected regional traffic patterns/travel times. The volume of non-airport traffic using airport roadways is a function of the volume of traffic on the regional roadways, and the travel-time savings these vehicles would experience if they were able to use airport roadways as a shortcut. These time savings can be determined by comparing the travel times via airport roadways and those on alternative routes, knowing the forecast congestion and travel times on these routes as forecast by regional travel models or other sources.

3.2.2 Off-Airport Origin and Destination Points (Trip Distribution)

With respect to analyses of airport terminal area and curbside traffic, trip distribution refers to the proportion of airport-generated trips originating (or terminating) in each portion of the airport's catchment area (by zip code, traffic analysis zone, cardinal direction, or other indicator) and consequently the path followed to enter and exit the airport. The paths passengers follow are a function of the travel mode they chose, where they enter airport property, and their on-airport destinations. These locations (or the distribution of these locations) are a function of airline passenger trip purpose, place of residency, regional land use patterns, the regional highway network, existing and forecast roadway congestion/travel times, the availability of public transit, and other factors.

At small airports, having a single primary access road that terminates at an intersection or interchange with a regional arterial highway, trip distribution refers to the proportion of airport-generated trips entering (or exiting) the airport from each cardinal direction and turning left, right, or through to enter or exit the airport. At airports reviewing new or expanded public transit service, analyses of passenger and employee trip distributions can indicate the volume of airport-generated trips that originate or terminate near (i.e., within walking distance of) a transit stop and comprise the likely market for transit.

Many large airports have multiple entrance/exit points—one serving the terminal area and separate entrances/exits for aircraft maintenance centers, general aviation terminals, military bases, or other land uses. Although the volume of traffic using each entrance/exit can often be determined by the land use(s) served by the specific entrance/exit, large airports may have multiple connections to the regional roadway system, where the use of each is determined by regional travel patterns (or a combination of regional travel patterns and the on-airport destination). For airports with multiple connections to the regional roadway system, it is necessary to know the routes passengers and employees follow when traveling to and from the airport in order to analyze (1) the intersections or junctions of the airport access roadways and regional roadway network, (2) traffic volumes on airport roadways associated with specific connections to the regional

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roadway network, (3) the effect of airport traffic on the regional roadway network, and (4) potential transit ridership.

At airports having multiple entry/exit points serving the terminal area (or other major land use), passengers typically select the most convenient entry/exit point, which generally implies the point that minimizes travel time. It is possible to estimate the proportion (and thereby the volume) of vehicles using each entry and exit point by determining (1) the actual locations where airline passengers, visitors, and employees begin their trips to the airport (or end their trips from the airport) or the distribution of these locations and (2) the most logical routes used by passengers from each of these origin or destination points.

At many airports, fewer than 30% of all trips begin/end in the downtown area, with the remainder arriving from or going to places of residency and employment or leisure destinations distributed throughout the region. A planner familiar with the regional highway network can determine the most likely routes from the primary regional origin and destination points. In addition, these data (or trip distributions) can be obtained from surveys of airline passengers or, when such data are not available, from the local metropolitan planning organization, which can provide information on future distributions of places of residence and employment, a description of the future regional transportation network, and the likely (minimum time) travel paths or approach/departure distributions.

3.2.3 Assigning Traffic Volumes to the Roadway Network (Trip Assignment)

Assigning the vehicle trips generated by airline passengers, visitors, employees, air cargo, and service/delivery vehicles to the on-airport roadway network requires information as to (1) where these vehicles enter or exit the airport, (2) their final and interim destination or origination points at the airport, and (3) the routes or paths available to these vehicles.

- **Airport entry and exit points.** The methodology for determining traffic volumes entering and exiting an airport at specific locations is provided earlier in this chapter (see Estimating Traffic Volumes [Trip Generation]).
- **Origin and destination points at the airport.** The methodology for determining the volumes of trips associated with specific on-airport origins and destinations is also provided in the previous section, "Estimating Traffic Volumes (Trip Generation)."
- **Travel paths.** Typically, on a regional roadway network, motorists can select from several alternative travel paths. Thus, a sophisticated traffic assignment procedure is required to allocate these vehicle trips among the available travel paths (i.e., to assign the vehicle trips to the regional roadway network) and, if desired, allocate trips to alternative routes, as primary routes become congested and travel times decrease. In comparison, at an airport, there is generally only a single logical travel path available for airline passengers and visitors, employees, and air cargo vehicles. Thus, the traffic assignment process is much simpler at airports.

As discussed earlier, at most airports there is only one travel path available between the airport entry and exit points and the primary origin/destination points. For example, at most airports, there is only one route connecting the airport entrance/exit and the terminal curbside areas, public parking areas, or rental car ready/return areas. Exceptions include those airports having several entrances/exits used by airline passengers, or having multiple terminal buildings served by separate roadways. Some large airports provide internal bypass roads allowing motorists to avoid slow-moving traffic at curbsides or other areas of potential congestion.

Generally, most motorists at an airport follow the guide signs directing them to the major on-airport destinations. Furthermore, most motorists will follow the prescribed routes, even if they

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become congested, and typically deviate to a different route only if directed to do so by a traffic control officer or, in some instances, their in-vehicle navigation system. Most employees and service vehicle drivers follow the quickest route, unless they are prohibited from using specific roads, or tolls or fees are associated with the use of specific routes.

The travel paths of originating airline passengers can be determined using the information presented in Table 3-2 (revised for the specific characteristics of the airline passengers and the airport being analyzed), and the travel paths of terminating airline passengers can be determined using similar information. As noted, care must be taken when assigning trips made by passengers who use multiple travel modes (e.g., those who park in a remote parking lot and also use a courtesy vehicle) or multiple legs (e.g., those who go to the curb and then to parking).

For example, assuming that 100 vehicle trips per hour are generated by originating airline passengers at an airport; 65% of these trips are generated by private vehicles; 30% of those private vehicles go to the curb and then go to parking, where they remain for their trip duration; and 80% arrive from the east and 20% arrive from the west; these assumptions result in 20 vehicle trips by private vehicles using both the curb and daily parking ($100 \times 65\% \times 30\%$), of which 16 vehicles enter from the east and 4 enter from the west.

The trip assignment process for airport roadways requires (1) repeating this calculation for every combination of travel mode, circulation path, and regional approach/departure path; (2) assigning these vehicle trips to the corresponding roadway links; and (3) finally determining the sum of all vehicle trips assigned to each roadway link. The sum of the vehicle trips on each roadway link represents the estimated traffic volume on that link. Travel forecasting software or spreadsheet analyses are frequently used to perform this repetitive process, particularly when traffic forecasts are being prepared for large airport roadway networks, and for multiple future years. The use of these methods allows planners to readily test the implications of alternative assumptions regarding mode choice, travel paths, or airline passenger activity patterns, as well as saving time and effort.

3.2.4 Challenges with Estimating Roadway Traffic Volumes

As noted, several challenges are associated with estimating roadway traffic volumes—either existing or future—using the traditional four-step travel forecasting techniques. Key challenges encountered by most airport operators include the following.

Lack of Data on Airline Passengers

Most airport operators do not conduct regular surveys of the travel modes used by airline passengers, the occupancies of vehicles transporting airline passengers, their lead and lag times, or their on-airport circulation patterns (e.g., the percent using parking or curbside areas). It is estimated that fewer than 20 U.S. airport operators regularly conduct surveys of airline passengers to gather reliable data indicating the passenger's travel modes, circulation patterns, and vehicle occupancies. Furthermore, even at those airports where passenger surveys are conducted, the surveys typically focus on enplaning passengers and gathering data during multiple hours and days with relatively few samples gathered during peak periods. Therefore, survey data may not capture mode-choice variances between departing and arriving passengers (e.g., drivers picking up arriving passengers may be more likely to use short-term parking and meet their passengers in the terminal whereas drivers dropping off departing passengers may be more likely to drop off the passengers at the curbside) nor mode-choice variances by time of day (e.g., passengers departing on early morning flights may be less likely to use public transit if transit service does not operate early enough to deliver them to the airport sufficiently ahead of their scheduled departure time).

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The percent of connecting passengers typically varies throughout the day, particularly at airports serving as connecting hubs. At connecting hubs, as many as 80% of the passengers may be making connections (and not using ground access) during mid-day peaks, while only 20% of the passengers may be making connections during the morning departure peak and evening arrival peaks.

Lack of Data on Hourly Passenger Volumes

Many airport operators do not have accurate data on hour-by-hour originating/terminating airline passenger numbers. At many airports, for planning purposes, hourly airline passenger numbers are calculated using (1) reported aircraft arrival and departure schedules, (2) aircraft sizes (and corresponding seat capacities) to determine the number of available seats per hour (or other time increment), (3) assumed load factors (by airline)—the portion of seats occupied by passengers, and (4) the assumed portion of originating or terminating passengers (by airline). A minor difference in the estimated load factor or the proportion of enplaned/deplaned passengers in the peak hour (or peak month) can lead to significant differences in the numbers of peak-hour passengers. Furthermore, although planners recognize that aircraft load factors vary throughout the day, by day of the week, and seasonally, typically, a single load factor is applied to all aircraft of a given airline. Similarly, while the percentage of passengers who originate or terminate at an airport may vary significantly throughout the day or seasonally, typically only a single originating/terminating factor is applied to all passengers of a given airline.

Lack of Data on Airport Employees

As previously noted, most airport operators have little or no data regarding the number of employees reporting to work on a daily basis, and less data on the work schedules, hour-by-hour arrival/departure patterns, and travel modes used by these employees. Few, if any, airport operators have forecasts of future employment that are considered to be as reliable as the available forecasts of airline passengers.

Lack of Data on Air Cargo and Service/Delivery Trips

As noted earlier, additional research is required on air cargo and service/delivery vehicle trips. At most airports, little data are available on the existing numbers of trips generated by these land uses, and no reliable method exists for forecasting future trips.

Effort Needed to Gather Required Data

Comprehensive surveys of originating and terminating airline passengers can be costly and time consuming to plan, authorize, and conduct, with several months required to review and summarize the resulting data before they are available for release to others.

Resulting Accuracy

As noted, forecasts of the traffic volumes generated by airline passengers are often prepared in substantially more detail than forecasts of traffic generated by employees, air cargo, or service/deliveries. However, although traffic generated by airline passengers may account for over 70% of the traffic during the peak hour, it typically represents less than half of the daily traffic generated by an airport. The costs and time required to gather the airline passenger data needed to forecast airline passenger vehicle trips should be compared with the benefits (i.e., anticipated level of accuracy).

3.3 Estimating Future Airport Roadway Traffic Volumes—Alternative Approach

An alternative approach to estimating future airport roadway traffic volumes involves determining existing traffic volumes on each roadway segment (or major segments) and applying a growth factor to the peak-hour volume to represent future conditions. This alternative approach is commonly called the "growth factor method." It is suitable for quick analyses of airport curbside and terminal area roadway operations for planning purposes. Compared to the four-step

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forecasting approach, this approach can be applied relatively quickly and inexpensively. The growth factor method requires (1) determining the existing peak-hour(s) roadway traffic volumes on each roadway segment or major segments, (2) developing growth factors, (3) adjusting the forecast growth to reflect changes in travel modes, and then (4) multiplying the existing peak roadway traffic volumes by the selected growth factor to develop an approximation of future conditions.

3.3.1 Growth Factor Method for Estimating Future Traffic Volumes

A growth factor is the ratio between traffic volumes during the current peak hour and during the peak hour to be analyzed. A growth factor can be developed by determining the ratio between (1) the forecast total annual airline passenger volumes (enplaned plus deplaned passengers) for the future year to be analyzed and (2) the equivalent existing airline passenger volumes. Seasonal growth factors can be developed to adjust for peak-month traffic operations using data commonly available at most airports. For example, seasonal factors can be developed using the ratio of parking revenues (or, preferably, public parking transactions) during the peak month to the revenues during the current month or the ratio of month-to-month airline passenger numbers.

Using judgment, the growth factor can be adjusted to reflect changes in travel modes (e.g., anticipated future of AVs or transit services) by passengers and employees, or possible changes in trip distribution to reflect new access routes, new or improved public transit services, or regional transportation improvements.

3.3.2 Challenges with Use of the Growth Factor Method

The major challenge with using the growth factor method is that it is relatively simplistic. This method assumes that existing patterns of activity and circulation will remain generally unchanged throughout the forecast period. This method also may not account for changes that may result from

- New land uses on or near the airport that could affect the paths that passengers and employees follow when entering or exiting the airport.
- Changes in choices of travel modes, parking facilities, or circulation paths that may result from new or improved public transportation services, changes in parking facilities or parking rates, or increases or decreases in the propensity of motorists to use curbside roadways. However, as noted above, judgment or available data can be applied to the calculated growth factor to account for these changes, particularly if the changes are expected to be relatively minor.
- Changes in the proportion of airline passengers during the future peak month, peak day, or peak hour, although again these changes could be compensated for by adjusting the growth factor appropriately. For example, if the peak hour is expected to account for a smaller proportion of daily traffic due to anticipated changes in airline schedules or a "flattening" of the peak due to increased traffic volumes, the growth factor could be adjusted accordingly.
- Changes in the roadway network on or near the airport. For example, the construction of a new major regional highway may affect how vehicles approach the airport and turning movement patterns at the airport entry/exit. Similarly, a new or modified airport roadway could alter internal traffic circulation and merging or weaving patterns at the airport.

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CHAPTER 4

Analyzing Airport Terminal Area Roadways

This chapter presents an overview of terminal area roadway analyses. It presents level-of-service definitions applicable to airport roadways and describes methods for estimating the capacity and levels of service. Chapter 5 presents comparable methods for analyzing curbside roadways.

As described earlier, a hierarchy of analytical methods—including quick-estimation, macroscopic, and microsimulation methods—for analyzing airport terminal area roadway and weaving section operations is proposed. The appropriate analytical method will evolve as a project proceeds from concept to final design and as more time and data become available to support the analyses.

This chapter presents the suggested quick-estimation methods for analysis of airport roadways with uninterrupted flows, signalized roadways, and airport roadway weaving sections; the macroscopic method for analyzing low-speed roadway weaving areas commonly found on airports; and an overview of the use of microsimulation methods.

The macroscopic methods and performance measures presented in the HCM are considered applicable for analyses of airport roadways with uninterrupted traffic flows and unsignalized and signalized intersections, but not for analyses of low-speed roadway weaving areas. It is suggested that the method presented in the section "Macroscopic Method for Analyzing Airport Roadway Weaving Areas" be used when macroscopic analyses of airport weaving areas are required, and that the methods presented in HCM be used for macroscopic analyses of all other airport roadways. The methods in this chapter are based on those presented in the HCM. The latest edition of the HCM should be used for analysis, recognizing that judgment may be needed in adapting subsequent updates to the HCM to airport conditions.

The methods and data presented in this chapter represent the best available information concerning airport roadway operations and the consensus of the research team, the project panel, and other reviewers at the time this Guide was prepared. It is suggested that additional research be conducted on low-speed weaving areas and maximum service rates for airport roadways.

4.1 Quality-of-Service Definitions for Airport Terminal Area Roadways

The key performance measures defining the quality of service of an airport terminal area roadway, including its sufficiency, are as follows:

- Average travel speed of through vehicles, which determines travel time.
- Traffic density, which determines the ability of motorists to easily maneuver into and out of travel lanes.
- Maximum volume-to-capacity (v/c) ratio, which along with through movement travel speed, indicates how close the roadway is to breakdown and is useful for determining other performance measures such as queue length and delays.

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- Duration of congestion and length of queues.
- Travel-time reliability reflecting the distribution of trip times over an extended period.

With the exception of the weaving analysis discussed in this chapter, the definitions, metrics, and procedures presented in the HCM are applicable to airport roadways with uninterrupted operations (e.g., freeways and other grade-separated facilities) and roadways that are governed by intersections (e.g., signalized intersections, stop-controlled intersections, and roundabouts).

The weaving analysis methods presented in the HCM method are primarily oriented toward operations on freeways and were developed using data primarily from freeways. At airports, weaving often takes place on roadway segments designed for speeds that are much slower than those on freeways or even on major arterial streets. As a result, while the weaving theory and methods presented in the HCM (and subsequent updates) may be applicable to airport roadways, the metrics defining quality of service do not apply. Consequently, subsequent portions of this chapter present alternative metrics for the low-speed weaving that occurs on airport roadways.

4.2 Quick-Estimation Methods for Analyzing Airport Roadway Operations

This section presents quick-estimation methods for analyzing uninterrupted flows, signalized roadways, and airport roadway weaving sections.

4.2.1 Quick-Estimation Method for Uninterrupted Flows on Airport Roadways

Quick-estimation methods are most appropriate for "sizing" a roadway in the early stages of planning and the design process when little has been decided (or is known) about the details of the required roadway. Such methods are suitable for use when preparing planning studies to size or evaluate a roadway and identify points of existing or future constraints.

Table 4-1, which is adapted from the HCM (Exhibits 12-16 and 12-18), presents the maximum service flow rate and adjusted flow rates for multilane roadways with uninterrupted flows. The adjusted flow rates represent the maximum flow rates of typical airport access and circulation roadways and were calculated assuming that (1) heavy vehicles—primarily single-unit trucks and courtesy vehicles—comprise less than 15% of the traffic volume on the access roadways and (2) a high proportion of drivers are infrequent users of the airport roadways and are likely to be unfamiliar with them. The free-flow speeds can be approximated by the posted speed limits on the roadway section unless drivers regularly exceed the posted speed limit, in which case the free-flow speed can be approximated by the average operating speed of the vehicles on the roadway.

These adjusted flow rates are also based on the following assumptions:

- Travel lanes are at least 12 feet wide.
- Lateral clearances (e.g., distance from walls, abutments, or other physical obstacles) are at least 6 feet on both the left and right sides of the roadway.
- Any vertical grades are less than 0.25 mile in length or less than 3% (i.e., rises less than 3 feet per every 100 feet of length).
- The roadways operate in one direction only, or for two-way roadways, at least two travel lanes are provided in each direction, separated by a median.
- There are fewer than 10 driveways or other access points per mile along the airport roadway.

If the roadway being evaluated falls significantly outside the lane width, lateral clearance, and percent of truck use, and varies from the other factors listed above, then the traffic volume thresholds presented in Table 4-1 may not be accurate. In addition, given that airport roadways

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Table 4-1. Levels of service for airport terminal area access and circulation roadways.

	Under Capacity			Near Capacity	At Capacity
	A	B	C	D	E
Criteria	Free-flow speed = 50 mph				
Maximum speed (mph)	50	50	50	50	46
Maximum volume/capacity ratio	0.27	0.43	0.62	0.89	1.00
Maximum service flow (passenger cars/hour/lane)	560	910	1,310	1,860	2,100
Maximum flow (vehicles/hour/lane)	460	740	1,070	1,520	1,710
	Free-flow speed = 45 mph				
Maximum speed (mph)	45	45	45	45	42
Maximum volume/capacity ratio	0.26	0.42	0.61	0.82	1.00
Maximum service flow (passenger cars/hour/lane)	500	810	1,170	1,570	1,910
Maximum flow (vehicles/hour/lane)	410	660	960	1,280	1,560
	Free-flow speed = 40 mph				
Maximum speed (mph)	40	40	40	39	36
Maximum volume/capacity ratio	0.26	0.43	0.62	0.82	1.00
Maximum service flow (passenger cars/hour/lane)	440	720	1,040	1,390	1,690
Maximum flow (vehicles/hour/lane)	360	590	850	1,130	1,380
	Free-flow speed = 35 mph				
Maximum speed (mph)	35	35	35	34	30
Maximum volume/capacity ratio	0.25	0.41	0.59	0.78	1.00
Maximum service flow (passenger cars/hour/lane)	390	630	910	1,210	1,550
Maximum flow (vehicles/hour/lane)	320	510	740	990	1,270
	Free-flow speed = 30 mph				
Maximum speed (mph)	30	30	30	29	26
Maximum volume/capacity ratio	0.22	0.37	0.53	0.73	1.00
Maximum service flow (passenger cars/hour/lane)	320	530	760	1,050	1,430
Maximum flow (vehicles/hour/lane)	260	430	620	860	1,170
	Free-flow speed = 25 mph				
Maximum speed (mph)	25	25	25	24	20
Maximum volume/capacity ratio	0.22	0.33	0.48	0.66	1.00
Maximum service flow (passenger cars/hour/lane)	280	430	620	860	1,300
Maximum flow (vehicles/hour/lane)	230	350	510	700	1,060

may experience a high share of infrequent users, all lanes on a given roadway segment may not be used equally. For example, if on a multilane roadway, signs indicate that certain facilities will be on the left, drivers may shift into the left lane well in advance of the actual exit. As a result, the combined lanes of the roadway may not achieve their full capacity or maximum service flow rate because many drivers would be unwilling to use each available lane. In such situations, a more detailed macroscopic analysis using procedures described in the HCM may be necessary to determine the maximum service volume for the facility.

If the lane width, lateral clearance, percent of truck use, and other factors described above are applicable to the roadway being analyzed, then the information in Table 4-1 should be applied as follows:

1. Determine the free-flow speed for the roadway. The free-flow speed is usually determined by measuring the mean speed of traffic under very light flow conditions. However, the posted speed limit can be used as an approximation of the free-flow speed.
2. Determine the target roadway sufficiency. The target is determined by individual airport operators (or local agencies) reflecting their individual policies and standards. If such a

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standard or policy is lacking, "near capacity" (equivalent to LOS D in the HCM) is a common standard for urban roadways, although many urban agencies have adopted "at capacity" (equivalent to LOS E in the HCM) as a standard. "Below capacity" (equivalent to LOS C in the HCM) is considered the common standard for planning new airport facilities, while at large-hub airports, "near capacity" is sometimes considered to be acceptable on existing roadways during peak periods.

3. Using Table 4-1, select the appropriate free-flow speed and the column with the desired level of service. The maximum flow provides the maximum traffic per hour per lane that the roadway can serve in one direction.

For example, if the free-flow speed is 50 mph and the target roadway sufficiency is "near capacity," then the maximum desirable flow rate for a two-lane, one-way road would be 3,040 vehicles per hour (twice 1,520).

4.2.2 Quick-Estimation Method for Signalized Roadways

The HCM (Volume 4, Chapter 31) presents a planning-level analysis intended to support simplified and approximate analyses of signalized intersections for motorized vehicles. This planning-level analysis has its roots in critical movement analysis. This method involves the following steps:

- **Step 1:** Identify the lane geometry.
- **Step 2:** Identify the hourly volumes, including left-turn, through, and right-turn volumes for each intersection approach.
- **Step 3:** Identify the signal phasing (i.e., which movements operate concurrently).
- **Step 4:** Perform left-turn check to determine the probability of each critical approach volume clearing the identified opposing or conflicting left-turn volume.
- **Step 5:** Assign lane volumes.
- **Step 6:** Identify critical volumes by identifying the conflicting or opposing traffic volumes (on a per lane basis) having the highest total volumes for each signal phase.
- **Step 7:** Sum the critical volumes.
- **Step 8:** Determine the intersection level of service.

4.3 Macroscopic Method for Analyzing Airport Roadway Weaving Areas

The HCM provides methodologies for evaluating traffic operations on airport roadways. However, the HCM methodologies were not designed to evaluate weaving conditions for low-speed airport roadways (speed limits of 30 mph or slower). These limits on applicability are commonly included in software that applies the HCM method, prohibiting the user from applying the software to weaving sections with free-flow speeds lower than 35 mph.

Consequently, a separate weaving analysis without the limitation on low free-flow speeds was developed and incorporated into a macroscopic model—QATAR. QATAR includes components that provide information about low-speed weaving and curbside roadway operations given certain inputs. The low-speed weaving operations are described in this section. The curbside operations components are described in Chapter 5.

QATAR uses the weaving analysis calculations and methodology presented in Chapter 13 of the HCM, for one-sided and two-sided weaving and applies these calculations to roadways having free-flow speeds slower than the lower bound of speeds (free-flow speeds less than 35 mph) presented in the HCM, 6th Edition. Three modifications were made to the HCM weaving method to extend its application to lower-speed roadway sections. First, the minimum speed for

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weaving traffic was reduced from 15 mph in the HCM to 10 mph. Second, the HCM level-of-service thresholds were converted to sufficiency thresholds for consistency with other recommended analyses on airport roadways, and a special set of sufficiency threshold traffic densities was developed for application to weaving sections on low-speed airport roadways. Third, as an input in determining the capacity of the weaving segment, maximum service flow rates for basic freeway segments under base conditions were extrapolated to correspond to input free-flow speeds (i.e., less than 55 mph).

The HCM presents macroscopic methods for analyzing airport roadway operations away from the terminal area. These methods, if adjusted for the factors used to develop Table 4-1 (e.g., heavy vehicles and roadway geometry), are applicable to analysis of airport roadways with uninterrupted traffic flows and flows at signalized intersections, roundabouts, and stop-controlled intersections.

4.3.1 Use of HCM Weaving Analysis Procedure

The HCM weaving analysis procedure involves the following seven steps, which are described in this section:

- **Step 1:** Collect and input roadway weaving section lane geometry, lane designations, free-flow speed, and peak-hour volumes.
- **Step 2:** Adjust the mixed flow traffic volumes to equivalent passenger-car volumes (adjust for percent of heavy vehicles, driver familiarity, and peak-hour factor).
- **Step 3:** Determine configuration characteristic, which is based on lane changes of weaving movements.
- **Step 4:** Determine the maximum weaving length, if weaving analysis is appropriate.
- **Step 5:** Determine the weaving section capacity.
- **Step 6:** Determine lane-changing rates.
- **Step 7:** Determine the average speed of weaving and non-weaving vehicles.

An eighth step in the HCM, determining the level of service, has been replaced in this report with determining the sufficiency of the weaving segment.

The remaining paragraphs of this section describe these steps in more detail with the recommended modifications for applying this analysis to weaving sections of low-speed airport roadways. Equations and additional detail on these steps are provided in the HCM.

Collect and Input Data

The analyst must collect data on existing and/or forecast peak-hour traffic volumes for each leg of the weaving section. The traffic data should include a peak-hour factor and percent of heavy vehicles. The peak-hour factor is the ratio of the total peak-hour flow rate (in vehicles per hour, vph) divided by the peak 15-minute flow rate within the peak hour (converted to vph).

The free-flow speed or posted speed limit should be observed (or estimated in the case of a new design or planning study).

The proposed (or existing) lane geometry must be identified (number of lanes on each leg, number of lanes in the weaving section, lane striping showing how the lanes on each leg transition to and from the lanes in the weaving section, and the length of the weaving section).

Adjust Flow Rates

Mixed (passenger cars, trucks, buses, etc.) flow rates should be converted to equivalent passenger-car rates by accounting for a peak-hour factor and the presence of heavy vehicles. The HCM

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no longer includes a driver familiarity factor, and it combines all heavy vehicle types into a single classification.

The user has two options for entering traffic volumes through the weaving segment. The first option is to enter actual origin and destination counts (or projected volumes) on the weaving section, and the second option is to enter approach and departure volumes, and then use QATAR to estimate the weaving volumes in the segment.

Determine Weaving Configuration

Several key parameters characterize the configuration of a weaving segment. The first step is to determine whether the roadway being analyzed is a one-sided ramp weave or a two-sided weave (graphical illustrations are provided in QATAR as well as in Figure 4-1; a photo of a weaving configuration is provided as Figure 4-2).

The key variables in subsequent steps of the methodology, for both types of weaving configurations are

LC_{MIN} = minimum rate at which weaving vehicles must change lanes to successfully complete all weaving maneuvers (lc/hr).
N_{WL} = number of lanes from which weaving maneuvers may be made with either one lane change or no lane changes. For one-sided weaving, this value is either 2 or 3, and for two-sided weaving, this value is always 0 by definition.

For a one-sided weaving segment, the two weaving movements are the ramp-to-freeway and freeway-to-ramp flows. For the purposes of this analysis, the "freeway" flows are the mainline flows, and the "ramp" flows are those flows that merge into or diverge from the mainline flow. The following values are established:

LC_{RF} = minimum number of lane changes that must be made by one ramp-to-freeway vehicle to successfully execute the desired maneuver.
LC_{FR} = minimum number of lane changes that must be made by one freeway-to-ramp vehicle to successfully execute the desired maneuver.
v_{RF} = ramp-to-freeway demand flow rate in weaving segment, passenger car (pc)/hr.
v_{FR} = freeway-to-ramp demand flow rate in weaving segment, pc/hr.

For a two-sided weaving segment, only the ramp-to-ramp movement is functionally "weaving." The following values are established:

LC_{RR} = minimum number of lane changes that must be made by one ramp-to-ramp vehicle to successfully execute the desired maneuver.
v_{RR} = ramp-to-ramp demand flow rate in weaving segment, pc/hr.

Determine Maximum Weaving Length

The concept of "maximum length" (L_{MAX}) of a weaving segment is critical to the methodology. Strictly defined, the maximum length is the length beyond which weaving turbulence no longer affects operations within the segment, or alternatively, no longer affects the capacity of the weaving segment.

If the length of the weaving segment is greater than or equal to L_{MAX}, then this weaving analysis methodology is not appropriate, and the weaving segment should be analyzed as separate merge, diverge, and basic segments as appropriate.

Determine Capacity of Weaving Segment

Weaving capacity is determined by two methods: density and weaving demand flows. The final capacity is the smaller of the results of the two methods.

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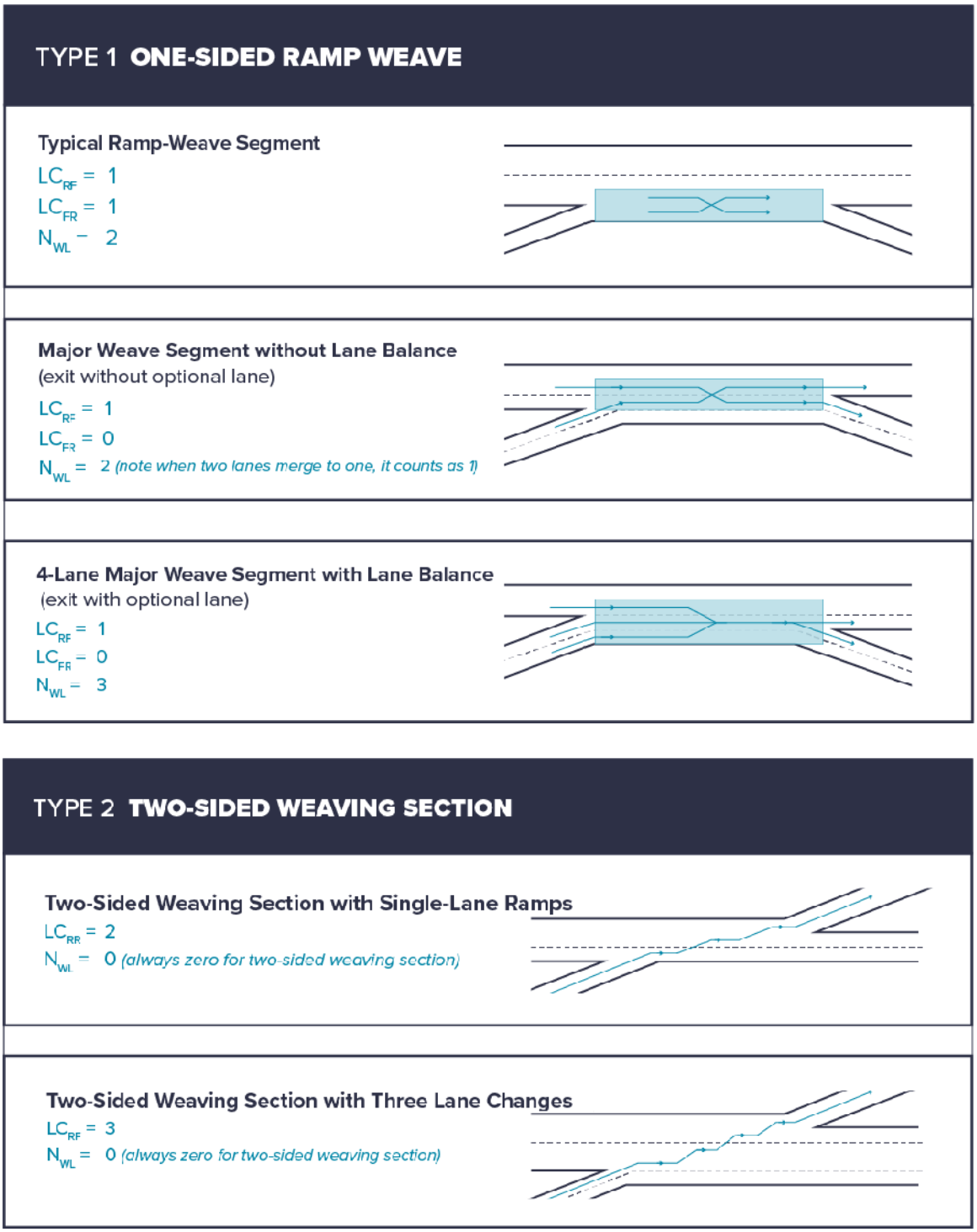
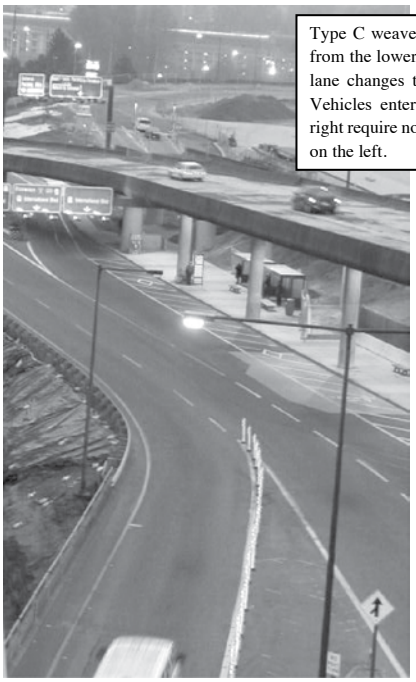


Figure 4-1. Example airport roadway weaving configurations.

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Type C weave. Vehicles entering from the lower left must make two lane changes to exit on the right. Vehicles entering from the lower right require no lane changes to exit on the left.

Figure 4-2. Example weaving configuration.

Determine Lane-Change Rates

The equivalent hourly rate at which weaving and non-weaving vehicles make lane changes within the weaving segment is a direct measure of turbulence in the flow of traffic (i.e., when vehicles exhibit irregular and apparently random fluctuations in speed). It is also a key determinant of speeds and densities within the segment, which ultimately determine the existing or anticipated level of service.

Determine Average Speeds of Weaving and Non-Weaving Vehicles in Weaving Segment

The methodology is used to compute the average speed of weaving vehicles in a weaving segment and the average speed of non-weaving vehicles in a weaving segment. For airport roadways, a minimum average speed (in miles per hour) of weaving vehicles expected in the weaving segment of $S_{MIN} = 10$ mph is recommended.

Note that, usually, the non-weaving speed should be modestly faster than the weaving speed. However, the developers of the HCM weaving methodology believe that it is acceptable for the non-weaving speed to be slightly slower than the weaving speed in some cases. If the analyst finds that the non-weaving speed is more than 3 mph to 5 mph below that of the weaving speed, then it is recommended that the analyst recompute the weaving speed using a lower S_{MIN} of 5 mph (instead of 10 mph). The analyst should be aware that these extrapolations of the HCM weaving methodology to airport roadways are very approximate; field conditions should be verified where possible.

The average speed of all vehicles in a weaving segment is computed using the average speeds for weaving and non-weaving vehicles.

Determine Sufficiency

The sufficiency of a weaving segment is related to the density in the segment. This is similar in concept to the use of level of service in the HCM for freeway segments.

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Table 4-2. Sufficiency criteria for weaving segments.

Sufficiency	Freeway weaving segments (pc/mi/ln)	Collector-distributor roadways (pc/mi/ln)	Airport low-speed roadways (pc/mi/ln)
Below capacity	28	32	40
Near capacity	35	36	50
At capacity	43	40	60
Over capacity	> 43, or v/c>1.0	> 40, or v/c>1.0	> 60, or v/c>1.0

Source: Adapted from HCM, Exhibit 13-6.
Notes: pc/mi/ln = passenger cars per mile per lane.
If the density exceeds the level-of-service threshold, then the roadway is over capacity.

Density is used to look up sufficiency in Table 4-2. A special set of density thresholds has been developed for weaving on low-speed airport roadways. Airport operators may choose their own thresholds based on local experience and perceptions of quality of service.

4.3.2 Caveats

Without more extensive research, it is impossible to know with certainty whether or not the results of the low-speed weaving macroscopic model presented in this section are accurate, but the results can provide an initial indication of whether a weaving section with certain parameters might operate successfully or not.

The results of the low-speed weaving analysis method and the revised metrics appear to (1) correlate reasonably well with the observations of airport roadway weaving operations conducted as part of the original *ACRP Report 40* research, and (2) produce results suitable for planning-level analyses of low-speed airport roadway weaving operations. While low speeds can be entered as inputs to most microsimulation models, it is not known whether the resulting modeled traffic flows represent actual traffic operation patterns under those conditions. Few, if any, studies have been conducted on the low-speed weaving conditions typical of airport roadways to allow full verification of the suggested low-speed weaving analysis method outputs. Significantly more observations at numerous locations are required to provide a basis for analysis of low-speed roadway weaving operations that is consistent with the level of analytical precision of the HCM or any similar document.

The proposed low-speed weaving method is not intended to serve as a basis for any of the following:

- Design of a new low-speed roadway weaving section,
- Design of modifications to an existing low-speed weaving section,
- A definitive operational analysis of an existing or proposed weaving section, or
- The assessment of the level of safety afforded by an existing roadway.

Under the above conditions, microsimulation models may be more appropriate for evaluating traffic operations.

4.4 Use of Microsimulation Methods

Microsimulation modeling is an analytical process that uses sophisticated computer programs to analyze traffic operations for complex roadway systems and support the design of roadway facilities. In microsimulation modeling, individual simulated vehicles are assigned characteristics, such as a destination, vehicle performance capabilities, and driver behavioral profiles. Each "vehicle" then travels through a computerized roadway network, and various aspects of its performance are recorded during its simulated trip based on its interactions with other vehicles and

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traffic controls. These performance statistics can be summarized in many ways, including performance measures commonly used by traffic engineers and transportation planners (e.g., delays, travel times, travel speeds, and queue lengths).

Some aspects of roadway systems, such as intersections controlled by isolated or coordinated traffic signals, can be analyzed using simpler techniques than microsimulation. The use of microsimulation models can be beneficial in other roadway environments, including those with complex traffic movements, such as weaving operations where some vehicles are entering, some are exiting, and some are traveling through the weaving sections.

Many airport roadway systems are sufficiently complex to warrant the use of microsimulation. The use of microsimulation models should be considered if simpler analytical tools and methodologies do not yield reasonable results, provide sufficient detail, or cannot be used because the roadway configuration or operating conditions are outside the range of those addressed in the HCM. However, the use of microsimulation models and analyses of traffic using these models is relatively complex. Training in the use of the specific model and experience in traffic engineering are required to fully understand the simulation process so that appropriate inputs are used and the outputs are interpreted correctly. Most microsimulation software packages require significant time to learn, as well. In addition, microsimulation requires input data that may be unavailable for the airport roadway to be analyzed, or the available data may not be sufficiently accurate or reliable to justify the effort required to develop and calibrate the model.

Suggested guidelines on when microsimulation is probably not needed are as follows:

1. Signalized or unsignalized intersections. These can usually be analyzed using methodologies in the HCM (Chapter 31). If exclusive turn lane storage area overflows, through movement queues that block access to exclusive turn lanes, or both are a significant problem, then HCM methodologies may yield optimistic estimates of signal performance. In this case, microsimulation modeling may yield more accurate results.
2. Roadway segments having few or no driveways or intersecting side roads.
3. Weaving segments with two entries and two exits and a reasonable distance (e.g., at least 500 feet) between the entrance to and exit from the segment and free-flow speeds of 35 mph or greater.
4. If the use of a simpler technique—even if the inputs are outside of the recommended ranges—yields outputs that are consistent with observed conditions (e.g., traffic seems very congested, and use of the HCM methodology yields LOS E or F).
5. Cases where inputs have a high degree of uncertainty, such as when future forecast volumes are at best approximate. In these cases, the apparent precision of microsimulation may not produce results that are any more accurate than simpler techniques that can be used with less time and effort.

Guidelines regarding when to consider microsimulation:

1. Signalized or unsignalized intersections that have more than four legs or are oriented in atypical ways. The analyst should initially attempt to use HCM methodologies and then consider microsimulation modeling if the inputs required to evaluate the roadway segments do not correspond to the HCM analysis structure.
2. Airport roadway segments with the number of lanes changing along the length of the segment and with multiple, and possibly unusual, orientations of driveways or intersecting side roads.
3. Multilane airport roadway segments where distribution of vehicles between the lanes is strongly influenced by which lane serves exits to downstream destinations.
4. If the access or circulation roadways contain roundabouts. The HCM provides guidelines on the analysis of roundabouts.

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5. Weaving segments that do not fit the orientation of the weaving analysis in the HCM. This could mean more than two entries or exits, dimensions or speeds outside the bounds defined in the HCM, signals or stop signs within the weaving segment, etc.
6. If simpler techniques are used to analyze what appear to be sufficiently simple facilities, but the results indicate operations that are much worse or much better than those observed.
7. When congestion on one roadway section causes queues or backups that extend back and interfere with operations on an upstream critical roadway section.
8. When congestion on one roadway section significantly restricts the volume of vehicles that can arrive at a downstream critical location.
9. When comparing the congestion resulting from different improvement options for situations where it is not possible to design sufficient capacity to eliminate significant congestion. In this case, comparisons are typically made of the extent of the congestion (duration and length of queues) produced by the various improvement options. Microsimulation modeling is the best available tool for making these comparisons.

FHWA's *Traffic Analysis Toolbox III: Guidelines for Applying Traffic Microsimulation Modeling Software 2019 Update to the 2004 Version* (Publication Number FHWA-HOP-18-036, April 2019) provides additional information on the use and application of microsimulation. This document is available at https://ops.fhwa.dot.gov/publications/fhwahop18036/index.htm.

4.5 Other Performance Measures

At some airports, the adequacy of a roadway or curbside area has been defined by the length of time a motorist requires to enter and exit the terminal area. Microsimulation models can be used to establish a baseline condition and compare the baseline travel time (or a predetermined acceptable travel time) with the travel times resulting from different levels of traffic demand and access and circulation roadway configurations. However, it is difficult to accurately estimate these travel times and queues without the aid of microsimulation models because of the relatively short distances being analyzed and the difficulty in estimating queue lengths through other means.

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CHAPTER 5

Evaluating Airport Curbside Operations

This chapter presents measures of curbside roadway performance, definitions of curbside roadway sufficiency, and a hierarchy of analytical methods for estimating curbside roadway capacities and sufficiency. It also describes use of a macroscopic method, QATAR, for analysis of airport curbside roadways, and explains the use of this method. Appendix C documents the queuing theory and assumptions used in QATAR.

In evaluating airport curbside roadway operations, analyses of both the curbside lanes (where motorists stop to pick up or drop off passengers) and the adjacent through lanes are required. As described in Chapter 2, these analyses are necessary because double- or triple-parked vehicles impede or delay the flow of vehicles in the adjacent through lanes. As a result, the capacity of the through lanes decreases as demand for curbside space approaches or exceeds the capacity of a curbside roadway segment, causing double- or triple-parking.

Throughout this chapter, the term "parked vehicle" refers to a vehicle that has come to a complete stop and remains stopped to allow the loading or unloading of passengers and their baggage. Vehicles on curbside roadways are not "parked" in the same sense as vehicles in a parking lot or an on-street parking space because these parked vehicles may not be left unattended on airport curbsides. Within the airport industry, vehicles stopped or standing at curbsides are commonly referred to as parked vehicles.

As described in more detail later in this chapter, the capacity of curbside pickup and drop-off areas depends on the number of lanes airport management allows to be used for vehicles to stop, load, or unload. For example, at airports where double-parking is prohibited, curbside capacity equals the effective length of the lane next to the curb. At airports where double-parking is allowed, curbside capacity equals twice the effective length of this lane. Effective curb length differs from the actual curb length in that it only includes space where motorists can stop, load, or unload and excludes areas reserved for other uses (e.g., crosswalks, disabled motorists, or airport police vehicles) or where motorists prefer not to stop (e.g., adjacent to columns, along sharp curves, or uncovered areas distant from the nearest terminal doors).

In this chapter, methods of estimating the volumes, capacities, and sufficiency of the curbside lanes and the through lanes are presented separately. However, when estimating airport curbside roadway capacities and sufficiency, it is necessary to consider the operations of both the curbside lanes (the lanes predominately used for passenger loading or unloading) and the through lanes concurrently because the capacity and sufficiency of an airport curbside roadway system is determined by the component that has the lowest capacity or provides the poorest sufficiency.

The methods and data presented in this chapter represent the best available information concerning airport roadway operations and the consensus of the research team, the project panel, and prior research. It is suggested that additional research be conducted to refine the estimated airport curbside performance.

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5.1 Performance Measures

The curbside utilization ratio is the recommended performance measure for airport curbside roadways. The curbside utilization ratio represents the ratio of the demand for curbside parking to its capacity and indicates the ability of a roadway to accommodate existing or projected requirements for vehicles loading or unloading at the curbside. It also indicates if spare capacity is available to serve additional demand and surges in demand.

As noted in prior chapters, roadway and curbside capacities are typically analyzed for the peak hour or design hour of a facility. For airport roadways, it is suggested that the design hour be a typical busy hour on the peak day of the week during the peak month. This suggestion is in contrast to planning for airfield and other airport facilities, which often considers the peak hour of an average day during the peak month. This approach recognizes there may be a limited number of times during the year when demand is higher, but constructing facilities to accommodate such infrequent occurrences may not be cost-effective. Other potential sources for a design hour could include the 30th busiest hour in the year (a peak period frequently selected in highway and transportation planning) and the 95th percentile hourly volume during the year, among others. Selection of a design hour may reflect a clock-hour or a busy 60-minute period (such as the busiest four consecutive 15-minute periods during the peak day of the peak month).

Typically, a curbside utilization ratio of 1.30 or less (65% of the combined capacity of the inner and second curbside lanes) is a desirable planning target for new curbside roadways. A curbside utilization ratio of 1.70 (85% of the combined capacity of the inner and second curbside lanes) is acceptable for existing facilities, recognizing that during peak hours and days of the year, demand will exceed capacity. However, individual airport operator policies regarding parking in multiple lanes may dictate different curbside utilization ratio planning targets.

The curbside utilization ratio is used as an indicator of curbside sufficiency, which provides an overall indication of the quality of the experiences of drivers and passengers using the curbside roadway. A curbside sufficiency of "under capacity" is a desirable planning target for a medium- or small-hub airport, both for the design of new curbside roadways and for analyzing an existing facility. This target should result in acceptable conditions during most of the year with a limited number of hours experiencing poorer conditions (such as around holidays). For a large-hub airport, a curbside sufficiency of "under capacity" is a desirable planning standard for the design of new curbside roadways. However, airport management could deem a curbside sufficiency of "near capacity" to be acceptable for design or existing operations depending on the desired customer experience, incremental cost to achieve an improved curbside utilization, the remaining useful life of the terminal and adjacent roadways, or number of hours per year during which "near capacity" conditions occur. A curbside sufficiency of "near capacity" may be considered acceptable for an existing curbside roadway at a large-hub airport, recognizing that on some peak days of the year, the level of service may decrease to "at capacity" or "over capacity." Two separate sufficiency measures are used: roadway sufficiency for through traffic and curbside sufficiency for curbside loading/unloading traffic.

When additional performance measures, as described below, are required to supplement the curbside utilization ratio, the analysis is conducted using a microsimulation model. Such supplemental measures cannot be accurately determined without the use of a microsimulation model, either quantitatively or in the field (i.e., they are difficult to quantify using field surveys). For example, the use of a microsimulation model would help document the ability of an existing curbside roadway to accommodate future demand or quantify the benefits resulting from alternative curbside improvement options. A microsimulation model also captures the interaction effects between adjacent zones, such as queue spillback from one zone to the next or demand starvation downstream of a queuing bottleneck. These effects are not captured in simple macroscopic methods.

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Supplemental performance measures include

- **Number of vehicles parked in the second and third lanes.** The number of through lanes blocked by parked or parking vehicles (and the proportion of the modeled hour during which this blockage occurs) is an indicator of the extent of roadway congestion. It is also an indirect indication of the ability of motorists to enter/exit and stop at their preferred curbside locations since it is difficult for motorists stopped in the curb lane to exit when triple-parking occurs without the intervention of traffic control officers.
- **Queue length.** Queue length is the number of vehicles waiting to enter the curbside roadway or a specific curbside parking area expressed in terms of the distance that the vehicle queue extends back from the curbside parking area or point of congestion. Queue lengths are estimated for different levels of probable occurrence. The mean queue length has a 50% probability of being exceeded sometime during the hour. The 95% queue length has a 5% probability of being exceeded. The 95% queue length is typically used for design purposes.
- **Queuing duration.** The queuing duration (in minutes) indicates how long the congestion will last, and is useful for comparing two potential design solutions, neither of which completely eliminates queuing. Ideally, the queuing duration is zero for a new curbside roadway and less than 1 hour for an existing curbside roadway.
- **Average vehicle delay.** Average vehicle delay consists of two components—through traffic delay and curbside loading/unloading delay.
- **Through traffic delay.** Through traffic delay is the extra amount of time, due to traffic congestion, required for a vehicle to traverse the entire curb length. To determine through traffic delay, the unimpeded travel time for through traffic on the curbside roadway is subtracted from the actual travel time to obtain the amount of through traffic delay per vehicle. In the design of a new curbside roadway, the delay to through traffic should ideally be near zero outside of crosswalks. For existing roadways, delays of up to 15 seconds per vehicle outside of crosswalks may be acceptable, recognizing that the delays could be significantly higher on peak days of the year. The acceptable amount of delay for through vehicles should be set by the airport operator based on the design of the landside circulation system and the number of other delays experienced by through vehicles on other portions of the roadway circulation system. For example, if through vehicles must pass several curbside loading/unloading areas, then delays at each curbside area are cumulative and may be less tolerable.
- **Curbside loading/unloading delay.** Curbside loading/unloading delay is the extra amount of time, due to traffic congestion, a vehicle requires to pull into a curbside stall, load or unload passengers, and exit. The minimum time necessary to drop off or pick up a passenger during uncongested periods (i.e., the average dwell time) should be subtracted from the total average observed time to obtain the amount of curbside loading/unloading delay. Curbside delays of up to 30 seconds are acceptable when designing a new roadway. Delays of up to 60 seconds per vehicle are acceptable for existing roadways.

As shown by the checkmarks in Table 5-1, use of these performance measures requires different analysis methods. When analyzing curbside roadways using microsimulation models, it is possible

Table 5-1. Recommended airport curbside performance measures.

Performance measure	Quick estimation	Macroscopic analysis	Microsimulation analysis
Curbside utilization ratio	✓	✓	✓
Number of vehicles parked in second and third lanes			✓
Queue length			✓
Queuing duration			✓
Average vehicle delay			✓

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to consider the number of vehicles parked in the second and third lanes, the length and duration of curbside queues, and average vehicle speeds (or delays). Without the aid of microsimulation models, it is difficult to accurately estimate vehicle parking patterns, travel times and delays, and queue lengths because of the relatively short distances on curbside roadways being analyzed and the difficulty estimating queue lengths through other means. Microsimulation also allows assessments of how traffic congestion on one curbside roadway link can impact upstream facilities or meter traffic bound for downstream facilities. When curbside roadways are being analyzed using the quick-estimation or macroscopic methods described in this chapter, the appropriate performance measures are curbside utilization and the corresponding levels of service. Use of the supplemental performance measures "through traffic delay" and "curbside loading/unloading delay" is not recommended due to the complexity of determining and confirming these delays.

5.2 Sufficiency Definitions for Airport Curbside Roadways

The primary element defining the sufficiency of an airport curbside roadway is the ability of motorists to enter and exit the curbside space of their choice (e.g., one near their airline door or another chosen destination). As roadway demand and congestion increase, motorists are required to stop in spaces farther away from their preferred destination. The motorist is required to either stop in a downstream or upstream curbside space, double-park, or, in an extreme case, circle past the curbside area multiple times while searching for an empty space.

The key performance measures defining the sufficiency of an airport curbside roadway are the

- Number of vehicles parked or stopped in the curbside lane, and the percent double- or triple-parked, or otherwise stopped, in a position that interferes with the flow of traffic in adjacent lanes. This number of parked vehicles is a function of curbside demand versus available capacity.
- Length and duration of queues at the entrance to the curbside area.
- Average delay encountered by private and commercial vehicles entering and exiting the curbside areas.
- Curbside utilization ratio, which is a comparison of the length of the vehicles stopped along the curbside and the effective length of the curbside (i.e., the total length less the space occupied by crosswalks or other areas in which vehicles, or certain classes of vehicles, cannot stop).

As stated, most of these measures are obtainable only through microsimulation modeling. Therefore, sufficiency definitions for airport curbside roadways shown in Figure 5-1 and presented in Table 5-2 are based on curbside utilization ratios. These definitions and ratios were validated using focus groups of airline passengers, airport landside managers, and commercial vehicle operators conducted as part of the research presented in *ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations*.

ACRP Report 40 used level-of-service definitions on a scale of LOS A to F similar to those used in the HCM for a variety of facilities. To simplify the method and avoid confusion with the HCM measures based on quality of service, this report uses a sufficiency rating similar to that used in the HCM for planning analyses of signalized intersections.

5.3 Estimating Airport Curbside Roadway Traffic Volumes

Curbside roadway traffic volumes can be estimated using the same methods used to estimate airport terminal area roadway traffic (see Chapter 3): the traditional four-step travel forecasting method and the growth factor method. The key differences between estimating terminal area

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WELL UNDER CAPACITY



UNDER CAPACITY



NEAR CAPACITY



AT CAPACITY



OVER CAPACITY



Note. These sufficiency ratings are comparable to the level-of-service grades used in the HCM: Well under capacity—LOS A, Under capacity—LOS B and C, Near capacity—LOS D, At capacity—LOS E, and Over capacity—LOS F.

Figure 5-1. Curbside levels of service.

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Table 5-2. Sufficiency criteria for airport curbside roadways.

Criteria	Airport Curbside Sufficiency					
	Under Capacity			Near Capacity	At Capacity	Over Capacity
When double- (and triple-) parking is allowed at the curbside						
Maximum demand for curbside standing or parking/effective curbside length[a]	0.90	1.10	1.30	1.70	2.00	>2.00
Maximum service flow rate						
5-lane curbside roadway, 4 approach lanes (vph)	3,400	3,205	2,945	2,295	1,790	Up to 1,790
5-lane curbside roadway, 3 approach lanes (vph)	2,850	2,690	2,445	1,820	1,385	Up to 1,385
4-lane curbside roadway, 3 approach lanes (vph)	2,790	2,545	2,225	1,540	1,155	Up to 1,155
4-lane curbside roadway, 2 approach lanes (vph)	2,545	2,335	2,065	1,485	1,145	Up to 1,145
3-lane curbside roadway, 2 approach lanes (vph)	1,675	1,375	1,155	905	810	Up to 810
When double-parking is prohibited at the curbside						
Maximum demand for curbside standing or parking/effective curbside length[a]	0.70	0.85	1.00	1.20	1.35	>1.35
Maximum service flow rate						
3-lane curbside roadway (vph)	2,480	2,295	2,005	1,560	1,285	Up to 1,285
Maximum through-lane volume/capacity ratio	0.25	0.40	0.60	0.80	1.00	1.00

[a] The ratio between the calculated curbside demand and the available effective curbside length.

roadway traffic and curbside roadway traffic include the need to prepare the following for curbside roadway traffic:

- **Separate estimates of vehicles stopping in a curbside lane and through traffic vehicles.** At small airports with a single terminal building and a short curbside area (e.g., less than 500 feet in length), the volume of through vehicles may equal the volume of vehicles stopping at the curbside. However, these volumes may differ at airports having (1) multiple terminal buildings or large concourses served by a common roadway, (2) a curbside area with inner and outer curbside roadways separated by a raised island with midpoint entrances and exits, or (3) curbside roadways that are used by non-curbside traffic (e.g., vehicles entering or exiting parking areas, rental car areas, or other facilities).
- **Separate analyses of the departures curbside and arrivals curbside roadways.** It is necessary to analyze these curbside areas separately because the departures and arrivals peak periods at an airport (and thus peak periods of curbside demand) occur during different hours of the day, and vehicle dwell times and space allocations (the proportion of curb length assigned to individual classes of vehicles) differ significantly at the departures and arrivals curbside areas, as described in subsequent sections of this chapter. At airports with dual-level curbside roadways, separate analyses of each level are required. At airports with a single-level curbside roadway, analyses of the peak periods for originating, terminating, and total passengers (originating plus terminating) are required.
- **Separate analyses of terminal-adjacent versus island (or parallel) curbsides.** The volume and mix of vehicles on center-island curbsides typically vary from those at the curbsides adjacent to the terminal building, depending on the assignment of vehicles or allocation of space to each curbside and curbside roadway. For example, certain classes of vehicles may be required to drop off and pick up passengers at a specific location, or through traffic may be directed to use certain lanes.

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- **Analysis of impacts of pedestrian crossings.** At-grade pedestrian crossings may connect the terminal building with center-island curbside areas or adjacent parking facilities. At-grade crosswalks reduce both the effective curb length and the roadway capacity. The proportion of time that a crosswalk is occupied and blocked to vehicle traffic is a function of (1) the volume of pedestrians using a crosswalk, which varies according to its location and the facilities it connects; (2) the time passengers require to walk across the roadway (e.g., at a design walking speed of 3.5 feet per second per the FHWA *Manual on Uniform Traffic Control Devices*); and (3) whether vehicular and pedestrian traffic at the crosswalk is controlled by a signal, stop sign, traffic control officer, or is uncontrolled. While signalized pedestrian controls can be coordinated using a common background cycle to minimize stops, some signalized crosswalks, as well as crosswalks controlled by traffic control officers, or unsignalized crossings, are not coordinated and thus may create multiple stops for through vehicles.
- **Separate analyses for each class of vehicle.** Private vehicles, taxicabs, limousines, TNCs, door-to-door vans, courtesy vehicles, and charter buses/vans each have different dwell times, required vehicle stall lengths, and maneuvering capabilities. Furthermore, each service provided by these vehicles may have different operational methods and be governed by different airport regulations. For example, on an arrivals curbside, an airport operator may permit taxicabs, certain TNCs, or valet-parked vehicles to stand at the curbside for 30 minutes or more to ensure that waiting taxicabs, TNCs, and valet-parked vehicles are available for arriving customers. At some airports, customers using valet parking may drop off and pick up their car at the curbside, with vehicles temporarily stored at this location. Space for valet operations should be analyzed separately. Airport staff may allow charter buses to remain at the curbside for 10 to 15 minutes to ensure that all members of a large party have claimed their bags and boarded the vehicle, but may only allow hotel/motel courtesy vehicles to stop while actively boarding passengers. It is expected that many of these classes (e.g., taxicabs, TNCs, valet-parked vehicles, and courtesy vehicles) will eventually be operated using AVs, which could result in an additional class of service(s) with different operational methods that are governed by different airport regulations.
- **Separate estimates of traffic volumes for each terminal building or concourse.** The peak periods of activity for each airline serving an airport may occur during different hours of the day. At airports with multiple terminals or large concourse(s) dominated by a single airline, the largest traffic volumes (and curbside area requirements) may occur during a different hour (or different 15-minute period) at each terminal or near each concourse. The average vehicle dwell times at terminals primarily serving international passengers are longer than those at other terminals—about 30% longer at drop-off curbsides and 45% longer at pickup curbsides based upon data provided by San Francisco International Airport. In addition, motorists prefer to stop at the curbside area nearest the doors (or skycap podiums) serving their airline (or that of the passenger they are transporting). Thus, demands are not distributed uniformly along the length of a curbside—particularly at airports with multiple terminals or large concourses—but are concentrated at the curbside areas corresponding to the airlines serving the largest volume of passengers during the peak period.

  As a result, at airports with several terminals or multiple concourses, the traffic volumes and curbside area requirements that correspond to (or are generated by) each terminal or concourse should be estimated separately. These estimates can be prepared by allocating the total peak-hour traffic volumes to each curbside area according to the percentage of total demand served by each area during the peak hour. The percentage of total demand served by each area can be estimated by analyzing (in decreasing order of reliability) the proportion of (1) peak-period originating (or terminating) passengers served by each terminal building or concourse; (2) the number of scheduled aircraft seats served by a terminal or concourse during the peak period; (3) TSA screenings occurring, assuming that there are separate checkpoints for each terminal or concourse; or (4) the number of aircraft gates served by each concourse.

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If the data are available from passenger surveys or other sources, it is preferable to prepare separate estimates of the traffic volumes occurring at each terminal curbside area by type of vehicle. This is because, as described in prior chapters, the travel mode choices and arrival/departure patterns of passengers on each airline may differ depending on the type of flight (e.g., domestic, international, or low-cost carrier), typical trip purpose (routes primarily serving business travelers versus those serving vacation or non-business travelers), a passenger's place of residency (local resident versus non-resident), party size/presence of persons not accompanying the passenger aboard the aircraft, use and availability of parking, and other factors.

5.4 Estimating Airport Curbside Roadway Capacity and Sufficiency

Estimating airport curbside roadway capacities and sufficiency requires analyses of both the curbside lanes and the through lanes because the number of vehicles stopped in the curbside lanes affects the flow of vehicles in the through lanes; as curbside lanes approach or exceed capacity, the capacity of the adjacent through lanes is reduced.

5.4.1 Establishing Curbside Lane Capacity

Curbside lane capacity is typically estimated in terms of the area (and the number of lanes) that the stopped vehicles may occupy while loading or unloading. Since vehicles stop in a nose-to-tail manner at most airports, this area is described as the effective length of curb measured in linear feet. As described previously, effective length is defined as the total length of the lane minus (1) any space unavailable for public use because it is reserved for crosswalks, disabled motorists, or specific classes of vehicles (e.g., taxicabs or public buses) and (2) space located beyond the ends of the terminal building or adjacent to columns or other physical barriers that discourage its use by motorists because passengers cannot easily open their doors or easily enter/exit a vehicle.

The number of stopped vehicles that the curbside lane(s) can accommodate (i.e., the capacity of the curbside lanes) varies depending on

- The number of lanes in which airport operators allow vehicles to routinely stop to load or unload passengers and their baggage. Airport operators establish specific policies concerning double-parking that reflect the width of their curbside lanes, enforcement policies and capabilities, customer service, and use by private and/or commercial vehicles.
- The amount of curbside length occupied by a single stopped vehicle. To allow sufficient space to maneuver in and out of curbside loading or unloading spaces, drivers leave space between their vehicles and the vehicles immediately in front of and behind them. For example, if a single vehicle is 18 feet long, the total space required to accommodate it is about 25 feet. Thus, the curbside length occupied by a single vehicle is a function of the typical vehicle length, its maneuverability, and how close drivers are willing to park adjacent to vehicles.

In practice, the number of stopped vehicles is not distributed uniformly along the entire effective curbside length, as motorists prefer to stop at the first or initial doorway serving their airline, near skycap/baggage drop-off stands, or other preferred locations. Similarly, the number of stopped vehicles is not distributed uniformly throughout the peak hour, instead, traffic volumes typically fluctuate in 5- to 15-minute bursts, particularly at smaller airports, reflecting airline schedules.

Airports Where Double-Parking Is Prohibited

At airports where double-parking is prohibited, the number of vehicles that can be accommodated in the curbside lane is equal to the effective length of a single curbside lane divided by the

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average length occupied by a single parked vehicle. Some airport operators restrict curbside parking or standing to a single lane for operational reasons (e.g., a narrow curbside roadway or curbsides used exclusively by commercial vehicles where double-parking is prohibited).

This description of the number of vehicles that can be accommodated in the curbside lane also applies to curbside roadways with a maximum of three lanes. This is because on a curbside roadway with three lanes only a single through lane would be available if double-parking were to occur, which would lead to frequent bottlenecks (e.g., when a double-parked vehicle or an open door of such a vehicle intrudes into the third lane). Thus, a single through/maneuvering lane for a significant portion of the curbside length is considered unacceptable, and double-parking is generally not tolerated on curbside roadways with a maximum of three lanes.

Airports Where Double-Parking Is Allowed

At airports where double-parking is allowed on the curbside roadways, the number of vehicles that can be accommodated at the curbside is equal to twice the effective curbside length divided by the average length occupied by a single parked vehicle. At airports where double-parking is regularly allowed, pavement markings typically have been installed designating the lane next to the sidewalk plus the adjacent lane for passenger drop-off or pickup, or where enforcement policies allowing double-parking have been established.

On roadways where double-parking is allowed, if the roadway were operating at full capacity, the stopped vehicles would likely not be evenly distributed along the length of the two curbside lanes; some motorists would choose to triple-park next to the most desirable doorways or other locations.

Additional Considerations

At airports with inner and outer curbside areas available for use by private vehicles, these areas have different effective capacities, even if they are the same length. Motorists prefer to stop at the most convenient space available (e.g., the inner curbside lane), even if they observe downstream congestion or delays on this roadway. Thus, if private vehicles are directed to both the inner and outer curbside areas it is necessary to consider the uneven distribution of demand between the two roadways, or alternatively, "discount" the capacity of the outer, less convenient curbside area. If one curbside is allocated to private vehicles and the second is allocated to commercial vehicles, such discounting is not required.

For example, motorists approaching the departures curbside at some airports can use either the curbside area adjacent to the terminal building or an alternative curbside area located within the adjacent parking garage. Passengers using the alternative curbside are provided with a grade-separated path to/from the terminal building and are offered skycap service on certain airlines. Notwithstanding the good access, good directional signage, and amenities available, motorists are reluctant to use the curbside area within the parking garage, even when the curbside area adjacent to the terminal is congested.

Consequently, it is suggested that, when calculating the capacity of a similar, remotely located curbside area, it is necessary to adjust (or discount) the actual length of curb space within a garage (or other supplemental location) to determine its effective capacity. This adjustment is necessary because, if both the primary and supplemental curbsides are allocated for private vehicle use, the supplemental curbside will provide less capacity (even though it may be the same length) than curb space adjacent to the terminal building because it attracts fewer passengers. This capacity adjustment factor is similar conceptually to the adjustments for lane utilization at various types of intersections as presented in the HCM. Again, if commercial vehicles are required to use a remotely located curbside (or passenger drop-off/pickup) area, it is not necessary to adjust its capacity; in this case, the use of this area is mandatory, not elective.

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No published research provides guidelines on this adjustment factor, but the factor appears to vary according to the traffic queues caused by downstream congestion, local enforcement policies, availability of skycap service and dynamic signage, and the demographics of the passenger market (e.g., the proportion of frequent travelers or those traveling primarily with carry-on baggage). It is suggested that analyses be guided by field observations of existing conditions, which would reflect the unique characteristics of the airport and its passengers. If field data are unavailable, it is suggested that the capacity of the supplemental curb space located in a garage be reduced by 50% and that the capacity of an outer curbside be reduced by 20% to 30%.

Alternative Curbside Configurations

It is assumed in the previous discussions that vehicles stop in the curbside lane(s) in nose-to-tail configuration. However, at some airports, the curbside areas are configured with pull-through spaces or 45-degree stalls. (See Chapter 2 for illustrations of alternative curbside configurations.) In such configurations, the curbside capacity is the number of individual parking spaces as opposed to a curbside length. Such configurations also may result in dwell times and through-lane capacities that are different from those discussed in the following section.

5.4.2 Calculating Curbside Lane Requirements

Quick-Estimation Method

This method is appropriate for use during the early planning and design stages for a new curbside when little is known about the details of the curbside design or layout. This method is used to compute the curb length required to serve a given demand, but it does not provide specific results on performance, such as average delay or queuing probability.

A curbside lane can be considered as a series of stopping spaces, each capable of accommodating one vehicle. The average number of vehicles each space can serve during a given time period is inversely proportional to the average length of time (referred to as the vehicle dwell time) a vehicle occupies a space. For example, if the average vehicle dwell time is 3 minutes, then each curbside space can accommodate, on average, 20 vehicles per hour. If the peak-hour volume is 160 vehicles, then (with the assumed average dwell time of 3 minutes per vehicle), the required curbside length is equivalent to eight spaces or 200 linear feet (assuming an average space length of 25 feet for illustrative purposes). This can be represented mathematically as

$$R_a = V * D_i / 60 * L$$

where

R_a = the average curbside length required to accommodate the vehicles stopping at a curbside area.
V = the hourly volume of vehicles stopping at a curbside area.
D_i = the average vehicle dwell time for vehicle type i (in minutes).
L = the average vehicle stall length.

This formula represents a condition where a single class of vehicles (vehicle type i) is using a curbside area (e.g., a curbside serving private vehicles exclusively), or where the requirements are developed assuming that all vehicles can be represented using average dwell times and a single stall length. More accurate estimates can be developed by considering, separately for each class of vehicle, the hourly volume, the distribution of dwell times (rather than average dwell time), and average vehicle length. Additional accuracy can result from consideration of the peak periods within the peak hour (e.g., analysis of the peak 15 or 20 minutes) and the non-uniform distribution of demand along the curbside lane caused by a concentration of traffic at specific

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airline doors or other attraction points. The non-uniform arrival rate and distribution of vehicles can be reflected using statistical factors (e.g., a Poisson distribution).

Vehicle dwell time is one of the key factors determining demand for curbside facilities. Dwell times are affected by

- Enforcement policies of dwell time limits (i.e., strict enforcement leads to shorter dwell times).
- Local driver behavior (i.e., do drivers double-park in a way that allows other motorists to easily enter and exit the lane immediately adjacent to the terminal).
- Party sizes and amount of luggage (i.e., larger party sizes and passengers with more luggage take longer to load and unload).
- The time required for a customer of a vehicle-for-hire (e.g., taxicabs, limousines, and TNCs) to complete the payment transaction for the ride.

When conducting curbside operations or planning studies, an airport would ideally be able to conduct dwell time measurements to identify values specific to their passengers and policies. In the absence of such data, an analyst may wish to select a value based on those presented in Table 5-3, which summarizes dwell time measurements collected at selected airports.

In addition, because dwell times can reflect enforcement policies, which impact both dwell times and driver behavior (i.e., enforcement staff can intervene to allow vehicles to quickly enter or exit curbside spaces), an analyst may wish to use professional judgment and assume future dwell times that reflect different enforcement policies and thus, different dwell times. For example, an airport currently experiencing 3-minute average dwell times for passenger pickup may wish to assume a shorter dwell time (e.g., 90 seconds, or a value at the lower end of those observed at other airports) when planning a new terminal on the assumption that the airport would be able to achieve shorter (but still realistically achievable) dwell times through enforcement.

Table 5-4 presents data—gathered at the airports serving Memphis, Nashville, Portland, Raleigh-Durham, and San Diego—used to calculate curbside lane requirements by class of vehicle, the application of a Poisson distribution (or adjustment) factor, and the resulting curbside requirements. The table presents examples of typical curbside dwell times and vehicle stall lengths based on observations of curbside roadway operations at the airports, the estimated curbside requirements (i.e., design length) for five zones (two zones on the enplaning curbside and two zones plus a courtesy vehicle lane on the deplaning curbside). A comparison of the estimated requirements with the available curb length yields utilization factors for each of the five zones. As shown, two of the zones are substantially over capacity as evidenced by the utilization factors over 2.0.

Table 5-3. Average dwell times observed at selected airports.

Mode	Average Dwell Times (s)									
	Departures					Arrivals				
	Memphis (2008)	Nashville (2017)	Portland (2013)	Raleigh-Durham (2017)	San Diego (2012)	Memphis (2008)	Nashville (2017)	Portland (2013)	Raleigh-Durham (2017)	San Diego (2012)
Private vehicles	101	86	100	98	105	92	74	81	118	89
TNCs	—	51	—	55	—	—	109	—	71	—
Taxicabs	112	113	139	94	71	—		—	—	—
Limousines		—	73	59	105	—		—	—	—
Courtesy vans[a]	35 – 67	—	43 – 89	69	—	—	—	—	127	—
Scheduled buses		—	—	134	—	—	—	—	200	—

Note: -- = Data not collected or vehicles are allowed to wait for customers at the curb (such as in a taxicab queue).
[a]Range reflects differences between vans operated by off-airport parking companies (low end of range) versus those operated by hotels/motels (upper end of range).

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The quick-estimation method involves the following steps:

- **Step 1:** Determine peak-hour traffic volume from field survey or estimates of future traffic.
- **Step 2:** Determine the vehicle mix. If vehicle mix is unknown, assume that private vehicles represent 55% to 80% of the total traffic volumes (with small- and medium-hub airports typically having a higher proportion of private vehicles), TNCs represent 15% to 20%, taxicabs and limousines represent up to 5%, courtesy vehicles represent 5% to 10%, and vans/buses/public transit represent 5% or less.
- **Step 3:** Determine the average vehicle stall length. Use the default values shown in Table 5-4 or the QATAR model (see Figure 5-2) or measure representative values, particularly for unusual vehicles (e.g., motorcycles, three-wheeled vehicles) or atypical parking configurations.

Table 5-4. Estimate of terminal building curbside requirements—sample calculation.

Mode	Hourly Volume (vph)	Average Curbside Dwell Time (minutes)	Required Curbside Stalls	Required Design Stalls[a]	Vehicle Stall Length (feet)	Design Length (feet)	Existing Curb Length (feet)	Curbside Utilization Factor
Enplaning Level, North								
Private Vehicles	621	3	31.1	40	25	1,000		
Taxicabs	52	2	1.7	4	25	100		
Limousines	9	2.5	0.4	1	30	30		
Door-to-door Vans[b]	38	3	1.9	4	30	120		
Courtesy Vans[b]	24	4	1.6	4	30	120		
Scheduled Buses[b]	10	5	0.8	2	50	100		
Total	**754**					**1,470**	**600**	**2.45**
Enplaning Level, South								
Private Vehicles	363	3	18.2	25	25	625		
Taxicabs	35	2	1.2	3	25	75		
Limousines	6	2.5	0.3	1	30	30		
Door-to-door Vans[b]	38	3	1.9	4	30	120		
Courtesy Vans[b]	24	4	1.6	4	30	120		
Scheduled Buses[b]	10	5	0.8	2	50	100		
Total	**476**					**1,070**	**830**	**1.29**
Deplaning Level, North								
Private Vehicles	580	5.2	50.3	62	25	1,550		
Limousines	5	5.2	0.4	2	30	60		
Total	**585**					**1,610**	**535**	**3.01**
Deplaning Level, South								
Private Vehicles	345	5.2	29.9	39	25	975		
Limousines	4	5.2	0.3	1	30	30		
Total	**349**					**1,005**	**780**	**1.29**
Deplaning Level, Courtesy Vehicle Lane								
Courtesy Vehicles[b]	223	1	3.7	7	30	210	300	0.70

[a]Represents calculated stall requirements adjusted to reflect random arrival of vehicles and non-uniform distribution of traffic volumes and demands using Poisson statistical probability factors.
[b]Assumes that this mode makes a single stop at the curbside.

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Quick Analysis Tool for Airport Roadways

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STEP 2: Enter vehicle classification, curbside parking lengths, and dwell times

Previous | Next | Export to File

Enter adjustment factor

Regional Adjustment Factor	95%
Percent of 1st lane full when next vehicle double parks	80%
Percent of 2nd lane full when next vehicle double parks	80%

Input Legend

	User Input
	Default Value
	Calculated Value

Select preferred data entry option

Specify roadway traffic volumes by entering:	
The total volumes of vehicles using the roadway, in vehicles per hour, and the percentage of total volume in each vehicle class.	◉
The volume of vehicles, in vehicles per hour, for each vehicle class.	○

Enter the total volume of traffic using the roadway

Total roadway traffic volume in vehicles per hour (vph)	-

This table provides an input for the vehicle class information required for the analysis. Users can enter vehicle class names, specify the volumes as direct values or percentages of total roadway volumes, and input parking lengths and dwell times.

Reset To Defaults

VEHICLE VOLUMES PERCENTAGE OF TOTAL VOLUME BY CLASS ARRIVALS ROADWAY			VEHICLE LENGTHS AND DWELL TIMES ARRIVALS ROADWAY			
Vehicle class	Vehicles per hour (vph)	Percent of total roadway traffic volume	Default Vehicle Parking Length (feet)	Optional User Override Vehicle Parking Length (feet)	Default Average dwell time (minutes)	Optional User Override dwell time (minutes)
Private vehicles	-	0.0%	25.0		1.60	
Taxis -- On demand	-	0.0%	20.0		1.90	
Taxis -- Pre-reserved	-	0.0%	20.0		1.90	
Limousines -- On demand	-	0.0%	30.0		5.50	
Limousines -- Pre-reserved	-	0.0%	30.0		5.50	
TNCs -- On demand	-	0.0%	30.0		2.70	
TNCs -- PIN System	-	0.0%	30.0		2.70	
Door-to-door vans	-	0.0%	30.0		2.70	
Courtesy vehicles	-	0.0%	30.0		2.80	
Courtesy bus	-	0.0%	50.0		3.50	
Charter vans	-	0.0%	30.0		2.70	
Charter minibus	-	0.0%	30.0		2.80	
Charter bus	-	0.0%	50.0		3.50	
Scheduled bus	-	0.0%	50.0		3.50	
Service and delivery vehicles	-	0.0%	30.0		2.70	
Airport operated vehicles	-	0.0%	25.0		1.60	
Autonomous shuttle vehicle	-	0.0%	30.0		2.80	
Autonomous passenger vehicle	-	0.0%	25.0		1.60	
	-	0.0%				
Total	-	0.0%				

Figure 5-2. Example of QATAR input sheet.

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- **Step 4:** Determine vehicle dwell times using field measurements or the default dwell times shown in Table 5-4 or the QATAR model.
- **Step 5:** Calculate curbside stall requirements that are equal to the volume multiplied by vehicle dwell times divided by 60 minutes.
- **Step 6:** Determine curbside design stall requirements that are equal to the curbside stall requirements times a probabilistic factor applied to the total curbside stall requirements (if a mixed-use curbside such as a typical departures curbside) or to an individual class of vehicles (if curb space is allocated to this classification), ranging from 3.0 for curbside stall requirements of fewer than 5 curbside stalls to 1.2 for curbside stall requirements of 100 or more. (This assumes the peak requirements for all vehicle classes occur at the same time, which is reasonable if the curbside predominantly serves private vehicles, TNCs, and taxicabs. If a large volume of traffic is composed of buses, vans, or other vehicles or is composed of many individual curbside zones, it may be appropriate to analyze each class separately to determine the average number of stalls for each class, and then apply the probabilistic factor to the sum.)
- **Step 7:** Determine curbside design length that is equal to the number of design stalls times the average vehicle stall length.
- **Step 8:** Calculate the curbside utilization ratio that is equal to the curbside design length divided by the existing curb capacity (or effective length) considering whether double-parking is allowed by the airport operator. As defined previously in this chapter, a curbside utilization ratio equal to or less than 1.3 is considered acceptable for a new design, while a curbside utilization ratio equal to or less than 1.7 is considered acceptable for existing curbside roadways.

Macroscopic Method

Alternatively, the curbside lane can be considered a series of processing points (or servers) and traditional queuing analyses can be used to calculate the capacity of individual servers and the total capacity of the curbside lane. The macroscopic method (QATAR) described in the upcoming section, "Analytical Framework Hierarchy for Airport Curbside Roadways," uses queuing analysis to estimate curbside capacity.

The following subsections describe the calculations of through-lane capacity and curbside capacity.

5.4.3 Calculating Through-Lane Requirements

The requirements for curbside roadway through lanes depend on the areas they serve. At airports with a single terminal building and a short curbside area, the volume of through vehicles may equal the volume of vehicles stopping at the curbside. As discussed in previous chapters, factors that may result in higher volumes of traffic in the through lanes include vehicles bypassing a curbside area (1) that does not serve their airline (e.g., a different terminal building or major concourse); (2) that is reserved for other classes of vehicles (e.g., authorized commercial vehicles); or (3) to enter or exit parking, rental car, or other land uses not related to curbside activities. As noted, bypass traffic proceeding to another terminal (as opposed to through traffic proceeding to a downstream portion of the curbside lane) may represent a significant portion of the total curbside roadway traffic volume. When these conditions occur, it is necessary to use the methods described in Chapter 4 to estimate the volume of traffic associated with the alternative land uses and/or to assign traffic volumes to each curbside roadway section (or airline) and class of vehicle.

The capacity of a curbside roadway through lane is measured using methods similar to those described in Chapter 4 for other airport terminal area roadways, adjusted to account for the presence of double- or triple-parked vehicles. As noted previously, double- and triple-parked vehicles block or delay the movement of vehicles in through lanes because through traffic must decelerate and maneuver around these stopped vehicles. As a result, through-lane capacity decreases when

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curbside lane demand exceeds the available capacity of a specific curbside segment (as opposed to the entire curbside length), and vehicles are double- or triple-parked.

The reduction in through-lane capacity resulting from increased curbside lane demand can be estimated using commercially available microsimulation models capable of simulating airport curbside roadways or using QATAR (discussed later in this chapter). Alternatively, the approximations shown in Table 5-2 can be used to estimate curbside roadway lane capacities.

Curbside roadway capacity must also be reduced when at-grade pedestrian crosswalks are present. As described above, the extent of the capacity reduction is a function of the volume of pedestrians crossing the roadway since the amount of time motorists must wait for pedestrians increases with pedestrian traffic. For example, if a crosswalk is controlled by a traffic signal, and if the signal allocates 25% of the green time during each hour to pedestrians, then the capacity of the curbside roadway would be 25% less than if there were no crosswalk. If, instead of a signal, crosswalk operations are controlled by a traffic control officer, then a similar approximation can be made by observing curbside roadway operations. If the crosswalk is uncontrolled, then the behavior of motorists (do they stop when a pedestrian enters a crosswalk?) and the volume of pedestrians need to be considered. For curbsides with multiple crosswalks, the combined impact of all crosswalks should be considered.

5.4.4 Additional Considerations in Estimating Commercial Ground Transportation Vehicle Curbside Requirements

The analytical methods used to estimate curbside traffic volumes presented in Chapter 4 are applicable to private vehicles and commercial ground transportation vehicles, the volumes of which can be directly correlated to airline passenger demand (e.g., limousines, taxicabs, and door-to-door vans dropping off passengers). However, these analytical methods are not applicable to vehicles that are allowed to remain at the curbside for extended periods (e.g., taxicabs, door-to-door vans, and TNCs operating in a demand-responsive mode standing in queues waiting to pick up passengers) or that operate on a scheduled or de facto scheduled basis (e.g., courtesy vehicles that generally operate on fixed headways regardless of the number of passengers transported).

Allocation of Curb Space

Generally, airport operators do not reserve space for commercial ground transportation vehicles dropping off airline passengers, with the exception of vehicles, such as public buses, that drop off and pick up passengers at the same curbside space. The amount of space allocated to commercial ground transportation vehicles picking up passengers is generally determined by airport management considering such factors as

- **Customer expectations.** Deplaning airline passengers generally expect on-demand taxicabs as well as TNCs to be available immediately adjacent to the baggage claim area or visible from the exit doors. Passengers who have reserved luxury limousines or use curbside valet parking expect a higher level of service than those choosing public transportation (e.g., baggage assistance, shorter walking times, minimal wait time).
- **Operational needs.** To minimize the wait times of deplaning passengers, taxicabs are generally allowed to wait at the deplaning curbside area in queues of 3 to 10 vehicles. The number of taxicabs in the queue is a function of airport policy, the proximity of a taxicab hold area (where additional taxicabs may wait until dispatched to the curb), and the availability of curb space. At airports where TNCs are permitted to operate using a PIN code issued to customers or a similar system, the amount of space allocated to TNCs is a function of the volume of customers, the number of authorized TNCs, and airport policies.
- **Space requirements.** In analyzing the amount of space to be allocated to each class of commercial vehicle operator (e.g., hotel/motel courtesy vehicles), the number of vehicles that will use

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the space concurrently (which is based on the number of operators and the frequency with which they serve the airport) and the permitted vehicle dwell times and vehicle sizes must be considered.

- **Vehicle maneuverability.** In determining the amount of curb space to be allocated to each class of commercial vehicle operator, consideration should be given to the maneuverability requirement of the vehicles used (e.g., vans, minibuses, or full-size buses) and, if appropriate, requirements of access to baggage compartments or baggage trucks. For example, a 45-foot-long full-size bus requires about 60 feet to stop parallel to a curb space, while a 60-foot articulated bus requires 90 feet or more. If a bus has an under-the-floor baggage storage compartment, curb spaces should be configured so that columns, sign poles, or other obstacles do not interfere with the opening of the baggage compartment.
- **Vertical clearances.** The ability of a full-size bus or other large vehicle to use a curbside area may be limited by the vertical clearance available (including low-hanging signs or drainage structures). For example, the minimum vertical clearances required are 13 feet for a full-size bus, 11.5 feet for the shuttle buses used by rental car companies, and 9 feet to 10 feet for courtesy vans serving hotels/motels. These dimensions can vary for those vehicles using compressed natural gas, having rooftop air conditioners, or having rooftop antennae. Some multilevel roadways cannot accommodate full-size buses or over-the-road coaches used by charter bus operators.
- **Competition.** Commercial vehicle operators compete with private vehicles, other operators providing the same service, and operators providing services that are perceived as being similar (e.g., taxicab and TNC operators). Each commercial vehicle operator generally wishes to be assigned space nearest the busiest terminal exit doors or space that is equivalent to or near the space provided to their competitors to maintain a "level playing field."
- **Airport management policy.** Some airport operators have policies that encourage the use of public transportation and, thus, assign public transit vehicles the most convenient or most visible curb space.
- **Revenues generated by commercial vehicle operations.** Airport operators receive significant revenues from public parking and rental car concessions. As such, the courtesy vehicles serving on-airport parking lots and rental car facilities may be assigned higher priorities than other courtesy vehicles, including those serving privately operated parking or rental car facilities located off-airport.

Number of Curbside Stops Made by Commercial Vehicles

An additional factor to be considered when estimating the curbside roadway lane requirements of commercial vehicles is the number of stops each vehicle makes. For example, a single courtesy vehicle or public bus may stop two or more times along a terminal curbside, depending on the length of the curb and airport policies. The calculation of curbside lane requirements for each courtesy vehicle, for example, must be adjusted to account for the number of stops.

5.5 Analytical Framework Hierarchy for Airport Curbside Roadways

Airport curbside roadway operations—particularly the reduction in through-lane capacity that results from increased curbside lane demand—can be analyzed using the quick-estimation method described below, the macroscopic method (QATAR) described in subsequent sections, or commercially available microsimulation methods used to simulate airport curbside roadways.

5.5.1 Quick-Estimation Method

The quick-estimation method is used to measure both the curbside utilization ratio (i.e., the ratio between curbside demand and curbside capacity) and the maximum through capacity for five-, four-, and three-lane curbside roadways. The sufficiency for a curbside roadway system is defined as the worst result of the curbside lane sufficiency and through-lane sufficiency.

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Estimates of the maximum flow rates (i.e., service flow rates) on curbside roadways at each sufficiency level can be determined using the data provided in Table 5-2. These data were established from observations of curbside traffic flows conducted as part of prior research and analyses of curbside roadway traffic flows conducted using microsimulation of airport roadway traffic. Figure 5-3 depicts the relationship between curbside roadway traffic flow rates and utilization factors for five-, four-, and three-lane curbside roadways.

Since as used in Table 5-2, "capacity" varies depending on whether an airport operator allows vehicles to double-park, the policy of the airport being analyzed should be reviewed.

To establish the sufficiency for a given curbside demand and traffic volume, the data in Table 5-2 should be used as follows:

1. Calculate the curbside utilization ratio.
2. Select the corresponding curbside utilization ratio for the curbside lane as shown in Table 5-2, rounding up to the next nearest value, and note the corresponding sufficiency level. For example, for a four-lane curbside roadway with a calculated ratio of 0.6, the sufficiency level is "under capacity."
3. Determine the through-lane sufficiency for the through lanes by (1) selecting the maximum service flow rate row in the table corresponding to the appropriate number of lanes on the entire curbside roadway (include all curbside lanes and through lanes), and (2) comparing this rate to the volume of traffic on the curbside to calculate the volume/capacity ratio. For example, the roadway capacity is 2,680 vehicles per hour for a four-lane roadway with curbside lanes operating at a sufficiency of "under capacity." If this roadway were serving 2,500 vehicles per hour, it would have a v/c ratio of 2,500/2,680 or 0.93.

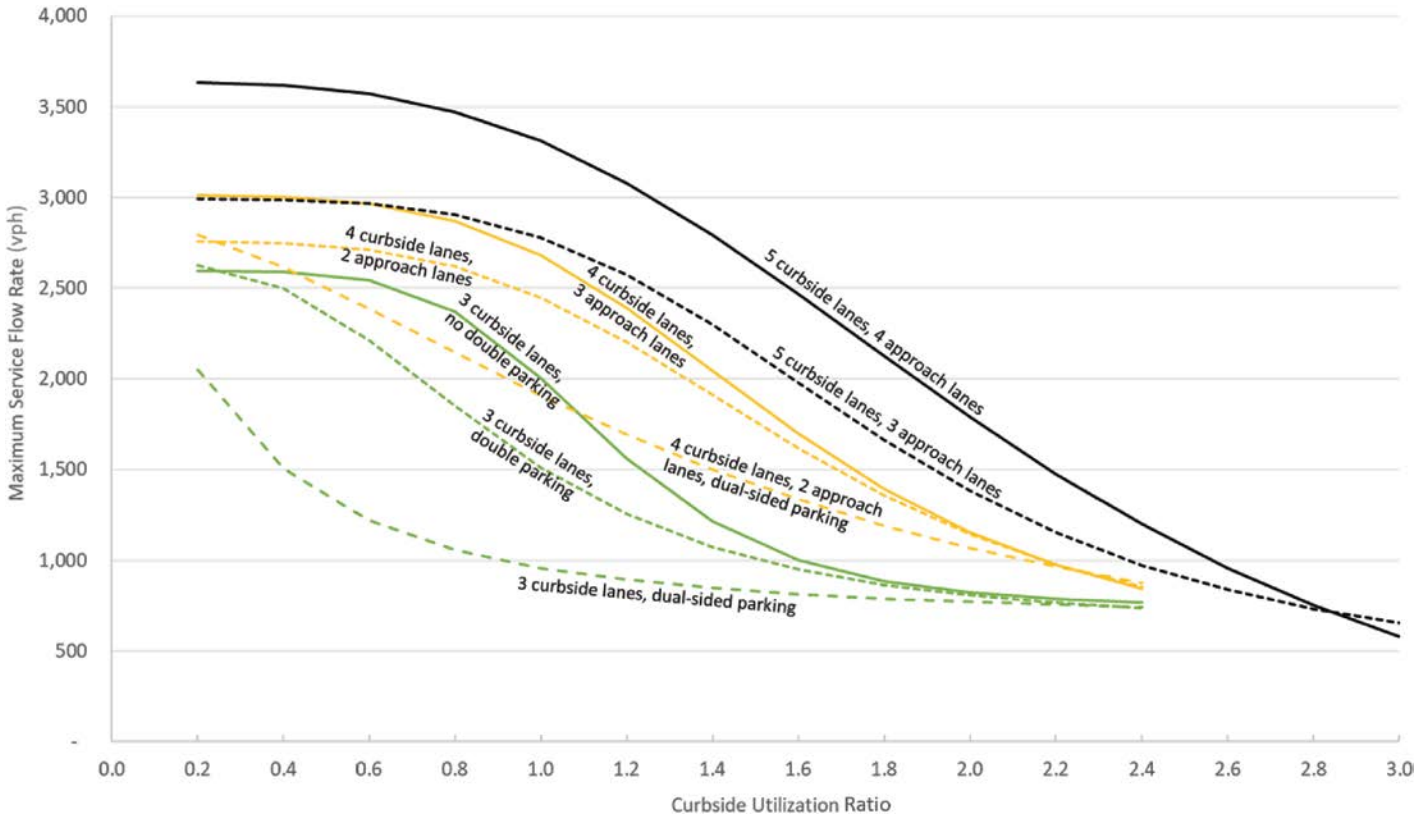


Figure 5-3. Curbside roadway capacity reduction curves.

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4. Use Table 5-2 to determine the through-lane sufficiency that corresponds to the calculated v/c ratio for the through lanes. A v/c ratio of 0.93 corresponds to a sufficiency of "at capacity."
5. The sufficiency for the entire curbside roadway system is determined by the component—either the curbside lane or the through lanes—with the poorest sufficiency. Considering the previously mentioned example, if the curbside utilization factor corresponds to "under capacity" and the peak-hour traffic volume, when compared to the through-lane maximum service flow rate, corresponds to "at capacity," the sufficiency for the curbside roadway system is "at capacity."

The maximum service flow rates shown in Table 5-2 apply to all vehicles on the curbside roadway, including those stopped in the curbside lane. These flow rates need not be adjusted for heavy vehicles or driver familiarity because they were developed from observations of traffic operations on airport curbside roadways.

5.5.2 Macroscopic Model—Quick Analysis Tool for Airport Roadways

Developed through prior research and updated as part of this research project, QATAR allows airport planners and operators to determine the ability of a curbside roadway to accommodate changes in traffic volumes, airline passenger activity, vehicle mix, curbside allocation plans, and curbside enforcement levels. QATAR also allows the user to observe how airport curbside roadway levels of service are expected to vary as these input factors change. Appendix C presents additional information on the methodology and mathematics used in QATAR.

In the analysis procedure used in QATAR, it is assumed that (1) vehicles begin to double-park and potentially triple-park, if allowed, as the number of vehicles stopping in a zone approaches the zone's capacity (or length), and (2) the capacity of the adjacent maneuver and travel lanes decreases as the number of double- and triple-parked vehicles increases. The propensity of arriving vehicles to double-park (reflecting the percentage of occupied curbside spaces) can be modified by the QATAR user to reflect local conditions and policies.

Using a multiserver (or multi-channel) queuing model, QATAR calculates

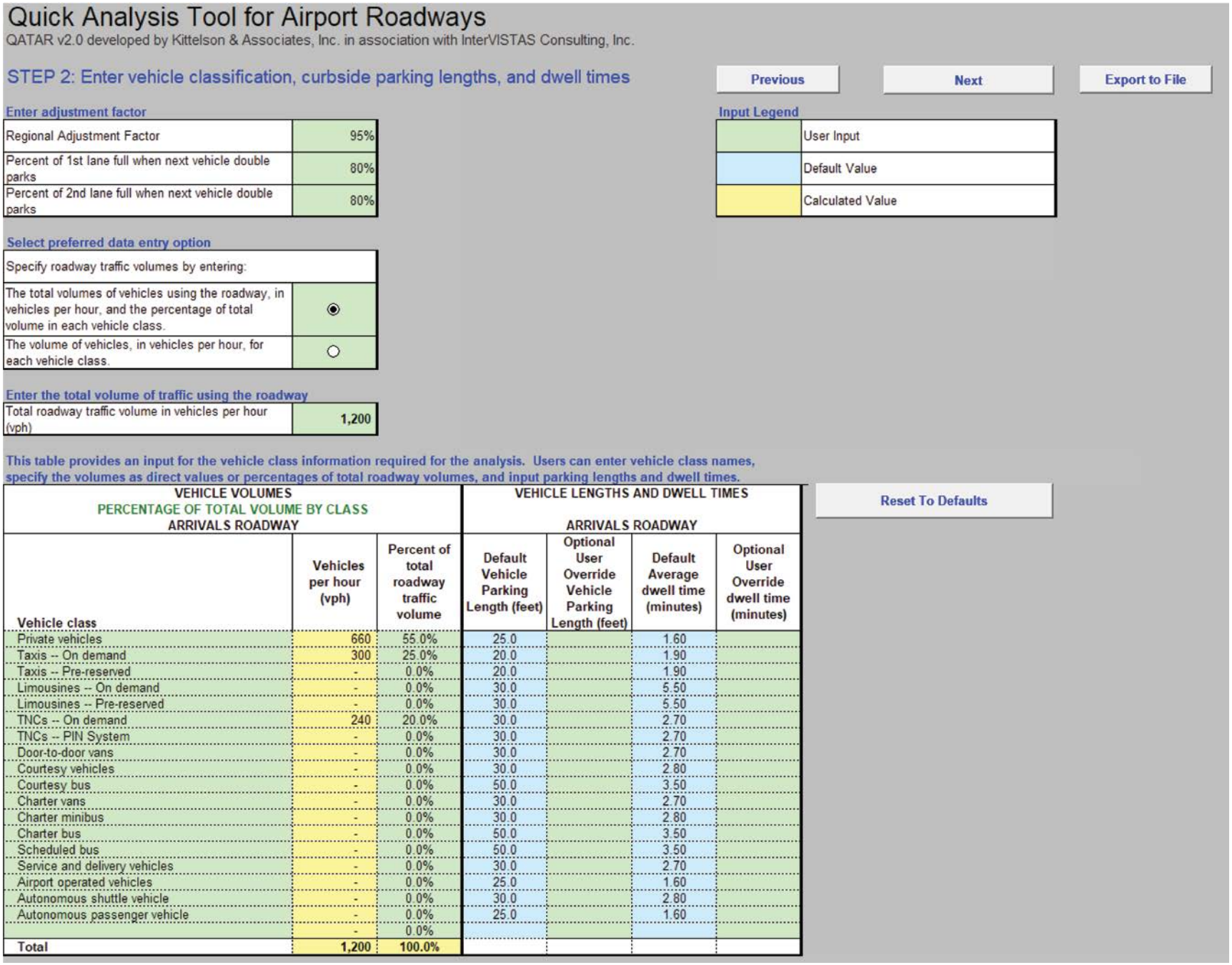
- The number of vehicles stopping in each curbside zone to drop off or pick up passengers. The number of spaces occupied simultaneously (assuming a 95% probability), when compared to the number of available spaces, defines the level of service for the curbside lane.
- The number of through vehicles proceeding to/from adjacent zones. The capacity of the through lanes is determined by the total number of travel lanes and the curbside utilization ratio. The number of through vehicles plus the number of parking vehicles, when compared to the capacity of the available through and maneuvering lanes, defines the sufficiency of the bypass lanes. As described previously, an increase in the curbside utilization ratio (i.e., an increase in the amount of double- and triple-parking) causes a reduction in the capacity of the through lanes.
- The peaking characteristics of the roadways, assuming that the volumes will not be exceeded 95% of the time during the analysis period. Traffic volumes on curbside roadways are not uniform throughout an hour-long period, or other analysis period, and peak periods of activity or microbursts of traffic occur frequently.

Inputs

Figure 5-4 and Figure 5-5 present examples of QATAR input sheets (including the suggested default values for dwell times and vehicle stall lengths). As shown, the following information is required to use QATAR:

- **Curbside geometry**—The physical characteristics of the curbside, including length, number of lanes, driver-side parking versus passenger-side parking, and number of roadway lanes approaching the curbside area. The curbside can be divided into a series of zones, each with its own characteristics.

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Quick Analysis Tool for Airport Roadways

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STEP 2: Enter vehicle classification, curbside parking lengths, and dwell times

Previous | Next | Export to File

Enter adjustment factor

Regional Adjustment Factor	95%
Percent of 1st lane full when next vehicle double parks	80%
Percent of 2nd lane full when next vehicle double parks	80%

Input Legend

	User Input
	Default Value
	Calculated Value

Select preferred data entry option

Specify roadway traffic volumes by entering:	
The total volumes of vehicles using the roadway, in vehicles per hour, and the percentage of total volume in each vehicle class.	◉
The volume of vehicles, in vehicles per hour, for each vehicle class.	○

Enter the total volume of traffic using the roadway

Total roadway traffic volume in vehicles per hour (vph)	1,200

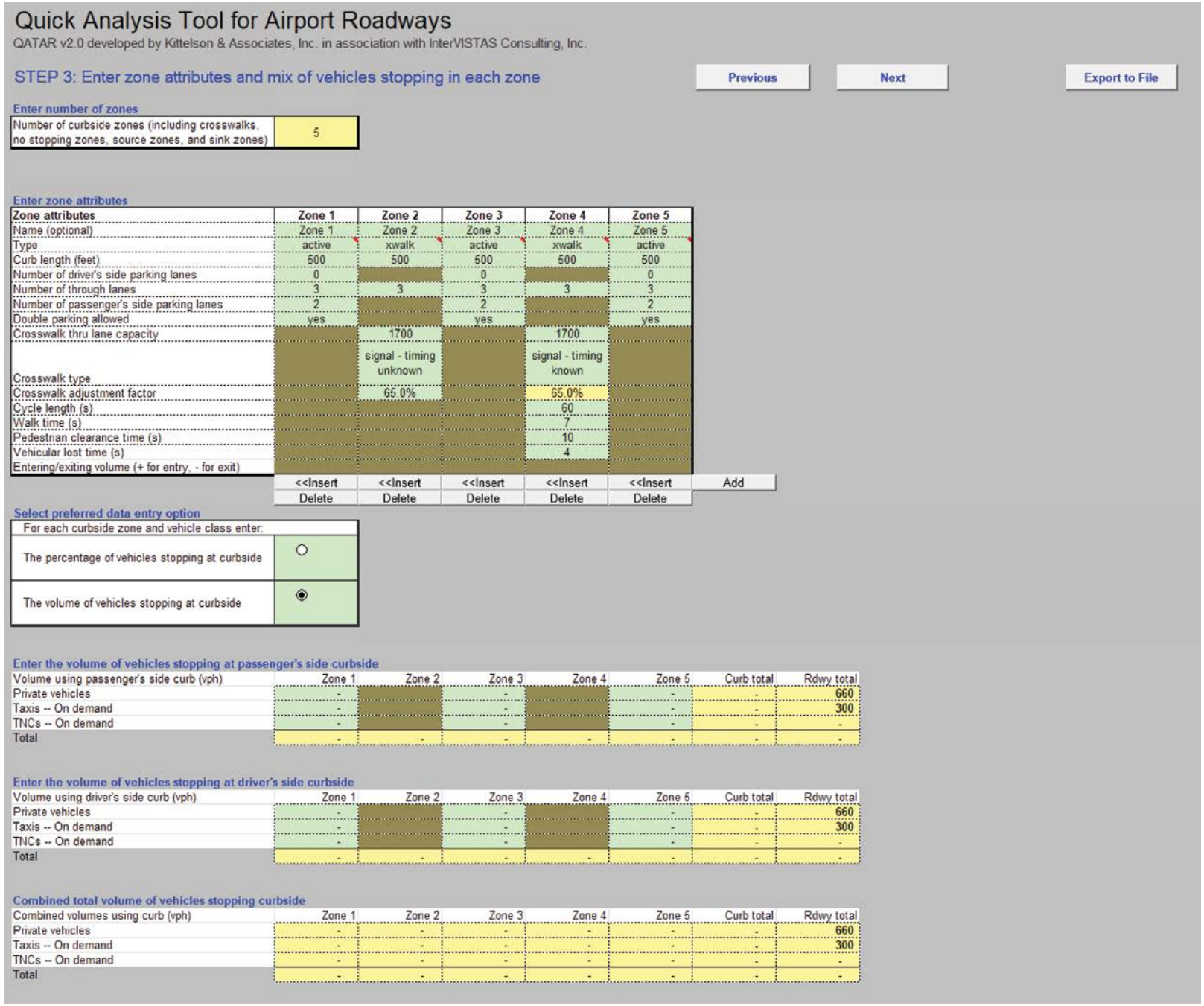
This table provides an input for the vehicle class information required for the analysis. Users can enter vehicle class names, specify the volumes as direct values or percentages of total roadway volumes, and input parking lengths and dwell times.

Reset To Defaults

VEHICLE VOLUMES PERCENTAGE OF TOTAL VOLUME BY CLASS ARRIVALS ROADWAY			VEHICLE LENGTHS AND DWELL TIMES ARRIVALS ROADWAY			
Vehicle class	Vehicles per hour (vph)	Percent of total roadway traffic volume	Default Vehicle Parking Length (feet)	Optional User Override Vehicle Parking Length (feet)	Default Average dwell time (minutes)	Optional User Override dwell time (minutes)
Private vehicles	660	55.0%	25.0		1.60	
Taxis -- On demand	300	25.0%	20.0		1.90	
Taxis -- Pre-reserved	-	0.0%	20.0		1.90	
Limousines -- On demand	-	0.0%	30.0		5.50	
Limousines -- Pre-reserved	-	0.0%	30.0		5.50	
TNCs -- On demand	240	20.0%	30.0		2.70	
TNCs -- PIN System	-	0.0%	30.0		2.70	
Door-to-door vans	-	0.0%	30.0		2.70	
Courtesy vehicles	-	0.0%	30.0		2.80	
Courtesy bus	-	0.0%	50.0		3.50	
Charter vans	-	0.0%	30.0		2.70	
Charter minibus	-	0.0%	30.0		2.80	
Charter bus	-	0.0%	50.0		3.50	
Scheduled bus	-	0.0%	50.0		3.50	
Service and delivery vehicles	-	0.0%	30.0		2.70	
Airport operated vehicles	-	0.0%	25.0		1.60	
Autonomous shuttle vehicle	-	0.0%	30.0		2.80	
Autonomous passenger vehicle	-	0.0%	25.0		1.60	
	-	0.0%				
Total	1,200	100.0%				

Figure 5-4. Example QATAR input sheet—Step 2.

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Quick Analysis Tool for Airport Roadways

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STEP 3: Enter zone attributes and mix of vehicles stopping in each zone

Previous | Next | Export to File

Enter number of zones

Number of curbside zones (including crosswalks, no stopping zones, source zones, and sink zones)	5

Enter zone attributes

Zone attributes	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Name (optional)	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Type	active	xwalk	active	xwalk	active
Curb length (feet)	500	500	500	500	500
Number of driver's side parking lanes	0		0		0
Number of through lanes	3	3	3	3	3
Number of passenger's side parking lanes	2		2		2
Double parking allowed	yes		yes		yes
Crosswalk thru lane capacity		1700		1700	
Crosswalk type		signal - timing unknown		signal - timing known	
Crosswalk adjustment factor		65.0%		65.0%	
Cycle length (s)				60	
Walk time (s)				7	
Pedestrian clearance time (s)				10	
Vehicular lost time (s)				4	
Entering/exiting volume (+ for entry, - for exit)					
	<<Insert Delete	<<Insert Delete	<<Insert Delete	<<Insert Delete	<<Insert Delete

Add

Select preferred data entry option

For each curbside zone and vehicle class enter:	
The percentage of vehicles stopping at curbside	○
The volume of vehicles stopping at curbside	◉

Enter the volume of vehicles stopping at passenger's side curbside

Volume using passenger's side curb (vph)	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Curb total	Rdwy total
Private vehicles	-		-		-	-	660
Taxis -- On demand	-		-		-	-	300
TNCs -- On demand	-		-		-	-	-
Total	-	-	-	-	-	-	-

Enter the volume of vehicles stopping at driver's side curbside

Volume using driver's side curb (vph)	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Curb total	Rdwy total
Private vehicles	-		-		-	-	660
Taxis -- On demand	-		-		-	-	300
TNCs -- On demand	-		-		-	-	-
Total	-	-	-	-	-	-	-

Combined total volume of vehicles stopping curbside

Combined volumes using curb (vph)	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Curb total	Rdwy total
Private vehicles	-	-	-	-	-	-	660
Taxis -- On demand	-	-	-	-	-	-	300
TNCs -- On demand	-	-	-	-	-	-	-
Total	-	-	-	-	-	-	-

Figure 5-5. Example QATAR input sheet—Step 3.

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- **Hourly traffic volumes**—The hourly volume of vehicles entering the curbside.
- **Through vs. curbside traffic volumes**—The proportion of vehicles using the roadway that stop at the curbside. If the user has divided the curbside into zones, the proportion (or volume) of vehicles stopping in each zone is required.
- **Vehicle mix**—The mix (i.e., classification) of vehicles in the traffic stream entering the curbside (either the actual volume or the percent of vehicles by vehicle classification). If the user has divided the curbside into zones, the proportion (or volume) of vehicles by classification stopping in each zone is required, or the user can determine that the proportion is constant in each zone.
- **Dwell times**—The user can accept the default values in QATAR or enter vehicle dwell times by vehicle classification.
- **Vehicle stall length**—The user can accept the default values in QATAR or enter vehicle stall lengths by vehicle classification.
- **Adjustment factors**—The user can enter adjustment factors in QATAR to reflect the effect of pedestrian crosswalks, regional conditions/driver behavior, and a weighting/calibration factor.

Outputs

Figure 5-6 presents an example of a QATAR output sheet.

As shown, QATAR yields the following outputs:

- **Sufficiency**—A graphic depicting the sufficiency for the curbside areas and roadway through lanes in each zone.
- **Volume/capacity ratio**—A tabular presentation of the volume/capacity ratio for the through lanes in each zone.
- **Curbside utilization ratio**—A tabular presentation of the curbside utilization ratio for the curbside area in each zone.

In some cases, the capacity of the roadway approaching the curbside may dictate the capacity of the curbside roadway segment. For example, the capacity of a five-lane curbside section with a two-lane approach roadway may, during periods of low curbside demand, be governed by the ability of the approach roadway to deliver vehicles to the curbside.

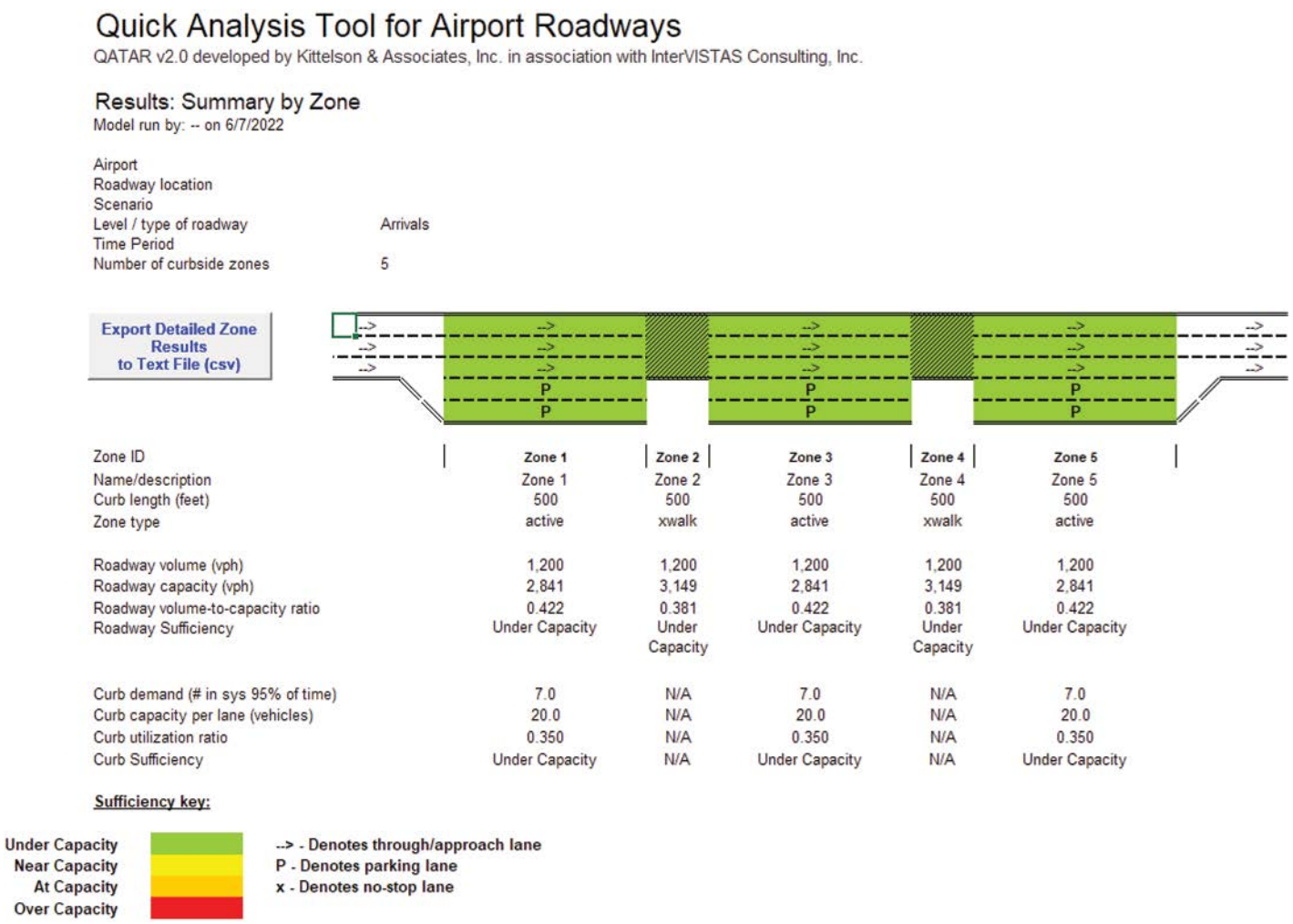
Limitations of the Analysis Tool

QATAR is used to analyze the macroscopic flow of vehicles but not the operation of individual vehicles (as would a roadway traffic microsimulation model). As such, QATAR does not

- Replicate or analyze operations, such as individual vehicles maneuvering into or out of curbside spaces, improperly parked vehicles, vehicle acceleration/deceleration characteristics, or how these characteristics vary by vehicle size or type.
- Analyze how roadway congestion or queues affect traffic operations in the zones located upstream of those being analyzed or meter (i.e., restrict) the flow of vehicles into downstream zones. The QATAR model looks at each zone independently and does not model interactions between zones.
- Analyze the performance zones that operate in a linear queue fed by dispatch from an offsite hold lot, such as a taxi queue or TNC PIN-system queue.
- Represent pedestrians crossing a curbside roadway (properly or improperly) or vehicle delays caused by pedestrian activity other than to allow the user to estimate the approximate decrease in roadway capacity.
- Evaluate the potential capacity decreases of specific curbside geometries. Rather, a single, continuous, linear curbside roadway is assumed in the model. If the curbside roadway consists of one or more parallel curbside roadways, QATAR should be used to analyze each parallel curbside roadway separately.

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Quick Analysis Tool for Airport Roadways

QATAR v2.0 developed by Kittelson & Associates, Inc. in association with InterVISTAS Consulting, Inc.

Results: Summary by Zone

Model run by: -- on 6/7/2022

Airport	
Roadway location	
Scenario	
Level / type of roadway	Arrivals
Time Period	
Number of curbside zones	5

Export Detailed Zone Results to Text File (csv)

Zone ID	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Name/description	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Curb length (feet)	500	500	500	500	500
Zone type	active	xwalk	active	xwalk	active
Roadway volume (vph)	1,200	1,200	1,200	1,200	1,200
Roadway capacity (vph)	2,841	3,149	2,841	3,149	2,841
Roadway volume-to-capacity ratio	0.422	0.381	0.422	0.381	0.422
Roadway Sufficiency	Under Capacity	Under Capacity	Under Capacity	Under Capacity	Under Capacity
Curb demand (# in sys 95% of time)	7.0	N/A	7.0	N/A	7.0
Curb capacity per lane (vehicles)	20.0	N/A	20.0	N/A	20.0
Curb utilization ratio	0.350	N/A	0.350	N/A	0.350
Curb Sufficiency	Under Capacity	N/A	Under Capacity	N/A	Under Capacity

Sufficiency key:

Under Capacity
Near Capacity
At Capacity
Over Capacity

--> - Denotes through/approach lane
P - Denotes parking lane
x - Denotes no-stop lane

Figure 5-6. Example QATAR output.

As such, QATAR produces an approximation of airport curbside roadway operations intended for planning purposes. If more detailed analyses are desired, including the modeling of interaction effects between zones, the user is encouraged to use a microsimulation model capable of simulating airport curbside traffic operations.

Interpreting the Results

Although it provides an approximation of airport curbside roadway operations, QATAR allows a user to identify which aspect of a curbside roadway (insufficient parking capacity or insufficient roadway capacity) is likely the cause of poor conditions. With that, QATAR can then be used to quickly test numerous alternative curb allocations, dwell time assumptions, and other factors to determine their potential benefits.

Certain vehicles (e.g., courtesy vehicles or door-to-door vans) may make multiple stops along the terminal curbside area, especially at large airports. Vehicles making multiple stops can be represented properly (using Option C—one of the available input sheet options in QATAR) because the total volumes of vehicles stopping in each zone need not equate to the total curbside roadway traffic. However, with Option C, QATAR requires percentages of vehicles to sum to 100% and vehicles making multiple stops may not be accurately represented, particularly if they account for a significant percentage of the total vehicles entering the roadway.

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5.5.3 Use of Microsimulation Models

Guidelines

Chapter 4 provides guidelines on the use of microsimulation models for analysis of airport roadways.

It is suggested that the capability of a software package be confirmed prior to considering its use in analyzing airport curbside roadway operations, as some packages do not accurately simulate the parking or maneuvering movements that occur on an airport curbside roadway.

The following guidelines are provided for calibrating a microsimulation model for airport curbside roadways:

- If double- or triple-parking is allowed, verify that the model correctly predicts the average number of double- and triple-parkers during the peak hour (compare one-hour model simulation to one-hour field counts).
- If queuing occurs on the existing curbside roadway, count the throughput in the through lanes and the number of vehicles processed per hour in the curbside lanes under such congested conditions.
- Validate through-lane flow rates. Enter demands into the simulation model and verify that the maximum through-lane flow rate for the peak hour predicted by the model matches the field counts. Adjust mean headways in the model until the model through-lane volumes match the field counts. A difference of 5% to 10% between model through-lane volumes and field counts is acceptable.
- Validate curbside processing capacity. Enter demands in the simulation model and compare the curbside processing rate to field counts. Adjust average dwell times in the model until the processing rate over the peak hour matches the field counts.

These guidelines are in addition to guidance published elsewhere (see FHWA guide on microsimulation model validation).

Curbside Performance Measures for Analyses Performed Using Microsimulation

The performance measures presented in Table 5-1 are intended to help select the appropriate curbside analysis method. When curbside roadways are analyzed using microsimulation methods, the performance measures presented in Table 5-2 can be used to compare curbside roadway alternatives in the context of level of service.

The measures listed in Table 5-1 do not directly correspond to quantitative values equaling a specific level of service. For example, duration of queuing is a potentially useful measure in the context of comparing alternatives (e.g., if one curbside roadway alternative would result in 2 hours of queuing, while another would result in 1 hour of queuing), but the magnitude of the queuing itself could be relatively minor, so reporting a result of "under capacity" for one alternative and a result of "near capacity" for the other could be misleading. Similarly, the queue length measure can provide an easy way to compare alternatives, but a relatively long queue could be a better condition than a relatively short queue if the rate at which vehicles are served at the curbside is relatively high for the alternative with the longer queue.

Together, length of vehicle queues and average speed—two measures that are typically microsimulation software outputs—can provide a time in queue measure that can be used to compare and evaluate analyses of curbside roadway prepared using microsimulation models. Because of the wide range of motorist expectations regarding traffic conditions when they arrive at an airport curbside, a range of thresholds for time in queue between acceptable and unacceptable operations were identified, with unacceptable operations corresponding to the threshold

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of capacity (volume-to-capacity ratio equal to 1). For the lowest of these thresholds, the time in queue was identified as 50 seconds. This time (50 seconds) is consistent with the delay threshold for unsignalized intersections operating at capacity and considered to be a reasonable lower threshold. For context, consider a small-hub airport, such as Billings Logan International Airport. Most of the time, there is no queue leading to this airport's curbside, even during peak periods in bad weather. If a queue did develop such that motorists would have to be in the queue for 60 seconds, it would seem unacceptable in that context.

For the upper bound of acceptable/unacceptable thresholds, a comment expressed in at least one focus group conducted as part of prior research—moving is acceptable, not moving is not acceptable—was used. From a motorist's perspective, it would seem as if a queue were not moving if a person could walk faster than the vehicles were moving. Using an arbitrary queue length of 1 mile and brisk walking speeds of 3 to 4 mph, the time spent in such a queue would be between 15 and 20 minutes. The 20-minute time in queue appears to be a reasonable upper bound for a threshold between acceptable and unacceptable (anecdotal experience suggests that queues of this length likely occur at large airports somewhat regularly). This time in queue is not intended to represent the longest queue time during the busiest days of a year, when delays may be even greater. Also, higher values of time in queue could be used by airport operators who observe higher thresholds at their locations.

Service thresholds corresponding to conditions that are "well under capacity" (the equivalent of the HCM's LOS A) have also been defined. It is suggested that time in queue should not be zero, but should seem to a motorist as if it were nearly zero. A simple way to identify this lowest value would be to take 10% of the "at capacity" value. For the "at capacity" threshold of 60 seconds, a time in queue of 6 seconds or less would correspond to "well under capacity"—from a practical perspective, that would essentially mean no queue or perhaps one vehicle waiting, which is consistent with the original basis for this threshold. With an "at capacity" threshold set at 20 minutes, the "well under capacity" time in queue would, therefore, be 120 seconds. Although 120 seconds in a queue seems high compared to, for example, a signalized intersection delay, for a motorist approaching a curbside anticipating a wait of up to 20 minutes, a 2-minute wait would seem remarkably short.

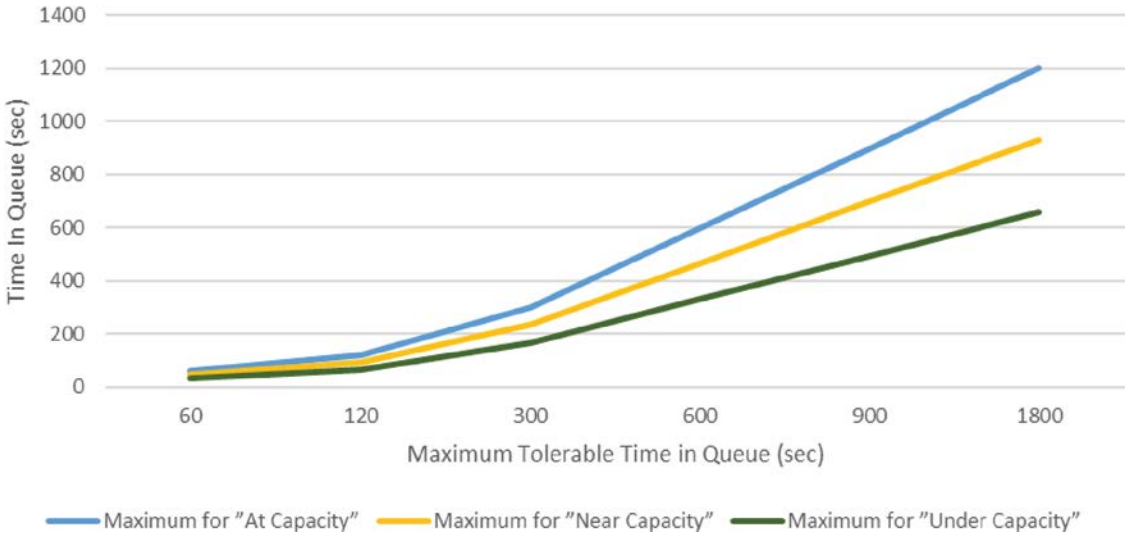
Once the upper and lower level-of-service bounds are identified, the values for the other sufficiency levels can be calculated using a straight-line projection between the two points, with the "under capacity" level representing the midpoint between "well under capacity" and "at capacity." The results of these estimates, assumptions, and calculations are presented in Table 5-5. The information can also be presented in graph form, as shown in Figure 5-7. As noted, the values of the time in queue can easily be extrapolated upward from the 20-minute level to any value.

Table 5-5. Time spent in queue for various sufficiency levels.

	Small-hub and smaller medium-hub airports[a]			Large medium-hub and large-hub airports[a]		
	Given maximum acceptable time spent in queue in seconds[a]					
Sufficiency	60	120	300	600	900	1,200
Maximum for "at capacity"	60	120	300	600	900	1,200
Maximum for "near capacity"	47	93	233	465	698	930
Maximum for "under capacity"	33	66	165	330	495	660

Note: Input data are to be taken from microsimulation modeling output.
[a]Analyst must first select a value for the maximum acceptable time spent in queue for the subject airport. Then, using queue length and average speed outputs from the microsimulation model, the level of service can be identified.

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Note: Input data to be taken from microsimulation modeling output. Analyst must first select a value for the maximum acceptable time spent in queue for the subject airport. Then, using queue length and average speed outputs from the microsimulation model, the sufficiency can be identified.

Figure 5-7. Time spent in queue for sufficiency levels, large medium-hub and large-hub airports.

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CHAPTER 6

Improving Airport Curbside and Terminal Area Roadway Operations

This chapter presents examples of commonly occurring airport curbside and roadway operational problems and potential improvement measures.

Analyses and evaluations of airport curbside and terminal area roadways generally involve the following steps:

1. **Identify the problem(s).** Problem identification includes determining the causes of existing congestion, delays, imbalances in demand, and/or whether the existing (or proposed) roadway network can accommodate anticipated future requirements.
2. **Document goals and objectives.** Documenting the relevant goals and objectives of airport management (and other stakeholders) with respect to roadway operations is a key step in the analysis and evaluation process. The relevant objectives may include such broad categories as the following:
   - Provide safe and secure operations for airport users.
   - Provide desired levels of customer service for airline passengers, visitors, employees, and other airport users.
   - Accommodate existing and future requirements.
   - Accommodate regional mobility needs/encourage the use of public transportation.
   - Support local, regional, and/or state air quality goals.
   - Support the airport's ability to maintain or enhance airfield capacity by ensuring that changes to roadways and curbsides do not negatively affect airfield operations or capacities.
   - Maintain and enhance the net revenues generated by the airport.
   - Ensure accessibility for those with disabilities.

   Detailed descriptions and definitions of goals will allow the development of airport-specific objectives that can be used to compare and evaluate alternative improvement measures.
3. **Identify and develop potential improvements.** The potential improvement measures described in this chapter can serve as a starting point for improvements that address commonly occurring airport curbside and terminal area roadway operations. The following sections present commonly occurring problems and potential improvement measures for terminal area roadways and curbsides, respectively.
4. **Evaluate the potential improvements.** The alternative analytical methods, described in previous chapters, can be used to quantify the changes expected to result from the potential improvements, assess their advantages and disadvantages, and identify the preferred improvement(s).
5. **Reach consensus on the preferred improvement.** A key step in the implementation process is to build consensus supporting the selection and implementation of the preferred alternative. An evaluation process that quantifies the extent to which the potential improvement would support the stated goals and objectives of airport management (and other stakeholders) provides a foundation for achieving consensus.

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6. **Implement the preferred solution.** This step could involve design and construction activities, operational improvements, or changes in airport management policies.

6.1 Typical Terminal Area Roadway Problems

Operational and physical problems can adversely affect the ability of terminal area roadways to accommodate traffic efficiently and safely. In this section, 10 types of deficiencies that may occur in an airport environment are identified. These deficiencies can typically result in queues or delays; many airport roadways exhibit one or more of the deficiencies described below. Possible strategies to address these problems are presented in Chapter 7.

6.1.1 Insufficient Roadway Capacity

A roadway has insufficient capacity if, during the analysis period, the roadway operates at "under capacity" or worse (see Chapters 4 and 5 for an explanation of sufficiency categories). "Under capacity" refers to congested roadways and is an unacceptable basis for planning airport roadways. Specific implications of insufficient roadway capacity include (1) congested roadway sections with queues extending to upstream roadways, (2) motorists experiencing frequent congestion and significant delays, and (3) a generally unsatisfactory airport experience.

6.1.2 Insufficient Merging Capacity

Insufficient merging capacity results when a roadway does not provide sufficient capacity at points where two or more streams of traffic combine into a single stream. This deficiency results in roadway delays, congestion, and traffic queues extending back from the merge point. Merge segment capacity is determined by (1) the volume of entering traffic, (2) the operating speeds, and (3) the number of lanes upstream and downstream.

6.1.3 Inadequate Weaving Distance

Inadequate weaving distance results when a roadway does not provide sufficient length or travel lanes to accommodate the traffic volumes at the point where two or more streams of traffic traveling in the same direction cross or merge, causing vehicles to decelerate (or stop) while waiting for adequate gaps in the traffic stream. This deficiency results in (1) vehicle delays and queues, (2) higher accident rates, and (3) slower speeds and flow rates. Factors influencing required weaving distances are operating speeds, traffic volumes (merging, weaving, and flowing through the segment), and the number of lanes that vehicles must cross to complete the desired maneuver.

6.1.4 Lane Imbalance

Lane imbalance results when a roadway segment, before a diverge or after a merge, contains two (or more) fewer lanes than the combined total number of lanes entering or exiting the segment. For example, at a point where two three-lane roadways merge, the downstream segment must consist of at least five lanes, or a lane imbalance will result. At a point where a roadway diverges into two two-lane roadways, the upstream segment (prior to the diverge) must consist of at least three lanes. A lane imbalance can cause (1) increased delays, (2) sudden diverge or weave maneuvers, (3) increases in the required roadway weaving distances (e.g., the number of lanes to be crossed), and (4) higher accident rates. Proper lane balance helps reduce or avoid forced merges, weaves, and sudden maneuvers and is typically accomplished by providing a lane that allows motorists access to both exit routes (e.g., an "either-or" lane) or by extending a lane downstream

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past the merge/diverge point and then dropping the lane using design guidelines appropriate for the roadway speed (e.g., taper distances).

6.1.5 Directional Information Overload

Directional or wayfinding information overload occurs when more information (or decisions) is presented to a motorist than the motorist can read, comprehend, and react to in the available time (and distance). This overload causes drivers to weave suddenly, miss exits, make sudden or erroneous movements, or, in extreme cases, stop in the roadway (or on the shoulder) to read the signage.

It is desirable to avoid presenting more than two decisions or more than four lines of text on each directional sign. If more than four lines of text must be used on one sign at an airport, it is necessary to prioritize the information and avoid using unfamiliar or inconsistent terms.

6.1.6 Insufficient Decision-Making Distance

Insufficient decision-making distance is defined as an insufficient distance (or time) for motorists to read, comprehend, and react to information regarding a decision that must be made. This situation causes drivers to weave suddenly, miss exits, make wrong turns, or, in extreme cases, stop in the roadway to read the message or back up to the decision point. Factors contributing to providing the necessary decision-making distance include (1) travel speed, (2) message content, (3) visibility of the decision point, and (4) visibility of the directional signage.

6.1.7 Insufficient Queuing Space

Queuing space represents the area required to accommodate vehicles stopped at an entrance (or exit) to a parking lot or other facility, traffic signal or turn lane, or vehicle inspection area so that vehicles in the queue do not interfere with traffic flow on the adjacent roadway or travel lanes. For example, a parking facility entrance should have sufficient space to accommodate vehicles queuing at the ticket-issuing machines without having the queue extend onto the adjacent roadway.

6.1.8 Unexpected Lane Drops/Inadequate Taper Lengths

Unexpected lane drops and inadequate taper lengths (the distance required to introduce a new lane or drop an existing lane) result when a through lane unexpectedly ends and motorists are required to unexpectedly merge quickly into an adjacent lane. Required taper lengths, which vary according to roadway operating speeds, are intended to allow sufficient distance for lane channelization and vehicle merging. Unexpected lane drops reduce roadway capacity and travel speeds, as motorists who become "trapped" in a lane are required to merge quickly (interfering with the flow of other vehicles in adjacent lanes).

6.1.9 Unexpected Transition from High-Speed to Low-Speed Roadway Environment

Some motorists do not realize they need to slow down as they exit from a regional roadway (which may operate at more than 55 mph) and approach a terminal area roadway (which may operate at less than 30 mph) until they encounter a sharp curve at the entrance to the terminal or vehicles stopped in the roadway. This situation is particularly true at airports where a limited-access highway, designed to freeway standards and capable of accommodating freeway speeds, connects the regional roadway network with the terminal area roadways (see Figure 6-1).

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Figure 6-1. Transition from high-speed to low-speed airport roadways at Baltimore/Washington International Thurgood Marshall Airport.

Motorists may be provided few visual clues that the driving environment is changing and requires them to decelerate. In addition, speed limit signs may get lost among the many other signs and distractions associated with roadways approaching an airport terminal.

This transition is compounded by the reduction in roadway capacity that accompanies the speed reduction: a three-lane access roadway operating at 55 mph (or more) has more capacity than a three-lane curbside roadway operating at 30 mph (or less). If the traffic volume on the access roadway is the same as that on the curbside roadway, it is necessary to provide additional travel lanes on the curbside roadway to compensate for the reduction in travel speed. Often, the volumes are not constant, as some traffic exits for non-terminal area destinations, such as parking and rental car facilities.

6.1.10 Missing Movements

Missing movements are defined as a desired travel path or traffic movement that is not provided on an airport roadway network. If a movement is missing, motorists may need to exit and re-enter the airport or travel extra distance. For example, at most major airports, motorists can proceed directly (1) from the enplaning curbside to short-duration parking and (2) from short-duration parking to the deplaning curbside without leaving the terminal area. The absence of roadway segments providing these direct movements increases traffic demand on the return-to-terminal roadways and vehicle miles of travel.

6.2 Typical Curbside Roadway Problems

Operational and physical problems can adversely affect the ability of airport curbside roadways to accommodate traffic safely and efficiently. Typical problems include those listed below. Possible strategies to address these problems are presented in Chapter 7.

6.2.1 Insufficient Curbside Roadway Capacity

Insufficient curbside roadway capacity exists when curbside requirements (lengths) are greater than 1.3 times the usable curbside length and/or the through lanes on a curbside roadway operate at LOS C or worse. When curbside demand exceeds available capacity, motorists experience

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significant delays and queues, as evidenced by high levels of double- and triple-parking throughout the entire curbside, which, in turn, reduces the capacity and travel speeds of the curbside roadway through lanes. As noted previously, curbside roadways must provide both adequate curbside length (stopping space) as well as adequate throughput capacities. Any deficiencies in one area will adversely affect the other. Factors that contribute to a lack of curbside roadway capacity are described in the following paragraphs.

6.2.2 Imbalances in Demand

Imbalances in curbside demand occur when the total length of curbside space available is sufficient to accommodate curbside demand, but most of the demand occurs at and overloads the capacity of one segment of the total curbside area. For example, if the airline(s) on one concourse serves most of the peak-hour airline passenger activity, then curbside traffic will be concentrated at the doors leading to the portions of the terminal building occupied by this airline(s), leaving the remainder of the curbside areas empty or underutilized. It is generally not feasible to relocate the assigned airline ticket counter or baggage claim area locations solely to balance curbside demand.

6.2.3 Insufficient Number of Travel Lanes

A curbside roadway that does not provide sufficient capacity to accommodate existing or future requirements at LOS C or better typically has an insufficient number of travel lanes. Generally, curbside roadways with four lanes or more provide sufficient capacity because two travel lanes remain available even when double-parking occurs. Congestion and delays may occur frequently on curbside roadways having three lanes or fewer as any double-parking severely restricts through traffic. Similar restrictions also occur when vehicles are allowed to stop on the inner and outer lanes of a four-lane curbside, leaving only the two center lanes for through traffic.

6.2.4 Pedestrian Crosswalks and Pedestrian Activity

Pedestrians crossing a curbside roadway restrict the volumes of through traffic that can be accommodated. Delays caused by crosswalks are related to (1) the volume of pedestrians walking across a curbside roadway, (2) the proportion of time that pedestrians occupy a crosswalk (properly or improperly), and (3) the number of crosswalks located at curbside. Traffic flows and safety can also be adversely affected by pedestrians stepping into the roadway to avoid columns or other obstacles, hail vehicles, or board/alight from a vehicle stopped in a through lane.

6.2.5 Driveways Serving Adjacent Land Uses

Driveways serving adjacent land uses (e.g., parking lots or rental car ready/return areas) may impede the flow of curbside traffic when vehicles in the lane farthest from the terminal decelerate (or accelerate) as they enter (or exit) the driveways serving the adjacent land uses. Vehicle queues formed at the entrances to these land uses may extend back onto the curbside roadways.

6.2.6 Insufficient Curb Length

Insufficient curb lengths result from curbside demand that is greater than 1.3 times the usable curbside length, which also occurs when there is significant double-parking.

6.2.7 Inefficient Allocation of Curb Space

Inefficient allocation of curb space results where the total available space is adequate to accommodate demand, but the available space has been divided into (or allocated among) many

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categories of ground transportation services such that some categories are allocated more curb space than required while others are not allocated enough. This situation may occur when curb space is allocated to vehicles that rarely serve the airport (e.g., charter buses), demands have changed as a result of the introduction of new services, or the space has been broken into small segments that do not correspond to the operational or maneuverability needs of the assigned class of ground transportation service.

Similarly, inefficient allocation of curb space results when the amount of curb associated with a specific airline does not match the relative share of passenger traffic served by that airline.

6.2.8 Unusable Curbside Roadway Geometry

Unusable curbside roadway geometries exist when vehicles cannot stop to load or unload passengers because of the curved alignment of the roadway, narrow sidewalks, or other physical obstructions. Many terminals have curved curbside roadways, but generally, the radii of these roads are very large and motorists do not perceive that they have stopped along a curvilinear section. However, some curbside roadways have small radii or tight curves that hinder a motorist's ability to park parallel to the sidewalk or to enter or exit this space. Motorists may be unable to park adjacent to curbsides having narrow sidewalks (e.g., the ends of island curbside areas) or columns (or other obstacles) adjacent to the roadways. Large bollards, which are sometimes placed on terminal building sidewalks to protect motorists and the adjacent terminal, may interfere with the ability of motorists to open/close their doors or enter/exit their vehicles.

Narrow sidewalks may also force pedestrians to step into the roadway (with their baggage) to bypass columns, queues of passengers formed at skycap positions or benches, or other obstacles.

6.2.9 Excessive Dwell Times

Excessive dwell times result when vehicles (either private or commercial vehicles) are allowed to remain at the curbside when not actively loading or unloading passengers. In the case of some commercial vehicle providers, excessive dwell times are permitted by airport rules and regulations. Excessive dwell times may reflect insufficient police presence or visibility, or permissive airport policies, and may occur even if the dwell times of most vehicles are within reasonable limits, and fewer than 10% of vehicles remain at the curbside for excessive periods.

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CHAPTER 7

Innovative Strategies

The following sections present the information practitioners need to understand, evaluate, and implement the traffic demand and congestion management strategies best suited for their airports. These sections describe strategies applicable to all commercial-service airports recognizing that they each have different or distinguishing characteristics. Included in each strategy are detailed descriptions of the benefits, disadvantages, capital and operating costs, implementation schedule, potential phasing, flexibility to adapt to unanticipated demands and operations, policy and regulatory issues, and stakeholder impacts and coordination. Descriptions of innovative case studies and technologies are also presented. These strategies are intended to be readily implementable, as opposed to those that require large capital investments and are disruptive to airport operations (e.g., widening or lengthening an elevated curbside roadway or extending rail service to the terminal).

There are three primary strategy categories: Enforcement, Traffic Demand, and Congestion Management Strategies. There are four complementary and/or stand-alone strategy categories: Customer Service, Sustainability, Accommodating TNCs, and Accommodating AVs. Furthermore, numerous supporting technologies are available to enhance these primary and complemental strategies. These supporting technologies are capable of monitoring traffic volumes and locations (by type of vehicle or individual vehicle), automatically dispatching taxicabs and other on-demand services, supporting curbside enforcement by directing officers to vehicles needing attention, guiding motorists by displaying changeable directional or parking information, and can enhance the primary and complemental strategies in other ways as well.

7.1 Enforcement Strategies

Enforcement of rules and regulations governing airport roadway traffic and curbside operations is a key element of almost all strategies. Without enforcement, some motorists are likely to disobey regulations governing traffic operations and add to roadway congestion. If a few motorists do so, additional drivers may perceive that they too can safely ignore rules and regulations, leading to traffic congestion and delays. This section discusses the programs and technologies to support the personnel responsible for enforcement. Examples include

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.2. Control of crosswalks—officers versus signals
- 7.1.3. Measures to support curbside enforcement: warning messages, ticketing, and towing
- 7.1.4. Regulatory sign placement and conformance

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7.1.1 Enforcement of Airport Roadways and Curbsides

Key considerations related to enforcing airport roadways include

1. Determining the number of officers needed to manage terminal curbside areas and enforce traffic rules.
2. Using law enforcement officers and traffic control officers.
3. Using technology in enforcement.
4. Balancing customer experience and strict enforcement.
5. Enforcing consistently and actively during peak and non-peak periods.

Number of Officers Required to Manage Terminal Curbside Areas and Enforce Traffic Rules

The number of officers required to manage terminal curbside areas and enforce traffic rules varies among airports reflecting multiple factors including

- The amount of curbside traffic during the peak period, and the composition of this traffic. Typically, managing commercial ground transportation vehicles and enforcing traffic rules requires fewer officers than managing the equivalent number of private vehicles.
- The number of curbside levels and terminal buildings. More officers can be assigned to the busy levels or terminal areas as warranted as opposed to having the same number of officers at all locations.
- The terminal curbside configuration (e.g., parallel curbsides separated by a raised island versus a single curbside). Patrolling a curbside roadway consisting of an inner and outer curbside requires fewer officers than if the same total length of space were on a single curbside. The allocation of one of the parallel curbs to commercial ground transportation can also impact the number of enforcement officers needed.
- The use of foot patrols versus motorized patrols. Motorized patrols are more visible to motorists, convey greater authority than foot patrols, and allow officers to respond to violators and roadway congestion more quickly. Typically, fewer officers are required with motorized patrols because officers can oversee more of the curbside and roadway areas compared with foot patrols. However, motorized patrols limit opportunities for one-on-one interaction with customers. Additionally, motorized patrols take physical space that could be otherwise used for other vehicle types and are typically unable to backtrack easily, whereas foot patrols can operate on the sidewalk or between travel lanes and can be more mobile in all directions.
- The relative adherence of the local population to posted regulations. In some communities, it is common for motorists to stop for pedestrians crossing a street, to allow gaps for weaving traffic, or to allow traffic to enter from a side street. In other communities, many motorists drive more aggressively.
- Curbside use tolerated in the past. The management of some airports has allowed motorists to dwell at the curbside for excessive periods or stop in through lanes while waiting for arriving passengers. Changing past behavior patterns can be challenging and require greater effort by enforcement staff.

Information reported by selected airports indicates that, depending upon the previously listed factors and other considerations, one officer can patrol between 400 to 800 linear feet of terminal curbside. For example, an airport with about 7,500 linear feet of curbside divided among two levels reported having 15 officers patrolling the curbsides during peak periods, or about one officer for every 500 feet. These officers are in addition to those required to patrol non-terminal area roadways or provide security elsewhere at the airport.

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Using Law Enforcement Officers and Traffic Control Officers

Traffic flow management and enforcement responsibilities can be assigned to law enforcement officers (LEOs), who are sworn officers; traffic control officers (TCOs), who are civilian officers; or more frequently a combination of the two. Use of LEOs and TCOs varies widely among airports, reflecting each airport's prioritization of customer service and enforcement, budget, configuration, governance, and other factors.

LEOs are public sector employees sworn in as police officers, sheriffs, or other positions to enforce the laws in the jurisdiction they serve. An LEO is licensed to carry a weapon, arrest violators, issue tickets, and perform other duties. At an airport, the responsibilities of LEOs may include enforcing airport rules and regulations and federal security directives (e.g., supporting TSA staff), responding to airport emergencies, conducting patrols and investigations, and managing traffic flow. LEOs working at airports may be (1) a member of the airport staff who reports to the airport director or (2) a member of a municipal police unit or patrol district assigned to the airport who reports to the municipal chief of police while coordinating with airport staff.

TCOs are public or private sector employees whose responsibilities at an airport may include managing roadway traffic flows, enforcing curbside parking and commercial ground transportation regulations, and performing airfield access gate controls and badge checks (a TCO is shown in Figure 7-1). TCOs are not licensed to carry a weapon, arrest individuals, intercede in fights or disputes among drivers or customers, or ticket moving vehicles. When enforcement of such actions is necessary, a TCO must request support from a LEO. At airports, TCOs may be employed directly by an airport operator or by a private contractor retained to enforce terminal curbside regulations, manage parking facilities, provide customer assistance, or perform other duties. When a private contractor is responsible for providing TCOs, often these services are included in a contract for broader services such as parking management.



Figure 7-1. TCO at Salt Lake City International Airport.

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Decisions regarding the assignment of LEOs and TCOs are influenced by differences in

- **Costs of wages and benefits.** The loaded costs (salaries plus benefits) of an LEO are generally more than those of a TCO particularly if the LEO is an employee of a municipal police force. The loaded costs of an LEO may be 1.5 to more than 2.0 times greater than those of a TCO depending on whether the TCO is employed directly by the airport or by a private contractor. (Private contractors typically offer employees fewer benefits than airports.)
- **Authorized tasks/need for back-up.** As noted, the enforcement powers of TCOs are generally limited. When TCOs are responsible for traffic management and curbside enforcement, the support of LEOs is required to issue tickets to motorists in moving vehicles (versus citations for disobeying no stopping/standing regulations), intercede in fights, arrest individuals, and conduct other tasks that TCOs are not permitted to perform. As such, it is necessary for TCOs to have the support of LEOs, usually accessed through the landside operations team, and for LEOs to be readily available and responsive.
- **Conflicting responsibilities/ready availability of staff for traffic control.** The availability of LEOs to perform airport roadway and curbside enforcement during busy hours may be limited because of their multiple responsibilities, and which responsibilities are considered by their supervisors to have higher priority. LEOs whose supervisors are not located at the airport (i.e., those who are "downtown") and are less familiar with the airport's needs may have less availability for roadway traffic enforcement duties compared to LEOs whose supervisors report to the airport director. In contrast, normally the primary responsibility of TCOs is traffic enforcement and their availability is determined by the airport director or her/his representative.
- **Level of attention.** Some airport staff report that some LEOs, especially more senior officers, prefer to perform tasks that reflect their police training and background rather than manage curbside traffic. In contrast, TCOs recognize that their primary responsibility is traffic management. Some airport staff reported that it can be challenging to encourage LEOs to enforce curbside operations in accord with airport customer service goals.

Using Technology in Enforcement

The technology needed to support the enforcement of airport roadways and curbsides is rapidly improving and evolving. Some airports have started to explore the use of radar, Light Detection and Ranging (LiDAR), license plate recognition (LPR), video processing, and other spatial sensing technologies that are employed to monitor traffic on highways and the use of curb spaces along urban streets. These technologies allow for the accurate tracking of the volumes, paths, and stopping locations of individual vehicles and pedestrians. Using these technologies, it is possible to determine when and where individual vehicles stop, monitor their dwell times, and inform officers of violators requiring attention in support of ticketing or administrative actions.

Such technologies allow for and support improved roadway curbside management, and, by potentially reducing the number of required officers, the efficient enforcement of these facilities. In some environments, such technology, supported by variable messaging signs, may replace enforcement staff during non-peak periods.

Balancing Customer Experience and Strict Enforcement

"Being as polite as one can won't make everyone happy. Drivers don't have a choice; they can either move or be ticketed."

Airport operators seek to enhance the experience of their customers, including those who are dropping off and picking up passengers. They also seek to minimize traffic congestion and ensure that motorists can easily access curb spaces. The airport operator's desire to enhance the experience of their customers may suggest allowing motorists to linger at the curbsides, even when they are not actively picking up a passenger. The desire to minimize traffic congestion may suggest strict enforcement of posted regulations (e.g., those prohibiting standing or parking) and ticketing or towing motorists violating these regulations. Lax enforcement of motorists can

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result in traffic congestion and delays. Motorists subject to strict enforcement may complain to airport management (or others) about being unfairly "harassed." To find a balance between these two extremes, airports can

The landside staff at Sacramento International Airport report they have had tremendous results when hiring traffic enforcement staff who have a customer service background versus a strictly security guard background. Staff with a customer service background are able to move traffic using a softer and more gentle approach. An example cited was to train civilian staff to tell motorists, "I need you to do XXX," rather than to demand "You need to do XXX."

- **Hire staff with customer service backgrounds.** To manage roadway and curbside traffic, airports are increasingly employing staff who possess customer service skills or training, rather than hiring experienced enforcement staff. Staff with customer service skills tend to be more persuasive than authoritative. At some airports (e.g., Portland International) such staff are referred to as passenger service agents rather than TCOs. The focus is on skills in persuading the public to follow rules rather than on more traditional security experience.
- **Train enforcement staff in customer service and traffic management.** Airports surveyed report using certified instructors to provide customer service training to enforcement staff, as well as training in how to direct and manage vehicles. Officers are told how to approach motorists, instruct motorists to move, direct motorists to cell phone waiting areas (or other locations), respond to unruly or argumentative drivers, and when to request support from LEOs, if needed.
- **Consider the appearance of TCOs.** To enhance their authority and gain the respect of motorists, it is recommended that TCOs be dressed like police officers (e.g., black, dark blue, or gray uniforms with caps). In an attempt to enhance customer experience, one Florida airport provided uniforms consisting of Bermuda shorts, khaki shirts/blouses, and pith helmets, which resulted in motorists not heeding the officers' directives.
- **Rotate staff.** At airports serving small communities, drivers may recognize enforcement officers, who may be acquaintances or distant relatives. Commercial ground transportation vehicle drivers, who serve an airport frequently, begin to recognize enforcement officers and may become friends with some. In both cases, motorists and commercial drivers may request special treatment, leading to claims of favoritism or uneven enforcement. Airports suggest rotating staff, and having them assigned to different posts and shifts to minimize or avoid such situations. Airports are best served by implementing strong accountability for any sort of favoritism, actual or perceived.
- **Find the most effective form of patrol.** Some airports supplement foot patrols of the curbsides with motorized patrols. The use of three-wheeled parking enforcement vehicles, as shown in Figure 7-2, can be effective and less expensive than the use of full-size automobiles. Three-wheeled parking enforcement vehicles allow the enforcement officers to patrol long curbside areas quickly, are very maneuverable, provide access to a bull horn or siren to encourage



Source: International Parking & Mobility Institute https://www.parking-mobility.org/2021/03/09/member-news-westward-industries-sfo/

Figure 7-2. Three-wheeled parking enforcement vehicles.

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motorists to move, and support the impression of parking enforcement. However, safety can be a concern as drivers, focused on finding an empty space or waiting passengers, may not see or expect the smaller vehicle.

Enforcing Consistently and Actively During Peak and Non-Peak Periods

It is recommended that airports enforce roadways, particularly curbside areas, consistently and actively throughout the entire day.

Consistent enforcement of curbside regulations, during both peak and non-peak periods, is necessary. A baseline of acceptable behavior is recommended. An example would be to not allow any vehicle to be unattended at any time, regardless of roadway conditions. This kind of "black and white" policy leaves no room for interpretation or negotiation. If motorists are allowed to linger at the curbside during off-peak periods when the curbside is virtually empty (or leave their vehicle unattended), they may fail to understand why the same behavior is considered unacceptable during peak periods. Experience suggests that motorists will argue with enforcement officers, explaining that they were allowed to park here recently and should be allowed to do so again. This can be particularly challenging at airports with pronounced peaks and periods when there are very few vehicles stopped at the curbside.

Active enforcement implies the visible and regular appearance of officers during peak and off-peak periods. Unfortunately, if officers are not present, some motorists will dwell at the curb for longer than necessary or attempt to leave their vehicles unattended. For example, if an officer is responsible for both curbside enforcement and patrolling inside the terminal (or performing some other non-curbside duty), the officer may tend to remain in the terminal when traffic is light. However, the officer's absence will lead to motorists violating posted regulations and believing they can do so at other times.

Enforcement should be supplemented with well-placed signage with clear messaging. Airport curbsides are frequently inundated with static messaging signage. Dwell signage should be given priority and reflect the message enforcement officers will be communicating.

Summary—Enforcement of Airport Roadways and Curbsides

Intended Goals

- Effective enforcement
- Improved traffic operations/reduced roadway and curbside congestion

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X	X	X	X	

  - Primarily patrol vehicles, security booths, and security cameras if used solely to supplement roadway and curbside enforcement.
  - Dependent on size of curbs managed by staff.

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
			X	X	X	X

  - Operations and maintenance costs will vary based upon the number of required officers, the mix of LEOs and TCOs, and the wages and benefits offered to these officers.
  - The cost range provided assumes LEOs patrolling about 400k/curbside positions ($30/hr) and TCOs patrolling about 215k/curbside positions ($20/hr).

Implementation Considerations

- **Implementation schedule:** If the airport is transitioning from only using LEOs to only using TCOs, the implementation schedule would depend on current staffing; time needed for transition to a different mix of officers, if desired; and time for recruiting and training. The implementation schedule is estimated to be less than 1 year unless additional staff are needed. Transitioning to a blend of officers could require less time. Implementing technology solutions includes the procurement time and calibration of the system, similar to calibrating LPR systems in the parking facilities.
- **Ability to implement in phases:** Yes.
- **Ease of enforcement:** Enhances and supports enforcement.
- **Likely political influence:** In some communities, there has been opposition to LEOs (sworn officers) being replaced by TCOs (civilian officers) by the unions representing the LEOs. However, typically the LEOs are assigned other duties and not replaced.

Complementary Strategies and Supporting Technologies

- 7.1.3. Measures to support curbside enforcement: warning messages, ticketing, and towing
- 7.1.4. Regulatory sign placement and conformance
- 7.2.3. Encourage the use of cell phone lots
- 7.2.9. Charging commercial vehicle operators headway fees or higher fees for curbside pickup
- 7.3.3. Raised crosswalk/speed platform
- 7.3.8. Using space detectors supplemented by flashing beacons to encourage efficient use of curb space
- 7.3.11. Modifying placement or use of regulatory and warning signs
- 7.3.12. Using variable message signs to encourage/direct traffic to less busy curbsides
- 7.6. Accommodating transportation network companies

Examples

- Airports that have implemented this strategy include Palm Beach International (certified instructor), Dallas/Fort Worth and Sacramento International (customer service training), and San Francisco International Airport (parking enforcement vehicles).

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7.1.2 Control of Crosswalks—Officers versus Signals

Management of pedestrian movements is a critical part of managing roadside congestion. Pedestrians crossing roadways between parking facilities and the terminal or between terminal islands and the terminal exacerbate congestion and increase the risk of accidents with injuries on the roadways. The proper use of crosswalks is directly related to enforcement or perceived enforcement of pedestrians and motorists. In the absence of positive controls, pedestrians will tend to cross curbside (or other) roadways at will and possibly jaywalk, ignoring traffic rules and designated crosswalks. To enhance the safety of both pedestrians and motorists, airports install fences and other barriers to encourage pedestrian use of designated crosswalks. Requiring pedestrians to use crosswalks helps motorists to see and avoid crossing pedestrians while simultaneously looking for waiting passengers and empty curbside spaces and attempting to avoid other vehicles.

The need for traffic controls at crosswalks depends on the volume of pedestrians and vehicles, the width of the roadway, the adequacy of the pedestrian refuge area (to ensure waiting pedestrians do not spill into traffic lanes), and the ability of motorists to see pedestrians in the crosswalk (e.g., this can be affected by the level of illumination and nearby columns obstructing visibility). Crosswalks connecting terminal buildings and parking facilities typically have more pedestrian traffic than do those only connecting the terminal and a center-island curbside.

Vehicular traffic at pedestrian crosswalks located on curbside (or other) roadways can be uncontrolled or controlled using stop signs, pedestrian-actuated signals, or officers. Key aspects of these options are described in the following:

- Uncontrolled crosswalks are often used when vehicular and pedestrian traffic volumes are low (i.e., there are frequent gaps in vehicular traffic allowing pedestrians to cross), the crosswalk and pedestrians in the crosswalk can be easily seen by drivers, and in communities where motorists commonly stop for crossing pedestrians. (The visibility of pedestrians using crosswalks can be enhanced and traffic speeds reduced through the use of raised crosswalks—see Section 7.3.3.)
- Stop signs may be used to control crosswalks, with the signs often placed in the middle of crosswalks to improve their visibility, as shown in Figure 7-3. The benefit of stop signs is that they provide positive traffic controls, are easily recognized, and thus enhance pedestrian safety at a much lower cost than crosswalk signals. The disadvantage of stop signs is that they require every motorist to come to a full stop, whether or not there is a pedestrian in the crosswalk,



Figure 7-3. Sign in the middle of the crosswalk at Reno-Tahoe International Airport.

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thus delaying traffic and adding to congestion. Stop signs are less effective in high-congestion environments where drivers are already distracted. Stop signs do not support the rapid movement of vehicles through the intersection, possibly adding to congestion.

Salt Lake City International Airport has its crosswalk signals timed so that when the button is pressed, the lights turn red and a walk signal is displayed almost immediately, most of the time. Staff set a minimum green time (20 seconds) for cars. Airport staff report that this discourages jaywalking against the signal since pedestrians see lights changing as soon as they push the button.

- Pedestrian-actuated crosswalk signals ("Walk/Don't Walk") at airports are normally actuated by waiting pedestrians through the use of push buttons, thermal imaging devices, or motion detectors. At airports, pedestrian-actuated signals are normally used rather than signals operated on fixed time cycles. The primary benefits of pedestrian-actuated signals (compared to stop signs) are that motorists encounter fewer delays since (1) they only need to stop when there are waiting pedestrians and (2) there are larger gaps between crossing pedestrians as the signals promote pedestrian platooning (as opposed to motorists having to stop due to the random and infrequent arrival of pedestrians). Disadvantages of crosswalk signals are (1) normally, vehicular traffic is stopped for a fixed length of time, regardless of the volume of crossing pedestrians; (2) the operations or timings of adjacent signals are rarely coordinated, requiring motorists to potentially stop at each crosswalk they encounter, much like they do at a series of stop signs; and (3) pedestrians may cross during the "Don't Walk" phase, particularly if they do not see approaching vehicles. An example of a signalized crosswalk is shown in Figure 7-4.
- Officers may be used to control crosswalks during busy periods, and with the crosswalks either uncontrolled or controlled by temporary stop signs during other hours. This schedule allows the officers to be used only when needed and can result in cost savings. The benefits of crosswalk enforcement by officers (whether TCOs or LEOs) are that they can (1) provide positive, visible enforcement of pedestrian and vehicular traffic; (2) require pedestrians to wait on the sidewalk, promoting platooning; (3) balance vehicular and pedestrian operations, responding to variations in traffic volumes; (4) stop vehicular traffic to provide pedestrians additional crossing time if needed (e.g., disabled or elderly passengers or passengers with small children or several large pieces of baggage); and (5) assist or direct motorists entering or exiting curbside spaces. As with the other options used to control pedestrian flows at a crosswalk, the use of officers may result in motorists being required to stop at multiple crosswalks.
- Costs are a key consideration when evaluating the use of signs, signals, or officers to control crosswalks:
  - Stop signs are the least expensive alternative, are readily available, and require little/no maintenance, other than replacement if damaged. However, information density is an important consideration when planning the use of additional static signage.
  - Activated signals require electrical power, a secure/vandal-free installation, regular inspection to ensure they are running well, and often a lengthy process in a larger airport for permitting.



Figure 7-4. Signalized crosswalk at Salt Lake City International Airport.

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Performance of maintenance and repairs requires specialized skills, which many airports outsource to private contractors or a municipal maintenance group to ensure a 24/7 response.
– Officers, due to the ongoing expense of their wages and benefits, are by far the most expensive of the alternatives presented.

Summary—Control of Crosswalks—Officers versus Signals

Intended Goals

- Improved pedestrian safety
- Improved traffic operations/reduced curbside roadway congestion and delays
- Ability to support traffic management (assuming officers are used to control crosswalks)

Applicability

- Applicable to all airports having crosswalks, but particularly those with heavily used crosswalks (e.g., connecting the terminal and parking facility)

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X	X	X		

  – Stop signs.
  – Pedestrian-actuated traffic signals.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
			X	X	X	

  – Signal operations and maintenance.
  – Officers (depends on the use of sworn/LEOs and/or civilian TCOs, the number of crosswalks, and how many hours/week they are staffed).
  – Dependent on local conditions for permits and local labor.
  – The cost range provided assumes LEOs patrolling about 400k/curbside positions ($30/hr) and TCOs patrolling about 215k/curbside positions ($20/hr).

Implementation Considerations

- **Implementation schedule:** Signs can be installed in 1 month or less. Actuated signals often require a minimum of 12 months to design, procure, and install. Assignment of officers may take 3 to 6 months, depending on staff availability, time needed for training, scheduling requirements, and budget.
- **Ability to implement in phases:** Yes.
- **Ease of enforcement:** Easy.
- **Likely political influence:** None.

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Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.4. Regulatory sign placement and conformance
- 7.3.1. Consolidating, coordination, or removal of crosswalks
- 7.3.2. Changing pedestrian crosswalk controls
- 7.3.3. Raised crosswalks/speed platforms

Examples

- Many airports, including those serving Madison, Reno, Salt Lake City, San Francisco, and others.

7.1.3 Measures to Support Curbside Enforcement: Warning Messages, Ticketing, and Towing

Enforcement officers have several tools and technologies to manage curbside traffic. These include

- **Persuade motorists.** Officers can remind drivers that the curbsides are to be used only for the active loading and unloading of passengers. For example, they can tell drivers "I need for you to move," a phrase recommended by an airport promoting customer service. Additionally, airports can persuade motorists by having an automated announcement play regularly to remind drivers that curbsides are for active loading and unloading of passengers.
- **Provide alternatives.** Officers can direct motorists to a cell phone waiting area. To help motorists find the lots, some airports have handouts with driving directions that officers can give to drivers. Other airports have added symbols (e.g., a small white triangle) to the appropriate directional signs, enabling officers to instruct drivers to simply follow this symbol. The use of this symbol avoids the need to add additional text to directional signs that may already contain too much text. (Section 7.2.3 provides additional information about the use of cell phone lots.)
- **Issue parking passes.** At Phoenix Sky Harbor International Airport, 1-hour "parking passes" can be provided to waiting drivers, especially those who appear to have arrived early. The passes allow drivers to park for up to 1 hour at no cost in an adjacent parking structure. To ensure the proper use of these passes, curbside staff only receive 10 passes at a time and must sign the passes upon issuing them. Airport staff report issuing about 10 passes on a typical weekday and as many as 50 to 60 passes on a busy holiday, such as Thanksgiving.
- **Issue passes to accommodate people with disabilities.** At Phoenix Sky Harbor International Airport, airport staff managing curbside traffic operations can issue special permits to motorists transporting passengers with disabilities, as shown in Figure 7-5. These permits allow the driver to enter the terminal building to assist their passengers. Before issuing the permits, airport staff must inspect the vehicle, notify their supervisor, and sign the permit.
- **Issue warning notices.** At one airport, enforcement staff issue warning notices that look similar to tickets. The notice indicates that the motorist left the vehicle unattended, remained at the curb too long, or otherwise was in violation of airport regulations. While these warning notices have no legal authority, they are considered more customer friendly than a ticket and serve to warn motorists. Officers will issue a ticket if, upon their return, they see that the vehicle is still at the curb despite the motorist having received a warning notice.
- **Issue tickets or citations.** Airport enforcement staff may issue tickets or citations for non-moving violations to motorists who have violated airport regulations. These tickets are the equivalent of parking tickets and carry fines that vary based on the motorist's offense. Tickets must be legally enforceable, supported by municipal codes or other laws, and posted signs.

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Source: Phoenix Sky Harbor International Airport

Figure 7-5. Example of a special needs permit.

- **Towing vehicles.** Airports may tow or threaten to tow vehicles, that are stopped or standing at the curbside in violation of airport regulations. Some airports actively tow ticketed vehicles; others threaten to do so (especially if issuing tickets does not encourage the desired behavior by motorists), or simply have a tow truck parked at the curbside entrance (see Figure 7-6) at all times or only during peak periods to serve as a warning.



***Figure 7-6. Tow trucks stationed on terminal curbsides at Baltimore/Washington International Thurgood Marshall Airport* (left) *and Salt Lake City International Airport* (right).**

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Figure 7-7. Portable ticket writing machine at Salt Lake City International Airport.

- **Handheld ticket writing equipment.** Enforcement officers can be furnished with handheld devices (or devices worn from their belts) that automate and simplify ticket writing, as opposed to having a ticket book and having to look up applicable violation codes. Figure 7-7 depicts a portable ticket writer and printer.

Summary—Measures to Support Curbside Enforcement: Warning Messages, Ticketing, and Towing

Intended Goals

- Improved traffic operations/reduced curbside roadway congestion and delays
- Support for traffic management and enforcement

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X	X			

  - Regulatory signage fabrication and installation.
  - Staging area for vehicles towed for violation.

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  - Handheld ticket writing equipment.
  - Dependent on the ticketing management system selected, the number of ticketing units, and the functionality of the system (invoicing, reporting).

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X	X			

  - Issuance of passes, permits, or warning.
  - Labor costs or the value of foregone parking revenues if parking passes are issued.
  - Costs of towing, which may be contracted out to a local towing company or third party.
  - Dependent on the cost and number of tow vehicles, the hours the vehicles are staffed, and the cost for development of a staging area for towed vehicles.

Implementation Considerations

- **Implementation schedule:** Passes, permits, and warning notices can be implemented in 1 month or less if supported by existing local codes.
- **Ability to implement in phases:** Yes.
- **Ease of enforcement:** Supports other strategies.
- **Likely political influence:** None.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.2. Control of crosswalks—officers versus signals
- 7.1.4. Regulatory sign placement and conformance
- 7.2.3. Encourage the use of cell phone lots
- 7.2.4. Encourage motorists dropping off or picking up passengers to use hourly parking
- 7.3.7. Charging commercial vehicles for excessive number of trips or dwell times
- 7.3.8. Using space detectors supplemented by flashing beacons to encourage efficient use of curb space
- 7.3.9. Alternative curbside layouts including the use of sawtooth or angled spaces, pull-through spaces, or permitting double-parking
- 7.3.11. Modifying placement or use of regulatory and warning signs
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes

Examples

- Airports that have implemented this strategy include those serving Palm Beach, Phoenix, Sacramento, Salt Lake City, and others.

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7.1.4 Regulatory Sign Placement and Conformance

Effective curbside traffic management and enforcement requires that motorists be made aware that parking and standing are prohibited, and, further, that the signs prohibiting these actions be legally enforceable. Best practices for these signs include conformance with municipal or state vehicle codes. Normally, active loading and unloading at the curbside are permitted as long as the driver remains with the vehicle. Depending on the local codes, a sign that says

- "No parking" allows motorists to stop to load or unload passengers, but drivers may not leave a vehicle unattended while waiting for arriving passengers.
- "No standing" allows passengers to stop to load or unload passengers, but not wait for arriving passengers.
- "No stopping" means that motorists cannot stop in the area for any reason.

The signs typically contain text saying "No Standing or Parking" to ensure that they are enforceable. At some airports, curbside regulatory signs include supplementary text such as "No Unattended Vehicles," "Do Not Leave Vehicle Unattended," and "Tow-Away Zone," as shown in Figure 7-8. The use of "Tow-Away Zone" is suggested only when tow trucks are available and used to support curbside enforcement.

These regulatory signs must be visible to motorists but mounted so as not to interfere with pedestrian circulation along the curbside sidewalk or those passengers boarding and alighting vehicles. FHWA's *Manual on Uniform Traffic Control Devices* recommends that regulatory signs be placed 2 feet from the curbside face and that the bottom of the sign be a minimum of 5 feet above the sidewalk surface.



Source (top photos): Dallas/Fort Worth International Airport.

***Figure 7-8. Regulatory signs at Dallas/Fort Worth International Airport* (top) *and Salt Lake City International Airport* (bottom).**

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Summary—Regulatory Sign Placement and Conformance

Intended Goals

- Improved traffic operations/reduced curbside roadway congestion and delays
- Support for traffic management and enforcement

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Fabrication and installation of regulatory signs.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – None, other than labor costs of sign installation.

Implementation Considerations

- **Implementation schedule:** Normally regulatory signs can be installed by airport maintenance staff and either fabricated by airport maintenance or purchased from a vendor.
- **Ease of enforcement:** Supports other strategies.
- **Likely political influence:** None.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.3. Measures to support curbside enforcement: Warning messages, ticketing, and towing
- 7.3.11. Modifying placement or use of regulatory and warning signs

Examples

- Airports that have implemented this strategy include those serving Dallas/Fort Worth, Phoenix, and many others.

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7.2 Traffic Demand Management Strategies

These strategies are intended to reduce traffic on airport curbside and other roadways by diverting motorists to alternate drop-off and pickup sites, encouraging the use of high-occupancy travel modes, or limiting commercial vehicle operations. Many of these strategies help to manage traffic demands and improve roadway capacities by promoting use of underutilized areas or by limiting the time vehicles are allowed to remain on the curbside. These strategies include

- 7.2.1. Moving commercial vehicle passenger boarding areas to an available remote location
- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.2.3. Encouraging the use of cell phone lots
- 7.2.4. Encouraging motorists dropping off or picking up passengers to use hourly parking
- 7.2.5. Encouraging use of long-duration parking areas rather than drop-off or pickup at the terminal curbs
- 7.2.6. Encouraging the use of transit or high-occupancy travel modes
- 7.2.7. Locating employee parking outside of the terminal area
- 7.2.8. Charging private motorists access fees
- 7.2.9. Charging commercial vehicle operators headway fees or higher fees for curbside pickup

7.2.1 Moving Commercial Vehicle Passenger Boarding Areas to an Available Remote Location

One method of controlling traffic demand at the curbsides is to divert some commercial vehicles away from the terminal curbs to a remote location, which may include

- Diverting TNCs (and/or other commercial vehicles) to a remote curb or remote lot.
- Diverting all commercial vehicles to ground transportation centers (or an alternative passenger drop-off or pickup area).
- Requiring that passengers use a shuttle or automated people mover (APM) to travel to remote boarding areas.

Diverting TNCs to a Remote Curb

Dwell times for TNC pickups can be longer than those for private vehicles due to the time needed for customers to find their driver, particularly during peak congested periods or at night.

This strategy locates TNC loading and/or unloading zones at a site(s) away from the terminal building curbsides but within walking distance of the terminal(s). The advantage of this strategy is that it diverts some vehicles away from the terminal curb roadway(s), reducing the overall demand for frequently congested areas. It can also improve the organization and allocation of commercial vehicle boarding areas, making it easier for customers to find the correct vehicle. While a remote location could provide for both passenger loading and unloading, only passenger boarding is provided at those airports with remote facilities; passenger unloading continues to occur at the terminal curbside. This is because it is simpler to identify, manage, and enforce regulations for TNCs picking up customers than doing so for TNCs dropping off customers. An example of an alternative TNC boarding area, LAX-it at Los Angeles International Airport, is shown in Figure 7-9.

Section 7.6 provides more detailed information on strategies to accommodate TNCs.

Diverting All Commercial Vehicles to Ground Transportation Centers

Another strategy is to divert all commercial vehicles to a ground transportation center located within walking distance of the terminal. This diverts traffic away from the terminal curb(s) and roadway. This strategy is particularly effective when there are numerous frequently or regularly scheduled commercial vehicles and limited terminal curb length. Delivery services and hotel/off-airport shuttle buses will require additional focus.

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Figure 7-9. LAX-it at Los Angeles International Airport.

Requiring Use of a Shuttle or Automated People Mover to Travel to Boarding Areas

In congested terminal cores at large-hub airports, it may be desirable to develop a remote commercial vehicle boarding area (and potentially drop-off area) at a site that is not within walking distance of the terminal core. This requires that passengers be transported in shuttles or an APM between the remote boarding area and the terminal building(s). If the remote site provides for passenger drop-off, consideration should be given to providing remote baggage check-in facilities so passengers need not bring their checked baggage onto the shuttle or APM. (Currently, it is not practical to provide remote baggage claim facilities unless passengers are required to first claim and then re-check their baggage in the terminal.) Airports considering this approach should carefully consider both initial design and development costs as well as long-term operation and maintenance costs and the customer experience of those passengers required to use the shuttle/APM.

Summary—Moving Commercial Vehicle Passenger Boarding Areas to an Available Remote Location

Intended Goals

- Divert commercial traffic away from terminal curbs

Applicability

- Applicable to airports with significant commercial vehicle traffic

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X		X	X	

  - Design and development of remote commercial vehicle curbs.
  - Design, development, and/or procurement of shuttle or APM (if applicable).
  - Depends on whether a new facility requiring busing would be needed or whether an existing facility that is walkable can be used. Excludes the cost of APMs.

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X				X	X

  – Operations and maintenance costs will vary based on the approach used. APMs and shuttle operations will require significant ongoing operating and maintenance costs.

Implementation Considerations

- **Implementation schedule:** Depends on approach implemented and existing airport facilities. Can be less than a year (e.g., if an existing parking facility is converted to a remote passenger drop-off or pickup zone) to multiple years (e.g., if a ground transportation center is to be incorporated into a new structure or if construction of an APM is required).
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.6. Accommodating TNCs

Examples

- Examples of TNC/rideshare-only curbs currently occur at the airports serving Boston (Central Parking Garage), Las Vegas (in garages), San Francisco (on the roof of the central garage), and Sacramento (which has remote curbs for each terminal serving TNCs).
- Consolidated commercial vehicle curbs/ground transportation centers are currently in use at airports such as Detroit Metropolitan Airport (both terminals), Harrisburg, Minneapolis, San Jose, and Seattle. At San Diego International Airport there are remote curbside pickup zones in the surface lot of Terminal 1 and the transportation center located under the departure curb for Terminal 2.
- Examples of the required use of a shuttle or APM can be found at the airports serving Los Angeles (LAX-IT lot, which also accommodates taxicabs and is located in a surface lot adjacent to Terminal 1), Newark, and Tampa. Two intermodal transportation facilities at Los Angeles International Airport (to open after 2023) will each consolidate parking, ground transportation providers, and remote passenger drop-off/pickup functions.

7.2.2 Moving Private Vehicle Pickup and/or Drop-Off Areas to an Available Remote Site

Curbside traffic demands can be controlled by diverting private vehicles from the terminal curb to a remote location. The remote pickup/drop-off area can be supplemental to the terminal curbside or replace the traditional terminal curb. An example of supplemental curbside is shown in Figure 7-10. Diversion of private vehicles to a remote location works well if it is an optional supplement to curbside drop-off and pickup, but this diversion may encounter resistance from customers if its use is mandatory.

This strategy requires consideration of unassisted walking distance and use of a shuttle or APM.

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Figure 7-10. LAX P3 garage supplemental curbside.

Walking Distance

Private vehicle pickup and drop-off areas can be located within walking distance of the terminal doorway or entrance. By locating a parking facility adjacent to the terminal building, the need for a supplemental shuttle, moving walkway, or APM can be avoided. Passenger pickup and drop-off zones within a parking facility can be offered by charging customers

- A fee, identical to or higher than those paid by parkers, for access to a premium, convenient pickup and drop-off location within the parking facility. This alternative produces additional airport revenues.
- No fee for access to (1) a nested zone within the parking facility having separate entrance(s) and exit(s) or (2) an entire parking facility (if customers are offered free parking for their first 30 minutes or other duration).

Using a Shuttle or Automated People Mover to the Remote Pickup/Drop-Off Areas

In congested terminal cores, it may be desirable to develop a remote private vehicle pickup/drop-off area that is within walking distance of the terminal(s), with passengers transported between the terminal and remote boarding area in shuttles or APMs. Airports considering this approach should carefully consider both initial design and development costs, long-term operation and maintenance costs, and the customer experience of those passengers required to use the shuttle/APM.

Summary—Moving Private Vehicle Pickup and/or Drop-Off Areas to an Available Remote Site

Intended Goals

- Divert private vehicle traffic away from terminal curbs and potentially off-airport roadways (even if temporarily)

Applicability

- Applicable to airports with significant vehicle traffic

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X		X	X	

  - Design and development of remote private vehicle drop-off and pickup zones.
  - Design, development, and/or procurement of shuttle or automated people mover (if applicable).

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X		X	X	

  – Operations and maintenance costs will vary based on the approach used. An extensive communication plan should be factored into costs.

Implementation Considerations

- **Implementation schedule:** Depends on the approach implemented and existing airport facilities. Can be less than a year to multiple years.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.2.1. Moving commercial vehicle passenger boarding areas to an available remote location

Examples

Remote passenger curbs exist in various forms at different airports. Sometimes they are simply a "Kiss and Fly" curb for private vehicles, and other times, remote passenger curbs have a fully integrated commercial curb operation as well. Some examples follow:

- Chicago O'Hare International Airport currently has a Kiss and Fly curb adjacent to the rental car center, which also includes various curbs for ground transportation and economy parking, all located at the last stop of the APM.
- Los Angeles International Airport will open two intermodal transportation facilities, which will both allow for private vehicle drop-off in combination with various ground transportation functions and economy parking, all connected to the terminals by a future APM line.
- Los Angeles International Airport uses the first floor of the P3 parking garage as a remote passenger drop-off/pickup area across from Terminal 3 and the Tom Bradley International Terminal.
- San Francisco International Airport has a Kiss and Fly curb adjacent to its rental car center where passengers can walk to the APM stop to get to the terminals or other garages.
- Phoenix Sky Harbor Airport's 44th Street Station, which has a Kiss and Fly curb, is connected to the terminals with an APM.
- Tampa International Airport has Express Curbsides, passenger drop-off and pickup areas for passengers without checked baggage that are located on the red and blue sides of the terminal. Travelers can take nearby escalators directly to the APMs leading to the aircraft gates.

7.2.3 Encouraging the Use of Cell Phone Lots

Cell phone lots are free parking lots typically located along the approach to the terminal roadway or near the terminal, where motorists in private vehicles can wait for arriving passengers

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to land and call for pickup. When adopted, they reduce the number of vehicles circulating on terminal loop roadways and reduce congestion along the arrivals curb. To encourage use of the cell phone lot, the cell phone lot should be planned considering the following:

- Convenient access (prior to the terminal core)
- Amenities
- Customer service and safety

Convenient Access

The cell phone lot should be convenient to use and easily identifiable, with clear access points. Ideally, a cell phone lot would be located on the airport's primary entry roadway or just adjacent to it. The cell phone lot should also provide waiting motorists convenient access to information regarding airline arrival status and updates on roadway congestion that can be communicated to passengers before they arrive at the curbs. This can be done with a dynamic sign located in advance of the cell phone lot and terminal curbs, access to an airport's website, or by providing the toll-free telephone numbers of individual airlines.

Amenities

Consider providing amenities for those utilizing cell phone lots. Amenities may include

- Large (multiple) flight identification displays
- Access to food trucks/coffee/restaurants
- Vending machines
- Wi-Fi
- Playgrounds
- Bathrooms
- Car wash
- Gas stations
- Electric vehicle charging

Amenities encourage drivers to use the cell phone lot and wait there, rather than at the terminal curb.

Customer Service and Safety

Ensuring customer service and safety is paramount to encouraging cell phone lot use. Airport staff should ensure that

- Proper security measures are provided, including security cameras and signs indicating the lot is regularly monitored by airport security vehicles. The airport should consider having a security vehicle present on the lot during peak times.
- Adequate lighting levels are provided throughout the cell phone lot to promote safety.
- All amenities are working properly and clean.
- Airport staff or contractors visit to see if waiting drivers need assistance. Through the airport's parking operator, motorists can be offered assistance with changing flat tires and jump-starting cars with dead batteries. Motorists can also be provided with small amounts of gas.

Summary—Encouraging the Use of Cell Phone Lots

Intended Goals

- Divert private vehicle traffic away from terminal curbs toward cell phone waiting lots

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Applicability

- Applicable to medium- and large-hub airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X				

  – Design and development of cell phone lots and amenities.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X					

  – Operations and maintenance costs will vary based on the approach used.

Implementation Considerations

- **Implementation schedule:** Depends on approach implemented and existing airport facilities. Can be less than a year to multiple years.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site

Examples

Nearly every medium- and large-hub airport in the United States has some form of a cell phone lot. Many larger small-hub airports also have them due to traffic congestion issues associated with outsized peaking that happens in the mornings for departures and in the evenings for arrivals. Cell phone lots (if positioned just off the entry roadway) can be very advantageous to airports looking to keep vehicles away from the curb. Some specific examples include the following:

- Fort Lauderdale-Hollywood International Airport, Salt Lake City International Airport, and St. Louis Lambert International Airport. These airports use dynamic signage to convey flight status.
- Austin-Bergstrom International Airport, which developed restrooms and has a food truck program.
- Tampa International Airport, which has a food truck program at its cell phone lot and offers free Wi-Fi.
- San Diego International Airport, which has portable restrooms in its cell phone lot, located just off Harbor Drive prior to the airport entry.

ACRP Synthesis 62: Cell Phone Lots at Airports presents information about cell phone lots to help airport staff evaluate their benefits and costs.

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7.2.4 Encouraging Motorists Dropping Off or Picking Up Passengers to Use Hourly Parking

There are multiple benefits resulting from increased use of hourly (or close-in) parking for passenger drop-off and pickup. The first is the additional revenue generated by customers. The second benefit is the reduced congestion at the terminal curb and on the terminal roadways. To encourage use of hourly parking airport managers should consider

- Parking rates or costs
- Distance to terminal
- Amenities
- Public communication and wayfinding

Parking Rates

To encourage use of hourly parking lots, airport staff should ensure that they are not prohibitively expensive. Lower prices, while generating lower revenues, may increase use of the parking facilities by motorists and, consequently, the effectiveness and efficiency of the roadway system. In some markets (i.e., those with significant curb congestion and/or low hourly lot usage), it may be appropriate to provide a grace period during which meeter/greeters would not be charged for using hourly parking or other facilities adjacent to the terminal building.

Distance to Terminal

Parking areas intended to attract motorists dropping off and picking up passengers should be located close to the terminal to maximize their desirability. It is suggested that such parking areas be within a convenient walking distance (e.g., less than 500 feet), offer weather protection or climate controls, and be accessible to those with disabilities. It may be ideal to have the parking area equidistant to the arrival and departure doors or, in some instances, to have separate lots catering to arrivals and departures.

Amenities

To encourage motorists dropping off or picking up passengers to use hourly parking facilities, it is suggested that airports consider providing amenities to attract customers including

- Free electric vehicle charging
- Valet parking
- Luggage cart pickup/drop-off
- Baggage check-in
- Wi-Fi
- Concessions in the form of food trucks and/or vending machines
- Designated smoking areas
- Playgrounds/family areas

Summary—Encouraging Motorists Dropping Off or Picking Up Passengers to Use Hourly Parking

Intended Goals

- Divert private vehicle traffic away from terminal curbs toward hourly parking lots

Applicability

- Applicable to medium- and large-hub airports

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Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  - Design and development of lots and procurement of amenities.
  - Changing parking access and revenue control system rate table and signage.
  - Marketing expenses.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  - Operations and maintenance costs will vary based on the approach used. One consideration would be the conversion of an existing parking asset into the hourly function—an analysis of the cost implications and/or cost benefit would need to be accomplished.

Implementation Considerations

- **Implementation schedule:** Depends on the approach implemented and existing airport facilities. Can be less than a year to multiple years (for large-scale construction projects).
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies:

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.2.3. Encouraging the use of cell phone lots
- 7.2.8. Charging private motorists access fees
- 7.5.1. Encouraging the use of transit or high-occupancy travel modes
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes

Examples

Many airports operate their adjacent parking lots and/or the first level of garages as hourly lots for those motorists who prefer to park and accompany or meet passengers in the terminal or simply wait in the parking facility. Some examples include

- Sacramento International Airport, which provides an hourly surface lot adjacent to Terminal B and utilizes the first floor of Garage A as an hourly lot.
- Denver International Airport, which has extensive concessions options such as coffee shops and restaurants prior to security checkpoints for departure.

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- Detroit Metropolitan Airport, which has coffee shops located at baggage claim and gift shops located at the international arrivals waiting area.
- Orlando Sanford International Airport, which provides free parking for the first 15 minutes in the hourly lot.

ACRP Report 24: Guidebook for Evaluating Airport Parking Strategies and Supporting Technologies describes additional strategies and amenities to encourage the use of parking facilities.

7.2.5 Encouraging Use of Long-Duration Parking Areas Rather than Drop-Off or Pickup at the Terminal Curbs

There are multiple benefits resulting from the increased use of long-duration parking for passenger drop-off and pickup. The first is the additional revenue generated by customers. The second benefit is the reduced congestion at the terminal curb and on the terminal roadways. To encourage use of long-term parking consider

- Parking rates or costs
- Ease of parking compared to amount of traffic congestion and delays in the terminal core
- Connectivity to the terminal
- Public communication and wayfinding

Parking Rates

To encourage use of long-term parking facilities, airports should ensure that they are not prohibitively expensive. Lower prices, while generating lower revenues, may increase use of the long-term parking facility and, consequently, the effectiveness and efficiency of the roadway system. Adoption of a grace period for the long-term lot (such as 60 minutes) may encourage motorist use of long-term parking facilities and thereby reduce roadway traffic and congestion.

Ease of Parking

To encourage customer use of long-term parking facilities and to reduce roadway congestion, airport staff should ensure that customers can readily find available parking spaces and recognize the cost savings resulting from the use of long-duration facilities. Parking availability and rates should be communicated via airport websites, mobile applications, and signage displayed prior to the parking entrance. The facilities should utilize smart parking technologies to help drivers find parking spots efficiently. This may include use of the airport's parking application and/or dynamic signage that indicates space availability by individual aisle and by space (i.e., single-space parking detection systems).

Connectivity to the Terminal

Customers can be encouraged to use long-term parking if they are provided quick and convenient access from the parking facility to the airport terminal. Where long-term parking is relatively close to the terminal, convenient access can be provided using elevated pedestrian bridges or tunnels incorporating moving walkways to reduce unassisted walking distances. Where the long-term parking facilities are located farther away from the terminal, convenient access can be provided using regular and frequent shuttle service (circulating within the parking facility) or APMs with stations integrated into the parking facility.

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Summary—Encouraging Use of Long-Duration Parking Areas Rather than Drop-Off or Pickup at the Terminal Curbs

Intended Goals

- Encourage use of long-term parking instead of curbside pickup/drop-off

Applicability

- Applicable to small-, medium-, and large-hub airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Design and development of parking facilities. Procurement and operation of shuttles.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Operations and maintenance costs will vary based on the approach used.

Implementation Considerations

- **Implementation schedule:** Depends on the approach implemented and existing airport facilities. Can be less than a year to multiple years, particularly when providing a new pedestrian linkage or APM.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.2.1. Moving commercial vehicle passenger boarding areas to an available remote location
- 7.2.4. Encouraging motorists dropping off or picking up passengers to use hourly parking
- 7.2.8. Charging private motorists access fees
- 7.5.1. Encouraging the use of transit or high-occupancy travel modes

Examples

- Detroit Metropolitan Airport has walking access between long-term parking and the terminal via bridge systems.
- Louisville International Airport discounts its long-term garage for the first few hours while all other long-term parking assets have a daily rate. The garage is connected directly to the terminal allowing easy access.

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- Los Angeles International Airport discounts the first 60 minutes in its terminal area garages to allow for meeter/greeters to park and get off the roadways. The fee increases for every subsequent 30 minutes until the parking duration reaches the maximum daily rate. Short-term parking rates in the economy lot are designed to encourage meeter/greeter use.

7.2.6 Encouraging the Use of Transit or High-Occupancy Travel Modes

Encouraging the use of transit and high-occupancy travel modes by passengers and employees working at the airport can be an effective way to manage traffic demand and reduce roadway congestion. Transit, including scheduled buses and trains, is capable of transporting many passengers and employees with minimal impact on roadways. There are many factors that impact the decision of passengers and employees to use transit or other high-occupancy travel modes including the availability of convenient transit at the non-airport trip end (i.e., near one's place of residence or workplace), frequency of transit service, transit fares, and the familiarity of passengers and employees with transit service (and how to use it). However, airport staff can influence only a few of the factors affecting a passenger's or employee's decision to use transit or other high-occupancy travel mode. These factors include

- Walking distance/level changes
- Ease of wayfinding
- Marketing/advertising
- Remote parking

Walking Distance/Level Changes

The distances passengers—particularly passengers with several pieces of baggage or with large family groups—have to walk to/from a transit station or stop (at both ends of their trips) and the number of level changes they encounter influence their choice of access modes. Ideally, passengers would have a short walk from their transit station or stop to the airport security checkpoint and would not need to change levels. Passengers may be discouraged from using transit if they need to walk long distances, must change travel modes (i.e., use a bus or people mover to travel between the transit station and airport terminal), or if there is a fare for this intermediate/connecting mode.

Ease of Wayfinding

To ensure that transit is easily identifiable at the airport, references to transit should be displayed on wayfinding signs, ensuring passengers can see the available transit services as well as the areas/destinations served, schedule, and fares. Non-resident passengers may be unfamiliar with destinations served, how to purchase a fare card, or the frequency of departures from the airport terminal. Passengers may be discouraged from using transit if they encounter long waits (e.g., a bus departing every 60 minutes).

Marketing/Advertising

To promote the use of transit, airport staff may wish to consider advertising the availability of transit service to/from the airport at the airport as well as at transit stations/stops throughout the region. These advertisements should describe the available schedules, routes, and prices to access the airport. Such advertisements may be beneficial during peak holiday seasons or summer vacations when there are more cost-sensitive, leisure passengers traveling through the airport.

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Remote Parking

Airport staff may wish to consider coordinating with local transit operators to allow passengers and employees to park at a discount rate to encourage the use of transit and reduce airport roadway congestion. Furthermore, the use of parking at transit centers allows these facilities to operate as off-airport parking lots, allowing passengers and employees to park off-airport and take transit to/from the airport.

Section 7.5.1. provides additional information regarding measures to encourage the use of transit or high-occupancy travel modes.

Summary—Encouraging the Use of Transit or High-Occupancy Travel Modes

Intended Goals

- Encourage use of transit or high-occupancy travel modes instead of curbside pickup/drop off

Applicability

- Applicable to medium- and large-hub airports having good rail or scheduled bus service

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X					

  - Design and development of transit services and stations.
  - Includes marketing promotional materials and changes to parking cost. Excludes physical changes to the terminal building/transit stations.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X					

  - Operations and maintenance costs will vary based on the approach used.

Implementation Considerations

- **Implementation schedule:** Depends on approach implemented and existing airport facilities. Can be less than a year for the initiation of bus service to multiple years for the initiation of rail service.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site

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- 7.2.8. Charging private motorists access fees
- 7.4.1. Providing convenient, enclosed passenger waiting areas
- 7.4.2. Informing waiting passengers of bus and shuttle arrival times
- 7.4.3. Presenting transit schedules, fares, and route information to passengers
- 7.4.4. Providing passenger service associates at curbsides
- 7.4.5. Enhancing safety and convenience for pedestrians crossing roadways and waiting for transportation
- 7.5.1. Encouraging the use of transit or high-occupancy travel modes

Examples

- Washington Reagan National Airport has a direct connection to Washington Metrorail, with an elevated walkway connecting the transit station to the terminal level where security checkpoints are located. Some airlines offer baggage check-in along this path, allowing transit riders to avoid going up to the ticket counters.
- San Francisco International Airport's International Terminal has direct access to BART (Bay Area Rapid Transit). Passengers traveling to the airport's other three terminals must use an APM.
- Salt Lake City International Airport has a Trax light rail station incorporated into the parking garage with grade-separated access provided between the terminal and rail station.
- JFK International Airport's AirTrain people mover provides a connection (for a fare) between the airport and the nearby Jamaica station, where passengers can transfer to the Long Island Railroad and several New York City Metropolitan Transportation Authority subway lines. AirTrain also connects to the Howard Beach subway station.
- Newark Liberty International Airport's AirTrain people mover provides direct access to a nearby rail station serving New Jersey Transit and Amtrak rail services.
- Los Angeles International Airport's planned APM will offer direct access to LA Metro, and the current G Bus provides access to LA Metro.
- San Diego International Airport provides shuttles to the Old Town Transit Hub and to the Palm Street light rail station.
- Chicago O'Hare International Airport is the terminus of the CTA Blue Line, accessible from all terminals.
- St. Louis Lambert Field is the terminus of the MetroLink Line.
- Cleveland Hopkins Airport is the western terminus of the RTA Red Line.
- Other airports with direct rail service include those serving Atlanta, Dallas/Fort Worth, Denver, Minneapolis, Philadelphia, Portland, and Seattle.
- Airports where rail service is available via an intermediate shuttle or people mover include those serving Baltimore, Boston, Burbank, Fort Lauderdale, Miami, Oakland, Milwaukee, Phoenix, Providence, and San Jose.

ACRP Report 4: Ground Access to Major Airports by Public Transportation describes programs to encourage the use of transit by airport passengers and employees.

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7.2.7 Locating Employee Parking Outside of the Terminal Area

By locating employee parking outside of the terminal area, airports can reduce traffic volumes and levels of congestion on the airport terminal area roads. Employees working for the airport, airlines, and other tenants may park away from the airport terminal and be transported in shuttles or APMs. In some cases, the airlines and other tenants lease space from an off-airport parking business and have their employees shuttled to/from the terminals.

Some airports require all persons working in the terminal area to park remotely except for airline station managers (or other selected senior staff) and employees of U.S. Customs and Border Protection or other federal agencies.

Summary—Locating Employee Parking Outside of the Terminal Area

Intended Goals

- Remove employee traffic from terminal area roadways

Applicability

- Applicable to medium- and large-hub airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
					X	

  - Design and development of parking facilities, if necessary.
  - Assumes construction of new employee lot with dedicated busing.
- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
						X

  - Operations and maintenance costs of shuttle services.
  - Assumes construction of new employee lot with dedicated busing.

Implementation Considerations

- **Implementation schedule:** Depends on approach implemented and existing airport facilities. Can be less than a year to multiple years.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.2.6. Encouraging the use of transit or high-occupancy travel modes

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Examples

- Tenants at Detroit Metropolitan and Newark Liberty International airports have entered into agreements to park at off-airport parking providers and have established remote surface parking for employees.
- Los Angeles International Airport provides employee parking in the remote East, West, and South lots with shuttle access.
- Sacramento International Airport has remote employee parking lots.
- San Francisco International Airport provides a remote surface lot, D, as well as a remote parking structure for employees who either use an exclusive shuttle bus or walk to the AirTrain APM, depending on where the employees park.
- Boston Logan International Airport provides employee parking in a multilevel structure located in an adjacent city.
- Denver International Airport provides security screening for employees working at the Airside Parking Lot (one of two remote parking lots) allowing employees to be transported directly to the secure airside portion of the terminal or concourses.

7.2.8 Charging Private Motorists Access Fees

The airport operator may charge motorists in private vehicles fees to access the airport or specific portions of the airport such as the terminal curb or alternative drop-off or pickup area. This strategy can reduce terminal area traffic and roadway congestion by encouraging passengers traveling in private vehicles to be dropped off or picked up at a remote location or take a shared ride or transit to the airport. This strategy is best implemented in concert with other strategies, such as remote parking and providing good transit and/or high-occupancy vehicle service at the airport.

Summary—Charging Private Motorists Access Fees

Intended Goals

- Reduce terminal curbside congestion and manage roadway traffic demands

Applicability

- Applicable to medium- and large-hub airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
					X	X

  - Design and development of payment facilities and procurement of equipment.
  - Depends on the extent of coverage and technology used. Administration is ~$100K. $500K for a 20-camera system.

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X	X			

  – Operations and maintenance costs including collection of fees at access control booths, pay stations, or through automated billing (i.e., use of a Radio-Frequency-Identification system linked to regional toll road system).

Implementation Considerations

- **Implementation schedule:** Depends on approach implemented and existing airport facilities. Can be less than a year to multiple years.
- **Stakeholder concerns:** Yes, likely concerns or opposition from local residents and elected officials.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.2.4. Encouraging motorists dropping off or picking up passengers to use hourly parking
- 7.5.1. Encouraging the use of transit or high-occupancy travel modes
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes

Examples

- Although the staff of several U.S. airports are evaluating charging motorists in private vehicles a fee to access the terminal roadways, as many airports in Europe have done, only Dallas/Fort Worth International Airport was doing so when this report was prepared.
- Since it opened in 1974, Dallas/Fort Worth International Airport has charged all motorists a fee to enter the airport terminal area, with access controlled by entry/exit plazas located at both the north and south ends of International Parkway, the spine road providing access to all the airport's terminal buildings and parking structures. Motorists use toll tags issued by the North Texas Tollway Authority, credit cards, or cash to pay duration-based fees at the exit plazas. In addition to reducing terminal area traffic and roadway congestion, the airport's access fees also discourage cut-through traffic (i.e., non-airport motorists using the airport's roadways to avoid congestion on adjacent regional highways).
- When it first opened, Denver International Airport charged motorists duration-based airport access fees for access to the terminal area, but these fees and the toll plazas were subsequently removed due to customer complaints and operational challenges.

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7.2.9 Charging Commercial Vehicle Operators Headway Fees or Higher Fees for Curbside Pickup

The impacts of commercial ground transportation vehicles on terminal roadways can be managed by charging the operators of these vehicles higher fees for passenger drop-off or pickup in preferential areas and/or for use of the curbsides during peak periods of demand (or lower fees for the use of secondary drop-off or pickup areas). For example, TNCs, or other commercial ground transportation providers, entering premium curb space near the terminal building entrance during peak times could be charged a higher fee. Such strategies rely on the existence of preferential drop-off and pickup areas (i.e., those areas that customers and providers prefer) and a determination of peak demand times. The premium fee could be combined with a secondary passenger pickup site (or ground transportation center) to promote motorist use of a secondary location. In "Solving Airport Congestion Through Curb Pricing" (*Medium*, August 18, 2019, https://medium.com/p/62fb91e13751), Bakari Brock notes that in a pilot program at the San Francisco International Airport, a modest curb pricing fee for TNCs resulted in a 10% trip shift to less congested curbs. Increased fees suggested that this trip diversion could increase to 45%. While some TNC providers expressed support for such a program, it would require additional training and coordination to educate drivers and riders about the additional prices and the applicable times and areas.

Alternative strategies include (1) imposing demand management fees to limit the number of trips commercial ground transportation operators can make during a given period (e.g., per month) or (2) imposing interval or headway fees to limit the headways between successive vehicles that one provider operates. The use of interval or headway fees is intended to discourage "bump-and-run" operations whereby a commercial vehicle operator continuously has a courtesy vehicle parked at the terminal curbside.

Summary—Charging Commercial Vehicle Operators Headway Fees or Higher Fees for Curbside Pickup

Intended Goals

- Accommodate TNC operations at airports—premium curb pricing
- Reduce courtesy vehicle trips—demand management fee or interval fees

Applicability

- Medium- and large-hub airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X				

  - Design and development of remote vehicle curbs. Design, development, and/or procurement of shuttle or APM (if applicable).
  - Design and development of ground transportation management system to monitor the number of monthly trips made by each provider and/or the number of trips made per hour/intervals between successive vehicles.
  - Implementation likely to require public and airport oversight authority review.

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Operations and maintenance costs will vary based on the approach used.

Implementation Considerations

- **Implementation schedule:** Depends on approach implemented and existing airport facilities. Can be less than a year to multiple years.
- **Ability to implement in phases:** Yes.
- **Politics:** Will require close coordination with and education of TNC operators and may result in unfavorable comments from customers, depending on the convenience of the alternative boarding area.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.2.1. Moving commercial vehicle passenger boarding areas to an available remote location
- 7.3.7. Charging commercial vehicles for excessive number of trips or dwell times
- 7.7. Accommodating autonomous vehicles

Examples

- San Francisco International Airport tested curbside pricing and provided preferred curb locations for pooled TNCs.
- Boston Logan International Airport provided a reduced fee for pooled TNCs.
- Los Angeles International Airport currently allows limousines and black-car-designated rideshare providers to pick up passengers at the curb for a premium fee. The airport previously imposed a cap on the number of monthly courtesy vehicle trips each rental car operator could make.
- Raleigh-Durham International Airport imposes interval fees.

Section 7.3.7 provides additional information about the use of interval fees and other demand management fees to reduce excessive vehicle trips. Section 7.6 provides additional information about management of TNCs.

7.3 Congestion Management Strategies

Airport terminal area roadways and curbsides are often space constrained (due to adjacent structures or facilities), limiting opportunities to use inexpensive solutions to address congestion. The following strategies are intended to improve operations and reduce congestion by using management tools to increase the available capacity of an existing roadway or curbside, thus avoiding or deferring large capital investments:

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- 7.3.1. Consolidating, coordinating, or removing crosswalks
- 7.3.2. Changing pedestrian crosswalk controls
- 7.3.3. Raised crosswalk/speed platform
- 7.3.4. Revising the location or amount of curb space allocated to specific mode(s)
- 7.3.5. Implementing virtual taxicab queue/automated taxicab dispatching
- 7.3.6. Modifying/lengthening weaving area
- 7.3.7. Charging commercial vehicles for excessive number of trips or dwell times
- 7.3.8. Using space detectors supplemented by flashing beacons to encourage efficient use of curb space
- 7.3.9. Alternative curbside layouts including the use of sawtooth or angled spaces, pull-through spaces, or permitting double-parking
- 7.3.10. Revising roadway and curbside signage/wayfinding
- 7.3.11. Modifying placement or use of regulatory and warning signs
- 7.3.12. Using variable message signs to encourage/direct traffic to less busy curbsides

7.3.1 Consolidating, Coordinating, or Removing Crosswalks

On curbside roadways, traffic controls at adjacent crosswalks are rarely coordinated, often causing motorists to stop at multiple crosswalks and leading to traffic congestion and delays. In addition, each crosswalk occupies one to three curb spaces (or between two to four curb spaces when there is a raised crosswalk or speed platform) assuming a single loading curbside (i.e., loading on passenger's side only).

While removal of crosswalks can reduce roadway congestion and improve the available length of effective curb space, crosswalk removal should be supported by effective wayfinding signage in the terminal to guide passengers to the remaining walkways and by fences or other barriers to discourage jaywalking. In the absence of barriers and enforcement, many passengers will take the most direct path between a terminal doorway and their destination on a center-island curbside or a parking facility on the opposite side of the roadway.

Analyses of pedestrian circulation paths and volumes are often performed to determine which crosswalks to remove. These analyses may consider the

- Volume of passengers inconvenienced by a crosswalk's removal. The volume of pedestrians using each crosswalk can be determined through surveys or estimated based on the land use and airline activity on either side of the crosswalk, that is, the number of pedestrian trips generated by (1) a parking structure, parking lot, center-island curbside, or other land use on the non-terminal side and (2) the proportion of airline passengers served by the airline(s) located at or near the terminal end of the crosswalk.
- Distances between adjacent crosswalks and the increase, if any, in walking distances resulting from a crosswalk's removal.
- Volume of curbside roadway traffic and the proportion of time that the crosswalk is occupied and motorists must stop while waiting for pedestrians to cross the road.

Construction of an elevated walkway above a curbside roadway or tunnel beneath the roadway allows for the removal of at-grade crosswalks. However, adding a grade-separated crossing to an existing terminal building is very expensive due to the costs of the enclosed, weather-protected bridge structure needed to span the roadway, or the costs of tunneling beneath an existing roadway, as well as the vertical circulation elements needed at both ends of the crossing (i.e., escalators plus elevators to accommodate passengers with wheeled baggage and disabilities). At airports having both at-grade crossings and grade-separated crossings, many pedestrians prefer to use the at-grade crossing rather than change building levels unless they are prevented from doing so (e.g., no visible doors leading into or out of the parking structure opposite the terminal).

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Summary—Consolidating, Coordinating, or Removing Crosswalks

Intended Goals

- Improved traffic operations/reduced curbside roadway congestion and delays
- Increased length of available, effective curb space

Applicability

- Applicable to all airports having crosswalks, but particularly those with multiple, lightly used crosswalks

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X		X			

  – Construction of pedestrian barriers or fences.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Minimal, other than enforcement.

Implementation Considerations

- **Implementation schedule:** Signs can be implemented in 2 to 3 months or less, depending on the time required for analysis and removal.
- **Ability to implement in phases:** Yes.
- **Ease of enforcement:** Easy.
- **Likely political influence:** None.

Complementary Strategies and Supporting Technologies

- 7.1.2. Control of crosswalks—officers versus signals
- 7.3.2. Changing pedestrian crosswalk controls
- 7.3.3. Raised crosswalk/speed platform

Examples

- Many large airports have elevated pedestrian crosswalks or tunnels beneath the curbside roadways linking a parking structure and terminal building. Examples include the airports serving Boston, Fort Lauderdale, Houston (both Hobby and Bush Intercontinental), Los Angeles, Minneapolis, and Salt Lake City. At some airports (e.g., Indianapolis International and Seattle-Tacoma International airports), the elevated crosswalk is below the upper-level, drop-off curbside roadway and above the lower-level pickup curbside

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roadway. Portland International and San Francisco International airports provide access between the parking garage and the terminal via both underground tunnels and overhead walkways.

7.3.2 Changing Pedestrian Crosswalk Controls

(Section 7.3.2 repeats much of the discussion in Section 7.1.2 because the strategy "control of crosswalks—officers versus signals" can be considered both an enforcement strategy and a congestion mitigation strategy.)

Vehicular traffic at pedestrian crosswalks located on curbside (or other) roadways can be uncontrolled or controlled using stop signs, pedestrian-actuated signals, or by officers. Key aspects of these options are described in the following:

- Uncontrolled crosswalks are often used when vehicular and pedestrian traffic volumes are low (i.e., there are frequent gaps in vehicular traffic allowing pedestrians to cross), the crosswalk and pedestrians in the crosswalk can be easily seen by drivers, and in communities where motorists commonly stop for crossing pedestrians. (The visibility of pedestrians using crosswalks can be enhanced and traffic speeds reduced through the use of raised crosswalks—see Section 7.3.3).
- Stop signs may be used to control crosswalks, with the signs often placed in the middle of crosswalks to improve their visibility. The benefit of stop signs is that they enhance pedestrian safety by providing positive traffic controls that are easily recognized. One disadvantage of stop signs is that they require every motorist to come to a full stop, whether or not there is a pedestrian in the crosswalk, thus delaying traffic and adding to congestion. Stop signs do not support the rapid movement of vehicles through the intersection. Also, stop signs are less effective in high-congestion environments where drivers are already distracted.
- Pedestrian-actuated crosswalk signals (Walk/Don't Walk) at airports are normally actuated by waiting pedestrians through the use of push buttons, thermal imaging devices, or motion detectors. At airports, pedestrian-actuated signals are normally used rather than signals operated on fixed time cycles. The primary benefits of pedestrian-actuated signals (compared to stop signs) are that motorists encounter fewer delays since (1) they only need to stop when there are waiting pedestrians, and (2) there are larger gaps between crossing pedestrians as the signals promote pedestrian platooning (as opposed to motorists having to stop due to the random and infrequent arrival of pedestrians). Disadvantages of crosswalk signals are (1) normally, vehicular traffic is stopped for a fixed length of time, regardless of the volume of crossing pedestrians, and (2) the operations or timings of adjacent signals are rarely coordinated, requiring motorists to potentially stop at each crosswalk they encounter, much like they do at a series of stop signs.
- Officers may be used to control crosswalks during busy periods, and with the crosswalks either uncontrolled or controlled by temporary stop signs during other hours. This schedule allows the officers to be used only when needed and can result in cost savings. The benefits of crosswalk enforcement by officers (whether TCOs or LEOs) are that they can (1) provide positive, visible enforcement of pedestrian and vehicular traffic; (2) require pedestrians to wait on the sidewalk, promoting platooning; (3) balance vehicular and pedestrian operations, responding to variations in traffic volumes; and (4) stop vehicular traffic to provide pedestrians additional crossing time if needed (e.g., disabled or elderly passengers, or passengers with small children or several large pieces of baggage). As with the other options used to control pedestrian flows at a crosswalk, the use of officers may result in motorists being required to stop at multiple crosswalks.

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Costs are a key consideration when evaluating the use of signs, signals, or officers to control crosswalks:

- Stop signs are the least expensive alternative, are readily available, and require little/no maintenance, other than replacement if damaged. However, information density is an important consideration when planning the use of additional static signage such as stop signs.
- Activated signals require electrical power, a secure/vandal-free installation, and regular inspection to ensure they are operating properly, and may require a lengthy permitting process in larger airports or some jurisdictions. Performance of maintenance and repairs requires specialized skills, which many airports outsource to private contractors or municipal maintenance agencies to ensure a 24/7 response.
- Officers, due to the ongoing expense of their wages and benefits, are by far the most expensive of the alternatives presented.

Summary—Changing Pedestrian Crosswalk Controls

Intended Goals

- Improved pedestrian safety and passenger convenience
- Improved traffic operations/reduced curbside roadway congestion and delays
- Ability to support traffic management (assuming officers are used to control crosswalks)

Applicability

- Applicable to all airports having crosswalks, but particularly those with heavily used crosswalks (e.g., connecting the terminal and parking facility)

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X	X	X		

  - Stop signs.
  - Pedestrian-actuated traffic signals.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
			X	X	X	

  - Signal operations and maintenance.
  - Officers (depends on the use of sworn/LEOs and/or civilian TCOs, the number of crosswalks, and how many hours/week they are staffed).
  - Dependent on local conditions for permits and local labor.
  - The cost range provided assumes LEOs patrolling about 400k/curbside positions ($30/hr) and TCOs patrolling about 215k/curbside positions ($20/hr).

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Implementation Considerations

- **Implementation schedule:** Signs can be implemented in 1 month or less. Actuated signals often require a minimum of 12 months to design, procure, and install. Assignment of officers may take 3 to 6 months, depending on staff availability, scheduling requirements, and budget.
- **Ability to implement in phases:** Yes.
- **Ease of enforcement:** Easy.
- **Likely political influence:** None.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.2. Control of crosswalks—officers versus signals
- 7.3.1. Consolidating, coordinating, or removing crosswalks
- 7.3.3. Raised crosswalk/speed platform

Examples

- Many airports have implemented this strategy.

7.3.3 Raised Crosswalk/Speed Platform

Raised crosswalks, also known as speed platforms, are traffic-calming devices that consist of a raised roadway section spanning the entire roadway width with sloping ramps for motorists approaching and leaving the raised section. The platform or crosswalk height is flush with the height of the adjacent sidewalks. The platform widths are designed so that the front and rear wheels of a vehicle are on top of the table at the same time, causing motorists to slow down.

The benefits of raised crosswalks include improved pedestrian safety because

- Traffic speeds are reduced by motorists slowing down at the crosswalk.
- Pedestrians in the crosswalk are more visible to drivers.
- Pedestrians in the crosswalk have an elevated view of approaching traffic.

In addition, raised crosswalks allow disabled pedestrians and those with wheeled baggage to avoid stepping down and up or navigating sloping ramps. Raised crosswalks address ADA requirements and assist passengers with wheeled baggage, eliminating the need for sloping ramps at either end of a crosswalk (although detectable warnings on both ends of a raised crosswalk are recommended).

The primary disadvantages of raised crosswalks are that they

- Remove more usable curb space than a standard crosswalk due to the sloping approach and departure ramps, which are typically about 6 to 10 feet in length, depending on the sidewalk height.
- Are potentially challenging to plow in areas with frequent snowfall.
- Require emergency vehicles to reduce speeds (but less than a speed bump).

An example of a raised crosswalk is portrayed in Figure 7-11.

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Figure 7-11. Raised crosswalk at Nashville International Airport.

Summary—Raised Crosswalk/Speed Platform

Intended Goals

- Improved safety for pedestrians crossing curbside roadways
- Aid disabled pedestrians and those with wheeled baggage
- Support traffic management and enforcement

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – According to the FHWA, raised crosswalks were reported to cost $8,000 to $30,000 each, with an average estimated cost of $8,170. These costs were based on 2013 estimates.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

Implementation Considerations

- **Implementation schedule:** Requires design and construction. Estimated to require 3 months.

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- **Ease of enforcement:** Supports enforcement of traffic speeds and pedestrian safety.
- **Likely political influence:** None.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.2. Control of crosswalks—officers versus signals
- 7.3.1. Consolidating, coordinating, or removing crosswalks
- 7.3.2. Changing pedestrian crosswalk controls

Examples

- Airports that have implemented this strategy include those serving Nashville, Phoenix, Portland, Raleigh, and Reno.

7.3.4 Revising the Location or Amount of Curb Space Allocated to Specific Mode(s)

Curbside traffic congestion can be managed and mitigated by modifying the location and/or amount of curb space assigned to each travel mode or transportation service. It is suggested that the allocation of curb space be reviewed regularly and adjusted as needed to reflect changes in traffic volumes, operations, and mix of vehicles. For example, a decline in the number of customers seeking on-demand taxicabs may warrant reducing the amount of curb space provided for waiting taxicabs and reallocating that space to a mode that has gained traffic over that time such as TNCs or rental car shuttles.

When allocating curb space, particularly the location and amount of space reserved for vehicles picking up passengers, airport managers consider the amount of curb space required by each travel mode or transportation service, the total amount of space required by all modes, and the total effective curb space available. They then prioritize the available space and assign space to each travel mode considering factors such as

- **Passenger needs.** The location of spaces reserved for passengers with disabilities is determined both by codes and by the volume of airport passengers having special needs. Airports serving many senior citizens may wish to provide more space than required by code.
- **Customer expectations.** Customers expect to find certain transportation services, such as taxicabs, near baggage claim areas. Customers who have reserved pre-arranged or luxury limousines expect to easily find their waiting vehicles and drivers.
- **Airport policies.** Some airports have a "transit first" policy and encourage customers to use public transit or other high-occupancy vehicles. At these airports, scheduled buses may be assigned the most convenient drop-off and pickup areas, space that is more convenient than the space assigned to private vehicles.
- **Airport revenues.** Airport management often assign higher-priority curb space (i.e., more convenient space) to transportation services from which the airport receives significant revenues. For example, on-airport parking and rental car shuttles may be assigned more convenient and visible curb space than transportation service operators that contribute less airport revenue.
- **Competition among operators.** For ground transportation providers that compete with one another (or perceive that they compete), airport management considers the proximity and relative convenience of the curb space assigned. For example, airport staff may attempt to provide comparable space to TNCs and taxicabs to avoid requiring customers of one service to walk

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past the pickup zone assigned to a competitor. Conversely, on-airport-parking shuttles may be assigned more convenient curb space than the shuttles operated by off-airport-parking businesses.

- **Operational needs.** Curb space allocations must allow adequate maneuvering space to accommodate the vehicles commonly used by the operators of each transportation service. For example, over-the-road motorcoaches commonly used by charter or tour bus operators require about 60 linear feet to enter and exit a parallel curbside area. Thus, the space allocated to this mode must be at least 60 feet in length, and if space for multiple vehicles is provided, the allocated space should be in increments of about 60 feet.

The QATAR model described in this Guide can be used to test alternative curb space allocations and compare how they impact curbside roadway operations and service levels.

Summary—Revising the Location or Amount of Curb Space Allocated to Specific Mode(s)

Intended Goals

- Improve curbside operations by promoting efficient use of passenger boarding areas
- Accommodate commercial ground transportation operators and private vehicles
- Respond to customer expectations and the needs of commercial ground transportation operators
- Support the goals of airport management (e.g., encourage use of public transit)

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X				

  - Primarily costs of reallocating curbside areas that may require relocation of signs and pavement markings.
  - Initial costs include restriping and signage. Costs will be significantly more if curbs have to be reconditioned for new service.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

Implementation Considerations

- **Implementation schedule:** Less than 2 months depending on time needed to modify signs and markings and coordinate with stakeholders.
- **Ability to implement in phases:** Possibly, but generally all boarding area space is reallocated at the same time.
- **Ease of enforcement:** Supports enforcement of curbsides.

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- **Likely political influence:** Operators who perceive they have been assigned less desirable space than their competitors.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.3. Measures to support curbside enforcement: warning messages, ticketing, and towing
- 7.1.4. Regulatory sign placement and conformance
- 7.2.1. Moving commercial vehicle passenger boarding areas to an available remote location
- 7.3.1. Consolidating, coordinating, or removing crosswalks
- 7.3.2. Changing pedestrian crosswalk controls
- 7.3.5. Implementing virtual taxicab queue/automated taxicab dispatching
- 7.3.9. Alternative curbside layouts including the use of sawtooth or angled spaces, pull-through spaces, or permitting double-parking
- 7.3.10. Revising roadway and curbside signage/wayfinding
- 7.3.11. Modifying placement or use of regulatory and warning signs

Examples

- Most airports regularly review curb space allocations and revise as needed.

7.3.5 Implementing Virtual Taxicab Queue/Automated Taxicab Dispatching

Curbside congestion can be reduced by right-sizing the amount of curb space allocated to taxicabs and other on-demand transportation services. The use of automated dispatching systems allows airports to reduce the space allocated to these services while maintaining desired customer service levels. The following paragraphs address the use of virtual queues and automated dispatching by airports to manage taxicab operations. Airports do not manage the dispatching of TNCs as each TNC controls its own virtual queue and automated dispatching system.

Automated taxicab dispatching and virtual queues replace traditional airport taxicab dispatching and starting. With traditional taxicab operations, drivers wait in hold lots (or staging areas) until airport staff or their representatives (e.g., a taxicab dispatcher) instruct the drivers to proceed to a passenger pickup zone, where the drivers wait in a queue until other airport staff (e.g., taxicab starters) assign them a waiting customer. Taxicab drivers frequently encounter long waits for airport customers, requiring airports to provide and maintain large hold lots to accommodate waiting taxicabs and provide restrooms, lounges, and other amenities desired by waiting drivers.

With a virtual queue, drivers using a smartphone can join a virtual queue of waiting taxicabs until they reach the head of the virtual queue. Depending on the size of the airport and the volume of taxicab activity, upon reaching the head of the virtual queue, drivers are instructed to proceed to

- A small taxicab hold lot (which can be reduced in size if it needs to accommodate fewer vehicles and provide fewer amenities because drivers are only in the lot briefly).

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- A taxicab stack, which is a short queue located away from the curbside but near the pickup area that allows drivers to respond to passenger requests more quickly than if drivers were traveling from a more distant hold lot. Drivers are then instructed to proceed from the stack to the pickup area when needed.
- The passenger pickup area.

A virtual queue supplements, but does not replace, the taxicab dispatcher and starter. In contrast, an automated dispatch system is capable of replacing both the dispatcher and starter by (1) monitoring the number of taxicabs waiting at each passenger pickup area, and (2) as a taxicab exits the pickup area notifying the next driver waiting in the virtual queue to proceed directly to the passenger pickup area(s) or taxicab stack. The number of taxicabs waiting at each passenger pickup area can be adjusted to respond to anticipated or actual passenger demands.

For taxicab drivers, the benefits of a virtual queue are that it allows them, using a smartphone, to (1) determine the number of taxicabs waiting for airport passengers ahead of them and estimate their wait time, (2) make an informed decision about when to work and join the queue if they choose to, (3) monitor their place in the queue, and (4) proceed to the airport when they approach the head of the queue.

For airport management, one benefit of a virtual queue is that it allows the taxicab hold lot to be drastically reduced in size or to be removed or used for other purposes. The additional benefits of an automated dispatch system are that it allows management to (1) minimize the curb space allocated for taxicab pickups while ensuring there are sufficient waiting taxis to accommodate arriving passengers; (2) reduce labor costs; and (3) better control short trips, pre-arranged pickups, and other non-standard fares.

Virtual queues and automated taxicab dispatch systems require the acquisition of a ground transportation management system (GTMS). Automated dispatch systems require vehicle detectors (e.g., gate arms, induction loops, Bluetooth, radio-frequency identification, or cameras) to be installed at the passenger boarding area to monitor the arrival and departure of authorized vehicles. Generally, airports either purchase off-the-shelf automated dispatch systems (which can be an add-on feature of an existing GTMS) or custom-built systems. Both require the preparation of requests for qualifications (RFQs) or requests for proposals (RFPs). Prior to issuing the RFQ or RFP, it is recommended that airport management

- Review existing taxicab regulations and policies and revise them as appropriate.
- Confirm desired dispatching policies such as response times.
- Establish the desired maximum and minimum number of waiting taxicabs at each passenger boarding area.
- Determine the availability of a site for a taxicab stack in proximity to the passenger boarding area.

Examples of automated dispatch displays are shown in Figure 7-12.

Summary—Implementing Virtual Taxicab Queue/Automated Taxicab Dispatching

Intended Goals

- Improve curbside operations by balancing available space and demands
- Improve roadway operations by reducing unnecessary trips

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- Accommodate taxicab operations and customers
- Support traffic management and enforcement

Applicability

- Applicable to medium- and large-hub airports with demand for taxicab service or many waiting taxicabs

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X	X	X	X	

  – $100,000 to more than $1,000,000 for GTMS; about $50,000 or less for virtual queue alone.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  – Ongoing costs of system maintenance and upgrades.
  – Dependent on the size of the location to be managed and the functionality to be provided, such as billing and account management.

Implementation Considerations

- **Implementation schedule:** 3 to 12 months, depending on system complexity and time required for procurement and hardware installation.
- **Ease of enforcement:** Supports enforcement of curbsides and taxicab operations.
- **Likely political influence:** Minimal. Requires taxicab drivers to have a credit card (for fee payment) and smartphone.

Complementary Strategies and Supporting Technologies

- 7.2.1. Moving commercial vehicle passenger boarding areas to an available remote location
- 7.2.9. Charging commercial vehicle operators headway fees or higher fees for curbside pickup
- 7.3.4. Revising the location or amount of curb space allocated to specific mode(s)
- 7.3.9. Alternative curbside layouts including the use of sawtooth or angled spaces, pull-through spaces, or permitting double-parking

Examples

- Airports with automated taxicab dispatch systems include those serving Minneapolis, Monterey, Philadelphia, Phoenix, San Francisco, San Jose, and Tampa.

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Source: Parkway Airport Services, November 2021

***Figure 7-12. Automatic dispatch display as seen by driver* (left). *Taxicab display in hold lot at Philadelphia International Airport* (right).**

7.3.6 Modifying/Lengthening Weaving Area

As described in prior chapters, weaving occurs when traffic on two roads traveling in the same direction crosses, requiring vehicles to merge and diverge in close proximity in order to execute lane changes. (Chapter 4 describes low-speed weaving operations on airport roadways and the use of QATAR to analyze these operations.) Congestion and delays occur when the volume of weaving traffic exceeds the capacity of the weaving segment, causing queues to form on the approach lanes.

The key factors determining the capacity of a weaving segment include the volume and speed of the weaving traffic, the number of lane changes needed, and the length of the weaving segment. Reducing roadway congestion in a weaving segment requires modifying one or more of these factors by

- **Directing traffic to an alternate route.** It may be possible to require motorists to follow an alternative but longer path. For example, at San Francisco International Airport, an unsatisfactory weaving operation was resolved by closing a roadway and requiring motorists exiting the domestic terminal curbsides to take a longer path to travel to the parking structure. When traffic queues begin to form at Los Angeles International Airport, airport staff can use a temporary gate to close a weaving segment exit, eliminating congestion but causing motorists to travel further. Denver and Portland both have closed return-to-terminal routes to push traffic further out. (See Figure 7-13.)
- **Reducing traffic speeds.** Traffic speeds in the weaving area may be reduced using traditional methods such as the posting and enforcing of regulatory signs.
- **Reducing the number of required lane changes.** The number of lanes motorists must cross to complete a weave can be reduced by
  - Merging traffic lanes before vehicles enter a roadway weaving segment using pavement markings, traffic delineators, signage, or a combination of these tools, assuming that the reduced number of entrance lanes provides adequate capacity.

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Figure 7-13. Weaving section at Denver International Airport resolved by closing roadway segment.

  - Directing motorists to the appropriate lane prior to their entering a weaving segment using directional signs and/or pavement markings and by separating weaving traffic from traffic that is not changing lanes.
- **Increasing the length of a weaving segment.** A weaving segment's length may be increased slightly by changing the pavement markings at the entrance and exit of the segment or minor changes (including use of the roadway shoulders) to the entrance and exit roadways. However, changes to weaving segments that involve roadway construction or the introduction of grade-separation (to eliminate a weave entirely) are not included in the management strategies presented in this Guide due to the large capital investments required.

Summary—Modifying/Lengthening Weaving Area

Intended Goals

- Improve operations in roadway weaving segments in order to reduce delays and improve traffic safety
- Improve overall roadway operations

Applicability

- Applicable to all airports having roadway weaving segments

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  - Costs vary depending on whether the airport is closing roadways or adding signs or markings or increasing the length of a weaving segment length.

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – None to minimal.

Implementation Considerations

- **Implementation schedule:** 1 to 6 months, depending on time required for analysis and installation of signs or pavement markings.
- **Ease of enforcement:** No enforcement needed unless roadway speeds are to be modified.
- **Likely political influence:** None.

Complementary Strategies and Supporting Technologies

- 7.3.10. Revising roadway and curbside signage/wayfinding
- 7.3.11. Modifying placement or use of regulatory and warning signs
- 7.3.12. Using variable message signs to encourage/direct traffic to less busy curbsides

Examples

- Los Angeles, San Francisco, Denver, and Portland airports.

7.3.7 Charging Commercial Vehicles for Excessive Number of Trips or Dwell Times

At some airports, operators of commercial ground transportation vehicles are charged fees (or fines), often referred to as demand management fees, in order to reduce roadway and curbside congestion. These fees are intended to

- Reduce traffic congestion on airport roadways.
- Encourage efficient use of passenger boarding areas.
- Improve air quality by reducing vehicle-generated emissions.
- Discourage operators from making non-essential trips.

Non-essential trips include those made by the operators of courtesy vehicles seeking to advertise or market their services by (1) circulating repeatedly past the terminal building curbsides or (2) ensuring that one of their courtesy vehicles is stopped at the curbside at all times (i.e., operating in a "bump-and-run" manner so that a vehicle remains at the curb until a subsequent vehicle arrives).

To reduce traffic congestion, encourage efficient use of the passenger pickup areas, and achieve other goals, airport managers

- Fine ground transportation operators whose vehicles remain at the passenger pickup area in excess of an established maximum dwell time (i.e., charge dwell time fees). The maximum dwell times normally vary based on vehicle size and type of service, with charter buses allowed to dwell for 15 to 20 minutes, while courtesy vehicles are allowed to dwell for 5 minutes or less. Typically, the amount of the fine is three to five times the per-trip fee such vehicles are required to pay. Taxicabs queued while waiting for arriving customers are not charged dwell time fees.

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- Place a cap on the number of courtesy vehicle trips made by each courtesy vehicle operator. For example, each hotel/motel may be limited to 1,500 courtesy vehicle trips per month and charged a fee for each additional or excess vehicle trip.
- Restrict the minimum time interval between the arrival of courtesy vehicles serving a business (e.g., there must be 15 minutes or more between successive courtesy vehicles operated by the same company) or limit the operator to four trips per hour. These fees are referred to as interval fees and are intended to discourage bump-and-run operations.

These restrictions require the use of GTMSs to monitor (1) the length of time vehicles remain at the boarding areas, (2) the volume of courtesy vehicle trips, (3) the time between successive courtesy vehicles, and (4) provide a basis for calculating fines or penalties. Monitoring dwell times requires vehicle detectors or antennae to be installed at the entrance and exit of the passenger boarding area. Only a single vehicle detector or antennae is required to monitor the number of monthly vehicle trips or the time interval between vehicles.

At some airports, rental car companies and other courtesy vehicle operators ignore these management fees in order to maintain corporate-mandated customer service levels. These companies consider the cost of dwell time fees and/or interval fees part of the "normal cost of doing business" at the airport and pay the associated fees. At these airports, dwell time and interval fees have proven to be less effective in achieving management's objectives.

Summary—Charging Commercial Vehicles for Excessive Number of Trips or Dwell Times

Intended Goals

- Improve curbside operations by promoting efficient use of the boarding areas and discouraging unnecessary trips
- Accommodate commercial ground transportation operators
- Support traffic management and enforcement

Applicability

- Applicable to all airports having GTMSs and significant courtesy trip volumes

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
		X	X	X		

  - Cost of GTMS.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  - Cost of monthly invoices or managing established accounts.
  - Depends on the number of vehicles/providers to be managed and the functionality to be provided such as billing and account management.

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Implementation Considerations

- **Implementation schedule:** 3 to 12 months if a new GTMS must be designed and installed, as well as time to coordinate with stakeholders. The benefit of this option is the ease of communicating the changes. This is easier than communicating changes to TNC drivers, where there is always a large number of first-time drivers in the system.
- **Ease of enforcement:** Supports enforcement of curbsides.
- **Likely political influence:** None.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.3.8. Using space detectors supplemented by flashing beacons to encourage efficient use of curb space

Examples

- Raleigh-Durham International Airport has established an interval fee.
- Airports having dwell time fees include those serving Minneapolis, Orlando, and Salt Lake.

7.3.8 Using Space Detectors Supplemented by Flashing Beacons to Encourage Efficient Use of Curb Space

Curbside congestion can hypothetically be reduced by implementing a space management system (using overhead detectors or cameras) similar to those utilized in "smart" parking garages. With this strategy, the signal or sign would note which spaces on the curb are available or are occupied to vehicles some distance away from approaching vehicles. Similar systems have been tested in urban areas (e.g., SFpark)

Use of space detectors could pose challenging if (1) the curbside boarding spaces are not clearly delineated (or striped) causing the signals not to detect stopped vehicles, and/or (2) use of the spaces is not actively enforced. An additional caveat to this system is the lack of control over the driver's instinctive behaviors, such as double- and triple-parking or clustering at the perceived "closest" door for the exiting or arriving passenger.

Summary—Using Space Detectors Supplemented by Flashing Beacons to Encourage Efficient Use of Curb Space

Intended Goals

- Improve utilization of the "full" curb
- Limit clustering earlier in the curb and distributing future "spaces"
- Limit double- and triple-parking

Applicability

- Applicable to large-hub and potentially medium-hub airports with very busy curbs that include heavy peaking activity, particularly those that have pull-through or angled spaces

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Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X	X	X			

  – Equipment installation for in-pavement or overhead space detectors.
  – Dependent on technology used and infrastructure (power/communications).

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  – Policing costs and equipment operations and maintenance.

Implementation Considerations

- **Implementation schedule:** It would likely take approximately 12 to 18 months (depending on the constructability of the rail that would run above the lane in question). Full procurement process from RFQ through design and construction is expected.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.3.9. Alternative curbside layouts including the use of sawtooth or angled spaces, pull-through spaces, or permitting double-parking

Examples

- This strategy was employed at Nashville International Airport, shown in Figure 7-14, but was removed by airport staff due to security concerns and to improve the flow of through traffic past the angled spaces.

7.3.9 Alternative Curbside Layouts Including the Use of Sawtooth or Angled Spaces, Pull-Through Spaces, or Permitting Double-Parking

Congestion on a linear (vehicles parked nose-to-tail) curbside roadway, particularly at pickup curbsides, can be reduced by using alternative curbside configurations such as angled or sawtooth spaces (either pull-through or back-out) or by allowing double-parking. These configurations can increase curb capacity, reduce passenger walking distances, and support enforcement.

Angled Spaces

Angled curbside spaces can (1) improve vehicle efficiency (i.e., the number of vehicles accommodated per square foot) compared with a linear curbside having three or more parallel lanes and (2) reduce passenger walking distances because more vehicles can be accommodated in the same curbside length. Airport curbsides can be configured using angled pull-through spaces or angled back-out spaces. An example configuration for angled pull-through spaces is shown in Figure 7-15.

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Figure 7-14. Pull-through angled spaces and overhead countdown clock previously used at Nashville International Airport.

To accommodate angled pull-through spaces, the curbside roadway needs to be at least 60 feet in width (the equivalent of about five parallel lanes). This provides sufficient width for the pull-through spaces plus at least one through lane for vehicles entering the spaces and a second through lane for vehicles exiting the spaces. Angled back-out spaces require approximately the same width because at least two through lanes are needed behind the spaces to allow motorists to back out safely. An example configuration of angled back-out spaces is shown in Figure 7-16.

There are challenges associated with pull-through and back-out spaces. The challenges with pull-through spaces include (1) the need for exiting vehicles to cross the paths of passengers walking between stopped vehicles and the terminal and (2) the tendency of motorists who are waiting for an empty space to double-park and possibly block through traffic. When double-parking is allowed to occur, vehicle queues can quickly form if other motorists cannot bypass the stopped vehicles. The challenge with back-out spaces is that drivers must back out into an active roadway while their ability to see oncoming traffic is constrained (i.e., they must back out into a blind spot), compromising roadway safety. Thus, both angled pull-through and back-out spaces

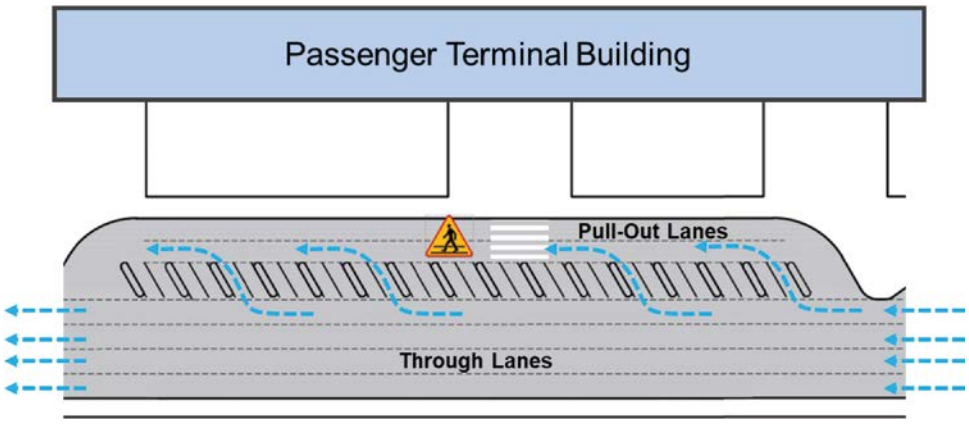


Figure 7-15. Example configuration for angled pull-through spaces.

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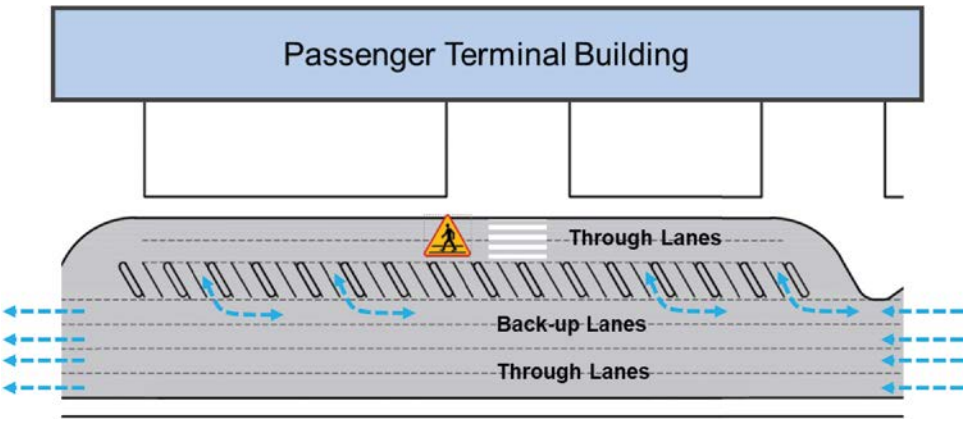


Figure 7-16. Example configuration for angled back-out spaces.

are more commonly found at passenger drop-off and pickup areas used exclusively by commercial ground transportation vehicles (e.g., a ground transportation center) or in a repurposed parking area.

Sawtooth Spaces

Sawtooth spaces allow buses and other large vehicles to park immediately adjacent to a curbside allowing passengers to step directly to/from the parked vehicle when boarding and alighting. A sawtooth space is shorter than the length of the space these vehicles require when parking along a traditional linear curbside. The disadvantages of sawtooth spaces are (1) the lack of flexibility—a space accommodates a single vehicle regardless of the vehicle's size whereas smaller vehicles require less linear curbside; and (2) like angled spaces, the delays to traffic in through lanes if vehicles waiting to enter a sawtooth space are allowed to double-park.

Designated Double-Wide Lanes

A double-wide lane is a curbside lane that is 20 to 24 feet wide (the equivalent of two travel lanes) separated by distinct pavement marking or striping from the adjacent through lanes. Motorists are allowed to double-park on the terminal side of a pavement marking but only moving traffic is allowed in the through lanes. Double-wide lanes facilitate enforcement by allowing double-parking in the designated area while discouraging motorists from blocking through lanes. A concern with double-wide lanes is that motorists loading or unloading adjacent to the curbside or inner lane can be "trapped" by vehicles stopped in the outer lane. As a result, motorists tend to only stop in the outer lane, leaving the inner curbside lane under-utilized.

Summary—Alternative Curbside Layouts Including the Use of Sawtooth or Angled Spaces, Pull-Through Spaces, or Permitting Double-Parking

Intended Goals

- Improve curb utilization by increasing efficiency
- Decrease policing requirements
- Create a safer environment for vehicles to enter, stop, and exit from the curbside

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Applicability

- Applicable on the curbside roadways at hub airports where there is adequate space (e.g., a curbside roadway width of 60 feet or more). Implementation at a ground transportation center or within a parking facility would be more likely at large- or medium-hub airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X	X	X	X	

  – Roadway paint and signage, depending on whether implementation simply requires new pavement markings or roadway reconstruction.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Minimal to none.

Implementation Considerations

- **Implementation schedule:** Really applies to the amount of time it would take to paint the pavement, adjust signage, and manage the change.

Complementary Strategies and Supporting Technologies

- 7.3.8. Using space detectors supplemented by flashing beacons to encourage efficient use of curb space

Examples

- St. Louis Lambert International Airport has angled pull-through curbs on the departures level. Several European airports (e.g., those serving Munich and Paris) have angled spaces.
- Sawtooth and angled parking are commonly used for commercial vehicle (e.g., bus and courtesy vehicle) parking in ground transportation centers, particularly where vehicles need not back out into active traffic lanes.
- Chapter 2 of this Guide provides additional information about alternative curbside configurations.

7.3.10 Revising Roadway and Curbside Signage/Wayfinding

Curbside congestion can be improved by revising roadway and curbside signage and wayfinding messages both by revising overhead signs and those mounted adjacent to roadways (i.e., column-mounted signs and graphics). Additionally, roadway signage that adds clarity via colors and clear messaging to better assist with the allocation of curbs and available spaces will

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benefit drivers who are already distracted by surrounding congestion. Roadway signage is most effective if the message starts before the vehicle enters the decision-making zone and is part of and consistent with the airport's signage and wayfinding master plan. Too many signs can lead to visual clutter, distract motorists, and make it challenging to find key signs. For this reason, advertisement or promotional banners are discouraged on airport roadways as such messaging can distract from directional and regulatory messages.

ACRP Report 52: Wayfinding and Signing Guidelines for Airport Terminals and Landside provides guidelines on the use, location, and design of signs including technologies, fonts, colors, and sizes. As noted in this resource, signs can be mounted on free-standing poles, overhead structures, or structural columns, or be displayed using column wraps or pavement markings.

Summary—Revising Roadway and Curbside Signage/Wayfinding

Intended Goals

- Improve visual identification of curb location/identifier
- Clear delineation of curb allocation via signage and roadway markings

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X	X	X	X		

  – Dependent on type of signage (static versus variable messaging) and location of signage.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  – Paint and signage replacement at regular intervals.

Implementation Considerations

- **Implementation schedule:** Implementation could take less than 6 months, depending on the number of signs and their size, the lead time needed to fabricate the signs, and if the signs are to be mounted to new structures or existing structures and the type of mounting used.
- **Signage.** Signage should be consistent with the airport's graphic standards manual and installed as part of an overall signage plan.

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- **New structural elements.** If new structural elements are required, lead times will need to be extended.

Complementary Strategies and Supporting Technologies

- This strategy for reducing curbside congestion should be implemented with most other strategies to ensure clear visual direction.
- Providing color coding to identify various areas/allocations is often effective.
- Reflective signs or illumination of signs should be considered for nighttime hours and lower-level curbsides.

Examples

- Column wrapping for mapping and direction provides assistance to passengers looking for clear directions around the airport (shown in Figure 7-17).
- Overhead signage (for drivers) based on key location identifiers is not typically advantageous. An example of typical overhead signage is shown in Figure 7-18.
- Roadway painting showing available curbside allocations (taxis, TNCs, other commercial ground transportation vehicles, and private vehicles).



Figure 7-17. Column wrapping at Dallas/Fort Worth International Airport.



Figure 7-18. Overhead signage at Reno-Tahoe International Airport.

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7.3.11 Modifying Placement or Use of Regulatory and Warning Signs

Curbside congestion can be reduced by airport staff by modifying the placement and/or use of regulatory and warning signs, but only if the message is supported by active and consistent enforcement. If private vehicle use of portions of the curbside is restricted, adjacent signage clearly identifying curbside availability and access should be provided.

Summary—Modifying Placement or Use of Regulatory and Warning Signs

Intended Goals

- Limit curbside access
- Eliminate specified curbside access

Applicability

- Applicable to all airports regardless of size

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Signage installation costs.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Minimal costs, typically maintenance and replacement of signage.

Implementation Considerations

- **Implementation schedule:** Implementation would likely take approximately 6 months (depending on signage construction).

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.3. Measures to support curbside enforcement: warning messages, ticketing, and towing
- 7.1.4. Regulatory sign placement and conformance
- 7.3.2. Changing pedestrian crosswalk controls
- 7.3.9. Alternative curbside layouts including the use of sawtooth or angled spaces, pull-through spaces, or permitting double-parking
- 7.3.12. Using variable message signs to encourage/direct traffic to less busy curbsides

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Examples

- All airports have signs designating which portions of the curbside are reserved for use by commercial ground transportation vehicles to guide the drivers of these vehicles and support enforcement.

7.3.12 Using Variable Message Signs to Encourage/Direct Traffic to Less Busy Curbsides

Variable message signs can be used to balance demand among available curbside areas by directing arriving motorists to less busy curbside levels (if the peak periods of demand for these curbsides do not coincide). Implementation requires placing variable message signs (using either permanently mounted or temporary trailer-mounted signs) on the approach roadways to inform entering motorists about the availability of alternative drop-off or pickup areas. An example of this is shown in Figure 7-19.

Web-based information about parking availability can be used to supplement variable message roadway signs and encourage motorists to use adjacent parking structures or surface lots.

Summary—Using Variable Message Signs to Encourage/Direct Traffic to Less Busy Curbsides

Intended Goals

- Fully utilize all available drop-off and pickup areas irrespective of their originally intended use
- Leverage underutilized assets
- Allow an airport to delay capital improvements

Applicability

- Applicable to any airport that has underutilized curbs during peak activity
- Very applicable at small- and medium-hub airports having significant directional peaks (e.g., large volumes of arriving passengers in the evening and departing passengers in the morning)
- Potentially applicable at large-hub airports with directional peaks and underused drop-off and pickup areas.

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X	X	X	X	

  - Installation of signage and supporting power and communications or cost for rental of trailers.
  - Dependent on the type of signs and the number of signs.

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  – Equipment operations and maintenance

Implementation Considerations

- **Implementation schedule:** Implementation would likely take approximately 6 to 12 months (depending on signage construction and permitting issues with any regulatory agencies).
- **Internal vertical circulation:** Passengers arriving at an airport with carry-on baggage can very easily be picked up on the departures level instead of at the arrivals level after deplaning. The same is not always as easy if the passenger has a checked bag. In this case, signage is needed to direct passengers from baggage claim to ticketing.
- **Peak-hour curbside use by commercial vehicles:** At some airports, commercial vehicles are directed to or required to use the opposite (Arrivals or Departures) curbside during peak hours of roadway congestion.

Complementary Strategies and Supporting Technologies

- 7.1.1. Enforcement of airport roadways and curbsides
- 7.1.3. Measures to support curbside enforcement: Warning messages, ticketing, and towing
- 7.2.4. Encouraging motorists dropping off or picking up passengers to use hourly parking
- 7.3.6. Modifying/lengthening weaving area
- 7.3.10. Revising roadway and curbside signage/wayfinding
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes

Examples

- Some airports have used both permanent and temporary signage to direct passengers to alternative curbs.
- Richmond International Airport has attempted to divert arriving passengers to the departures level via temporary signage.
- Sacramento International Airport has used temporary signage to divert passenger pickups to the departures level, as shown in Figure 7-20.
- Portland International Airport has the ability to direct motorists to alternative curbside levels using permanently mounted, dynamic signage located on its entry roadway (giving passengers enough time to process the option).
- Los Angeles International Airport has used temporary signage to divert passenger pickups from the Tom Bradley International Terminal curbs to curbs on the first floor of parking garage P3.
- San Francisco International Airport has directed customers and on-demand vehicles to the lower-level roadway during enplaning peak periods.

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Figure 7-19. Variable message sign directing passengers at Dallas/Fort Worth International Airport.



Source: Sacramento International Airport.

Figure 7-20. Sign directing arrival passengers to use the departures level.

7.4 Customer Service Strategies

It is important for an airport to provide good customer service because a customer's first and last experiences at the airport are usually the most memorable. The strategies discussed in this section are focused on reducing roadway and curbside congestion and encouraging the use of transit and high-occupancy vehicles by improving customer experience. The strategies are

- 7.4.1. Providing convenient, enclosed, passenger waiting areas
- 7.4.2. Informing waiting passengers of bus and shuttle arrival times
- 7.4.3. Presenting transit schedules, fares, and route information to passengers
- 7.4.4. Providing passenger service associates at curbsides
- 7.4.5. Enhancing safety and convenience for pedestrians crossing roadways and waiting for transportation

7.4.1 Providing Convenient, Enclosed, Passenger Waiting Areas

The customer experience of passengers waiting to be picked up can be enhanced by providing climate-controlled waiting areas that include seating, heating, and other amenities. These shelters can be located both at terminal curbsides and at remote/consolidated facilities. These shelters protect

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passengers from the elements while providing centralized locations that increase the efficiency of the curb.

Key considerations related to implementing passenger waiting enclosures include

- Location
- Amenities
- Maintenance

Location

The shelters should be near pickup locations—particularly on uncovered roadways—serving regularly scheduled buses, courtesy vehicles, and on-demand vehicles that are not standing at the curbsides such as taxicabs or TNCs. (Generally, shelters are not required at taxicab stands as normally there is a queue of waiting taxicabs, eliminating the need for passengers to wait for arriving vehicles.) Passengers waiting in the shelter should be able to clearly see approaching vehicles and other portions of the curbside. Depending on the curb length and number of areas allocated to buses, courtesy vehicles, and TNCs, one shelter positioned at a central location or several evenly spaced locations may be needed to minimize walking distances and provide sightlines. Generally, shelters should be located at least 3 feet back from the curb face, with a minimum setback of 10 feet from any crosswalks. Shelter locations should be evaluated for their impact on pedestrian traffic in the surrounding area.

Amenities

Shelters should provide weather protection against wind, snow, and blowing rain, with heating provided in some climates. Shelter sizes vary based on anticipated peak-period occupancy (i.e., occupancy depends on the estimated number of peak-period passengers waiting concurrently at each shelter for buses, courtesy vehicles, and TNCs) and the local climate. Shelters in more extreme climates need to provide greater capacity than those in more temperate climates due to the proportion of passengers expected to wait inside the structures. Additional guidelines may be available from local transit codes. Climate-controlled waiting areas should contain seating, with additional amenities considered to improve customer experience, including

- Heating elements in colder environments
- Proper ventilation in warm environments and protection from the sun
- Lighting levels meeting/exceeding local standards to ensure passenger safety and comfort
- Courtesy phones to connect with the service provider and airport security
- Dynamic signs/information stations with key contact and pricing information
- Variable message boards displaying the estimated arrival time of the next airport-operated parking shuttle(s) or public transit bus
- Seating bench
- Vending machines
- Charging stations
- Overhangs, leaning rails, and additional seating on the outside of shelters to meet peak demand

Maintenance

Shelters should be properly maintained to maintain desired levels of customer experience. Maintenance includes regular cleanings that avoid peak periods of use and upkeep to address general wear and tear on a monthly or annual basis. (Every customer service enhancement must be kept in working order and properly maintained or the amenity may negatively impact customer experience.) Any equipment associated with safety and security must be kept in working order as non-working equipment reflects a significant liability to the airport.

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Summary—Providing Convenient, Enclosed, Passenger Waiting Areas

Intended Goals

- Provide safe passenger waiting spaces
- Improve customer experience while waiting
- Provide additional amenities (if possible)

Applicability

- Applicable to all airports, depending on the anticipated number of waiting passengers and local climate (Small shelters similar to city bus stops are easily procured and affordable. A larger-scale brick-and-mortar facility with more amenities, which would likely be located at a remote transfer point or ground transportation center, is a more expensive option but may be warranted at larger airports, those airports with particularly warm or cold weather, or those wishing to consolidate passengers into fewer pickup points to improve shuttle service and reduce curbside needs.)

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X	X	X			

  - Design and development/procurement of climate-controlled booths or design and construction of a passenger building.
  - Depends on the number of areas, infrastructure (power/communications), and level of amenities.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  - Operations and maintenance costs will vary based on the number of required enclosures and the environments and amenities associated with them.

Implementation Considerations

- **Implementation schedule:** Depends on the type of structure planned and the level of amenities to be provided. Estimated to be a minimum of 6 months for procurement and installation of a standard bus shelter (depending on the amount of infrastructure such as power and communications that is needed) and up to 24 months for a brick-and-mortar solution.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.4.2. Informing waiting passengers of bus and shuttle arrival times
- 7.4.3. Presenting transit schedules, fares, and route information to passengers
- 7.4.4. Providing passenger service associates at curbsides

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- 7.4.5. Enhancing safety and convenience for pedestrians crossing roadways and waiting for transportation

Examples

- Detroit Metropolitan Airport's ground transportation curbs include a central conditioned space with seating and vending machines while also providing additional bus shelters on the curb with heating elements and seating.
- Los Angeles International Airport's temporary bus transfer provides a consolidated curb for regional buses with canopy coverage that includes seating. The former bus depot was a conditioned brick-and-mortar facility with seating and vending options.
- Sacramento International Airport provides bus shelters with seating at its two remote curbs but does not provide additional amenities.
- Chicago O'Hare International Airport's Multi-Modal Facility is the terminating stop for its people mover train and includes a central terminal for rental car and ground transportation customers as well as remote pickup/drop-off functions. The facility is conditioned against the elements associated with Chicago weather, has vending functions, and provides areas for charging and communication with ground transportation providers.
- Ted Stevens Anchorage International Airport provides a ground transportation lobby on the center island that can be accessed via an alternative underground route directly from the terminal building. The ground transportation lobby is shown in Figure 7-21.

7.4.2 Informing Waiting Passengers of Bus and Shuttle Arrival Times

It is helpful to display the expected arrival times of buses and airport shuttles to passengers awaiting these vehicles. Doing so improves the customer experience and reduces their anxieties. Key considerations related to informing waiting passengers include

- Information/signage
- Customer service kiosks
- Accuracy of information

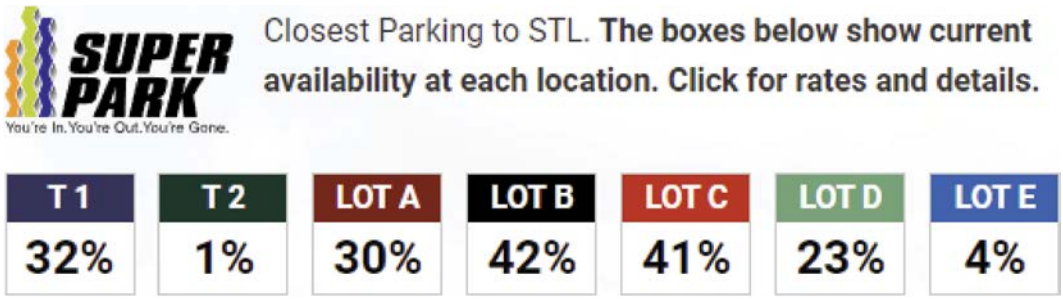


Source: Rob Stapleton, Jr. / Ted Stevens Anchorage International Airport

Figure 7-21. Ground transportation lobby at Ted Stevens Anchorage International Airport.

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Source: https://www.flystl.com/, screen capture: May 16, 2022.

Figure 7-22. St. Louis Lambert International Airport website showing parking availability.

Information/Signage

At key locations on the curb, message signs can be installed displaying regularly scheduled route and shuttle information. These can be standard static informational signs or dynamic displays, showing the arrival time of the next bus (or shuttle) and route information/destinations served. Some airports have found it beneficial to provide a web application, or "app," that includes detailed information about the ground access system, including information on parking occupancy and schedules for regular ground access services, such as scheduled buses, courtesy shuttles, and public transit. St. Louis Lambert International Airport shows current parking availability, shown in Figure 7-22.

These applications can also display the same real-time data as the dynamic signage along with live global positioning system (GPS) tracking on a map showing the location of shuttles, as shown in Figure 7-23.

Customer Service Kiosks

Staffed or automated customer service kiosks, located at the curbside or within the baggage claim areas, can be used to provide passengers with information such as schedules, fares, and destinations served by transit services and scheduled buses. Staff at these kiosks can also provide other information about the airport and its airlines, provide directions, and respond to questions from passengers and visitors. Indeed, most of the questions that the staff at ground transportation desks receive do not pertain to the airport's ground transportation services.

Accuracy of Information

It is important to maintain up-to-date information on transit and scheduled bus schedules and fares, particularly at airports with significant seasonal traffic and where transit schedules and services vary accordingly. This applies to both static and dynamic information sources, but dynamic displays are more easily updated. Airport staff should regularly contact transit authorities, scheduled bus operators, and other providers to ensure that all information is current. Information should be automated where possible to remove the opportunity for human error.

Summary—Informing Waiting Passengers of Bus and Shuttle Arrival Times

Intended Goals

- Improve customer experience while waiting

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Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  – Depends on the strategies being implemented. It is assumed that staffed customer service positions will have the highest cost due to employee salaries and benefits. Communication methods can have a significant impact on capital and recurring costs.
  – Depends on the number of stops, infrastructure needed (power/communications), and the functionality of the GPS.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X				

  – Operations and maintenance costs will vary based on the technologies used.

Implementation Considerations

- **Implementation schedule:** Depends on amenities to be provided. Estimated less than 1 year.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.4.1. Providing convenient, enclosed, passenger waiting areas
- 7.4.3. Presenting transit schedules, fares, and route information to passengers
- 7.4.4. Providing passenger service associates at curbsides
- 7.4.5. Enhancing safety and convenience for pedestrians crossing roadways and waiting for transportation

Examples

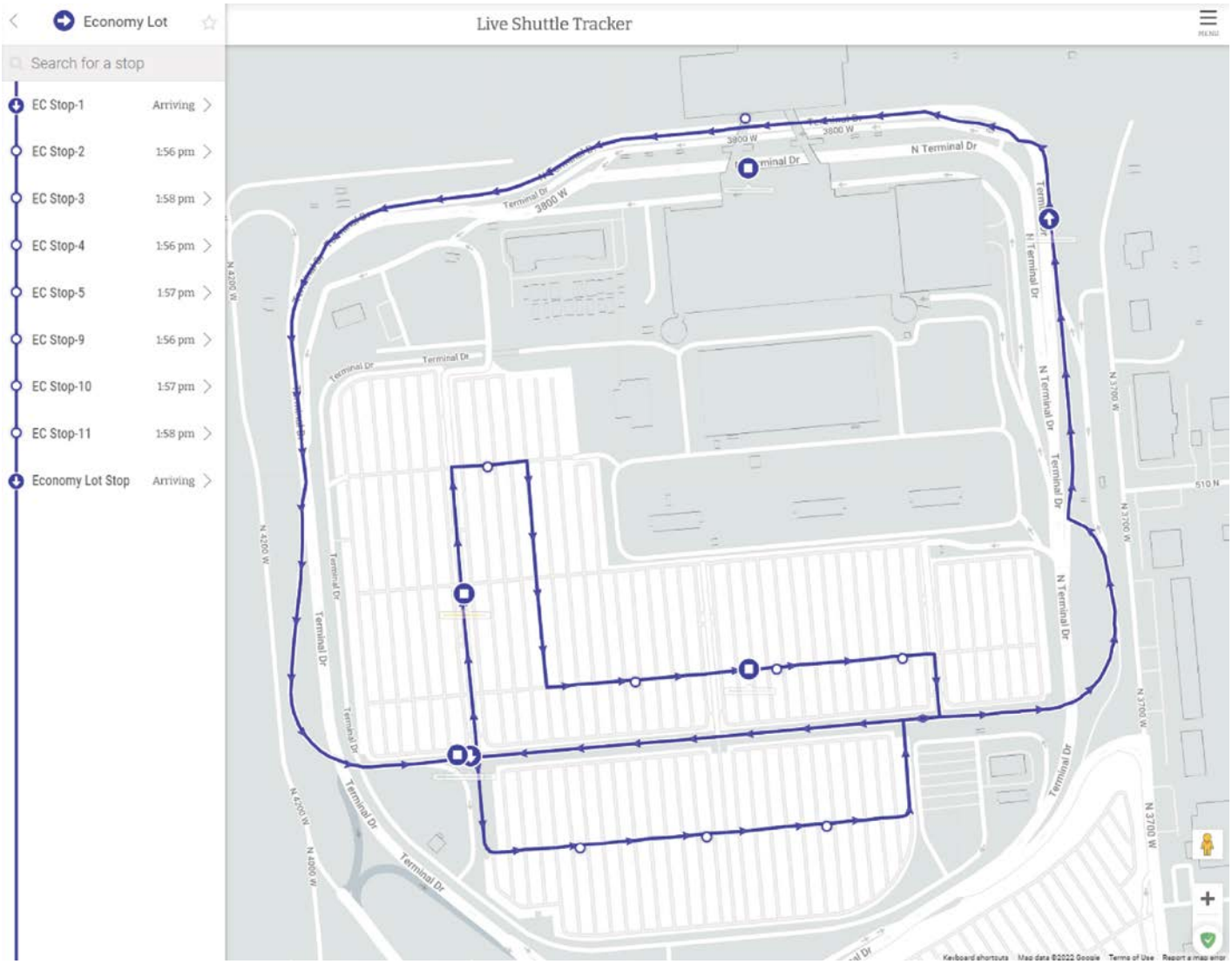
- St. Louis Lambert International Airport provides a website with real-time information on transit schedules and fares.
- Los Angeles International Airport provides ground transportation updates on its website, with additional static signage on the departures/arrivals curbs, as well as at the LAX-it lot, which consolidates all TNC and taxi traffic in one location.

7.4.3 Presenting Transit Schedules, Fares, and Route Information to Passengers

It is helpful to provide arriving passengers with information describing the departure times/schedules, fares, routes, and destinations served by public transit and scheduled buses at the airport.

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Source: slcairport.ridesystems.net, screen capture: May 16, 2022.

Figure 7-23. Salt Lake City International Airport live shuttle tracker website.

This information is particularly useful to non-resident passengers not familiar with the community. Other helpful information may include instructions on how to purchase a ticket/fare card.

Key considerations related to informing arriving passengers include:

- Signage
- Customer service kiosks
- Accuracy of information

Signage

At key locations on the curb, airports should provide displays containing fare and schedule information for scheduled routes. These can be standard signs or dynamic signs, showing the next arrival time and route. Some airports may find it beneficial to develop and provide a web application that includes detailed information on their ground access system, including information on schedules and fares for regular ground access services such as scheduled buses and rail services and regional shuttles.

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Customer Service Kiosks

Staffed or automated customer service kiosks placed in the baggage claim, curbside, or other areas can assist passengers by providing a variety of information, including regularly scheduled service information. Staffed or automated kiosks have higher costs than static displays, but they are more flexible and can provide a greater amount of information.

Current Information

Efforts should be made to maintain current information on schedules, fares, and routes/destinations, particularly at airports with significant seasonal traffic. Airports should regularly engage with operators of privately operated, scheduled bus service and public transit agencies to ensure that all information is current. Information displays should be automated where possible to remove the opportunity for human error.

Summary—Presenting Transit Schedules, Fares, and Route Information to Passengers

Intended Goals

- Improve arriving passenger access to ground transportation information

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X					

  - Varies depending on the number of stops, the infrastructure needed (power/communications), the type (static vs. variable message) and amount of signage, and/or the applications developed.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X					

  - Costs for static and dynamic signs and web applications include the salary and benefit costs of the responsible staff.
  - Costs for kiosks include the salaries and benefits of the staff working in the kiosks.

Implementation Considerations

- **Implementation schedule:** Depends on amenities to be provided. Estimated to be less than 1 year.
- **Ability to implement in phases:** Yes.

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Complementary Strategies and Supporting Technologies

- 7.4.1. Providing convenient, enclosed, passenger waiting areas
- 7.4.2. Informing waiting passengers of bus and shuttle arrival times
- 7.4.4. Providing passenger service associates at curbsides
- 7.4.5. Enhancing safety and convenience for pedestrians crossing roadways and waiting for transportation

Examples

Airports that have implemented these strategies include:

- St. Louis Lambert International Airport, which has a web application presenting transit schedules and fares as well as parking occupancies and costs.
- The Minneapolis-St. Paul International Airport Ground Transportation Center, which has an enclosed waiting area providing seating, schedule and fare information, and ticket counters for van and limousine services.
- Fort Lauderdale-Hollywood International Airport, which has a staffed ground transportation information desk.

7.4.4 Providing Passenger Service Associates at Curbsides

Prioritizing customer service by assigning airport representatives, such as guest service ambassadors, to the curbsides can increase the level of customer satisfaction. Staff may be airport staff, volunteers, or contracted. These representatives can (1) assist passengers with loading and unloading at the curb to gain efficiencies, (2) provide directions to support nearby signage or direct passengers to information displays, and (3) assist passengers with special needs or mobility challenges. There are a number of considerations that airports need to address when establishing guest services or ambassadors:

- Tasks for airport representatives
- Training
- Liability
- Cost

Tasks for Airport Representatives

Airport representatives may perform many customer service tasks, including

- Providing directions, locations of boarding areas for specific transportation services, and airline and transit schedule information.
- Monitoring areas for safety and security concerns (i.e., sharing concerns with enforcement officers).
- Assisting passengers with special needs or mobility challenges including helping with boarding/alighting, handling baggage and personal items, and sharing directions.
- Assisting with crowd control.
- Assisting in emergencies.

Training

As with all customer service positions, particular care should be taken to ensure airport representatives have sufficient and recurrent training. Training should include ADA assistance, security

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and safety standards, impacts of construction, seasonal changes impacting travelers, and participation in emergency drills (performing crowd control and working with enforcement officers and emergency services). Standard operating procedures should be documented by the airport and should be shared with tenants such as airlines and rental car agencies to ensure consistency and the highest level of customer service.

Liability

Consideration should be given to the liabilities associated with the tasks assigned to the airport representatives. Careful consideration should be given to the level of assistance being provided to disabled patrons and whether employees of the airport can physically touch a passenger's belongings.

Cost

The costs of providing guest services/airport representatives vary based on the number of employees needed. Some airports have outsourced this service by retaining a third-party contractor to provide and manage their curbside customer service representatives.

Summary—Providing Passenger Service Associates at Curbsides

Intended Goals

- Improve customer experience while waiting

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X					

  – Costs for communication infrastructure, if required.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
			X	X	X	X

  – Cost of labor, primarily salary and benefits, or cost of a third-party contractor via a time/materials agreement.

Implementation Considerations

- **Implementation schedule:** Depends on procurement. Estimated to be less than 1 year.
- **Ability to implement in phases:** No.

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Complementary Strategies and Supporting Technologies

- 7.4.1. Providing convenient, enclosed, passenger waiting areas
- 7.4.2. Informing waiting passengers of bus and shuttle arrival times
- 7.4.3. Presenting transit schedules, fares, and route information to passengers
- 7.4.5. Enhancing safety and convenience for pedestrians crossing roadways and waiting for transportation

Examples

- Fort Lauderdale-Hollywood International Airport uses a third-party vendor to provide curbside customer service agents. They are overseen by airport staff to ensure satisfactory levels of service are provided.
- Portland International Airport provides passenger service agents at the deplaning curbside to assist motorists and passengers.
- Several European airports have call boxes or buttons at the drop-off curbside allowing disabled passengers and others requiring assistance to summon a passenger service agent.

7.4.5 Enhancing Safety and Convenience for Pedestrians Crossing Roadways and Waiting for Transportation

Airports with multiple curbs or with parking facilities within walking distance may have pedestrian crossings on roadways. Enhancements start with effective wayfinding inside the airport and on the curbs. Proper crossings lead to more effective curbside roadways and increased passenger satisfaction, convenience, and safety. Some areas to consider are the following:

- Lighting
- Traffic controls
- Walkway widths
- Level changes and slopes

Lighting

As a general rule, pedestrian crossings and the areas leading up to them should be well lit and provide motorists with clear views of pedestrians and pedestrians with clear views of approaching vehicles. Roadways being crossed should also be well lit, allowing pedestrians ample time to assess traffic situations.

Traffic Controls

Various types of traffic controls are available to assist in pedestrian crossings. These include signage, signaling, and staffed crossings. More information on each of these types of controls and their effect on congestion are available in Section 7.3.2. Identifying the appropriate type of control is essential to balancing pedestrian and curb roadway levels of service, ensuring safe and efficient use.

Walkway Widths

Pedestrian crossings should be appropriately sized, generally in the range of 8 to 12 feet wide, to ensure crossings are safe and predictable. Crossings should be wide enough to accommodate

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peak pedestrian traffic levels without unreasonable vehicle stoppage time, but not so wide that drivers have difficulty identifying pedestrians entering the walkway.

Level Changes and Slopes

Grade-separated pedestrian crossings can improve customer experience and safety by separating pedestrians from the roadway. While grade-separated crossings allow for continuous flows of traffic and pedestrians without the risk of interaction, they are costly to construct and may not be appropriate for curbs of all passenger activity levels or terminal building configurations. Raised crosswalks/speed platforms are described in Section 7.3.3.

Summary—Enhancing Safety and Convenience for Pedestrians Crossing Roadways and Waiting for Transportation

Intended Goals

- Provide safe and efficient pedestrian walkways

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X	X			

  - Varies depending on the strategy deployed, use of elevated or below-grade pedestrian crossing, and terminal building configuration.
  - Dependent on the method used to enhance safety and the number of crossings affected.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X					

  - Varies depending on the strategy deployed.

Implementation Considerations

- **Implementation schedule:** Varies depending on the strategy deployed.
- **Ability to implement in phases:** Yes.

Complementary Strategies and Supporting Technologies

- 7.3.2. Changing pedestrian crosswalk controls
- 7.3.3. Raised crosswalk/speed platform

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Examples

Many airports throughout the United States have different strategies:

- Detroit Metropolitan Airport utilizes connecting bridges over the roadway linking the ground transportation center with the terminals.
- Many airports have elevated pedestrian bridges (e.g., those serving Boston, Fort Lauderdale, Los Angeles, San Jose, and Seattle), and others have pedestrian tunnels linking the terminals and adjacent parking structures (e.g., those serving Baltimore, Raleigh, and San Francisco).

7.5 Sustainability Strategies

Sustainability strategies strive to reduce or eliminate carbon emissions generated by landside traffic. These strategies include

- 7.5.1. Encouraging the use of transit or high-occupancy travel modes
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes
- 7.5.3. Promoting use of electric vehicles by private motorists and others

7.5.1 Encouraging the Use of Transit or High-Occupancy Travel Modes

ACRP Report 4: Ground Access to Major Airports by Public Transportation and *ACRP Report 35: Planning for Offsite Airport Terminals* provide further information on strategies to improve use of public transit and offsite passenger terminals by airport passengers and employees.

Curbside traffic congestion can be mitigated (1) by increasing the number of passengers using high-occupancy modes, which reduces the number of vehicles using curbside facilities, or (2) by moving traffic to drop-off and pickup sites located outside of congested areas, such as a remote lot or drop-off point. Policies to encourage passenger use of high-occupancy modes typically focus on making those services more attractive to passengers and/or making lower-occupancy modes (i.e., services carrying a single passenger or travel party) less attractive. For this discussion, high-occupancy modes are those that (1) carry passengers who otherwise would not be traveling together in a single vehicle and (2) provide transportation for passengers who typically do not consider alternative modes, such as charter buses or courtesy vehicles operated by off-airport parking facilities, hotels/motels, and rental car operators. Therefore, services addressed in this section include

- Shared TNCs (such as UberX Share and Lyft Line)
- Shared-ride vans
- Scheduled vans and buses that are privately operated
- Scheduled vans and buses that are publicly operated, including those operated by public transit agencies or airport operators
- Rail or other fixed guideway services

Increasing the Attractiveness of High-Occupancy Modes

The following strategies can increase the attractiveness of high-occupancy modes for airline passengers:

- **Convenient and attractive pickup/drop-off locations.** Locate passenger pickup and/or drop-off at locations that minimize unassisted walking distances and level changes between the vehicle's stop and airline ticket counters, security checkpoints, and baggage claim. Consider providing baggage drop-off or supplemental baggage drop-off services for passengers nearby.

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- **Locate fixed-rail station for easy access to the airport.** Minimize unassisted walking distances, level changes, and mode changes between the vehicle's stop and airline ticket counters, security checkpoints, and baggage claim areas. If track geometry and other considerations limit a close-in station, provide amenities at the rail station, such as baggage drop and walking assistance (e.g., moving walkways or APMs and motorized carts for disabled passengers).
- **Express services.** Provide commercial vehicle and rail routes that offer frequent, non-stop service between the airport and key regional destinations. The availability of economical long-term parking at those destinations further increases the attractiveness of such services.
- **Reduced trip fees for providers.** Reduce ground access fees (e.g., per-trip fees) for the providers of high-occupancy modes with the intent that the operators could, in turn, provide lower fares to passengers and/or increase the number of routes and/or frequencies of existing services.
- **Fare or operating cost subsidies.** Provide subsidies to reduce fares for passengers and employees (e.g., monthly passes) and/or increase the number of routes and/or frequencies of existing services.
- **Marketing.** Actively and visibly advertise and promote the availability of high-occupancy modes to serve airline passengers. Locations for such marketing include the airport's webpage, context-based ads on other webpages (e.g., results for a web search for "TNC fares to XYZ airport" would also display ads for high-occupancy modes serving XYZ airport), and signage in high-visibility areas (e.g., baggage claim and billboards along major access roadways).

Decreasing the Attractiveness of Lower-Occupancy Modes

Airline passenger use of high-occupancy access modes could be increased by decreasing the relative attractiveness of lower-occupancy modes that use the curbsides. Some strategies are

- **Less convenient pickup/drop-off locations.** As a corollary to designating convenient pickup/drop-off areas for high-occupancy modes, direct lower-occupancy modes to less convenient locations, such as to outer island curbsides or parking facilities (i.e., provide no terminal curbside for lower-occupancy services as is done at some European airports, such as London Heathrow's Terminal 5).
- **Increased trip fees.** Increase ground access fees for lower-occupancy commercial vehicles with the intent that the operators would, in turn, increase fares for their service.
- **Curbside access fees.** Implement curbside access fees for lower-occupancy services, including private vehicles, as is done in many airports in the United Kingdom.

Section 7.2.6. provides additional information regarding measures to encourage the use of transit or high-occupancy travel modes.

Summary—Encouraging the Use of Transit or High-Occupancy Travel Modes

Intended Goals

- Reduce curbside congestion by reducing the number of vehicle trips per airline passenger
- Reduce vehicle emissions by reducing the number of vehicle trips per airline passenger

Applicability

- Applicable to all airports

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Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X					

  – Varies widely depending on the specific strategy employed.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Operations and maintenance costs are mostly communication costs.

Implementation Considerations

- **Implementation schedule:** Varies depending on the strategy employed and the extent to which implementation depends on other parties.
- **Ability to implement in phases:** Yes.
- **Ease of enforcement:** Enforcement will typically be limited to a strategy that reassigns curb space to favor high-occupancy travel modes.
- **Likely political influence:** Varies depending on the strategy employed and support of the local community. Strategies that increase service levels and lower fares for high-occupancy modes may generate significant support whereas strategies that impose curbside access fees for private vehicles or increase per-trip fees for lower-occupancy commercial vehicle modes may generate significant opposition.
- **Potential complications:** For services such as TNCs, which can provide both single-party and shared-ride services in the same vehicle, it is necessary to obtain the cooperation of the providers to determine which service the vehicle is providing on a particular trip.

Complementary Strategies and Supporting Technologies

- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.2.8. Charging private motorists access fees
- 7.4.1. Providing convenient, enclosed, passenger waiting areas
- 7.4.2. Informing waiting passengers of bus and shuttle arrival times
- 7.4.3. Presenting transit schedules, fares, and route information to passengers
- 7.4.4. Providing passenger service associates at curbsides
- 7.4.5. Enhancing safety and convenience for pedestrians crossing roadways and waiting for transportation
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes
- Systems are being developed to support the ability of airport operators to charge motorists for terminal access (i.e., airport access fees). These systems use cameras and analytical software to identify the classification of each vehicle and the vehicle's activity on the curbside. When implemented, these systems will allow airport operators to allocate

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curb space and terminal roadways by use, automatically monitor their use, and, if desired, introduce variable pricing to incentivize the desired user behavior. It is anticipated that such systems will be available for use at U.S. airports by 2024 if not earlier.

Examples

Many airports promote use of high-occupancy modes by airline passengers. Some examples follow:

- **Pickup/drop-off proximity.** Airports that provide pickup/drop-off locations for high-occupancy services that are more convenient than for lower-occupancy services include
  - Washington Reagan National Airport, which assigns the closest pickup curb to commercial vehicles, including high-occupancy services.
  - Denver International Airport, which assigns the curbside at the baggage claim level to commercial vehicles, including high-occupancy modes (passengers getting picked up by private vehicles must change one level to reach their loading area).
  - Los Angeles International Airport, which has separated the transportation options between upper roadway, lower roadway (inside), and lower roadway (outside) to maximize flowthrough for each roadway and incentivize high-occupancy options with better service, as well as relocated TNC and taxi service areas (both pickup and drop-off) to a lower-density traffic area, reducing congestion on the terminal roadways by as much as 30%.
  - Numerous airports integrate rail stations into each terminal building or provide stations immediately adjacent to each terminal building, including London Heathrow, London Gatwick, Hartsfield-Atlanta, and Portland (Oregon).
- **Express services—airport-operated.** Boston Logan International Airport (Logan Express) and Los Angeles International Airport (LAX FlyAway) sponsor dedicated express bus service between a limited number of regional destinations and the airport, with no interim stops.
- **Express services—privately operated.** Many airports are served by private bus or van lines providing scheduled service, on fixed routes, between the airport and regional destinations. Examples of airports with such services include Chicago O'Hare International Airport, Newark Liberty International Airport, San Francisco International Airport, and Seattle-Tacoma International Airport.
- **Public transit express bus.** While many U.S. airports are served by public transit bus routes, very few have express routes dedicated to the airport. One example is Sacramento Regional Transit's Route 142, which connects a single downtown stop with the Sacramento International Airport terminals and has no other stops.
- **Public transit express rail.** While many airports are served by public transit rail routes, no U.S. airports have express routes dedicated to the airport. Overseas examples include the Heathrow Express (connecting London Heathrow International Airport to Paddington Station with no interim stops), Gatwick Express (connecting London Gatwick with Victoria Station with no interim stops), and the Union Pearson Express (connecting Toronto Pearson International Airport with Union Station with two interim stops).
- **Trip fees.** U.S. airports typically do not charge per-trip fees for vehicles operated by public transit agencies or by airports (such as the airport-operated express services previously described).

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- **Fare or operating cost subsidies.** San Jose International Airport subsidizes the operating costs of the Valley Transit Authority's Route 60 (the "Airport Flyer"), which connects three nearby transit stations with the airport.
- **Marketing.** Many airports promote use of high-occupancy modes, including
  - Oakland International Airport (radio ads promoting use of regional rail system for airport access).
  - San Francisco International Airport (website homepage Ground Transport menu lists "Public Transit," "Shuttles," and other commercial services, but does not list "parking" and provides no direct link to driving directions).
  - Seattle-Tacoma International Airport (website Ground Transportation page lists "Alternatives to Driving" before listing other options).

7.5.2 Encouraging Use of Long-Duration Parking Rather Than Drop-Off or Pickup Modes

ACRP Report 24: Guidebook for Evaluating Airport Parking Strategies and Supporting Technologies provides further information on products and strategies to improve the attractiveness of on-airport, long-duration parking facilities.

Curbside traffic congestion can be mitigated by increasing the number of passengers using long-duration parking facilities, which reduces the number of vehicles using curbside facilities. Furthermore, if passengers choose to use long-duration parking facilities instead of using curbside pickup and drop-off modes, the number of vehicle trips and vehicle miles traveled associated with their air trip can be cut in half. Long-duration parking also typically can generate more net revenue per passenger than pickup and drop-off modes.

Policies to encourage passenger use of long-duration parking focus on either making long-duration parking more attractive to passengers and/or making curbside pickup and drop-off modes less attractive.

Increasing the Attractiveness of Long-Duration Parking or Remote Options

The following strategies can increase the attractiveness of on-airport, long-duration parking options for airline passengers:

- **Price.** Price long-term parking below the estimated cost of travelers using low-occupancy options. While many airports offer a fixed daily price, regardless of parking duration, some airports offer discounts for stays of 1 day or longer to compete with round-trip fares offered by taxis, TNCs, and other commercial ground transportation services.
- **Availability.** Ensure customers can find a space in their desired facility upon their arrival at the airport. Methods include offering parking reservations for a specific space, facility, or zone within a facility, or providing space availability information on the airport's website.
- **Online booking systems.** Offer customers the ability to reserve and pay for a long-duration parking space in advance, which guarantees that their desired space will be available at a price they are willing to pay. Online booking systems can also be used to offer reduced prices.
- **Attractive products.** Offer specific parking products that use one or more of the strategies described in this section to target specific customer types (e.g., "Corporate Parking," which is a short shuttle ride to the terminal, guarantees space availability, and offers newspapers and bottled water on the shuttle bus).
- **Amenities.** Offer amenities, such as car washing, vehicle servicing, and concierge programs.
- **Shuttle service.** For remote parking facilities, offer high-quality shuttle service between the parking facility and the terminal. Such services can include high-frequency service, pickup

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and drop-off at the customer's vehicle, weather-protected shelters within the parking facility and at the terminal building, and drop-off and pickup points at curbside areas located close to terminal doorways (as opposed to on island curbsides).

- **Loyalty programs.** Offer customers a program that provides rewards (such as free parking days) based on their parking activity.
- **Marketing.** Actively promote the features and benefits of the on-airport, long-duration parking facilities. Such efforts include outreach to corporations to offer discounts or other benefits and use of search engine optimization to ensure web searches for parking at a given airport list the airport's parking webpage at the top.
- **Pickup/drop-off locations.** Designate pickup and drop-off locations for on-airport parking courtesy shuttles in areas that minimize unassisted walking distances and level changes between the vehicle's stop and airline ticket counters, security checkpoints, and baggage claim.
- **Reduced fees.** Reduce ground access and/or privilege fees for courtesy vehicles operated by off-airport parking operators with the intent that the operators could, in turn, provide lower prices to passengers.

Decreasing the Attractiveness of Curbside Pickup/Drop-off Modes

Airline passenger use of long-duration parking could be increased by strategies to decrease the relative attractiveness of curbside pickup/drop-off options:

- **Less convenient pickup/drop-off locations.** Direct pickup and drop-off services to less convenient locations such as outer island curbsides or parking facilities (i.e., provide no terminal curbsides).
- **Increased trip fees.** Increase ground access fees for vehicles using the terminal curbsides with the intent that the operators would, in turn, increase fares for their service.
- **Curbside access fees.** Implement curbside access fees for all modes, including private vehicles.
- **Introduce airport transportation options at the time an airline ticket is booked.** Educate the passenger about options for transportation to and from the airport earlier in the trip-planning process.

Section 7.2.5. provides additional information regarding measures to encourage the use of long-duration parking areas rather than drop-off or pickup at the terminal curbs.

Summary—Encouraging Use of Long-Duration Parking Rather Than Drop-Off or Pickup Modes

Intended Goals

- Reduce curbside congestion by reducing the number of curbside vehicle trips per airline passenger
- Reduce vehicle emissions by reducing the number of vehicle trips and vehicle miles traveled per airline passenger
- Improve the airline passenger's guest experience
- Increase net revenues per airline passenger

Applicability

- Applicable to all airports

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Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Costs of changing the Parking Access and Revenue Control System (PARCS) and communication costs.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  – Costs of changing the PARCS and communication costs.

Implementation Considerations

- **Implementation schedule:** Varies, depending on the strategy employed and the extent to which implementation depends on other parties. Strategies related to pricing can be implemented relatively quickly while those related to amenities can take longer.
- **Ability to implement in phases:** Yes.
- **Ease of enforcement:** Enforcement will typically be limited to strategies that reassign curb space to discourage pickup and drop-off services.
- **Likely political influence:** Varies, depending on the strategy employed. Strategies that increase service levels and lower long-duration parking prices may generate significant support whereas strategies that impose curbside access fees may generate significant opposition. Off-airport parking operators (who may have political influence as local businesses) may oppose reduced on-airport, long-duration parking fees because they would require off-airport facilities to reduce their prices (and profitability) to remain competitive.

Complementary Strategies and Supporting Technologies

- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.5.1. Encouraging the use of transit or high-occupancy travel modes
- Comprehensive commercial revenue management programs, which provide integration between pricing efforts of airport assets, flexibility to move vehicles between options quickly and efficiently, and required coordination between operations and marketing efforts

Examples

- **Price.** Many airports offer discounts for longer parking durations. As of January 2022, Seattle-Tacoma International Airport offered a weekly rate for garage parking that was five times less than the daily price. Airports with online booking systems (see the following) frequently offer discounts that may increase as parking durations increase.
- **Availability.** Airports offering guaranteed spaces through reservations or membership in a parking program include Minneapolis-St. Paul International Airport, Boston Logan

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International Airport, and San Francisco International Airport, among many others. Numerous airports provide space availability information on their websites, either as "Available/Unavailable," percent of spaces available, or a specific number of spaces available. An example is shown in Figure 7-24.

- **Online booking systems.** Numerous airports offer online booking systems that customers can use to reserve and pay for parking in advance of their trip. In the United States, such a system is offered at Dallas/Ft. Worth International Airport, Newark Liberty International Airport, Raleigh-Durham International Airport, San Francisco International Airport, and Seattle-Tacoma International Airport, among others.
- **Attractive products.** Many airports offer attractive products targeting specific customer segments. Selected examples include
  - Charlotte International Airport, which offers "Express Deck," a valet parking product provided at a remote parking garage.
  - Dallas/Fort Worth International Airport, which offers "Express," a remote parking product offering covered and uncovered spaces where the shuttle picks up and drops off customers at their vehicles.
  - Minneapolis-St. Paul International Airport, which offers "ePark Elite," a membership-based product guaranteeing an available space in a parking garage adjacent to the terminal building.
  - Los Angeles International Airport, which has integrated parking, commercial revenue management/pricing, and marketing operations at a level that allows weekly adjustments to pricing and occupancy goals.
- **Amenities.** Many airports offer amenities to patrons parking for long durations. Selected examples include
  - San Francisco International Airport, which offers in-space car washing (for an additional charge).
  - Spokane International Airport, which offers a free drive-through car wash for all parking customers.
  - Los Angeles International Airport, which offers more than 1,200 electric vehicle chargers for use across all of the parking facilities/parking products.
- **Shuttle service.** Tampa International Airport and Baltimore/Washington International Thurgood Marshall Airport offer shuttle services between remote parking structures and the terminals. Several airports offer "trunk-to-trunk" shuttle services that pick up and drop off customers at their individual parking spaces, allowing them to avoid walking to a shuttle bus stop.
- **Loyalty programs.** Many airports offer parking loyalty programs to encourage use of the airport's long-duration parking facilities, including (but not limited to) Dayton International Airport (DAYperks), Memphis International Airport (MEMPerks), Kansas City International Airport (KCI Rewards), and Savannah Hilton Head International Airport (SAV PARKING REWARDS).
- **Marketing.** Most U.S. airports actively promote and market their long-duration parking products. The vast majority of these efforts use online methods such as web-sites and email.

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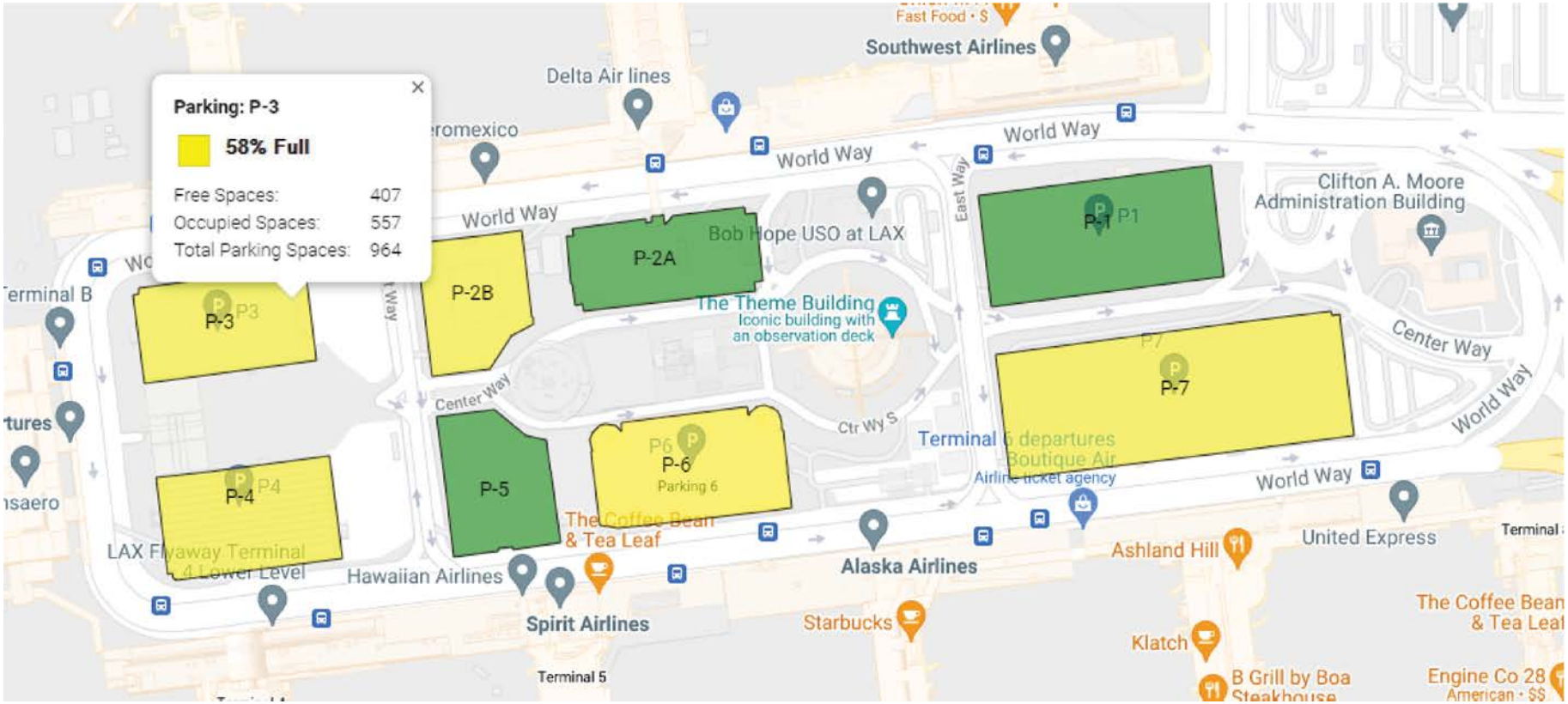


Figure 7-24. Parking availability information provided on Los Angeles International Airport's website (retrieved January 5, 2022).

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7.5.3 Promoting Use of Electric Vehicles by Private Motorists and Others

To reduce emissions of vehicles traveling to and from the airport, an airport operator can take action to encourage airport passengers and commercial vehicle operators to purchase electric vehicles (EVs) and use them for airport access. EV charging infrastructure is lagging far behind demand. Providing EV charging services impacts sustainability goals, revenue generation, and the traveler experience. Such actions include

- Providing EV charging stations in airport parking facilities, commercial vehicle staging lots, and elsewhere.
- Providing EV chargers in attractive parking spaces (such as those located closest to the terminal building) and in commercial vehicle staging lots as a revenue/guest experience enhancement.
- Offering free or discounted power through the EV chargers as an incentive to increase utilization in certain areas.

ACRP Synthesis 54: Electric Vehicle Charging Stations at Airport Parking Facilities provides further information on the technologies, business and policy issues, and planning considerations for airports considering EV chargers.

Summary—Promoting Use of Electric Vehicles by Private Motorists and Others

Intended Goals

- Reduce vehicle emissions by rewarding airline passengers and commercial vehicle operators who purchase and operate EVs
- Provide much-needed EV infrastructure for both short-term and long-term travelers and ground transportation operators
- Provide airports with a new, rapidly growing revenue stream

Applicability

- Applicable to all airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X	X	X	X		

  - Costs are impacted by demand for service and existing infrastructure.
  - The cost of chargers averages around $2K per unit. The cost of smaller installations can be low if existing power is available and chargers can be installed near power sources to limit conduit and trenching. The costs of larger installations may include increasing power to selected facilities and significant conduit and trenching. Improvements to power distribution may be mitigated with EV charger power management systems.
  - Regardless of the scope of implementation, marketing and signage must be modified to promote new or enhanced services.
  - Integration of the EV chargers into the PARCS can reduce operating costs and improve operational efficiencies.

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- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X	X					

  - Limited to ongoing maintenance of EV chargers and the cost of purchasing power (which could be passed through to customers using the EV chargers).
  - Transition of parking spaces from general/premium spaces to EV spaces may result in a decline in space occupancy until demand and awareness increase.

Implementation Considerations

- **Implementation schedule:** Assuming power and data are available (or close to) EV charger locations, installation can require between 5 and 60+ days, depending on the required infrastructure and the need for permits in larger airports.
- **Ability to implement in phases:** Yes; EV chargers can be installed one at a time.
- **Ease of enforcement:** If enforcement in public parking is performed manually, visual inspection is required to ensure non-EVs do not use the spaces. Because travelers using long-duration parking may be parked significantly longer than the time required to fully charge their EV, an airport may wish to deploy methods to move a charger once an EV is fully charged. Enforcement can be integrated with more advanced Parking Guidance Systems (PGS), which can help to detect when a space is not occupied by a qualified vehicle and can ensure that any administrative penalties are captured within the PARCS. As commercial vehicle operators typically remain in staging areas for relatively brief periods, Level 2 or Level 3 EV chargers may be warranted.
- **Likely political influence:** The airport's and the local community's commitment to sustainability will determine whether the system should be provided at no cost to users, whether it should be operated to recover costs, or whether it is expected to provide a revenue stream.

Complementary Strategies and Supporting Technologies

- 7.2.7. Locating employee parking outside of the terminal area
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes

Examples

- Many airports offer EV chargers in their public parking facilities. Some, such as San Francisco International Airport, locate the chargers on the ground floor of the parking garage, immediately adjacent to the pedestrian corridors connecting to the terminal buildings. Such spaces are designated as "Green Vehicle Parking."
- Most U.S. airports offering EV chargers offer the power for free, but new technology to integrate the systems with the PARCS for fee collection and the PGS to automate enforcement is influencing pricing strategies. Airports currently charging for use of EV chargers include Daniel K. Inouye (Honolulu) International Airport ($7.00 per use of a charging station for 24 hours), Los Angeles International Airport ($2.00 plus $0.45 per kilowatt-hour), and Toronto Pearson International Airport (CA$3.85 per 30 minutes use of a fast-charging station at the cell phone waiting lot and Greater Toronto Airports Authority Administration Building; free in other public parking facilities).

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7.6 Accommodating Transportation Network Companies

To accommodate the growing number of passengers using TNCs, airports have implemented strategies to facilitate TNC operations while minimizing the adverse impacts TNCs may have on airport roadway and curbside operations. Such strategies need to accommodate the changing needs of passengers, TNCs, and other ground transportation providers, as well as airport management. Solutions to mitigate TNC contributions to roadway operations and congestion require balancing the impacts on airport operations and traveler experiences and include

- 7.6.1. Establishing a geofence
- 7.6.2. Providing hold lots
- 7.6.3. Requiring use of remote boarding areas
- 7.6.4. Pooling and dispatching
- 7.6.5. Establishing differential prices for transportation network company passenger boarding areas
- 7.6.6. Improving the customer's ability to identify and locate a waiting vehicle
- 7.6.7. Dispatching of vehicles by transportation network companies

7.6.1 Establishing a Geofence

Geofencing is a technology that creates a virtual boundary around a defined geographic area allowing the detection of vehicles entering and exiting the geofenced area via GPS signals from a mobile phone or other device. The signals can be used to monitor and record the time, location, and travel direction of individual vehicle trips. Using these data, it is possible to record the volume of vehicle trips (to support charging per-trip fees, facility planning, or other purposes), ensure that only drivers waiting in hold sites receive dispatches, enforce and audit operations, and perform other tasks.

Airport geofences rely on signals from drivers' telephones. TNC drivers are incentivized to keep their phones on in order to be dispatched to customers. Geofences are less practical for managing courtesy vehicles, shuttles, taxicabs, and other vehicles as the drivers of these vehicles need not keep their telephone turned on to pick up customers or do business.

It is recommended that as part of their permitting process airports require TNCs to define geofences around the entire airport, designated hold lots, and assigned TNC customer drop-off and pickup areas. Use of these geofences:

- Ensures that only drivers waiting in a designated hold lot are dispatched (by the TNC) to waiting customers and ensures that drivers do not attempt to "cruise" along the curbsides seeking a customer. Drivers may not need to wait in a hold lot at those airports that authorize TNCs to (1) dispatch drivers in anticipation of customer requests or (2) dispatch drivers who have dropped off a customer at a nearby location without requiring the driver to enter the hold lot.
- Provides an accurate report of each vehicle trip including the time and location of the trip, the vehicle identification number, and whether the vehicle was entering or exiting the airport with a customer.
- Monitors the length of time a vehicle remains in specific areas. At airports that limit the length of time a TNC vehicle is allowed to wait in a specific location or passenger boarding area, this information can be used to mitigate congestion or curbside occupancies.

7.6.2 Providing Hold Lots

TNC drivers waiting for customers should be required to remain in hold lots that are located near designated passenger boarding areas so that drivers can respond promptly to customer

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requests, but the hold lots should be located away from the terminal curb and congested roadway areas. At most airports, TNCs are provided hold areas that are separated from those provided to waiting taxicabs and other providers of commercial ground transportation.

At many airports, TNC drivers remain in the hold lot for less than 30 minutes. At airports where driver wait times frequently exceed 60 minutes, it is recommended that airport staff work cooperatively with the TNCs to discourage long wait times. This is because long waits frequently result in driver requests for restrooms, Wi-Fi, vehicle charging, vending machines, and other amenities.

7.6.3 Requiring Use of Remote Boarding Areas

While curbside drop-off and pickup may provide the best customer experience, at some airports TNC boarding occurs at remote sites (i.e., not immediately adjacent to the terminal building). Airport staff require remote boarding of TNCs (and other vehicles) to mitigate roadway and curbside congestion. Typically, these prohibitions do not apply to TNC-affiliated vehicles operating as licensed limousines (e.g., Uber Black). Examples of airports where TNCs are precluded from picking up customers at the terminal curbside include

- Terminal B at Boston Logan International Airport, where pickup occurs in the adjacent central parking structure. A rendering of this is shown in Figure 7-25.
- Los Angeles International Airport, where pickup occurs in a remote parking lot (LAX-it) requiring customer use of a shuttle bus.
- Domestic terminals at San Francisco International Airport, where pickup occurs in the Central Garage.
- Seattle-Tacoma International Airport, where boarding occurs in a parking structure.
- Las Vegas McCarran International Airport, where boarding occurs in a parking structure.
- Portland International Airport, where boarding is located on Level 1 of the P1 Short-Term Parking garage. A schematic is shown in Figure 7-26.

7.6.4 Pooling and Dispatching

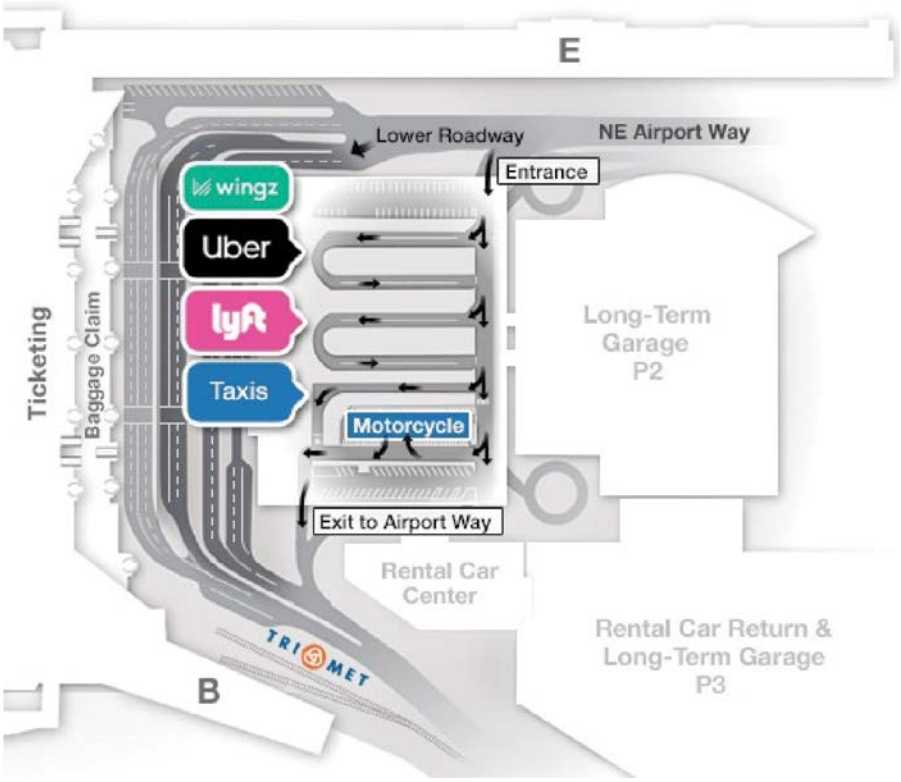
TNC services to airports include individual; small party; and shared ride, matched door-to-door transportation (similar to a carpool). In markets with high TNC adoption rates, TNCs appear to be mostly used by single-party or individual users. Many of these users appear to have



Source: Massport

Figure 7-25. Rendering of new pickup and drop-off areas at Boston Logan International Airport.

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Source: Portland International Airport.

Figure 7-26. Boarding area schematic at Portland International Airport.

come from shared-ride modes, which has resulted in increases in curbside congestion. Encouraging passenger use of pooling by TNC customers or matched door-to-door transport (e.g., UberX Share and Lyft Line) can result in an 11% reduction in the number of trips to an airport, thereby reducing curb congestion (K. Hermawan, "Transportation Network Companies' (TNC) Impacts and Potential on Airport Access," 2018. https://escholarship.org/uc/item/01m726rr). Achieving success with these programs would require coordination with TNCs and perhaps offering reduced per-trip fees to promote pooling.

Some airports allow TNC drivers to be dispatched to a waiting customer immediately after dropping off a customer (i.e., pre-dispatched). Pre-dispatching is used where customer drop-off occurs immediately prior to and on the same curbside roadway as customer pickup. Pre-dispatching improves the efficiency of TNC operations, leading to reduced roadway traffic volumes, and improved customer service (i.e., reduced wait times).

7.6.5 Establishing Differential Prices for Transportation Network Company Passenger Boarding Areas

Airports have tested pricing strategies that charge TNCs higher per-trip fees for the use of premium boarding areas (e.g., the terminal curbside) than they are charged for the use of an alternate boarding area (e.g., a nearby parking structure).

As part of a March 2019 pilot program at San Francisco International Airport, TNCs were charged a $3.00 price differential: $5.00 for use of the premium curbside boarding area versus $2.00 for use of the adjacent parking structure. This pricing strategy resulted in a 10% trip shift

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in TNC traffic from the terminal curbside to the remote boarding area. The differential pricing program at San Francisco International Airport was canceled after 3 months. While some TNCs have expressed support for such a differential fee program, successful implementation requires informing drivers and customers (many of whom travel from the airport infrequently) about the available benefits (reduced costs and potentially lower waiting times) as well as the inconvenience associated with longer walks.

7.6.6 Improving the Customer's Ability to Identify and Locate a Waiting Vehicle

It can be difficult for TNC customers to identify their drivers and for drivers to spot their customers. This can be a particular challenge when the curbside sidewalk and/or roadway is congested, at night if the roadway is poorly lit, or during inclement weather conditions. Identifying a vehicle can be challenging because many drivers operate cars that appear similar in shape and color. While the TNC app shows customers the license plate number of their driver's vehicle, it can be difficult for customers to read the plate as the vehicle approaches or while it is stopped between other vehicles.

While normally TNC dwell times are shorter than those of private vehicles, TNCs can experience longer dwell times when customers cannot find their driver or when drivers must call their customers to indicate the vehicle's location. These increased dwell times can contribute to and aggravate curbside congestion. Recognizing these challenges, TNCs and airport operators have developed and tested several solutions:

- Mounting signs high enough so that they are readily visible to customers and drivers.
- Having customers receive a colored signal on their app that matches the color displayed on a variable sign in the windshield of their assigned car. An example of such a sign is shown in Figure 7-27.
- Working with airports to divide large pickup zones into smaller subzones, allowing drivers to inform customers that they are waiting in a particular subzone. See Figure 7-28.

Airport staff could require TNCs to implement these or other solutions to improve customer and driver matching. Solutions that require providing all drivers with displays or other hardware may be challenging for TNCs to implement due to the frequent turnover of TNC drivers.

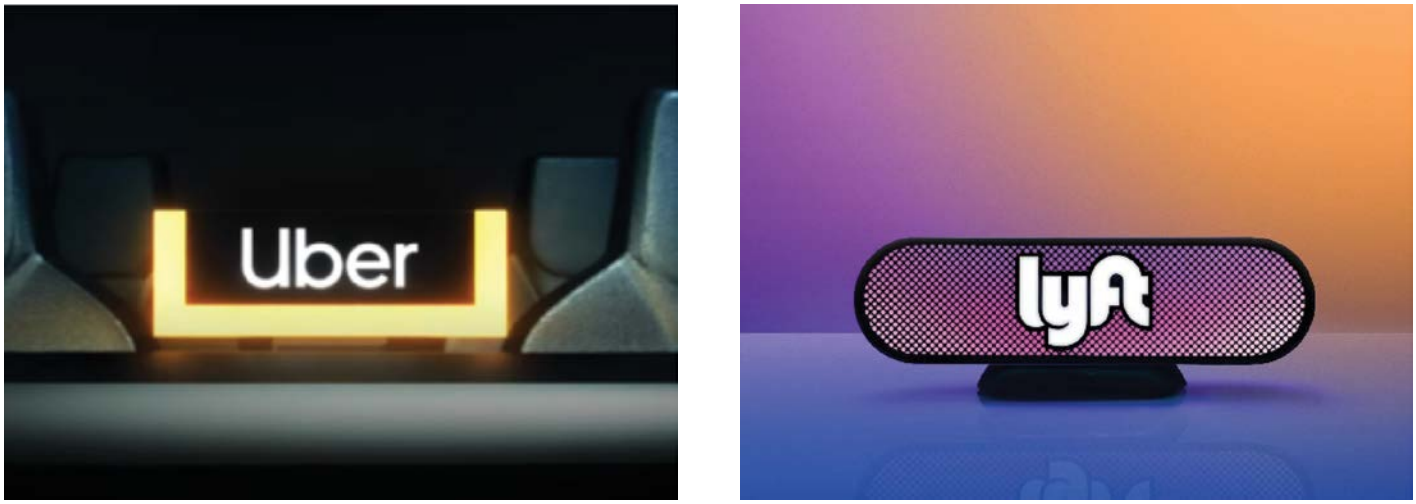


***Figure 7-27. Uber Beacon pilot program* (left) *and Lyft Amp pilot program* (right).**

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Figure 7-28. San Diego International Airport curbside Uber pickup location.

7.6.7 Dispatching of Vehicles by Transportation Network Companies

All major TNCs have implemented programs intended to reduce customer wait times for drivers and reduce curbside traffic volume. While Uber and Lyft both have similar programs, they refer to these programs using different names. Some of the programs include

- Dynamic pickups. At airports having subzones or marked columns, drivers are dispatched by the TNCs to specific curbside zones (or product areas) in order to balance demand among the available boarding areas.
- ExpressMatch or tiered queues. A limited number of drivers are positioned at the boarding area in anticipation of customer requests, with the number of vehicles located at the curb set in coordination with airport management. This results in the vehicles waiting for the customer rather than a customer waiting for a vehicle.
- Prematch or predispatch. A limited number of drivers are released from the hold lot in anticipation of customer requests. This results in slightly longer wait times for customers than ExpressMatch, but prevents empty vehicles standing at the curbside boarding areas.
- Rematch. Allows a driver who has completed a customer drop-off to be immediately dispatched to a customer awaiting pickup. This allows the driver to skip returning to the hold lot and waiting for a dispatch. Rematch reduces curbside traffic volumes as well as customer wait times.
- PIN dispatch. A limited number of drivers are repositioned from a waiting lot to the curb based on real-time demand forecasting. Drivers join a curbside queue (similar to a taxi stand) and are matched with a rider who is physically waiting in line.

A comparison of the different dispatching programs is shown in Table 7-1.

Table 7-1. Dispatching programs.

Dispatch type	Minimizes wait times?	Minimizes deadheading?	Allows dispersed pickup points?
Standard/waiting lot		N/A - Default Dispatch Type	
Rematch	Yes	Yes	Yes
Prematch/predispatch	Yes	No	Yes
Expressmatch/tiered queues	Yes	Yes (with Rematch enabled)	Yes
PIN dispatch	Yes	Yes (with Rematch enabled)	No (PIN dispatch requires a single consolidated pickup point)

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Summary—Accommodating Transportation Network Companies

Intended Goals

- Accommodate TNC operations at airports

Applicability

- Medium- and large-hub airports

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X	X			

  - Design and development of remote vehicle curbs including signage for drivers and customers.
  - Development of hold lots.
  - Dependent on GPS management system and functionality.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  - Operations and maintenance costs will vary based on the approach used.

Implementation Considerations

- **Implementation schedule:** Depends on the approach implemented and existing airport facilities. Can be less than 1 year to multiple years.
- **Ability to implement in phases:** Yes, for some of the strategies discussed in this section (e.g., vehicle identification).
- **Likely political influence:** Requires close coordination with TNC providers. Some strategies require cooperation from drivers, enforcement, and messaging to the public on the changes and the benefits.

Complementary Strategies and Supporting Technologies

- 7.2.1. Moving commercial vehicle passenger boarding areas to an available remote location
- 7.3.4. Revising the location or amount of curb space allocated to specific mode(s)

Examples

- Boston Logan International Airport
  - Uses geofencing for remote hold lots and consolidated TNC drop-off/pickup on outer curbs at each terminal.
  - Formerly provided a reduced fee to customers who utilized pooled TNCs.

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- Dulles International Airport uses remote hold lots.
- San Francisco International Airport
  - Currently uses geofencing for both its remote hold lot and consolidated drop-off and pickup on the roof of its central garage.
  - Previously utilized dynamic curbside pricing prior to development of a consolidated TNC drop-off/pickup location.
  - Formerly provided preferred curb locations for pooled TNCs.
- Los Angeles International Airport
  - Provides a consolidated, remotely located TNC/taxi passenger pickup area, referred to as LAX-it, which reduces TNC traffic on terminal roadways.
  - Allowed Black Car TNCs to pick up at terminals for an additional fee plus the posted parking rates.

7.7 Accommodating Autonomous Vehicles

AVs automate certain driving systems that do not require human input. SAE International has defined six levels of autonomy for vehicles, from Level 0 (no automation) to Level 5 (where the steering wheel becomes optional for the vehicle). The trend in car manufacturing is to achieve Level 5 autonomy for vehicles, where the steering wheel becomes optional. It is also anticipated that AVs will be electrically powered in the future.

At the time this research was conducted, multiple companies were testing AVs on public roads, but the timeline for the rollout of customer-ready AVs and widespread adoption of these vehicles continues to be uncertain. However, some researchers anticipate that

1. The functionality of AVs will be phased in, with the initial use at airports likely to be autonomous valet operations, i.e., moving a vehicle from a designated drop-off point inside a parking facility (or other reserved zone) to an individual storage or parking space. On the customer's return, the vehicle would be moved or summoned to a designated boarding area.
2. AVs will likely be introduced at airports earlier and at higher volumes than they will be in the surrounding region. This is because two operators with a significant presence at airports, rental car companies and TNCs, are expected to be early adopters of AVs.
3. Airports will benefit from the introduction of autonomous shuttle buses with operating characteristics (e.g., travel speed) that are superior to those of autonomous shuttle buses now used at some sites to connect passenger terminals with parking and other facilities.

The term AV, as used in this report, includes connected vehicles. Whereas AVs may be able to operate solely based on in-vehicle technologies, connected vehicles may operate driverless but do so based on communications with other vehicles, roadways, roadside infrastructure, and other elements.

In addition to traditional ground transportation modes, AVs are expected to facilitate the development of new access services, such as mobility as a service (MaaS). MaaS is the offering of multiple forms of transportation, such as TNCs, taxicabs, car sharing, car rentals, and public transit through a single gateway, such as a mobile application, and payment system. MaaS also allows travelers to seamlessly plan, book, and pay for multimodal travel. The introduction of AVs, which could allow a single vehicle to operate as more than one of those services (i.e., operate as a car-sharing company on one trip and as a TNC on the next), is expected to advance the transition to MaaS because it may blur the distinctions between traditional modes.

As of 2022, AVs are still being developed, and operations on public roadways are limited to vehicles being tested by system developers or vehicle manufacturers. Although the timeline for AV development and deployment is uncertain, once AVs do appear on airport roadways and curbsides, there will be a multi-year period where both AVs and human-driven vehicles are

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operating on-airport; however, AV operations may be limited to certain conditions or within limited geographic areas. Therefore, the strategies presented here for accommodating AVs on airport curbsides focus on this transition period.

- 7.7.1. Separating autonomous vehicles from non-autonomous vehicles
- 7.7.2. Updating airport rules and regulations
- 7.7.3. Preparing infrastructure

7.7.1 Separating Autonomous Vehicles from Non-Autonomous Vehicles

As AVs are currently still in the development and testing stage, their ability to interact with non-AVs is currently unknown. Given that airport curbside operations can be less predictable and more chaotic than typical on-street or highway operations (i.e., high turnover of vehicle loading/unloading spaces, frequent lane changes, frequent pedestrian/vehicle interactions, vehicles stopped in travel lanes or double-parked), it is possible that an AV's ability to successfully navigate the environment may be compromised. This uncertainty may be compounded in that the operation of traditional vehicles or non-AV operations on an airport curbside (e.g., double-parking, very low travel speeds as drivers search for waiting passengers) are often significantly different than typical street traffic. It is currently unknown how an AV will operate in atypical circumstances that could fall outside of the AV's normal operating parameters. Therefore, to ensure the safety of pedestrians and motorists and to minimize potential disruptions, airports may wish to separate AVs from non-AVs—particularly at passenger drop-off and pickup areas. This separation could be accomplished by (1) designating passenger drop-off and pickup areas (such as an island curb or zones within a parking facility) for exclusive use by AVs and (2) designating paths (potentially using roadways less frequented by non-AVs) that AVs must follow while on airport property.

7.7.2 Updating Airport Rules and Regulations

Even prior to the operation of AVs on an airport's roadways, airport management can update its rules and regulations so that AV-related elements are in place prior to AVs appearing in noticeable numbers. It is anticipated that federal, state, and other local jurisdictions, rather than airport operators, will provide the primary guidelines regarding the operations of AVs on public roadways and will address elements related to safety, vehicle licensing, insurance, liability, and law enforcement ability to control AVs as part of their traffic control and security role (i.e., the ability to quickly remove an empty AV from a curbside or other sensitive location). However, it is assumed that an airport will incorporate AV-related elements into its general rules regarding on-airport traffic operations as well as into its rules, regulations, and business arrangements with commercial ground transportation providers. AV-related items could include identification of operating areas (including required travel paths), dwell time limits on curbside or other designated passenger loading and unloading areas, rules regarding use of hold lots or other staging facilities, data-sharing requirements for AV operators, sustainability requirements for companies conducting business on airport property using AVs, and AV and passenger interactions with law enforcement (such as the ability to relocate/impound AVs that violate airport-specific rules and regulations).

7.7.3 Preparing Infrastructure

Despite uncertainty regarding the development and adoption of AVs, airports may wish to start adjusting their infrastructure to facilitate AV operations. It is likely that the FHWA will establish national standards regarding traffic control devices that support AV operations. On-airport public roadways will likely need to incorporate those standards. Such standards are expected to focus on pavement markings and traffic signals but will likely not deviate significantly from existing

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standards as long as human drivers continue driving on the same facilities. An early example of such standards is the 2018 adoption in California of 6-inch-wide lane striping (increased from 4 inches) to accommodate AVs.

To support on-airport AV operations, an airport may need to implement connected-vehicle infrastructure that would allow the roadway or parking facility to share information with individual AVs. Such information could include information on traffic conditions; alternative routes; and drop-off, pickup, and staging facility availability.

Airports should consider AVs when reconditioning parking facilities and roadways.

Summary—Accommodating Autonomous Vehicles

Intended Goals

- Enhance safety for AVs, human-driven vehicles, and pedestrians. Provide AVs with operating areas that are organized and less likely to present unfamiliar situations
- Maintain or enhance the efficiency of curbside operations by minimizing potential disruptions that could result from AV operations in mixed traffic and in unusual environments
- Provide flexibility to adapt as AVs gradually comprise a greater share of the vehicles operating on-airport
- Provide flexibility to adapt to future changes in the types of ground transportation services available at an airport, such as the introduction of MaaS and the potential that one AV could operate as one of several access modes

Applicability

- Applicable to all airports potentially experiencing AV operations on their roadways and curbsides

Estimated Costs

- **Capital:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
	X	X	X			

  - Roadway signage to indicate to non-AVs that certain roadways or facilities are restricted to AVs.
  - Signage for pedestrians to help them locate facilities reserved for AVs. Depending on the terminal configuration, such signs may be needed inside the terminal.
  - Hold lot amenities, such as EV charging equipment.

- **Operations and Maintenance:**

<$10K	$10K to $50K	$50K to $100K	$100K to $500K	$0.5M to $1M	$1M to $5M	>$5M
X						

  - Incremental and maintenance costs will likely be limited to maintenance for new hold lots, hold lot amenities, and signage.

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Implementation Considerations

The actions described in this section related to accommodating AVs on airport roadways and curbsides are non-specific and will likely be highly dependent on the physical configuration of an airport's roadways, its operating conditions, and its regulatory environment. Therefore, known implementation considerations are limited to

- **Time required to update rules and regulations.** Changes to airport rules and regulations, including those related to commercial ground transportation operations and business agreements, typically require several months to allow sufficient time for stakeholder and public consultation.
- **Enabling actions.** Changes to rules and regulations will likely need to precede changes to operations and physical facilities. Depending on how an airport chooses to accommodate AVs, it may need to develop AV staging facilities, including electric charging facilities and vehicle service support.
- **Evaluation of liability to the airport**. As the airport implements the different functionalities of AVs, airport management should weigh the liability associated with each new functionality.

Complementary Strategies and Supporting Technologies

- 7.2.1. Moving commercial vehicle passenger boarding areas to an available, remote location
- 7.2.2. Moving private vehicle pickup and/or drop-off areas to an available remote site
- 7.3.4. Revising the location or amount of curb space allocated to specific mode(s)
- 7.3.10. Revising roadway and curbside signage/wayfinding
- 7.5.1. Encouraging the use of transit or high-occupancy travel modes
- 7.5.2. Encouraging use of long-duration parking rather than drop-off or pickup modes

Examples

- In 2022, Dallas/Fort Worth International Airport initiated a 3- to 6-month demonstration test of AV technology composed of three subsystems: (1) low-speed vehicle automation for automated valet parking, (2) supervisory parking management for AVs in an exclusive zone, and (3) active digital curb management. These three interconnected subsystems were tested at Terminal A to provide an innovative solution to the congested curbside drop-off and pickup areas at Dallas/Fort Worth International Airport.
- The Port Authority of New York and New Jersey, as part of its ongoing maintenance strategy, is gradually installing larger signal control boxes at its airports and other facilities to provide room for equipment that may be needed in the future for communications with connected vehicles. Starting in 2018, the California Department of Transportation started using 6-inch-wide lane striping (increased from 4 inches) in anticipation of AVs.

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APPENDIX A

Glossary

Adjusted flow rates—The maximum rate of flow adjusted for traffic conditions, traffic composition, roadway geometry, and other factors.

Airfield licenses—Licenses or permits required to operate a vehicle on an airfield at major airports.

Airport curbside—The one-way roadways located immediately in front of the terminal buildings where vehicles stop to pick up and drop off airline passengers and their baggage.

Air taxi—A for-hire passenger or cargo aircraft that operates on an unscheduled basis.

Automatic traffic recorder machines—Equipment, often portable, that records the traffic volume crossing a tube or detector.

Automatic vehicle identification (AVI)—A system or Radio-Frequency-Identification equipment (i.e., vehicle-mounted tags or transponders) commonly used on roadways and bridges to collect tolls.

Auxiliary lane—A supplementary lane intended to facilitate weaving or merging vehicle movements.

Bypass lanes—Curbside lanes intended for use by vehicles bypassing or not stopping at a curbside section or zone.

Bypass vehicles—Vehicles traveling past, but not stopping in, a curbside section or zone including vehicles recirculating past the curbside, vehicles traveling to/from adjacent curbside zones, or service/delivery vehicles using the curbside roadway.

Cell phone lots or call-and-wait lots—Free parking lots, typically located away from the terminal area, provided for use by motorists waiting to pick up deplaning airline passengers.

Commercial vehicles—Vehicles transporting airline passengers (including taxicabs, limousines, courtesy vehicles, buses, and vans), driven by professional drivers, for which passengers pay a fee or for which the transportation is incidental to the service provided (e.g., a hotel courtesy vehicle).

Cost paths—A person's perception of the cost that would be incurred while traveling along a defined path or route, typically including their value of time.

Courtesy vehicle—Door-to-door, shared-ride transportation provided by the operators of hotels/motels, rental car companies, parking lots, and other services solely for their customers.

Critical movement analysis—Calculation of the lanes or movements requiring the most "green" time at a signalized intersection and, therefore, a method of estimating intersection capacity.

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Critical volumes—The volume or combination of volumes (e.g., conflicting movements) that produces the highest demand for an intersection lane or signal phase.

Curbside geometries—The horizontal and vertical alignment features of a curbside roadway including lane widths, grades, curvature, and crosswalks.

Customs and Border Protection (CBP)—The U.S. government agency responsible, among other duties, for inspection of international arriving passengers and goods to collect import duties and prevent importation of illegal goods.

Decision-making distance—The physical distance between successive decision points.

Decision points—The physical location where a driver must select between alternative paths or roadways.

Deplaned passengers—Passengers alighting from an aircraft at an airport including both connecting and terminating airline passengers.

Discount factor—An adjustment applied to reduce the effective capacity of curbsides having an unusual configuration, location, or operation.

Double-parking—Two or more parallel parked or standing vehicles.

Driver population factor—A factor applied to roadway capacities to reflect driver behavior and operating characteristics including familiarity with roadways, intersections, and traffic patterns.

Electronic ticketing kiosks—A self-serve machine used by airline passengers to print boarding passes and other documents.

Enplaned passengers—Passengers boarding an aircraft at an airport including both connecting and originating airline passengers.

Equivalent passenger-car volumes—A hypothetical passenger-car volume representing an actual volume composed of a mix of trucks, buses, and other vehicle sizes in the traffic stream.

Fixed-base operator—An aviation business that serves general aviation aircraft owners and operators with services including fueling, catering, aircraft maintenance, and storage.

Flattening the peak—A reduction in the proportion of demand occurring in a 15-minute or hourly interval due to demand management, changes in schedules, demands approaching capacity, or other causes.

Flow rate—As used in this Guide, the volume of vehicles or people passing a point per unit of time such as vehicles per hour.

Free-flow speed—The mean speed of traffic under very light flow conditions.

General aviation—All flights (or aircraft operations) other than scheduled/commercial-service or military flights.

Green time—The portion of a traffic control signal phase during which vehicles are permitted to flow (i.e., the green signal is displayed).

Growth factor—A factor applied to passenger or traffic volumes, for example, to adjust for anticipated future growth.

Heavy vehicle factor—A factor applied to roadway capacities to reflect the proportion of large vehicles (e.g., large trucks and buses) in the traffic stream.

Highway capacity analysis procedures—Analysis conducted using the procedures described in the *Highway Capacity Manual*.

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***Highway Capacity Manual* (HCM)**—The HCM, published by the Transportation Research Board (7th edition published in 2022).

Immigration and Customs Enforcement—The U.S. government agency responsible for, among other duties, inspection of international arriving passengers and crew prior to their entering the country.

Intelligent transportation system (ITS)—Information and communication technologies applied to transportation infrastructure and vehicles to improve operations, safety, and efficiency.

Lag time—The length of time after their flight's scheduled arrival time that a passenger gets to the airport curbside.

Landside circulation system—The airport roadway network providing for inbound and outbound traffic and the internal circulation of traffic between airport land uses.

Lane balance—The number of lanes entering and exiting a roadway merge or diverge point.

Lane geometry—The horizontal and vertical alignment features of a roadway or roadway lane including lane widths, grades, lengths, curvatures, tapers, and other physical features.

Lateness distribution—The distribution of passengers leaving an airport after the scheduled arrival time of their aircraft (i.e., a distribution of passenger lag times).

Lead time—The length of time in advance of their flight's scheduled departure time that a passenger arrives at the airport curbside (as used in this Guide).

Macroscopic models—Models or analytical procedures that consider the flow of vehicle streams (or other objects) rather than the flow of individual vehicles.

***Manual on Uniform Traffic Control Devices* (MUTCD)**—The principal standard governing the application, design, and placement of traffic control devices, published by the FHWA. See http://Mutcd.fhwa.dot.gov.

Maximum service flow—A maximum rate at which vehicles can transverse a point or short segment during a specified time period.

Merging capacity—Maximum service rate at a merge point.

Metropolitan planning organization (MPO)—A policy-making organization responsible for planning, analyses, and development of multimodal transportation facilities in a region or community.

Microsimulation—Models or analytical procedures that simulate the operation of individual vehicles (or other objects) on simulated roadway (or other) networks.

Mixed flow traffic volumes—A traffic flow consisting of multiple types of vehicles.

Operational characteristics—Traffic flow characteristics including speed, density, vehicle mix, and volumes.

Passenger load factor—A measure of the historical proportion of the available aircraft seats that are occupied.

Peak-hour factor—The relationship between the hourly volume and the maximum 15-minute flow rate within the hour.

Performance capabilities—As used in this Guide, refers to the performance capabilities of an individual vehicle or group of vehicles including acceleration, maneuverability, and turning radii.

Poisson distribution—A discrete probability distribution that expresses the probability of a number of events occurring in a fixed period of time.

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Remote curbside—A curbside located outside of the immediate area of the passenger terminal building such as in a parking structure, surface lot, or multimodal facility.

Rental car ready/return—The parking or storage area(s) to which rental car customers return "dirty" vehicles or pick up "ready" vehicles.

Signal phasing—The part of a traffic control signal time cycle allocated to any traffic movement receiving the right-of-way.

Skycap—A porter employed by an airline or airport to provide baggage service to passengers.

Steady-state performance—The traffic flow rates occurring on a roadway or intersection when there are no disruptions or interruptions to the traffic stream.

Terminal area roadway—The roadways serving the terminal building and surrounding areas including access, curbside, and circulation roadways.

Through vehicles—As used in this Guide, vehicles bypassing the curbside area or zone. (See bypass vehicles.)

Time path—A person's perception of the time that would be incurred while traveling along a defined path or route including time in motion, delays due to congestion, and waiting time.

Traffic controls—Devices directing vehicular and pedestrian traffic flows, particularly at conflict areas, including signals, signs, and pavement markings.

Transportation demand management—The application of policies and strategies to reduce travel demand or redistribute this demand in space or time.

Transportation network company (TNC)—A company providing pre-arranged transportation services for compensation using an online application or platform (such as smartphone applications) to connect drivers using their personal vehicles with passengers.

Trip generation rate—The number of vehicle or person trips generated by a household, zone, land use, or other facility generally during a daily or peak-period basis.

Triple-parking—Three or more parallel parked or standing vehicles.

Waybill—A list of passengers or goods being carried on a vehicle, often containing details about the arrival time, destination, and other information about the passengers or goods.

Weaving areas—The roadway segment in which two or more traffic streams traveling in the same general direction along a significant length of roadway cross without the aid of traffic control devices.

Weaving distance—The length of a weaving area or segment.

Weaving intensity factor—A measure of the influence of weaving activity on the average speed of both weaving and non-weaving vehicles.

Vehicle mix—The proportion of each type of vehicle (i.e., bus, car, van, truck) in a traffic stream.

Vehicle occupancy—The number of passengers (including the driver) in a vehicle.

Vehicle stall length—The length of curb space occupied by a standing vehicle including the distance required to maneuver into and out of the space.

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APPENDIX B

Bibliography

This document presents a bibliography of documents reviewed during the literature search for ACRP Project 07-17, "Update to ACRP Report 40: Airport Roadway Analysis and Curbside Congestion Management Strategies," with an emphasis on those documents that describe analytical procedures and performance measures and strategies for optimizing efficiency of airport curbside and terminal area roadway operations. The literature review that follows is organized around the following topics:

- Estimating Traffic Volumes
- Analyzing Curbside Roadways
- Strategies for Improving Operations and Reducing Delay
- Ground Transportation Modes: TNCs, Autonomous Vehicles, and Others
- Roadway Level-of-Service Metrics

This literature review builds upon that conducted in developing *ACRP Report 40*. This review updates the study to include analyses and best practices developed after the 2010 publication of *ACRP Report 40*.

The literature review was conducted by examining scholarly responses to *ACRP Report 40*. Additionally, the literature review team identified conference presentations and papers that provide case studies of curbside and terminal area roadways after the publication of *ACRP Report 40*. Lastly, the team identified analyses that addressed changing patterns in transportation modes associated with maturing transportation network companies (TNCs) and changing public behavior.

Estimating Traffic Volumes

Measuring airport roadway traffic volumes is often cost-prohibitive or impractical during planning studies. Therefore, methods to estimate existing, and project future, traffic volumes based on other sources are an important starting point for curbside and roadway analyses.

Duncan, Gavin, and Hugh Johnson. "Development and Application of a Dynamic Simulation Model for Airport Curbsides." San Francisco, CA, 2000.

This conference paper provides a methodology for modeling curbside traffic.

———. "Development and Application of a Dynamic Simulation Model for Airport Curbsides." Presented at the ASCE Specialty Conference: 26th International Air Transportation Conference, San Francisco, CA, June 19, 2000.

This presentation accompanied the conference paper of the same name. It provides additional graphics and examples.

Kim, Seong Ho, Won Ho Suh, and Jun Gin Kim. "Airport Access Road Traffic Demand Estimation," 764:1356–60. *Applied Mechanics and Materials*, 2015.

This paper provides a framework to estimate airport roadway traffic volumes using airline flight schedules.

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Kim, Seongho, Wonho Suh, and Jungin Kim. "Forecasting Hourly Traffic Volume: Incheon International Airport Access Road." *Scientia Iranica* 22, no. 4 (2015): 1510–16.

This paper uses Incheon International Airport as a case study to examine the relationship between traffic volume and airport operational efficiency. It further suggests a mathematical model that can be used to predict traffic volumes at various points of the airport's ground access network and then derive future levels of congestion at those points.

Analyzing Curbside Roadways

Curbside roadway analysis for airports requires specialized models. The following presentations and publications discuss the use of modeling in analyzing curbside roadways and provide case studies of existing curbside programs. Publications describing techniques developed after the publication of *ACRP Report 40* in 2010 are included.

Ceballos, Gustavo, and Alex Chambers. "Airport Curbside Operation: Macro Analysis Enhancement through Simulation." PTV User Group Meeting 2019, October 10, 2019.

This presentation makes a case for the use of VISSIM simulation models in analyzing curbside roadway capacity and throughput. Notably, it discusses the relationship between pedestrian crossings and curbside roadway level of service.

Chen, Yanyan, Nuo Zhang, Haoning Wu, Yao Lu, Liang Zhang, and Jun He. "Calculation Method of Traffic Capacity in Airport Curbside," 725–36. Springer, 2016.

Dallas/Fort Worth International Airport Parking Business Unit. "Dallas/Fort Worth International Airport Parking Business Unit." October 20, 2015.

This presentation describes the ground access and curbside programs at Dallas/Fort Worth International Airport.

Dickens, Calvin. "Reducing Commercial Vehicle Congestion: Curbside Usage at William P. Hobby Airport (HOU)." November 2006.

DiPrima, Chris. "Integrating Ride App Vehicles into the SFO Curbside Traffic Model." Presented at the ESRI User Conference, San Diego, July 11, 2019.

This presentation provides a methodology for using VISSIM microsimulation to analyze TNC data and curbside usage at San Francisco International Airport (SFO).

Duncan, Gavin. "Commercial Vehicles: Who's Curb Is It Anyway?" Presented at the Airport Ground Transportation Association Spring 2008 Meeting, Atlanta, GA, April 8, 2008.

———. "Curbside Operations: Challenges & Solutions." Presented at the AAAE Ground Transportation and Landside Management Workshop, November 10, 2004.

Evans, Alastair. "Managing the Curb through Simulation." n.d.

This presentation examines urban curb management and proposes simulation modeling of flexible-use curbs.

Galagedera, Sameera, H.R. Pasindu, and Saman Bandara. "Airport Curbside and Parking Area Operations at BIA—Analysis of User Behavior." *Engineer, Journal of the Institution of Engineers, Sri Lanka* 47 (October 27, 2014): 43–51. https://doi.org/10.4038/engineer.v47i4.6882

This paper evaluates vehicle operations and passenger behavior, using Bandaranaike International Airport (BIA) as a case study. It proposes methods to evaluate existing levels of service and suggests measures to improve operational efficiencies to meet future demand.

Harris, Tyler M., Mehdi Nourinejad, and Matthew J. Roorda. "A Mesoscopic Simulation Model for Airport Curbside Management." Edited by Richard S. Tay. *Journal of Advanced Transportation* 2017 (July 6, 2017): 4950425. https://doi.org/10.1155/2017/4950425

This article proposes a mesoscopic traffic model based on traffic conditions at Toronto Pearson International Airport, to address the deficiencies of *ACRP Report 40*'s QATAR model, but avoids detailed vehicle interaction inherent in microsimulation models. The conditions not discussed in *ACRP Report 40*, but addressed in this article, include double-parking, inclement weather effects reducing the ratio of available inner curb and supply of commercial vehicles, and a reduction of private vehicle dwell time associated with an increase in volume.

Houston Airport System. "IAH Curbside Operations." 2016.

This presentation provides an overview of the curbside operational structure at George Bush Intercontinental Airport (IAH).

Lockhart, Bob. "Evaluation of Landside and Parking Operations." 2010.

Mandle, Peter, and Gavin Duncan. "New Concepts for Analyzing Airport Roadway Operations and Level of Service." Presented at the 2010 Annual Meeting of the Transportation Research Board, January 11, 2010.

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Pasindu, H. R., Kaushan Devasurendra, and Sajith Udayanga. "Assessment of Curbside Roadway Level of Service of the Bandaranayke [*sic*] International Airport—Sri Lanka: A Comparison of the Analysis Tools in ACRP 40." Presented at the 11th Asia Pacific Transportation and the Environment Conference, 2018.

This paper presents a comparative analysis of theoretical results from *ACRP Report 40* and the real-world data from the Bandaranaike International Airport in Sri Lanka. It suggests alternative dwell times for different vehicle types. The authors also used a variation of the best-fit curve with higher predictive accuracy than *ACRP Report 40*.

San Francisco International Airport. "Ground Transportation Performance Measures." n.d.

Strategies for Improving Operations and Reducing Delay

Various new strategies have emerged in recent years to manage demand on airport roadways such as providing cell phone lots and commercial vehicle hold lots and adding mechanisms to collect fees from all curbside users. The strategies assessed include strategies for (1) enforcement, (2) traffic demand, (3) congestion management, (4) customer experience management, (5) sustainability, (6) TNC and autonomous vehicle accommodation, and (7) additional supporting technologies. These themes are broad and are touched on in works across all categories. For brevity, these have not been reiterated in this category for strategies for improving operations and reducing delay.

Biagi, Michael. "LAX Ground Transportation Program." November 2007.

Brock, Bakari. "Solving Airport Congestion through Curb Pricing." *Medium*, August 18, 2019. https://medium.com/sharing-the-ride-with-lyft/solving-airport-congestion-through-curb-pricing-62fb91e13751

This article describes curbside operation innovations introduced by the TNC Lyft and introduces a case for curb pricing. These innovations include rematch and predispatch to optimize the effectiveness of TNC vehicles in the airport's ground access system.

Collins, Jeff. "Managing Your Curbsides to Meet the Expectations of Your Customers, Security, Airport Management, and the Providers." 2017.

Duncan, Gavin. "What's New in Airport Curbside Management?" Presented at the ACI-NA Operations and Technical Affairs Conference, April 6, 2011.

This presentation provides an overview of emerging technologies and opportunities related to curb management.

Greco, Chet. "The FLL Airport Experience Using a 3rd Party Contractor for Curbside Management." San Francisco, CA, September 26, 2019.

This presentation provides a case study of using a 3rd party contractor for curbside management at FLL. It presents the challenges and best practices associated with managing various curbside elements.

Hutchison, D., Scott. "SFO Curbside Management: Taxi Dispatch & Shared Shuttle Management." Presented at the AAAE: Parking and Landside Management Workshop, November 6, 2013.

This presentation presents a case study of offsite queuing and hold lots at SFO.

Leutwiler, Mark. "Managing Your Curbs." 2017.

This presentation provides an overview of the curbside elements at Portland International Airport and suggests methods to balance demand management with customer service.

McCarter, Horace. "Curbside Management of Commercial Ground Transportation: Effective Third Party Management." Presented at the AAAE Meeting, November 8, 2006.

This presentation discusses the potential obstacles and benefits of utilizing 3rd party management at commercial burbs.

Mitman, Meghan, Steve Davis, Ingrid Ballus Armet, and Evan Knopf. *Curbside Management Practitioners Guide*. Institute of Transportation Engineers, 2018.

This document outlines decision-making frameworks for curbside management in urban locations. Additionally, it suggests best practices and performance measures for curbside management.

Moorthy, Aditi, Robert De Kleine, Gregory Keoleian, Jeremy Good, and Geoff Lewis. "Shared Autonomous Vehicles as a Sustainable Solution to the Last Mile Problem: A Case Study of Ann Arbor-Detroit Area." *SAE International Journal of Passenger Cars - Electronic and Electrical Systems* 10, no. 2 (March 28, 2017): 328–36. https://doi.org/10.4271/2017-01-1276.

This article examines the effects of autonomous vehicles and other transit modes on the last mile of traffic to and from airports through the lens of sustainability. It estimates the mode split associated with autonomous vehicles. It discusses funding implications and the potential additional cost factors associated with this mode type.

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Ground Transportation Modes: Transportation Network Companies, Autonomous Vehicles, and Others

For decades, the proportion of airline passengers arriving at and departing from airports remained relatively constant from one year to the next. However, changes in technology, particularly the market adoption of TNCs since 2015, have triggered a series of changes in airport ground access. The impacts that TNCs have had on airport landside operations and revenue have been profound, and there is therefore considerable research already in this area.

CAPA Centre for Aviation. Airports and Uber: Transportation Network Companies Now More Welcome at Airports, 2016.

Contreras, Seth D., and Alexander Paz. "The Effects of Ride-Hailing Companies on the Taxicab Industry in Las Vegas, Nevada." *Transportation Research Part A: Policy and Practice* 115 (2018): 63–70.

Conway, Alison, Camille Kamga, Anil Yazici, and Abhishek Singhal. "Challenges in Managing Centralized Taxi Dispatching at High-Volume Airports: Case Study of John F. Kennedy International Airport, New York City." *Transportation Research Record: Journal of the Transportation Research Board,* No. 2300, 2012, 83–90. https://doi.org/10.3141/23

Coogan, Matthew A., MarketSense Consulting LLC, and Jacobs Consultancy. *ACRP Report 4: Ground Access to Major Airports by Public Transportation*. Transportation Research Board of the National Academies, Washington, DC, 2008. https://doi.org/10.17226/13918

Gatien, Seth, John Gales, Ata Khan, and Ariel Yerushalami. "The Importance of Human Factors When Designing Airport Terminals Integrating Automated Modes of Transit," 597–602. Springer, 2020.

Gosling, Geoffrey David. *ACRP Synthesis 5: Airport Ground Access Mode Choice Models*. Transportation Research Board of the National Academies, Washington, DC, 2008. https://doi.org/10.17226/23106

This ACRP Synthesis outlines multinomial logit utility models and nested multinomial logit models. Others subsequently extended this methodology, particularly after market introduction of TNCs.

Henao, Alejandro, Joshua Sperling, Stanley E. Young, Garikapati Venu, and Yi Hou. "Airport Analyses Informing New Mobility Shifts: Opportunities to Adapt Energy-Efficient Mobility Services and Infrastructure: Preprint." 2018. https://www.osti.gov/biblio/1454744.

Hermawan, Karina. "Transportation Network Companies' (TNC) Impacts and Potential on Airport Access," 2018.

Dissertation describing TNC activity at airports with emphasis on activity at San Francisco International Airport (SFO) and Los Angeles International Airport (LAX). Multinomial and nested multinomial logit modeling were conducted to understand ground access mode choice at study airports.

Leiner, Craig, and Thomas Adler. *ACRP Research Report 215: Transportation Network Companies (TNCs): Impacts to Airport Revenues and Operations—Reference Guide*. Transportation Research Board, Washington, DC, 2020.

ACRP Research Report 215 explores the financial and operational challenges introduced to the airport landside environment by TNC market adoption. Best practices are presented for airport policy setting, including linking permitting to reporting and insurance requirements, and ground transportation fee setting.

L. E. K. Consulting. "The Future of Airport Ground Access: How Airports Can Respond to Disruption," n.d.

This report provides recommendations to meet changing modes of transport, particularly electric vehicles, TNCs, and vehicle sharing.

Mandle, Peter, and Stephanie Box. *ACRP Synthesis 84: Transportation Network Companies: Challenges and Opportunities for Airport Operators*. Transportation Research Board, Washington, DC, 2017.

One of the first publications to document the impact of TNCs on the airport environment, *ACRP Synthesis 84* provides an initial overview of shifting ground access modes and summarizes early market adoption rates at surveyed airports.

Meyer, Michael D., Simon M. Mosbah, and Louis D. Wolinetz. *ACRP Research Report 216: Guidebook for Assessing Collaborative Planning Efforts Among Airport and Public Planning Agencies.* Transportation Research Board, Washington, DC, 2020.

This report provides a general overview of ground access modes at airports.

Orrick, Phyllis, and Karen Trapenberg Frick. "Airports and Bicycles: What Are the Obstacles and Incentives for Operators to Improve Bicycle Access?" 2013.

Provides an overview of general planning considerations for bicycle transportation at airports, with emphasis on U.S. west coast, large-hub airports. The eight considerations assessed at each airport are (1) governance structure, (2) location, (3) access roads, (4) self-perceived environmental stewardship, (5) spending restrictions on non-aviation transportation improvements, (6) proximity to transit, (7) policies and mandates to reduce environmental impacts, and (8) land use constraints. The primary assessments were based on employee bicycle access, as that has been a focus for airport operators as a means to reduce single-occupancy vehicle traffic.

Passos, L., Rosaldo Rossetti, and Luís Reis. "Evaluation of Taxi Services on Airport Terminal's Curbside for Picking up Passengers." Proceedings of the 6th Iberian Conference on Information Systems and Technologies, CISTI 2011.

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Passos, Lúcio S., Zafeiris Kokkinogenis, Rosaldo J.F. Rossetti, and Joaquim Gabriel. "Multi-Resolution Simulation of Taxi Services on Airport Terminal's Curbside," 2361–66. IEEE, 2013.

Pitcher, Paige Marie. "Hit the Deck: Impacts of Autonomous Vehicle Technology on Parking and Commercial Real Estate." Massachusetts Institute of Technology, 2017.

This thesis models the effects of autonomous vehicles on the financial performance of urban parking garages. It suggests that parking garages hold a limited benefit in the short term, and recommends against investments in new parking facilities. This provides insight to potential changes in destination of traffic on terminal area roadways in light of autonomous vehicles.

Shapiro, Phillip. "Intermodal Ground Access to Airports: A Planning Guide—A Good Start." Presented at the Sixth TRB Conference on the Application of Transportation Planning Methods, Dearborn, MI, May 19–23, 1997.

Zhang, Yu, and Yuan Wang. "Implications of Autonomous Vehicles to Airport Terminal Planning and Design," n.d.

Estimates future impacts of autonomous vehicles to airport public parking activity and revenue, using the Tampa International Airport as a case study.

Roadway Level-of-Service Metrics

Traditionally, roadway level of service has been analyzed according to metrics defined by the Highway Capacity Manual (HCM). The airport roadway environment requires special consideration to address low-speed merging, diverging, and weaving sections on a network of predominantly one-way roadways.

Eibert, Shannon, Ian Girardeau, and Jamie Phillips. "Addressing Airport Congestion as Traffic Takes Off in the Age of Uber and Lyft." ACRP University Design Competition, April 29, 2019.

This report presents recommendations for reducing terminal curbside congestion in association with the increase in TNC mode share.

Mullins, Patricia. "Ground Transportation." 2017.

This presentation provides an overview of the ground transportation network at Memphis International Airport.

Pasindu, H. R., Kaushan Devasurendra, and Sajith Udayanga. "Assessment of Curbside Roadway Level of Service of the Bandaranayke [*sic*] International Airport—Sri Lanka: A Comparison of the Analysis Tools in ACRP 40." Presented at the 11th Asia Pacific Transportation and the Environment Conference, 2018.

Presents a comparative analysis of theoretical results from *ACRP Report 40* and the real-world data from the Bandaranaike International Airport in Sri Lanka. It suggests alternative dwell times for different vehicle types.

Udayanga, P. A. S., and H. R. Pasindu. "Incorporating User Characteristics for Level of Service Improvement at Airport Curbside and Roadside Operations at BIA." *Annual Sessions of IESL*, 2015, 69–75.

Compares roadway analysis results between *ACRP Report 40* and real-world data observed at the Bandaranaike International Airport in Sri Lanka. A new model was developed called the QAT-ACR, which is based on principles used in *ACRP Report 40*'s QATAR model and appears to primarily focus on single sections of curbside roadways.

Additional Materials

The following materials, while not directly addressing the focus areas, are relevant and include planning standards, best practices, and previous ACRP reports.

Aviation Department—Port Authority of New York and New Jersey. *Airport Planning Standards*. 3rd ed., 2018.

This document provides existing metrics and methods used in practical planning efforts by the Port Authority of New York and New Jersey. It is overarching and touches on all aspects of airport planning, including curbside and roadway planning. It also serves as a case study for the levels of service airports currently plan for.

Broward County Aviation Department. "Managing Your Curbsides Efficiently, Effectively, and Enthusiastically." 2017.

This presentation provides an overview of the Fort Lauderdale—Hollywood International Airport's existing curbside program. It includes a case study of 3rd party operations and their effect on customer service levels.

LeighFisher, Dowling Associates, Inc., JD Franz Research, Inc., and WILTEC. *ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations*. Transportation Research Board of the National Academies, Washington, DC, 2010.

Parizi, Mahmoud S., and John P. Braaksma. "Optimum Design of Airport Enplaning Curbside Areas." *Journal of Transportation Engineering*. 120, no. 4 (1994).

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APPENDIX C

Overview of the Curbside Analysis Methodology in the Quick Analysis Tool for Airport Roadways (QATAR)

Queuing Model Structure

The basic assumptions used in developing the Quick Analysis Tool for Airport Roadways (QATAR) curbside queuing model are presented in the following. Several excellent texts on queuing theory are recommended for further reading on this topic (Hillier and Lieberman 2015, Newell 1982). The field of queuing theory is well developed, with standardized terminology. This section introduces the standard queuing theory terminology and assumptions used in constructing the QATAR model.

A queuing system consists of a facility providing some form of service to a defined population. The key values used to define a queuing system are the arrival rate, service time, number of servers, system capacity, calling population, and queue discipline. The time between arrivals (inter-arrival time or arrival rate) of members into the queuing system is modeled as a probability distribution.

The roadway in front of the airport terminal represents the service facility. It is modeled as a multiserver facility, with each curbside space considered a server. The system capacity is assumed to be infinite. The size of the population can be either finite or infinite, but for analytical models, it is usually defined as infinite because of the difficulty of deriving analytical solutions for finite systems. The queue discipline is specified as first-in-first-out.

Queuing models have assumptions and restrictions on their application. Queuing models are derived assuming steady-state conditions. For example, at some airports, there are relatively few early morning flights and consequently few vehicles driving through the terminal curbside area. The system is in a transient state, and queues form and disappear quickly or sporadically. As more flights begin to depart (or arrive), there is a steady flow of vehicles in and out of the terminal area, and the condition during those times is more analogous to a steady-state condition. Another restriction for queuing models is that the utilization factor ρ must be less than 1. The utilization factor is defined as $\rho = \lambda/(s\mu)$, where λ is the mean arrival rate of customers, s is the number of servers in the service facility, and μ is the mean service rate of the overall system. For any model, if the mean arrival rate, λ, is greater than the mean service rate, μ, the queue would grow indefinitely.

It is well documented in operations research literature and commonly assumed in transportation applications that customers randomly arriving at a service facility have an arrival pattern that is assumed to be a Poisson distribution process. The assumption of a Poisson arrival process results in an exponential inter-arrival time distribution. Assuming an exponential service rate for an airport terminal curbside is also realistic because drop-off and pickup times vary around a mean value due to variations in how long it takes to complete the loading/unloading process (number of passengers, amount of luggage, etc.).

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Based on the above, the curbside can be modeled by a multiserver queuing system with a Poisson arrival process, a mean service rate that follows an exponential probability distribution, and an infinite calling population (an M/M/s model in queuing theory terminology). If the queue discipline is first-in-first-out, the analytical results for the basic outputs are given in Equation C-1 through Equation C-9 (Hillier and Lieberman 2015):

$$\lambda = \sum_i v_i \tag{C-1}$$

$$\mu = \frac{\sum_i \left(D_i * v_i\right)}{\sum_i v_i} \tag{C-2}$$

$$\rho = \frac{\lambda}{\mu s} \tag{C-3}$$

$$L_q = \frac{P_0 \left(\lambda/\mu\right)^s \rho}{s!\left(1-\rho\right)^2} \tag{C-4}$$

$$L = L_q + \left(\lambda/\mu\right) \tag{C-5}$$

$$W_q = L_q/\lambda \tag{C-6}$$

$$W = W_q + \left(1/\mu\right) \tag{C-7}$$

$$P_n = \begin{cases} \dfrac{\left(\lambda/\mu\right)^n}{n!} P_0 & \text{if } 0 \le n \le s \\ \dfrac{\left(\lambda/\mu\right)^n}{s!s^{(n-s)}} P_0 & \text{if } n \ge s \end{cases} \tag{C-8}$$

$$P_0 = 1 \Bigg/ \left[1 + \sum_{n=1}^{s-1} \frac{\left(\lambda/\mu\right)^n}{n!} + \frac{\left(\lambda/\mu\right)^s}{s!} \sum_{n=s}^{\infty} \left(\frac{\lambda}{s\mu}\right)^{n-s} \right] \tag{C-9}$$

where

s = number of servers in the service facility
λ = mean arrival rate of customers
v_i = flow rate of vehicle type i
μ = mean service rate of the overall system
D_i = dwell time of vehicle type i
ρ = utilization factor
L_q = number of customers in the queue
L = number of customers in the queuing system, equal to the customers in the queue plus the customers being served
W_q = expected waiting time in the queue
W = expected waiting time in the queuing system
P_n = probability that exactly n customers are in the queuing system

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Inputs and Intermediate Calculations

1. Curbside Configuration

The curbside may be specified using one of the following configurations:

- **Active:** In this configuration, the curbside is used for loading and unloading along its entire length, with drivers assuming any available space. QATAR models these operations using curbside capacity and through-lane capacity models.
- **Crosswalk:** Crosswalks may be uncontrolled, controlled by active warning devices, or controlled by a traffic control signal. Flagging operations can be approximated using a traffic control signal. QATAR models these operations using crosswalk capacity models.
- **Source/Sink:** These are curbside segments of zero length that reflect any entrance or exit volumes between segments.
- **Taxi/Transportation Network Company (TNC) queue:** These are curbside segments used by taxi queues or by TNC vehicles operating with a PIN system. QATAR does not model operations of these types of queuing systems, as they are typically actively managed and fed from an offsite hold lot.
- **Other:** These curbside segments are used by vehicles providing service to disabled passengers, service vehicles, emergency vehicles, or other types. QATAR does not model operations of these types of segments.

2. Hourly Vehicle Arrival Rate

The number of vehicles arriving at the zone per hour is the sum of the volume of all vehicle types approaching zone z, as given in Equation C-10.

$$WtdVehicleArrivalRate_z = \sum_i v_{z,i} \tag{C-10}$$

where

$WtdVehicleArrivalRate_z$ = weighted vehicle arrival rate for zone z [veh/h]
$v_{z,i}$ = flow rate of vehicle type i in zone z [veh/h]

3. Number of Equivalent Servers per Zone

The available curbside length divided by the average vehicle stall length provides an estimate of the number of equivalent servers in each lane within a zone. The number of equivalent servers multiplied by the number of available lanes provides an estimate of the total number of equivalent servers per zone. Because drivers may choose to park in lanes not intended for loading or unloading (e.g., by double-parking or triple-parking), the calculation of equivalent servers uses the total number of lanes on the roadway. These are given in Equation C-11 through Equation C-13.

$$WtdVehicleLength_z = \frac{\sum_i ParkingLength_i * v_{z,i}}{WtdVehicleArrivalRate_z} \tag{C-11}$$

$$CurblaneCapacity_z = \frac{CurbFrontage_z}{WtdVehicleLength_z} \tag{C-12}$$

$$DerivedNoOfServers_z = NumberOfLanes_z * CurblaneCapacity_z \tag{C-13}$$

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where

$WtdVehicleLength_z$ = weighted vehicle length for zone z [ft/veh]
$ParkingLength_i$ = parking length of vehicle type i [ft]
$v_{z,i}$ = flow rate of vehicle type i in zone z
$CurblaneCapacity_z$ = curb lane capacity for zone z [veh]
$CurbFrontage_z$ = length of curb frontage for zone z [ft]
$DerivedNoOfServers_z$ = equivalent parking spaces (servers) for zone z
$NumberOfLanes_z$ = total number of lanes (through and parking) for zone z

4. Service Rate

The service rate is computed as the weighted dwell time, as shown in Equation C-14 and Equation C-15. QATAR allows the modeling of individual vehicle types. In such a situation, the service rate is equal to the dwell time assumed for the specific vehicle type.

$$WtdDwellTime_z = \frac{\sum_i \left(D_i * v_{z,i}\right)}{WtdVehicleArrivalRate_z} \quad \text{(C-14)}$$

$$EstimatedServiceRate_z = \frac{60}{WtdDwellTime_z} \quad \text{(C-15)}$$

where

$EstimatedServiceRate_z$ = vehicle service rate for zone z [veh/h]
$WtdDwellTime_z$ = weighted dwell time zone z [min/veh]
D_i = dwell time of vehicle type i [min/veh]
$v_{z,i}$ = flow rate of vehicle type i in zone z [veh/h]
$WtdVehicleArrivalRate_z$ = weighted vehicle arrival rate for zone z [veh/h]

5. System Utilization Factor

The system utilization factor is determined by Equation C-16. This factor is used to determine the overall ability of the system to serve demand and is a core assumption of the queuing calculation method. A system utilization factor greater than 1 will result in an error message in QATAR. In such a situation, the number of vehicles attempting to load and/or unload exceeds the number of servers in a zone (i.e., every lane is fully occupied by vehicles attempting to load and/or unload).

$$SystemUtilFactor_z = \frac{WtdVehicleArrivalRate_z}{EstimatedServiceRate_z * DerivedNoOfServers_z} \quad \text{(C-16)}$$

where

$SystemUtilFactor_z$ = system utilization factor for zone z
$WtdVehicleArrivalRate_z$ = weighted vehicle arrival rate for zone z
$EstimatedServiceRate_z$ = estimated service rate for zone z
$DerivedNoOfServers_z$ = equivalent parking spaces (servers) for zone z

6. Number of Vehicles in System Within a Zone

Using the results from the previously presented equations and the assumptions inherent to the queuing model, it is possible to estimate the probability of having n vehicles in the system.

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In this model, probabilities are computed for values of n from 0 vehicles to 170 vehicles using Equation C-17 and Equation C-18 (Stewart 2009). The mathematical structure here is slightly different from that presented earlier but can be demonstrated to be mathematically equivalent.

$$P_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!} P_0 & \text{if } 0 \le n \le s \\ \dfrac{(\lambda/\mu)^n}{s! s^{(n-s)}} P_0 & \text{if } n \ge s \end{cases} \tag{C-17}$$

$$P_0 = \left[\sum_{s=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \frac{1}{1 - \lambda/(\mu s)} \right]^{-1} \tag{C-18}$$

where

s = number of servers in the service facility
λ = mean arrival rate of customers
μ = mean service rate of the overall system
P_n = probability that exactly n customers are in the queuing system

QATAR then incrementally calculates a cumulative density function using the sum of the probabilities from zero to n. The 95th percentile from the cumulative density function, $Pcntile95CarsInSys_z$, is an estimate of the maximum number of vehicles in the system 95 percent of the time; this value is used to determine the performance of the curbside for zone z.

7. Curbside Utilization Ratio

Parking activity in each lane is estimated based on the curbside utilization ratio. The curbside utilization ratio is calculated by comparing the total length of vehicles assumed to be parked within a zone simultaneously (based on the 95th percentile from the cumulative distribution function for that zone) with the curbside length available for parking in that zone, as shown in Equation C-19. This ratio is a measure of the use of a given parking space within a curb segment and typically ranges from zero to 3, with the latter suggesting the use of double-parking and triple-parking at the same curb segment. This equation is a practical estimate that compares a practical, design-level demand with the capacity of the curb lane; it differs from the system utilization factor described previously that determines the overall applicability of the macroscopic queuing model.

$$CurbUtilRatio_z = \frac{Pcntile95CarsInSys_z}{CurblaneCapacity_z} \tag{C-19}$$

where

$CurbUtilRatio_z$ = curbside utilization ratio for zone z
$Pcntile95CarsInSys_z$ = 95th-percentile number of vehicles in system for zone z
$CurblaneCapacity_z$ = curb lane capacity for zone z [veh]

8. Percent Occupancy in Lanes 1, 2, and 3

The model uses two inputs to determine the tendency for drivers to double-park and triple-park. If the zone only has three total lanes, the model assumes that parking only occurs in the first and second lanes.

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The model includes two parameters: T_{Lane2}, the proportion of the first lane that is filled before drivers start to park in the second lane, and T_{Lane3}, the proportion of the second lane that is filled before drivers start to park in the third lane. These parameters have a built-in default value for both inputs of 80%, which means that drivers park in the first lane until 80% of the first lane is occupied and that drivers will park in the third lane once 80% of the second lane is occupied. These values are approximations based on observations at multiple airports with multiple attraction points (e.g., doors, skycap positions) within one curbside zone. Due to the approximations used for estimating throughput capacity (discussed in the following), further refinement of these default values is unlikely to improve overall planning estimates. If more precision is needed, simulation is recommended.

For curbside zones with four or five total lanes, the percent occupancy of lanes 1, 2, and 3 used for parking ($P_{1,z}$, $P_{2,z}$, and $P_{3,z}$) is calculated for each zone z as shown in Table C-1.

For curbside zones with three total lanes and double-parking allowed, $P_{1,z}$ and $P_{2,z}$ are calculated for each zone z as shown in Table C-2.

For curbside zones with three total lanes, one parking lane, and double-parking prohibited, $P_{1,z}$ and $P_{2,z}$ are calculated for each zone z as shown in Table C-3.

Table C-1. Percent occupancy by lane for four or five total lanes, double-parking allowed.

Condition	Equations
$\boldsymbol{CurbUtilRatio_z \le T_{Lane2}}$	$\boldsymbol{P_{1,z} = CurbUtilRatio_z}$
$\boldsymbol{T_{Lane2} < CurbUtilRatio_z \le (1 + T_{Lane3})}$; $\boldsymbol{P_{1,z} \le 1.0}$	$\boldsymbol{P_{1,z} = T_{Lane2} + (CurbUtilRatio_z - T_{Lane2})/2}$ $\boldsymbol{P_{2,z} = (CurbUtilRatio_z - T_{Lane2})/2}$
$\boldsymbol{T_{Lane2} < CurbUtilRatio_z \le (1 + T_{Lane3})}$; $\boldsymbol{P_{1,z} > 1.0}$	$\boldsymbol{P_{1,z} = 1}$ $\boldsymbol{P_{2,z} = CurbUtilRatio_z - 1}$
$\boldsymbol{CurbUtilRatio_z > (1 + T_{Lane3})}$; $\boldsymbol{P_{2,z} \le 1.0}$	$\boldsymbol{P_{1,z} = 1}$ $\boldsymbol{P_{2,z} = (CurbUtilRatio_z - 1) + (CurbUtilRatio_z - (1 + T_{Lane3}))/2}$ $\boldsymbol{P_{3,z} = (CurbUtilRatio_z - (1 + T_{Lane3}))/2}$
$\boldsymbol{CurbUtilRatio_z > (1 + T_{Lane3})}$; $\boldsymbol{P_{2,z} > 1.0}$	$\boldsymbol{P_1 = 1}$ $\boldsymbol{P_2 = 1}$ $\boldsymbol{P_3 = CurbUtilRatio_z - 2}$

Table C-2. Percent occupancy by lane for three total lanes, double-parking allowed.

Condition	Equations
$\boldsymbol{CurbUtilRatio_z \le T_{Lane2}}$	$\boldsymbol{P_{1,z} = CurbUtilRatio_z}$
$\boldsymbol{CurbUtilRatio_z > T_{Lane2}; P_{1,z} \le 1.0}$	$\boldsymbol{P_{1,z} = T_{Lane2} + (CurbUtilRatio_z - T_{Lane2})/2}$ $\boldsymbol{P_{2,z} = (CurbUtilRatio_z - T_{Lane2})/2}$
$\boldsymbol{CurbUtilRatio_z > T_{Lane2}; P_{1,z} > 1.0}$	$\boldsymbol{P_{1,z} = 1}$ $\boldsymbol{P_{2,z} = CurbUtilRatio_z - 1}$

Table C-3. Percent occupancy by lane for three total lanes, one parking lane, double-parking prohibited.

Condition	Equations
$\boldsymbol{CurbUtilRatio_z \le 1}$	$\boldsymbol{P_{1,z} = CurbUtilRatio_z}$
$\boldsymbol{CurbUtilRatio_z > 1}$	$\boldsymbol{P_{1,z} = 1}$ $\boldsymbol{P_{2,z} = CurbUtilRatio_z - 1}$

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The total number of vehicles in each lane is calculated using Equation C-20 through Equation C-22.

$$NoOfVehiclesInCurbsideLane_z = P_{1,z} * CurblaneCapacity_z \quad \text{(C-20)}$$

$$NoOfVehiclesDoubleParked_z = P_{2,z} * CurblaneCapacity_z \quad \text{(C-21)}$$

$$NoOfVehiclesTripleParked_z = P_{3,z} * CurblaneCapacity_z \quad \text{(C-22)}$$

9. Curbside Sufficiency

The curbside sufficiency is based on comparing the curbside utilization ratio for each zone *z* to the thresholds given in Table C-4. These thresholds vary based on whether double-parking is allowed or prohibited.

10. Roadway Capacity

The roadway capacity, or through-lane capacity, of an airport curbside roadway is affected by curbside activity (i.e., vehicles stopping to load or unload passengers). As the curbside lanes become more congested, double-parking and potentially triple-parking will begin to block the roadway through lanes. Roadway capacity decreases continuously as curbside utilization increases. The decrease is more prominent when curbside activity reaches a level at which additional lanes are blocked, producing a non-linear slope to the curves. Capacity data are difficult to obtain due to the need for representative capacity conditions, which includes as a prerequisite a persistent queue upstream of the curbside section. In such cases, the roadway capacity is assumed to be the throughput on the roadway, measurable by standard industry traffic counting techniques.

Due to the significant downturn in air travel during the COVID-19 pandemic during the research for this project, no additional field data were collected. The research team used a limited amount of field data collected previously from the original research for *ACRP Report 40*, including data from Washington Dulles International Airport (IAD), San Francisco International Airport (SFO), and Oakland International Airport (OAK). To supplement the sparse field data available, the research team developed a VISSIM model to determine the curbside volume stopping in the parking lanes, which corresponds to various curbside utilization levels. By holding that level of curbside activity constant and increasing through traffic volumes during multiple simulation tests, the research team observed a progressive collapse of traffic flow, with a queue forming upstream of the curbside section. When the queue became persistent or continuously increasing, the roadway section was said to have reached capacity. The resulting throughput downstream of the modeled segment represents the capacity under those conditions. In addition, the research

Table C-4. Curbside sufficiency definitions.

Curbside Utilization Ratio for Zone z, $CurbUtilRatio_z$	Curbside Sufficiency
Double-Parking Allowed	
<= 1.30	Under Capacity
> 1.30 to <= 1.70	Near Capacity
> 1.70 to <= 2.00	At Capacity
> 2.00	Over Capacity
Double-Parking Prohibited	
<= 1.00	Under Capacity
> 1.00 to <=1.20	Near Capacity
> 1.20 to <=1.35	At Capacity
> 1.35	Over Capacity

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team applied considerable judgment in fitting the curves, based on experiences at airports throughout the United States and in other countries, including knowledge of the typical parking patterns around key features such as door locations.

The resulting complex, non-linear interactions can be described by the curves shown in Figure C-1. As noted previously, these curves were developed using a combination of the micro-simulation testing of hypothetical airport curbsides, actual but limited field observations, and research team judgment.

The curves in Figure C-1 are mathematically represented as follows in Equation C-23, with constants for various configurations given in Table C-5:

$$RdwyThruCapacity_z = \frac{ConstA * ConstB + ConstC * \left(CurbUtilRatio_z\right)^{ConstD}}{ConstB + \left(CurbUtilRatio_z\right)^{ConstD}} \quad \text{(C-23)}$$

where

$RdwyThruCapacity_z$ = roadway through capacity for zone z
$CurbUtilRatio_z$ = curbside utilization ratio for zone z
ConstA, ConstB, ConstC, ConstD = constants as given in Table C-5

Because model inputs are given in terms of driver's side parking, through lanes, and passenger's side parking, QATAR maps these configurations into one of the modeled configurations specified by total and approach lanes, as shown in Table C-6. Any configurations not included in Table C-6 are flagged to the user as not being supported by QATAR.

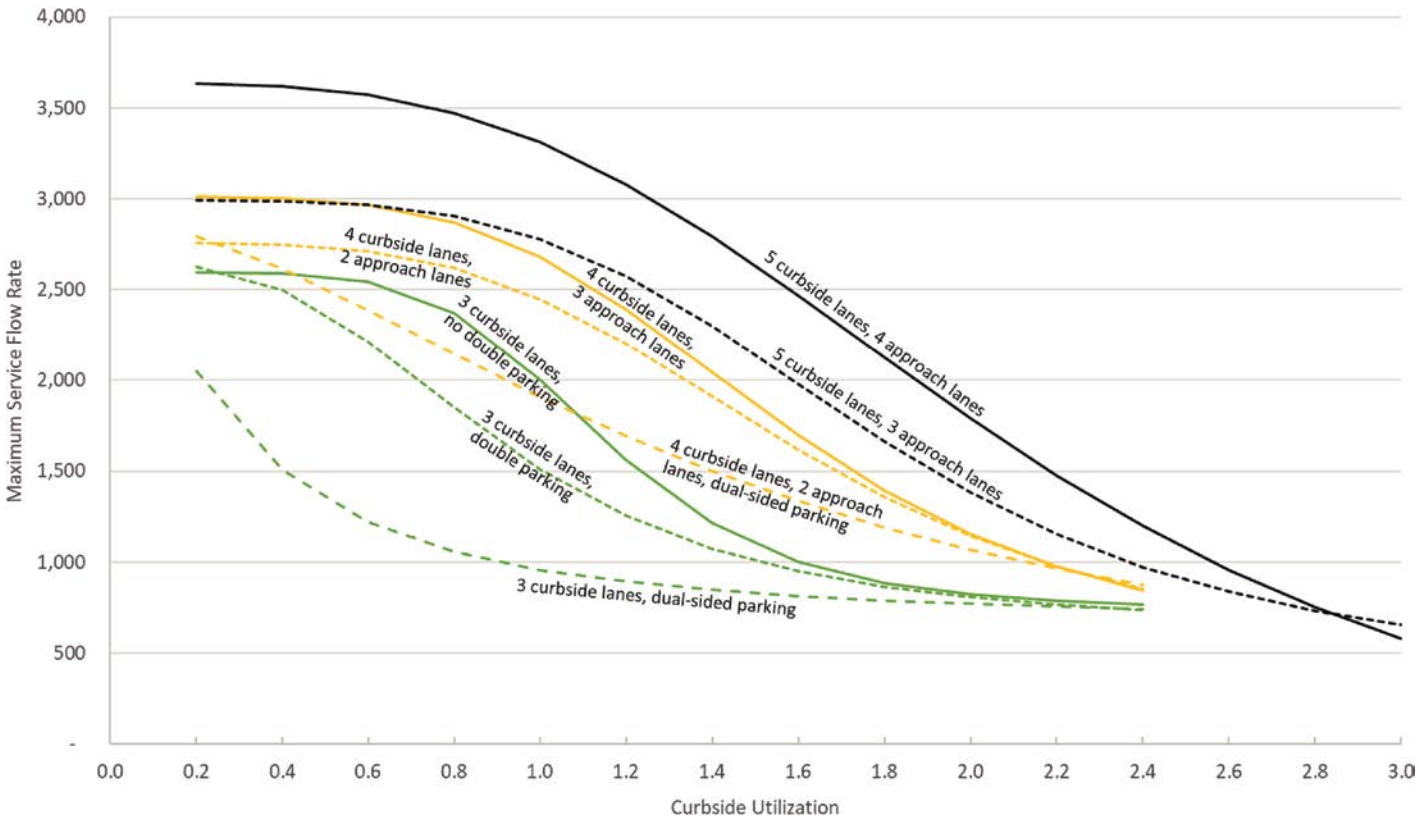


Figure C-1. Maximum through-lane service flow rate as function of curbside utilization ratio.

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Table C-5. Roadway through capacity model definitions.

Model	Total curbside lanes	Approach (through) lanes	Parking allowed on both sides?	Double-parking prohibited?	*ConstA*	*ConstB*	*ConstC*	*ConstD*
3,2 no dub	3	2	No	Yes	736.85	0.4672	2597.5	–5.4292
3,2 dub	3	2	No	No	638.72	1.291	2642.8	–3.0582
3,2 dual	3	2	Yes	No	650.81	5.9473	2791.65	–1.5025
4,2	4	2	No	No	454.79	0.15502	2759	–3.9169
4,2 dual	4	2	Yes	No	117.92	0.53449	2868	–1.8199
4,3	4	3	No	No	472.73	0.1506	3012.4	–4.1772
5,3	5	3	No	No	392.95	0.089126	2993.2	–4.184
5,4	5	4	No	No	–373.44	0.088833	3638	–3.2659

11. Adjusted Through-Lane Roadway Capacity

The lane capacity is multiplied by the Regional Adjustment Factor to obtain the adjusted through-lane capacity, as shown in Equation C-24.

$$RdwyThruCapacityAdjusted_z = RdwyThruCapacity_z * gRegionalAdjFactor \quad \text{(C-24)}$$

where

$RdwyThruCapacityAdjusted_z$ = adjusted capacity for zone z
$RdwyThruCapacity_z$ = estimated capacity for zone z
$gRegionalAdjFactor$ = regional adjustment factor

12. Roadway Volume-to-Capacity Ratio

The roadway volume-to-capacity (v/c) ratio is calculated by dividing the total volume of the study zone by its adjusted through-lane roadway capacity, as shown in Equation C-25.

$$RdwyVCRatio_z = \frac{TotRdwyVol_z}{RdwyThruCapacityAdjusted_z} \quad \text{(C-25)}$$

where

$RdwyVCRatio_z$ = roadway volume-to-capacity ratio for zone z
$TotRdwyVol_z$ = flow rate for zone z
$RdwyThruCapacityAdjusted_z$ = adjusted capacity for zone z

Table C-6. Mapping of inputs to through-lane models.

Driver-side parking	Through lanes	Passenger-side parking	Double-parking prohibited?	Through-lane model	Notes
0	2	1	Yes	3,2 no dub	NA
0	2	1	No	3,2 dub	NA
1	1	1	No	3,2 dual	Assumes 2 lane approach. Left approach lane becomes driver's side parking lane.
0	2	2	No	4,2	NA
1	2	1	No	4,2 dual	NA
0	3	1	No	4,3	NA
0	3	2	No	5,3	NA
0	4	1	No	5,4	NA

Note: NA = Not Applicable

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13. Roadway Sufficiency

The roadway sufficiency is based on comparing the roadway v/c ratio for each segment to the thresholds given in Table C-7.

14. Crosswalk Capacity with Traffic Signal Control or Traffic Control Officer Control

The methodologies presented in this section are derived from methods in the *Highway Capacity Manual* (Transportation Research Board 2022). Capacity under traffic signal control is governed by the saturation flow rate for the given lane model and the ratio of effective vehicular green time to the cycle length, using Equation C-26.

$$CCAF = \frac{c}{s} = \frac{g}{C} \tag{C-26}$$

where

$CCAF$ = crosswalk capacity adjustment factor
c = capacity [veh/hr/ln]
s = saturation flow rate [veh/hr/ln]
g = effective vehicular green time [s]
C = cycle length [s]

In the absence of known signal timing, a default value of $CCAF = 0.65$ may be assumed.

The effective g/C ratio, or the ratio of effective green time to the cycle length, can be calculated directly if signal timing for the pedestrian crosswalk is known. When traffic control officer control is used, the flagger behavior should be approximated as if it were controlled by a traffic control signal using default values unless field operations are known.

If pedestrian signal timing is known, the effective g/C ratio can be calculated using Equation C-27.

$$\frac{g}{C} = 1 - \frac{W + \frac{L}{S_p}}{C} \tag{C-27}$$

where

W = walk indication time [s, default = 10 s]
L = length of crosswalk [ft, default = 12 ft/ln]
S_p = pedestrian walking speed [ft/s, default = 3.5 ft/s]
C = cycle length [s, default = 60 s]

15. Crosswalk Capacity with No Control or Active Warning Devices

It is possible using HCM methods (Transportation Research Board 2022) to estimate the vehicular capacity across a crosswalk with no control or with only active warning devices, such as rectangular rapid flashing beacons. However, the analytical methods to do this require estimates

Table C-7. Roadway sufficiency thresholds.

Maximum Roadway v/c Ratio, $RdwyVCRatio_z$	Roadway Sufficiency
<= 0.60	Under Capacity
> 0.60 to 0.80	Near Capacity
> 0.80 to 1.00	At Capacity
> 1.0	Over Capacity

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of pedestrian volumes and their likelihood to cross in groups, as well as estimates of driver yielding behavior. Pedestrian volumes and their likelihood to cross in groups are sensitive to activity patterns at the airport and to the specific layout of the airport terminal area that draws pedestrians to cross at a particular location for a specific purpose (e.g., access to parking or courtesy shuttle buses). While estimates of yielding behavior are available for midblock crossings outside of airport environments, little research has been done on the yielding behavior observed within the airport terminal environment. As such, for planning purposes, a default overall crosswalk capacity adjustment factor of $CCAF = 0.65$ is suggested. If further detail on the pedestrian crossing patterns and driver yielding behavior is known, microsimulation can be used to estimate crosswalk capacity more accurately, as well as determine the effect of the crosswalk on upstream and downstream zones.

References

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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